



Catalog of Emission Lines in Astrophysical Objects

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Authors	Meinel, Aden B.; Aveni, Anthony F.; Stockton, Martha W.
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ASTROPHYSICAL
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CATALOG OF EMISSION LINES IN ASTROPHYSICAL OBJECTS

Aden B. Meinel
Anthony F. Aveni
Martha W. Stockton

SECOND EDITION

Optical Sciences Center and Steward Observatory
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PREFACE TO THE SECOND EDITION

This edition has been prepared in order to include more recent information than was available at the time of the first printing. New material is not incorporated into the text directly but is placed at the end of each table. It will be found on pages B-133, C-12, D-12, and E-8.

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INTRODUCTION

One of the great handicaps to the study and analysis of astrophysical emission objects is that the literature concerning the observation and identification of emission lines in astrophysical sources is so widely scattered. This volume represents our attempt to gather together the various publications, dating from 1930, that relate to these objects. The survey is complete to mid-1966, when compilation ceased. We publish the results in this preliminary form to make the entire subject available for critical analysis.

Included with our collected data on stellar spectra are data on contamination sources: air glow, city lights, aurora, and lightning. Not included are observed emission lines in the chromosphere and corona of the sun, for these are extensive enough to warrant a separate volume.

This publication comprises two principal sets of tables:

Tables A-1 to A-17, Observed Wavelengths, present the principal emission lines actually observed in individual classes of emission objects. Each entry represents a line reported by several independent workers, and the wavelength is the mean of all observations reported for that feature. Wavelengths were not included in this mean when it was not clear that the observer had published "observed wavelength" rather than "identification wavelength." The tables are of particular value in that regions that have been poorly observed are readily obvious.

Table B, Identifications and Intensities, gives the identifications that the various observers have proposed for the emissions listed in Tables A. It is obvious that different observers attribute the same feature on their spectrograms to different identifications. Thus, multiple identifications given for a single line may represent actual contributors to a wide line, or incorrect identities. We have made no attempt to arbitrate between "erroneous" and "probably correct" identifications. This table is arranged by wavelength so that the identifications of nearby wavelengths are readily seen. To facilitate this tabulation--and possible future revision--the data for this table were punched onto computer cards. They are thus available for further classification if desired.

Three supplementary tables follow those just described: *Table C* gives object identifications represented in each class; *Table D* gives wavelength range and instrumentation for each of the references cited; and *Table E* is a complete bibliography.

All the data in this volume are based on the most recent available studies of the particular objects. Because older studies were limited by low dispersion, they probably contain many incorrect identifications. Commonly, different authors disagree over the identification of the same apparent feature. We have listed these divergent identifications without interposing our own analysis in order to preserve the research usefulness of this compilation.

A frequent deficiency in published articles should be noted: It is often difficult to ascertain whether the researcher is giving the measured wavelength or the wavelength of the identification.

It has been our experience that the cited literature in the case of the airglow, aurora, and lightning contains many improbable identifications, no longer held by researchers. It is hoped that this compilation of astrophysical source wavelengths and identifications is not so faulty; however, even casual inspection from a perspective of astrophysical reasonability would indicate that, on close examination, an appreciable number of the identifications would be deleted. In order to avoid our personal bias on this matter we have not edited the literature, since one of the purposes of this compilation is to focus much needed attention on the subject areas.

One by-product of this compendium is the recognition of the need for observations of certain objects in both the ultraviolet and the infrared, especially with instruments of moderately high dispersion. In a very real sense we hope that this catalog will be rendered obsolete by intensive work on emission-line objects.

If this table has any general usefulness to the astronomical community, the authors would appreciate comments.

OBJECT CLASSES

For purposes of this catalog, the wide range of astrophysical emission objects has been grouped into a relatively few general classes. Some of these classes consist of several recognized generic subclasses, for within the literature there is frequent disagreement concerning the generic classification of objects. It seemed best to skirt these controversies by combining subclasses.

The resultant classifications are shown below. The categories used in the two principal tables differ slightly, for some subclasses are listed in Table A. In general, the classes are arranged in order of decreasing energy expenditure, roughly in order of excitation temperature. (The symbiotic stars are classed as objects of high excitation temperature, neglecting the fact that their spectra show molecular bands in absorption.)

Table A. Observed Wavelengths

1	Quasars
2	Nuclear emission (Seyfert) galaxies
3a	Supernovae (Type II)
b	Novae
c	Old novae
4	Nova-like stars (η Carinae class)
5	Wolf-Rayet N stars
6	Wolf-Rayet C stars
7	Planetary nebulae
8a	Gaseous nebulae (Orion nebula)
b	Supernova remnants (Crab nebula)
9	Symbiotic stars
10a	Oe-class stars
b	Be-class stars (P-Cygni class)
11	Ae-Fe-Ge stars
12a	T-Tauri stars
b	Nebulous late stars (Herbig-Haro objects)
13	Ke-Me-N stars
14	Airglow
15	City lights
16	Aurora
17	Lightning

Table B. Identifications and Intensities

1	Quasars
2	Nuclear emission galaxies
3	Novae
4	Nova-like stars
5	Wolf-Rayet N stars
6	Wolf-Rayet C stars
7	Planetary nebulae
8	Gaseous nebulae
9	Symbiotic stars
10	Oe-Be-P Cygni stars
11	Ae-Fe-Ge stars
12	T-Tauri stars
13	Ke-Me-N stars
14	Airglow
15	City lights
16	Aurora
17	Lightning

The general classes are discussed in the following pages.

Quasars.--The quasars are the quasi-stellar objects that show emission spectra. Because of the extreme red shift of their spectra, the wavelengths listed are those of the identifications. To accommodate these identifications, the tables of wavelength have been extended from 3000 Å, the atmospheric penetration limit, down to 1000 Å. None of the other objects listed in the tables can be observed at these shorter wavelengths from terrestrial observatories.

Nuclear emission galaxies.--The peculiar galaxies of the Seyfert type that show strong nuclear emission are included with nuclear emission galaxies because of their possible relationship to the quasi-stellar galaxies in which the nuclear emission dominates the spectrum.

Novae.--The wide range of nova spectra includes those of supernovae (Type II), novae, and old novae. Since the spectrum of a single nova may vary drastically during the course of the outburst, the listed intensities cannot be interpreted literally. Moreover, some novae have very broad emission lines, and many multiplets could be invoked in their identification. In the table of identifications and intensities (Table B) we have tried to indicate the several identifications that are ascribed to a single emission band by a B. Although all in the group could be contributing to the band, we feel that the proper interpretation is that only one or a few are actually responsible for the emission.

Nova-like stars.--Nova-like stars show variations of light on a recurring or nonrecurring basis. It is extremely difficult to differentiate between these objects and true novae. Since the lines in the nova-like spectra usually do not have the extreme widths as those of novae, their identifications are not quite so ambiguous. On the other hand, nova-like stars are relatively unobserved with the more modern spectrographic equipment, and the entire list of data may be superseded. McLaughlin's article* has been a most helpful guide in separating members of the nova and nova-like classes.

Wolf-Rayet stars.--Identifications of the lines in the Wolf-Rayet emission spectra are complicated, as for novae, by the extreme broadness of the emission features. We recognize the traditional subdivision into classes N (nitrogen sequence) and C (carbon sequence) although some objects show features common to both classes. It should be recognized that multiple identifications in the literature given for a single band represent possible contributors and that the

*McLaughlin, D. B., 1960, "The spectra of novae," Chap. 17 (pp. 585-652) in J. L. Greenstein, ed., *Stellar Atmospheres* (vol. 6 of *Stars and Stellar Systems*, ed. by G. P. Kuiper and B. M. Middlehurst), Chicago, U. of Chicago Press.

emission may in all probability be due to only one or two of those cited. It should also be noted that it is practically impossible to assign intensities to the observed lines, owing to the broadness and enhancement of features in Wolf-Rayet spectra.

Planetary nebulae.--The data for the planetary nebulae include all observations of planetaries, making no distinction between objects or zones in objects of higher or lower excitations.

Gaseous nebulae.--The Orion Nebula, which is bright enough to be studied with high dispersion, is taken as the prototype gaseous nebula. The principal sources of data are the recent publications by Bowen. A few older identifications found in the literature are included, but they are probably erroneous.

Symbiotic stars.--Symbiotic stars, which show simultaneously emission lines of high excitation and an absorption spectrum typical of a late spectral class, are given a separate class even though recent researches indicate that they are binary objects of extreme classes rather than a "single" star with an unusual atmosphere.

Oe-Be-P Cyg stars.--The fact that there is actually a smooth transition from nova-like star spectra through P Cygni through Be makes the Oe-Be-P Cygni subdivision quite arbitrary. All show to a degree the asymmetry of absorption edge on the emission profile. Moreover, at times Oe spectra can take on Wolf-Rayet characteristics; this behavior, however, may be due to the multiplicity of the O star(s) represented by the observable "star."

Ae-Fe-Ge stars.--The Ae-Fe-Ge stars are rare and represent a transitional group from stars showing high excitation lines to the T-Tauri objects, showing emission of metals in the first and second ionization stages. It is debatable whether these objects deserve a separate classification from the T-Tauri stars since the latter are variable in spectral class of the continuum over a wide range.

The Cepheid class objects, which show emission H lines, are not listed in our tables.

T-Tauri stars.--The T-Tauri class, defined by Joy* and Herbig,** consists of late type dwarfs showing fluorescent FeI emissions, the [S II] lines, Balmer series, and CaII, H and K in emission.

*Joy, A. H., 1945, "T Tauri variable stars," *Astrophys. J.* 102(2):168-195.

**Herbig, G. H., 1958, "Stars of types F, G, and K with emission lines; introductory report," *Etoiles a Raies d'Emission, Mém. Soc. Roy. Sci., U. Liège*, pp. 251-270.

Ke-Me-N stars.--Ke-Me-N stars are those very late class variable stars that show principally chromospheric emission characteristics.

Contamination sources.--In listing wavelengths and identifications for the contamination sources, we have ignored many weak features contained in some of the referenced papers. In general, all of the wavelengths listed herein are represented by conspicuous features in the photographic spectra of the airglow and aurora illustrated in Chamberlain.*

Airglow.--The airglow, or night sky, spectrum is a common contaminant in astrophysical spectra, especially of faint sources; hence a tabulation of these wavelengths is important. The listed lines in the red for the strong OH rotation-vibration bands refer to the individual lines. In the blue and ultraviolet there are many weak O₂ bands, and only the maxima of intensity are listed, with the letter V (violet) or R (red) to indicate the direction to which the band is shaded.

City lights.--This catalog would be incomplete without a list of the lines present in the glow from nearby cities, for even at the most remote observatories the spectra of faint sources seem to show the effects of emissions due to civilization. The actual intensities depend on the length of the air path traversed in reaching the telescope. Therefore, the intensities may be weighted to either the red or the blue of those shown here, which are observed within the Tucson metropolitan area.

Aurora.--The auroral spectrum is included in its grosser aspects. The presence of many molecular bands means that, with sufficient spectrograph dispersion and exposure time, many individual lines would be observed. However, since such practice is seldom used in studies of stellar spectra, only the principal atomic emissions and band maxima need be listed here. The presence of auroral contamination of astrophysical spectra should be obvious to even a casual observer without recourse to these wavelengths.

Lightning.--The spectrum of lightning seldom bothers the observer, except from distant storms. As is the case with city lights, the intensity depends on the length of the air path traversed in reaching the telescope. Thus, the relative spectrum of lightning is affected by atmospheric absorption (reddening) and occasional fluorescence, as in the case of the $\lambda 3914 \text{ N}_2^+$ band. Since the tables list intensities for *nearby* lightning events, the usual contamination to astrophysical spectra would be heavily weighted to the red and infrared.

*Chamberlain, J. W., "The blue airglow spectrum," *Astrophys. J.* 128(3):713-717, 1958.

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In preparation of the final copy, Miss Raquel Hanzi wrote the computer program to tabulate the data of Table B, and Mrs. Judy Langen spent innumerable hours at the keypunch, entering the data, and long hours subsequently in proofreading and pasting up the sheets.

Table A. OBSERVED WAVELENGTHS

Listed in the subtables below are the wavelengths of emission lines observed in spectra from each of the object classes. Only the wavelengths most frequently observed are listed. To decide which ones should be entered, we tabulated all of the reported wavelengths, then selected the ones appearing in several references. The listed value is the mean of the several reported wavelengths. Wavelengths were included in this mean only when it was clear that the observer had published "observed wavelength" rather than "identification wavelength." In some cases it was necessary to take into account the dispersion of the spectrograph used by an observer in deciding whether the reported wavelength was a blend and was represented by separate emissions by other observers, and so tabulated in our groupings.

All but Table A-11 give observed wavelengths; Table A-11 gives identification wavelengths. The strongest features are underlined. Some objects show multiple components for the strongest emission lines, but these special objects are not included in these lists.

A-1. Quasars

Observed zero-velocity wavelength; low dispersion.

1488	1663	2796	3133	3889	4959
1508	1909	2804	3203	3968	5007
1548	2424	2854	3346	4102	
1551	2470	2869	3426	4340	
1564	2506	2928	3727	4363	
1640	2626	2974	3869	4861	

A-2. Nuclear emission (Seyfert) galaxies

Low dispersion.

3312	3968	4659	5158	6087	6717
3346	<u>3970</u>	4686	5199	<u>6300</u>	<u>6730</u>
3425	<u>4068</u>	<u>4711</u>	5276	<u>6364</u>	<u>7136</u>
<u>3444</u>	<u>4076</u>	4740	5670	<u>6374</u>	7320
3727	<u>4101</u>	<u>4861</u>	5721	<u>6548</u>	
<u>3868</u>	<u>4340</u>	<u>4959</u>	5755	<u>6563</u>	
<u>3889</u>	<u>4363</u>	<u>5007</u>	5876	<u>6583</u>	

A-3a. Supernovae (Type II)

Low dispersion. All emissions are very broad, with widths ranging from 100 Å to more than 500 Å.

4040	5350	6060
4320	5570	6400
4620	<u>5770</u>	<u>6530</u>
4920	5890	
5200	<u>6220</u>	

A-3b. Novae

Nova spectra vary in width of emission feature from object to object and with time after outburst in any one object. The table below does not distinguish between these variables, but only reports measured wavelengths observed in the objects listed in Table C.

	3783.-	4087.2	4303.7	4554.7	4944.9
	3791.3	4098.0	4318.9	4585.2	4950.2
	3798.7	<u>4101.4</u>	<u>4341.6</u>	4603.5	<u>4959.0</u>
	3835.0	<u>4103.9</u>	<u>4352.4</u>	4607.-	<u>4970.0</u>
	3839.1	4110.8	4358.6	4614.-	5000.0
3311.6	3846.2	4119.0	<u>4363.2</u>	4620.0	<u>5007.6</u>
3329.-	3868.7	4127.-	<u>4386.2</u>	4629.1	<u>5017.1</u>
3342.7	<u>3890.2</u>	4143.2	4412.8	4634.3	5042.5
3382.8	3911.4	4169.-	4416.4	4640.5	5047.6
<u>3426.2</u>	3923.5	4173.1	4431.4	4656.0	5055.9
3444.1	3933.-	4178.6	4440.3	4686.1	5116.2
3471.1	3968.2	4190.4	4453.6	4714.6	5131.4
3587.1	<u>3970.9</u>	4199.2	4471.3	4728.6	5147.0
3664.0	3986.8	4206.-	4490.0	4750.2	5160.1
3711.8	3996.5	4232.4	4502.0	4774.-	5169.7
3725.1	4008.7	4243.1	4509.6	4813.6	5176.8
3734.5	4026.0	4266.6	4516.2	4823.7	5186.5
3750.5	<u>4037.5</u>	4275.5	4520.5	<u>4862.6</u>	5197.2
3759.4	4069.9	4287.0	4541.5	<u>4889.-</u>	5234.3
<u>3769.0</u>	4077.4	4296.2	4550.4	4923.3	5271.7

Novae, cont.

5276.0	5759.8	6348.8	7067.0	7999.-
5289.5	5820.6	6362.9	7087.-	8217.6
5303.2	5877.0	6371.1	7117.0	8233.0
5308.4	5910.-	6374.6	7137.5	8447.1
5316.5	5932.2	6416.7	7238.0	8550.5
5335.0	5958.3	6434.4	7250.-	8625.-
5363.3	5987.6	6456.2	7310.-	8687.-
5411.3	5991.0	6515.8	7325.5	
5426.6	6000.9	6561.6	7440.-	
5490.2	6027.7	6678.2	7468.-	
5497.4	6052.3	6702.0	7477.0	
5534.5	6087.4	6717.0	7485.5	
5577.2	6103.3	6730.0	7592.5	
5587.2	6149.2	6744.6	7680.-	
5614.-	6187.7	6770.-	7718.5	
5666.2	6232.3	6827.6	7776.0	
5677.4	6241.4	6855.4	7877.-	
5722.0	6248.2	6893.-	7886.0	
5730.-	6301.2	6916.5	7896.-	
5753.6	6317.9	7006.-	7989.0	

A-3c. Old Novae

Low dispersion.

	4026	4417	4652
	4070	4433-44 +	4686
	4102	4471	4722
	(4233) *	4510-14 +	4861
	(4244)	4571	4959
3727	4268	(4584)	5007
<u>3837</u>	4340	4630	
<u>3869</u>	(4359)	4637	
3889	4363	4640	
<u>3970</u>	4381	4645	

*() are emissions reported in some objects; all other lines appear to occur in numerous objects.

†Blend

A-4. Nova-like stars (η Carinae class)

In η Carinae, strong emissions may have several maxima arising from one atomic transition; identifications must take account of this duplicity of lines.

	3854.2	4063.9	4305.9	4485.0	4724.3
	3855.7	4067.0	4308.0	4489.0	4728.0
	3865.6	<u>4068.4</u>	4312.9	<u>4491.6</u>	<u>4731.4</u>
	3868.5	4076.0	4314.5	<u>4501.1</u>	4734.2
	3872.8	4097.3	<u>4319.4</u>	<u>4508.3</u>	4746.8
3376.2	3878.1	4100.2	4326.1	4515.3	4754.8
3393.6	3885.9	<u>4111.0</u>	4338.5	<u>4520.2</u>	4764.7
3474.9	3887.8	4114.3	<u>4341.3</u>	<u>4522.6</u>	4772.1
3481.8	3889.6	4119.9	<u>4346.8</u>	<u>4528.4</u>	4774.7
3493.9	<u>3900.2</u>	<u>4122.4</u>	<u>4351.8</u>	4533.2	<u>4798.2</u>
3502.4	3903.0	4128.2	4355.3	4534.1	4810.8
3513.7	3905.8	<u>4143.6</u>	4358.7	4541.6	4814.4
3538.6	3914.0	4146.3	<u>4361.8</u>	4545.3	<u>4824.3</u>
3559.3	3930.1	4152.3	4364.8	4549.4	4829.2
3576.3	3931.9	4163.3	4369.3	<u>4555.8</u>	4833.2
3584.9	3934.5	4171.5	4372.4	4558.8	4836.5
3628.8	3935.7	4173.3	4374.9	4563.7	4848.4
3677.5	<u>3938.1</u>	<u>4176.9</u>	4382.7	4571.3	4852.6
3685.1	3945.1	4178.8	4385.2	4576.4	4859.1
3697.3	3952.3	<u>4200.9</u>	<u>4387.6</u>	<u>4580.1</u>	<u>4862.6</u>
3703.8	3954.2	4210.9	4394.1	4583.7	4874.6
3713.8	3965.0	4214.1	4395.2	<u>4588.9</u>	<u>4885.5</u>
3721.2	3966.8	4225.0	4399.6	4592.3	4889.6
3726.8	3969.4	4229.3	4409.9	4596.1	<u>4898.7</u>
3735.3	<u>3970.9</u>	<u>4233.0</u>	<u>4413.7</u>	4616.7	<u>4905.3</u>
3749.0	3973.1	4243.9	4416.3	4618.8	4919.7
3760.7	3974.4	<u>4246.8</u>	<u>4419.0</u>	4620.4	<u>4923.9</u>
3761.9	3979.5	<u>4248.9</u>	4422.1	4623.0	<u>4930.0</u>
3764.2	3992.9	<u>4252.9</u>	4432.3	4629.3	4947.4
3769.4	<u>4002.3</u>	4258.2	4443.9	<u>4634.2</u>	<u>4950.7</u>
3783.0	4005.7	4262.0	4447.9	4635.6	4973.4
3798.2	4009.2	<u>4273.2</u>	4452.0	4639.6	4993.6
3800.4	4011.8	<u>4276.6</u>	<u>4454.7</u>	<u>4653.4</u>	5001.6
3805.9	4015.4	<u>4283.4</u>	4457.8	4657.7	5006.2
3817.5	4024.6	<u>4287.2</u>	4461.4	<u>4664.2</u>	<u>5018.8</u>
3819.3	<u>4025.9</u>	4289.6	4464.1	4666.9	5043.3
3824.9	<u>4028.3</u>	4293.3	4467.8	4670.2	5048.4
3835.7	4032.9	<u>4296.5</u>	<u>4471.4</u>	4697.0	5056.3
<u>3838.4</u>	4045.0	<u>4299.4</u>	<u>4474.8</u>	4701.4	5060.3
3849.7	4053.6	<u>4302.9</u>	<u>4481.1</u>	4712.9	5072.6

Nova-like stars, cont.

<u>5107.8</u>	<u>5534.8</u>	<u>6305.-</u>	<u>7281.0</u>	<u>8438.0</u>	<u>8838.9</u>
<u>5111.5</u>	<u>5551.4</u>	<u>6312.5</u>	<u>7286.8</u>	<u>8446.4</u>	<u>8862.6</u>
<u>5132.8</u>	<u>5557.1</u>	<u>6318.1</u>	<u>7307.7</u>	<u>8450.9</u>	<u>8875.8</u>
<u>5145.8</u>	<u>5580.7</u>	<u>6347.7</u>	<u>7320.8</u>	<u>8468.2</u>	<u>8891.6</u>
<u>5154.6</u>	<u>5587.0</u>	<u>6352.-</u>	<u>7323.4</u>	<u>8489.9</u>	<u>8906.4</u>
<u>5158.6</u>	<u>5601.1</u>	<u>6364.0</u>	<u>7347.7</u>	<u>8499.4</u>	<u>8912.4</u>
<u>5164.0</u>	<u>5614.4</u>	<u>6368.9</u>	<u>7377.6</u>	<u>8502.4</u>	<u>8916.3</u>
<u>5181.9</u>	<u>5628.0</u>	<u>6383.7</u>	<u>7388.0</u>	<u>8505.8</u>	<u>8926.5</u>
<u>5184.7</u>	<u>5642.8</u>	<u>6416.8</u>	<u>7411.4</u>	<u>8508.6</u>	<u>9015.0</u>
<u>5197.8</u>	<u>5650.3</u>	<u>6432.6</u>	<u>7452.4</u>	<u>8523.4</u>	<u>9122.6</u>
<u>5220.1</u>	<u>5657.1</u>	<u>6441.5</u>	<u>7462.4</u>	<u>8544.1</u>	
<u>5234.6</u>	<u>5673.4</u>	<u>6456.3</u>	<u>7468.1</u>	<u>8567.9</u>	
<u>5255.6</u>	<u>5695.2</u>	<u>6484.5</u>	<u>7513.4</u>	<u>8594.2</u>	
<u>5261.4</u>	<u>5719.5</u>	<u>6491.4</u>	<u>7686.3</u>	<u>8598.4</u>	
<u>5268.7</u>	<u>5726.4</u>	<u>6515.8</u>	<u>7711.6</u>	<u>8599.4</u>	
<u>5274.3</u>	<u>5748.2</u>	<u>6527.2</u>	<u>7764.6</u>	<u>8616.6</u>	
<u>5284.1</u>	<u>5754.6</u>	<u>6562.9</u>	<u>7866.5</u>	<u>8629.3</u>	
<u>5296.7</u>	<u>5835.6</u>	<u>6587.-</u>	<u>7917.7</u>	<u>8648.7</u>	
<u>5316.2</u>	<u>5875.4</u>	<u>6666.7</u>	<u>7970.1</u>	<u>8655.6</u>	
<u>5327.1</u>	<u>5886.6</u>	<u>6678.0</u>	<u>7975.6</u>	<u>8662.0</u>	
<u>5333.8</u>	<u>5894.1</u>	<u>6729.0</u>	<u>7999.8</u>	<u>8665.0</u>	
<u>5347.7</u>	<u>5899.8</u>	<u>6746.6</u>	<u>8110.3</u>	<u>8672.2</u>	
<u>5363.3</u>	<u>5914.2</u>	<u>6809.4</u>	<u>8125.3</u>	<u>8680.1</u>	
<u>5376.2</u>	<u>5957.1</u>	<u>6813.4</u>	<u>8157.3</u>	<u>8683.4</u>	
<u>5408.5</u>	<u>5979.4</u>	<u>6835.-</u>	<u>8216.3</u>	<u>8686.3</u>	
<u>5412.9</u>	<u>5991.6</u>	<u>6857.-</u>	<u>8281.1</u>	<u>8703.3</u>	
<u>5425.6</u>	<u>6044.2</u>	<u>6874.-</u>	<u>8287.3</u>	<u>8706.3</u>	
<u>5433.0</u>	<u>6084.2</u>	<u>6880.-</u>	<u>8306.5</u>	<u>8711.8</u>	
<u>5455.0</u>	<u>6130.5</u>	<u>6944.1</u>	<u>8308.2</u>	<u>8715.7</u>	
<u>5477.7</u>	<u>6148.8</u>	<u>7064.9</u>	<u>8357.0</u>	<u>8719.1</u>	
<u>5491.7</u>	<u>6188.4</u>	<u>7077.5</u>	<u>8374.2</u>	<u>8722.3</u>	
<u>5496.0</u>	<u>6229.4</u>	<u>7135.4</u>	<u>8392.3</u>	<u>8750.5</u>	
<u>5511.2</u>	<u>6239.3</u>	<u>7154.2</u>	<u>8410.0</u>	<u>8784.3</u>	
<u>5523.6</u>	<u>6247.6</u>	<u>7172.0</u>	<u>8413.2</u>	<u>8805.2</u>	
<u>5527.9</u>	<u>6291.-</u>	<u>7255.8</u>	<u>8423.8</u>	<u>8829.6</u>	

A-5. Wolf-Rayet N stars

The observed emission wavelengths may have appreciable shifts to the red because of the influence of absorption lines on the violet edge of the feature.

	<u>3889.4</u>	4336.2	<u>4618.5</u>	5841.-
	<u>3921.-</u>	4340.1	<u>4636.6</u>	<u>5876.2</u>
	<u>3928.4</u>	4347.4	<u>4687.0</u>	<u>6075.4</u>
	<u>3937.-</u>	4359.3	<u>4735.5</u>	6117.3
	<u>3942.-</u>	4380.5	<u>4777.3</u>	6174.3
3166.3	<u>3967.6</u>	4387.1	4797.4	6217.-
<u>3204.0</u>	<u>3998.9</u>	4391.6	<u>4860.4</u>	6233.1
<u>3266.5</u>	<u>4025.3</u>	4416.0	<u>4909.2</u>	6315.4
3346.8	<u>4056.8</u>	4455.2	4923.0	6335.-
3359.8	<u>4074.-</u>	<u>4471.5</u>	<u>4939.0</u>	6389.-
3399.1	4080.-	4480.5	5200.8	6407.0
3457.2	4088.6	4487.2	5412.5	6459.8
<u>3482.8</u>	<u>4099.9</u>	4496.-	5471.4	<u>6563.6</u>
<u>3616.1</u>	<u>4110.-</u>	4514.7	5482.5	<u>6679.2</u>
3704.2	<u>4115.4</u>	4519.8	5505.7	
3735.2	4121.5	4533.-	5662.1	
<u>3753.7</u>	<u>4200.4</u>	4541.9	5674.0	
<u>3807.4</u>	<u>4241.8</u>	4576.1	5800.0	
3829.9	4323.9	4598.0	<u>5806.3</u>	
3856.6	<u>4328.-</u>	<u>4603.0</u>	<u>5815.8</u>	

A-6. Wolf-Rayet C stars

The observed emission wavelengths may have appreciable shifts to the red because of the influence of absorption lines on the violet edge of the feature.

	3860.7	4292.0	5019.8	6747.3	8195.7
	3888.9	4293.7	5049.9	6782.9	<u>8242.9</u>
	3920.6	4304.5	5090.8	6853.5	<u>8263.0</u>
	<u>3929.7</u>	4317.6	5123.7	6861.4	8297.6
	<u>3935.9</u>	<u>4321.0</u>	5135.6	6898.3	8308.8
3127.9	3954.5	<u>4338.6</u>	5155.1	6942.2	8325.3
3143.4	3964.7	<u>4350.6</u>	5171.1	6970.8	<u>8341.4</u>
3167.1	<u>3969.9</u>	4360.7	5243.3	7031.4	8354.6
<u>3203.6</u>	<u>3983.5</u>	4367.9	5266.3	<u>7062.3</u>	8384.3
<u>3233.0</u>	3992.7	4376.2	5303.7	<u>7107.7</u>	8410.4
<u>3261.1</u>	3996.6	4381.5	5349.3	7131.2	8442.9
<u>3335.8</u>	4002.5	4388.2	5363.6	7156.4	8466.8
3348.8	4008.5	4399.2	<u>5413.4</u>	7181.8	<u>8499.6</u>
3358.6	<u>4026.3</u>	4416.0	<u>5440.6</u>	7232.2	<u>8572.0</u>
3382.7	<u>4031.5</u>	<u>4441.1</u>	5469.6	7283.2	8611.1
3399.5	4049.0	4455.6	5501.1	7323.0	8662.8
3410.8	<u>4056.3</u>	<u>4471.2</u>	5594.0	7345.6	
3431.6	<u>4068.0</u>	<u>4481.7</u>	5651.3	7376.9	
3447.0	<u>4070.9</u>	4507.9	5696.5	7420.0	
3464.8	<u>4076.2</u>	<u>4517.9</u>	5746.3	7432.9	
3494.2	<u>4089.6</u>	4543.3	5758.8	7481.7	
3506.5	<u>4100.5</u>	4555.1	5816.2	7517.2	
3561.7	<u>4112.7</u>	4557.0	<u>5875.6</u>	7500.6	
3582.9	4121.1	4585.2	<u>6038.6</u>	7586.8	
3607.8	4130.6	4593.8	6077.2	<u>7662.5</u>	
3634.0	4145.3	<u>4602.8</u>	6116.7	7682.1	
3645.4	4151.6	<u>4618.5</u>	6150.8	7707.0	
3688.7	4156.3	4633.6	6173.5	7728.1	
3707.7	<u>4163.2</u>	<u>4652.6</u>	6205.6	<u>7774.1</u>	
3717.1	<u>4171.6</u>	<u>4668.5</u>	6235.5	7783.8	
3724.7	4182.3	<u>4687.0</u>	6317.4	7797.8	
3728.7	<u>4187.9</u>	<u>4703.0</u>	6350.7	7859.2	
3735.9	<u>4199.1</u>	4713.1	6408.0	7879.0	
3756.4	<u>4209.1</u>	4737.6	6454.8	7913.3	
3763.5	4214.2	4787.1	6503.4	7942.3	
3770.6	4218.2	4798.1	6527.5	7964.8	
3784.8	<u>4228.0</u>	4804.6	<u>6578.8</u>	8019.0	
3798.0	<u>4247.7</u>	4814.6	<u>6615.9</u>	8068.9	
3814.6	4256.8	<u>4860.9</u>	<u>6679.0</u>	8142.0	
3836.0	<u>4267.3</u>	<u>4884.5</u>	<u>6737.2</u>	8164.1	

A-7. Planetary nebulae

	3353.3	3466.4	3592.9	3703.8	3778.9
	3354.9	3467.3	3595.2	<u>3705.0</u>	3781.7
	3357.6	3468.1	3596.5	3706.2	3783.5
	3362.2	3469.7	3597.5	3707.3	3784.9
	3367.2	3471.8	3607.2	3708.1	3786.1
3047.1	3370.1	3476.3	3611.8	3709.4	3787.8
<u>3076.9</u>	3372.0	3477.2	3613.6	3710.4	3791.3
3107.2	3379.3	3479.0	3618.6	<u>3712.0</u>	3792.4
3121.7	3381.1	3481.4	3620.4	<u>3713.0</u>	3795.4
<u>3127.6</u>	3385.5	3482.0	3622.3	3714.0	3796.3
3132.9	3387.2	3485.6	3634.3	3715.1	3797.8
<u>3187.8</u>	3392.7	3487.7	3650.0	<u>3718.1</u>	<u>3801.0</u>
3203.2	3394.2	3488.3	3652.2	3719.8	3803.7
<u>3206.0</u>	3396.7	3492.6	3654.0	3720.3	3805.1
3207.9	3403.5	3497.3	3657.7	<u>3721.9</u>	3805.8
3211.2	3405.7	3498.6	3658.6	3724.1	3806.6
3239.7	3406.6	3503.3	3659.4	<u>3726.0</u>	3810.6
3241.7	3407.9	3504.9	3660.3	<u>3728.7</u>	3812.9
3243.9	3409.6	3511.4	3661.2	<u>3730.6</u>	3813.5
3246.0	3411.7	3512.6	3662.2	3731.1	3815.3
3258.4	3413.7	3513.8	3663.4	3732.8	3819.6
3261.0	3415.2	3514.8	3664.7	<u>3734.3</u>	<u>3823.4</u>
3265.4	3420.0	3517.5	3666.1	<u>3736.8</u>	3829.9
3285.9	3424.0	3519.5	3667.7	<u>3737.5</u>	3831.8
3293.3	<u>3425.8</u>	3521.6	3669.4	3738.8	3833.7
3296.8	3427.1	3522.5	3671.5	3739.6	3835.4
<u>3299.3</u>	<u>3428.6</u>	3530.6	3673.7	3740.3	<u>3838.1</u>
<u>3301.5</u>	<u>3430.4</u>	3534.5	3676.4	3743.2	3839.5
3303.7	3431.1	3538.4	3677.8	3744.8	3842.0
3305.1	3432.0	3540.6	3679.4	3745.9	3843.0
3312.3	3433.4	3543.4	3681.4	3747.8	3849.-
<u>3322.0</u>	3439.9	3546.7	3682.8	3748.5	3851.2
3323.9	3441.8	3548.4	3685.2	<u>3750.1</u>	3853.5
3324.6	3443.2	3550.2	3686.8	<u>3754.6</u>	3856.0
3326.0	<u>3444.0</u>	3551.7	<u>3689.5</u>	<u>3755.9</u>	3857.6
3327.2	3446.0	3554.3	3690.1	3757.3	3858.1
3329.1	3447.5	3561.2	<u>3691.6</u>	<u>3759.8</u>	3860.4
3334.9	3450.3	3568.5	<u>3692.7</u>	<u>3762.4</u>	3862.5
3338.9	3451.1	3571.4	3694.3	3766.6	3864.6
<u>3340.8</u>	3453.0	3573.7	3695.5	3768.9	3866.7
3342.6	3454.4	3577.3	3697.2	3770.6	3868.8
<u>3345.9</u>	3456.9	3579.2	<u>3698.7</u>	<u>3772.5</u>	<u>3870.7</u>
<u>3347.9</u>	3460.9	3584.1	3699.5	3774.0	3871.6
3349.1	3464.5	3587.1	3700.6	3774.9	3874.7
3350.8	3465.9	3590.8	3702.6	3777.1	3876.8

Planetary nebulae, cont.

3878.4	4017.1	4119.2	4220.-	4346.9	4467.8
3887.5	4018.6	4120.7	4227.4	4348.2	4471.4
3888.9	4019.6	4122.5	4229.2	4349.5	<u>4477.7</u>
<u>3891.3</u>	4023.9	4125.3	4230.2	<u>4363.1</u>	4481.2
3895.5	<u>4026.1</u>	4128.8	4231.6	<u>4365.5</u>	4483.5
3905.1	4029.6	4129.8	4240.5	4366.8	4485.4
3911.9	4038.9	4132.9	4242.0	4368.5	4486.6
3918.7	4041.0	4134.6	4244.0	4372.0	4487.9
3919.2	4041.5	4135.8	4245.7	4373.3	4491.2
3920.7	4044.0	4136.4	4253.8	4375.9	<u>4511.0</u>
3923.4	4048.8	4140.0	4257.7	4378.3	4514.6
3926.6	4050.9	4143.6	4258.3	4379.2	4516.8
3929.2	4054.8	4145.6	4261.4	4380.3	4518.2
3933.4	4056.5	4146.2	<u>4267.1</u>	4382.1	4523.6
3935.7	4057.8	4147.8	<u>4269.5</u>	<u>4387.8</u>	4525.5
3937.6	4060.2	4148.9	4271.5	4389.9	4526.9
3943.7	4063.1	4150.6	4273.6	4390.5	4529.8
3948.0	<u>4068.6</u>	4152.6	4275.5	4391.9	4531.4
3953.4	<u>4070.0</u>	4153.3	4276.6	4394.0	4532.7
3954.5	4071.3	4155.5	4278.8	4400.2	4534.4
3956.7	4072.1	4156.3	4283.0	4403.9	4538.5
3958.4	4073.9	4159.2	4285.1	4409.3	4540.8
3960.9	4076.2	4160.8	4287.2	4411.1	4541.4
3961.3	<u>4078.7</u>	<u>4163.2</u>	4288.0	4413.2	<u>4542.8</u>
3962.0	4080.9	<u>4163.9</u>	4289.1	4414.9	4544.8
3962.8	4081.4	4166.8	4292.5	4417.0	4546.6
3964.8	4083.9	4167.8	4294.4	4422.4	4548.6
<u>3967.5</u>	4084.9	4169.0	4297.0	4428.5	4549.5
<u>3970.0</u>	4086.8	4171.3	4299.2	4430.2	4552.-
<u>3972.9</u>	4087.2	4174.6	4301.7	4431.1	4557.1
3976.4	4089.1	4177.4	4303.9	4432.6	4562.6
3983.2	4091.5	4179.1	4311.4	4434.3	<u>4571.0</u>
3984.5	4093.0	4180.9	4314.9	4437.5	<u>4583.7</u>
3985.5	4094.4	4183.0	4317.1	4440.4	4588.7
3989.4	<u>4097.3</u>	4183.9	4319.5	4442.4	4590.8
3992.9	<u>4100.1</u>	4184.2	4321.7	4445.9	4596.0
3995.0	<u>4101.7</u>	4185.5	4323.7	4447.8	4598.7
3997.6	<u>4103.3</u>	<u>4186.8</u>	4325.9	4452.6	4601.8
4000.5	<u>4104.7</u>	<u>4188.0</u>	4328.6	4454.1	4604.6
4004.4	4107.3	4189.8	4333.1	4456.6	4606.6
4007.9	4108.6	4194.8	<u>4338.9</u>	4458.7	4609.4
4008.5	4110.5	4195.6	<u>4340.6</u>	4461.4	4616.8
4009.3	4112.2	<u>4199.8</u>	<u>4343.0</u>	4462.4	4619.9
4011.6	4114.8	<u>4201.7</u>	<u>4343.9</u>	4463.2	4620.9
4014.6	4116.5	4217.3	4345.4	4465.1	4621.7

Planetary nebulae, cont.

4625.4	4715.7	4972.1	5621.-	6074.3	7005.7
4630.3	4724.7	5006.9	5630.9	6082.4	7065.3
4632.1	4725.7	5015.5	5669.-	6086.9	7135.8
4634.0	4733.4	5041.3	5677.0	6101.8	7170.6
4638.8	4740.1	5047.9	5680.-	6118.3	7177.8
4640.6	4748.0	5090.-	5696.1	6151.4	7237.3
4641.9	4754.0	5112.-	5721.1	6166.2	7262.8
4644.8	4754.9	5131.2	5726.1	6170.7	7281.3
4647.3	4759.3	5146.0	5754.6	6218.6	7319.9
4649.1	4761.3	5161.7	5776.4	6221.4	7330.2
4650.4	4762.1	5176.5	5786.6	6232.5	7378.-
4651.5	4764.0	5191.8	5790.8	6233.8	7530.5
4657.7	4765.6	5198.0	5794.6	6246.-	7587.-
4658.4	4768.6	5198.9	5801.4	6285.-	7592.8
4661.6	4779.7	5200.4	5806.3	6300.3	7726.5
4663.6	4780.9	5216.-	5812.1	6312.1	7751.0
4665.6	4781.4	5232.-	5820.1	6347.5	8045.6
4667.2	4782.4	5262.0	5828.6	6363.8	8196.6
4669.2	4793.4	5270.3	5836.5	6371.3	8236.6
4670.8	4805.1	5277.8	5847.1	6393.6	8446.-
4673.5	4813.7	5308.8	5858.3	6403.7	8502.5
4676.1	4861.3	5323.6	5862.3	6406.5	8545.5
4678.2	4877.9	5335.1	5864.8	6435.1	8598.4
4680.1	4879.2	5346.0	5867.8	6462.1	8665.0
4682.7	4880.7	5411.5	5875.4	6518.3	8750.5
4685.8	4894.8	5424.0	5885.9	6527.3	8862.2
4686.6	4899.5	5426.6	5891.0	6548.1	9014.9
4688.2	4909.0	5470.9	5894.0	6560.2	9069.-
4701.3	4921.9	5484.8	5896.9	6562.9	
4707.3	4931.2	5517.7	5913.5	6583.4	
4708.0	4934.6	5537.8	5932.2	6598.8	
4711.4	4944.0	5577.4	5953.0	6678.2	
4712.6	4950.5	5592.2	5977.0	6716.5	
4713.2	4959.1	5604.5	6004.8	6730.7	
4714.3	4959.9	5614.6	6037.2	6890.8	

A-8a. Gaseous nebulae (Orion Nebula)

	3451.9	<u>3664.7</u>	3829.8	4077.8	4332.6
	3453.3	<u>3666.1</u>	3831.8	4085.9	<u>4340.3</u>
	3465.9	<u>3667.7</u>	3833.6	4087.1	<u>4342.7</u>
	3471.9	<u>3669.5</u>	<u>3835.4</u>	4089.2	4344.0
	3477.9	<u>3671.5</u>	<u>3837.8</u>	4097.3	4345.5
<u>3187.7</u>	3478.9	<u>3673.8</u>	3838.4	4099.5	4346.8
<u>3214.1</u>	3484.8	<u>3676.4</u>	3853.9	<u>4101.6</u>	4347.5
3215.5	3486.0	<u>3678.3</u>	3856.1	<u>4105.0</u>	4349.4
3238.5	3487.7	<u>3679.4</u>	3857.2	4110.7	4351.3
3246.4	3496.1	<u>3682.8</u>	3862.7	4117.6	<u>4358.3</u>
3253.7	3497.3	3685.3	3867.7	4119.2	4359.4
3255.2	3498.6	<u>3686.8</u>	<u>3868.8</u>	<u>4120.8</u>	<u>4363.2</u>
3268.8	3500.0	<u>3691.6</u>	<u>3870.5</u>	<u>4131.7</u>	<u>4365.0</u>
3286.2	3505.4	<u>3694.2</u>	3871.8	4132.8	4367.0
3293.0	3509.3	<u>3697.2</u>	3882.5	<u>4143.7</u>	4368.4
3296.7	<u>3512.6</u>	<u>3703.9</u>	3884.8	4153.3	<u>4387.9</u>
3298.8	<u>3513.8</u>	<u>3705.0</u>	3886.0	4156.3	<u>4406.2</u>
3306.2	<u>3530.5</u>	<u>3709.5</u>	<u>3888.9</u>	4169.1	4409.1
3324.0	<u>3537.9</u>	<u>3711.1</u>	<u>3907.5</u>	4171.0	4413.9
3325.0	3539.2	<u>3716.2</u>	3918.9	4185.5	4415.0
3334.9	3540.9	3717.8	<u>3920.7</u>	4189.7	4416.3
3342.9	3553.0	<u>3721.9</u>	<u>3926.5</u>	4200.7	4416.9
3353.3	<u>3554.5</u>	<u>3726.1</u>	3928.5	4244.1	4420.9
3354.6	<u>3587.3</u>	<u>3728.8</u>	<u>3964.8</u>	4251.2	4421.8
3356.7	<u>3605.2</u>	<u>3732.9</u>	<u>3967.5</u>	4253.7	4423.1
3358.8	<u>3613.6</u>	<u>3734.3</u>	<u>3970.0</u>	4257.0	4437.7
3360.5	<u>3634.3</u>	<u>3746.0</u>	<u>3982.6</u>	4263.3	4446.9
3365.7	<u>3647.6</u>	3747.9	3983.7	<u>4267.1</u>	4447.8
3370.4	3652.0	<u>3750.1</u>	3985.9	<u>4277.0</u>	4448.5
3371.3	3654.7	<u>3756.2</u>	3993.2	4285.0	4452.3
3387.0	3656.0	3758.5	4002.9	4287.5	4458.1
3398.2	3656.7	3761.0	<u>4009.3</u>	4300.4	<u>4471.4</u>
3409.3	3657.3	<u>3770.6</u>	<u>4026.2</u>	4303.8	<u>4590.9</u>
3410.2	3658.0	<u>3777.0</u>	<u>4046.6</u>	4305.8	4598.4
3422.5	3658.7	3781.9	<u>4068.7</u>	4308.1	4607.1
3429.9	3659.4	3784.9	4069.8	4311.2	4621.4
3442.1	3660.3	<u>3797.9</u>	4070.9	4317.0	4628.5
3443.5	3661.2	<u>3805.7</u>	4072.1	4319.6	4630.6
3445.6	3662.2	3806.5	4075.8	4326.1	4634.0
<u>3447.6</u>	3663.4	<u>3819.6</u>	<u>4076.4</u>	4328.6	4637.8

Gaseous nebulae, cont.

4638.8	4754.9	5313.-	6085.-	<u>6583.6</u>	8600.-
4640.7	4769.6	5411.6	6102.-	<u>6590.6</u>	8647.-
4641.9	4774.4	5517.9	6180.-	6597.7	8724.-
<u>4649.2</u>	4776.7	<u>5536.9</u>	6205.-	<u>6678.0</u>	8824.-
<u>4650.8</u>	4777.9	<u>5577.-</u>	6246.-	<u>6716.4</u>	9000.-
4656.3	<u>4781.4</u>	5621.-	6285.-	<u>6730.8</u>	9069.4
4657.2	<u>4814.7</u>	5655.-	<u>6300.3</u>	<u>6763.-</u>	<u>9229.0</u>
<u>4658.2</u>	<u>4861.2</u>	5669.-	<u>6312.0</u>	7007.-	<u>9546.0</u>
<u>4661.7</u>	<u>4864.1</u>	5680.-	<u>6347.0</u>	<u>7065.3</u>	<u>9531.6</u>
4665.6	<u>4881.1</u>	5695.-	<u>6363.9</u>	<u>7135.8</u>	<u>10049.4</u>
4676.1	<u>4922.0</u>	5720.-	6371.3	<u>7234.5</u>	<u>10830.-</u>
4678.0	<u>4931.0</u>	5737.-	6398.-	<u>7254.4</u>	<u>10938.1</u>
4699.1	<u>4958.9</u>	<u>5754.9</u>	6410.6	<u>7281.3</u>	
<u>4701.7</u>	<u>5006.8</u>	<u>5777.-</u>	6435.-	<u>7319.9</u>	
<u>4705.5</u>	<u>5015.7</u>	5800.-	6464.-	<u>7330.2</u>	
4708.4	5147.5	5812.-	6492.-	7605.-	
4711.5	5156.4	5875.6	6518.0	7755.-	
<u>4713.2</u>	5183.-	5957.5	6529.-	8454.-	
<u>4733.9</u>	<u>5198.7</u>	5977.5	<u>6548.2</u>	8506.-	
<u>4740.3</u>	<u>5269.4</u>	<u>6045.8</u>	<u>6562.9</u>	8545.-	

A-8b. Supernova remnants (Crab nebula)

3727	4340	4959	6563
3869	4363	5007	6584
3889	4472	5875	6711
3967	4686	6300	6728
4100	4861	6548	

A-9. Symbiotic stars

	<u>3277.4</u>	<u>3601.0</u>	<u>3739.5</u>	<u>3827.8</u>	<u>3943.4</u>
	<u>3281.4</u>	<u>3609.7</u>	<u>3741.5</u>	<u>3828.5</u>	<u>3945.0</u>
	<u>3282.6</u>	<u>3613.8</u>	<u>3743.1</u>	<u>3829.5</u>	<u>3951.6</u>
	<u>3285.1</u>	<u>3626.2</u>	<u>3744.5</u>	<u>3832.3</u>	<u>3955.5</u>
	<u>3289.2</u>	<u>3632.7</u>	<u>3745.6</u>	<u>3833.4</u>	<u>3961.5</u>
<u>3102.7</u>	<u>3295.8</u>	<u>3634.3</u>	<u>3748.2</u>	<u>3835.3</u>	<u>3963.5</u>
<u>3104.9</u>	<u>3303.4</u>	<u>3660.8</u>	<u>3750.2</u>	<u>3838.3</u>	<u>3964.8</u>
<u>3108.8</u>	<u>3312.3</u>	<u>3662.7</u>	<u>3754.6</u>	<u>3839.5</u>	<u>3966.3</u>
<u>3114.2</u>	<u>3323.1</u>	<u>3664.6</u>	<u>3756.5</u>	<u>3840.8</u>	<u>3967.6</u>
<u>3116.5</u>	<u>3340.6</u>	<u>3666.0</u>	<u>3757.7</u>	<u>3843.0</u>	<u>3968.5</u>
<u>3121.6</u>	<u>3346.2</u>	<u>3667.7</u>	<u>3759.5</u>	<u>3848.2</u>	<u>3969.4</u>
<u>3123.8</u>	<u>3353.2</u>	<u>3669.5</u>	<u>3761.2</u>	<u>3849.6</u>	<u>3970.1</u>
<u>3125.8</u>	<u>3358.8</u>	<u>3671.4</u>	<u>3764.2</u>	<u>3850.4</u>	<u>3973.1</u>
<u>3128.8</u>	<u>3364.4</u>	<u>3673.7</u>	<u>3766.8</u>	<u>3853.6</u>	<u>3973.9</u>
<u>3132.9</u>	<u>3377.4</u>	<u>3676.4</u>	<u>3767.8</u>	<u>3856.0</u>	<u>3978.0</u>
<u>3135.3</u>	<u>3405.5</u>	<u>3677.4</u>	<u>3769.6</u>	<u>3859.6</u>	<u>3981.9</u>
<u>3144.8</u>	<u>3411.8</u>	<u>3679.3</u>	<u>3770.6</u>	<u>3861.2</u>	<u>3991.6</u>
<u>3148.4</u>	<u>3414.8</u>	<u>3682.6</u>	<u>3773.6</u>	<u>3862.5</u>	<u>3997.9</u>
<u>3153.4</u>	<u>3425.8</u>	<u>3685.1</u>	<u>3777.4</u>	<u>3865.4</u>	<u>4002.3</u>
<u>3162.4</u>	<u>3428.4</u>	<u>3686.8</u>	<u>3779.6</u>	<u>3867.3</u>	<u>4005.4</u>
					<u>4009.4</u>
<u>3166.8</u>	<u>3430.2</u>	<u>3688.5</u>	<u>3783.2</u>	<u>3868.6</u>	
<u>3176.9</u>	<u>3435.4</u>	<u>3691.5</u>	<u>3784.7</u>	<u>3869.4</u>	<u>4012.5</u>
<u>3179.0</u>	<u>3439.2</u>	<u>3694.1</u>	<u>3785.6</u>	<u>3871.8</u>	<u>4014.0</u>
<u>3183.0</u>	<u>3443.6</u>	<u>3697.1</u>	<u>3787.2</u>	<u>3872.5</u>	<u>4015.3</u>
<u>3185.5</u>	<u>3452.2</u>	<u>3700.0</u>	<u>3788.2</u>	<u>3874.8</u>	<u>4016.6</u>
<u>3187.8</u>	<u>3455.3</u>	<u>3703.6</u>	<u>3791.3</u>	<u>3878.3</u>	<u>4017.6</u>
<u>3190.6</u>	<u>3459.7</u>	<u>3705.0</u>	<u>3795.3</u>	<u>3882.2</u>	<u>4019.9</u>
<u>3193.2</u>	<u>3471.0</u>	<u>3705.6</u>	<u>3797.2</u>	<u>3883.7</u>	<u>4023.3</u>
<u>3196.1</u>	<u>3475.4</u>	<u>3707.5</u>	<u>3797.9</u>	<u>3886.3</u>	<u>4025.6</u>
<u>3198.9</u>	<u>3482.6</u>	<u>3709.9</u>	<u>3799.4</u>	<u>3887.3</u>	<u>4026.5</u>
<u>3203.0</u>	<u>3484.0</u>	<u>3712.0</u>	<u>3805.6</u>	<u>3888.8</u>	<u>4028.2</u>
<u>3210.3</u>	<u>3487.0</u>	<u>3713.4</u>	<u>3806.5</u>	<u>3891.4</u>	<u>4035.4</u>
<u>3213.3</u>	<u>3488.8</u>	<u>3715.0</u>	<u>3813.0</u>	<u>3895.6</u>	<u>4045.7</u>
<u>3217.0</u>	<u>3495.0</u>	<u>3716.4</u>	<u>3813.8</u>	<u>3900.1</u>	<u>4053.8</u>
<u>3223.9</u>	<u>3501.7</u>	<u>3718.6</u>	<u>3814.5</u>	<u>3903.2</u>	<u>4063.7</u>
<u>3227.7</u>	<u>3504.5</u>	<u>3719.8</u>	<u>3815.8</u>	<u>3905.4</u>	<u>4064.7</u>
<u>3231.6</u>	<u>3507.9</u>	<u>3721.8</u>	<u>3816.9</u>	<u>3913.9</u>	<u>4067.4</u>
<u>3233.3</u>	<u>3511.5</u>	<u>3724.1</u>	<u>3818.6</u>	<u>3918.7</u>	<u>4068.6</u>
<u>3238.5</u>	<u>3513.8</u>	<u>3725.3</u>	<u>3819.6</u>	<u>3920.9</u>	<u>4069.6</u>
<u>3243.5</u>	<u>3535.7</u>	<u>3726.9</u>	<u>3820.1</u>	<u>3923.2</u>	<u>4071.7</u>
<u>3247.4</u>	<u>3539.4</u>	<u>3728.5</u>	<u>3822.4</u>	<u>3926.5</u>	<u>4074.0</u>
<u>3256.1</u>	<u>3559.6</u>	<u>3733.2</u>	<u>3824.4</u>	<u>3930.2</u>	<u>4075.8</u>
<u>3258.8</u>	<u>3576.4</u>	<u>3734.4</u>	<u>3824.8</u>	<u>3933.7</u>	<u>4079.0</u>
<u>3271.4</u>	<u>3586.1</u>	<u>3735.7</u>	<u>3825.6</u>	<u>3935.9</u>	<u>4080.8</u>
<u>3273.8</u>	<u>3589.9</u>	<u>3737.0</u>	<u>3827.0</u>	<u>3938.3</u>	<u>4088.3</u>

Symbiotic stars, cont.

<u>4089.3</u>	4186.9	4320.9	4445.6	<u>4576.6</u>	4774.6
<u>4092.8</u>	4187.8	4330.4	4447.7	<u>4577.7</u>	4778.7
4094.3	4189.8	4331.4	4450.4	4580.0	4781.7
4096.6	4192.1	4337.6	4451.2	<u>4582.7</u>	4788.3
<u>4097.3</u>	4195.6	4338.6	<u>4457.9</u>	<u>4583.7</u>	4804.9
4100.1	4196.6	4339.4	4461.4	4588.1	4812.6
<u>4101.7</u>	<u>4199.9</u>	<u>4340.5</u>	4464.7	4589.5	4824.1
<u>4103.2</u>	4201.9	<u>4344.1</u>	<u>4468.4</u>	4590.8	4828.1
<u>4104.8</u>	4205.3	4345.7	<u>4470.1</u>	4595.9	4834.3
4105.7	4219.8	4348.3	<u>4471.5</u>	4602.0	<u>4861.3</u>
4106.6	4225.0	4349.4	<u>4472.5</u>	4606.6	4919.8
4115.4	4226.5	4350.5	<u>4477.6</u>	4618.6	<u>4922.4</u>
4116.2	4227.3	<u>4351.7</u>	4479.2	<u>4620.3</u>	<u>4923.9</u>
4117.7	4231.0	<u>4359.0</u>	<u>4481.0</u>	<u>4629.4</u>	4936.6
4118.5	<u>4233.2</u>	<u>4359.3</u>	<u>4482.2</u>	<u>4631.5</u>	4940.5
4119.2	4240.3	<u>4363.2</u>	4488.0	<u>4634.3</u>	4945.0
<u>4120.9</u>	4242.3	<u>4367.1</u>	<u>4489.0</u>	<u>4637.7</u>	4949.5
<u>4122.4</u>	4244.2	4368.2	<u>4491.4</u>	4638.8	<u>4958.9</u>
<u>4123.6</u>	4245.5	<u>4369.4</u>	<u>4501.2</u>	<u>4640.4</u>	<u>4985.6</u>
<u>4124.8</u>	4250.3	<u>4371.8</u>	<u>4507.0</u>	<u>4641.6</u>	4990.0
<u>4128.3</u>	4251.6	4374.6	<u>4508.6</u>	4643.0	4994.0
<u>4131.2</u>	4252.5	4376.8	<u>4509.6</u>	4644.2	5001.5
<u>4132.3</u>	4257.9	4377.5	4511.6	4645.7	5004.8
4138.2	4261.7	4378.6	<u>4514.4</u>	4646.6	<u>5007.0</u>
<u>4143.6</u>	4264.6	4379.7	<u>4515.4</u>	4647.3	<u>5010.5</u>
4144.6	<u>4267.1</u>	4383.4	4516.2	<u>4648.2</u>	5011.3
4145.7	<u>4271.7</u>	<u>4384.2</u>	4518.3	<u>4649.1</u>	5012.2
4151.6	4273.2	<u>4385.3</u>	<u>4520.6</u>	<u>4650.2</u>	<u>5015.7</u>
4153.5	4275.6	<u>4386.3</u>	<u>4522.7</u>	<u>4651.2</u>	<u>5017.0</u>
<u>4156.3</u>	<u>4276.7</u>	<u>4387.8</u>	<u>4523.6</u>	<u>4657.7</u>	<u>5018.5</u>
4161.5	<u>4277.7</u>	4390.5	4525.8	4661.2	5032.2
<u>4163.2</u>	<u>4287.3</u>	<u>4394.8</u>	<u>4528.9</u>	4663.5	<u>5040.7</u>
<u>4165.9</u>	<u>4289.9</u>	<u>4397.4</u>	<u>4534.3</u>	4666.7	<u>5042.4</u>
4169.0	<u>4293.7</u>	<u>4399.7</u>	<u>4540.7</u>	<u>4670.0</u>	<u>5047.9</u>
<u>4171.8</u>	4294.9	4404.8	<u>4541.5</u>	4676.0	<u>5058.6</u>
4173.4	<u>4296.4</u>	4406.2	<u>4549.4</u>	<u>4685.5</u>	5068.6
<u>4175.8</u>	<u>4297.8</u>	<u>4413.6</u>	<u>4551.0</u>	<u>4698.9</u>	5103.6
<u>4177.6</u>	4299.9	<u>4414.7</u>	4552.4	<u>4701.4</u>	5117.2
<u>4178.1</u>	4301.8	<u>4415.2</u>	4554.7	<u>4704.9</u>	5136.-
<u>4178.7</u>	<u>4303.3</u>	4416.2	<u>4555.9</u>	<u>4713.4</u>	5143.7
4179.7	<u>4307.6</u>	<u>4417.0</u>	<u>4558.3</u>	4725.2	5150.5
4180.3	<u>4313.0</u>	<u>4421.4</u>	<u>4563.8</u>	<u>4731.5</u>	5156.0
4181.5	<u>4314.5</u>	<u>4427.1</u>	<u>4568.3</u>	<u>4733.8</u>	5160.2
4183.2	<u>4316.2</u>	<u>4437.6</u>	<u>4571.6</u>	4755.6	5168.1
4185.4	<u>4319.4</u>	<u>4443.2</u>	<u>4574.8</u>	4769.2	5178.1

Symbiotic stars, cont.

5183.7	5415.3	5713.-	<u>6086.1</u>	6456.5
5197.6	5425.0	<u>5720.6</u>	<u>6089.2</u>	6491.3
5199.3	5432.5	<u>5724.2</u>	<u>6238.4</u>	6493.0
5200.3	5435.0	5732.-	6247.7	6516.0
5218.3	5450.0	5739.5	6300.4	6548.0
5226.7	5481.1	5746.6	6306.-	<u>6562.3</u>
<u>5235.2</u>	5486.9	<u>5753.5</u>	6311.0	<u>6584.-</u>
<u>5270.2</u>	5494.0	<u>5789.7</u>	6315.9	<u>6678.1</u>
5272.6	5508.3	5796.1	6318.2	<u>6683.4</u>
<u>5274.4</u>	5526.2	5801.3	6346.6	7066.-
5281.2	5531.9	5831.5	6350.3	7733.-
5283.9	<u>5535.2</u>	5843.9	6364.3	7772.-
5309.0	<u>5549.6</u>	5853.-	6369.4	7932.5
<u>5316.9</u>	5575.6	<u>5875.9</u>	<u>6370.9</u>	8200.6
<u>5334.-</u>	5577.8	<u>5904.-</u>	<u>6374.3</u>	8445.-
5337.5	5581.-	5920.4	6383.7	8665.7
5354.-	5593.0	5943.6	6385.5	8703.-
5362.8	5617.6	5971.1	6415.8	
5408.4	5673.8	5991.4	6432.6	
<u>5412.0</u>	5698.-	6002.8	6443.8	

A-10a. Oe-class stars

4485.4	4679.3	<u>6563.2</u>
4504.0	4685.9	
4541.-	<u>4708.6</u>	
4634.1	4855.9	
<u>4641.2</u>	5695.8	

A-10b. Be-class stars (P-Cygni class)

The absorption-emission profiles for most lines may result in the emission lines being systematically red-shifted in wavelength.

	3486.2	3684.2	3762.9	3849.6	3936.0
	3499.3	3685.3	3764.0	3850.4	3938.3
	3501.5	3687.0	3769.5	3853.7	3939.0
	3506.4	3691.6	<u>3770.6</u>	3856.1	3945.0
	3512.4	<u>3697.0</u>	<u>3774.7</u>	3859.9	3950.0
3008.-	3530.4	3700.4	3776.1	3862.6	3952.0
3013.0	3554.4	3703.9	3778.3	3963.8	3961.5
3023.6	3585.8	<u>3705.5</u>	3779.5	3865.6	<u>3964.5</u>
3136.3	3587.2	<u>3706.1</u>	3781.5	3872.7	<u>3966.0</u>
3173.7	3589.8	<u>3712.2</u>	3783.4	3878.0	<u>3970.3</u>
3175.7	3600.8	3713.0	3786.3	3882.3	3973.6
3177.8	3603.8	3715.3	3787.2	3886.3	3974.1
3184.8	3613.6	3720.0	3791.0	<u>3889.1</u>	3975.9
3187.4	3634.3	<u>3722.0</u>	<u>3798.0</u>	<u>3896.1</u>	3977.3
3204.5	3649.3	<u>3727.3</u>	<u>3813.4</u>	3899.8	3979.6
3212.0	3652.7	3732.8	3814.1	3900.6	3981.6
3215.3	3659.7	<u>3734.5</u>	3814.6	<u>3903.1</u>	3982.0
3266.6	3661.3	<u>3736.9</u>	3819.7	<u>3905.5</u>	3987.6
3275.9	3662.3	3738.3	<u>3820.4</u>	3906.8	<u>3993.1</u>
3280.8	3663.5	3741.7	<u>3821.9</u>	3908.0	<u>3995.0</u>
3288.5	3664.8	3745.7	3824.4	3913.5	3997.1
3291.8	3666.2	3746.6	3825.9	3914.4	4000.1
3305.0	3667.9	3748.5	3827.0	3916.4	<u>4002.3</u>
3322.0	3669.5	<u>3750.1</u>	3827.8	3918.7	<u>4002.9</u>
3324.9	3671.7	<u>3754.6</u>	3829.4	3920.3	4003.4
3329.3	3673.9	3755.5	3832.3	3922.8	4005.7
3338.8	3676.5	3757.7	3835.5	<u>3928.1</u>	<u>4009.0</u>
3354.3	3677.8	3759.3	<u>3838.3</u>	<u>3930.3</u>	<u>4012.4</u>
3396.5	3679.7	3761.4	3845.2	3932.0	4014.5
3405.5	3683.0	3761.8	3848.2	3933.6	4015.4

Be-class stars, cont.

4022.5	4136.0	<u>4238.9</u>	4338.6	4416.9	<u>4522.4</u>
4023.3	4137.8	<u>4242.5</u>	4340.6	4417.7	<u>4528.5</u>
4024.5	4138.1	<u>4244.0</u>	<u>4344.3</u>	4418.3	4529.5
4025.1	4139.1	<u>4244.8</u>	4346.8	4421.9	4533.9
<u>4025.9</u>	4140.1	<u>4246.3</u>	4350.8	4427.3	4534.0
4028.4	4143.8	4250.8	<u>4351.7</u>	4428.0	4535.3
4030.7	<u>4145.6</u>	4252.6	<u>4352.5</u>	4430.6	4539.6
4033.0	<u>4147.6</u>	4254.4	<u>4354.6</u>	4431.4	<u>4541.5</u>
4035.6	4155.3	4258.1	4357.6	4434.0	<u>4544.0</u>
<u>4036.7</u>	4160.6	<u>4260.5</u>	4358.5	4435.5	4545.1
4043.4	4161.6	4267.1	<u>4359.3</u>	4437.4	<u>4549.6</u>
<u>4046.8</u>	<u>4163.7</u>	4269.2	<u>4362.1</u>	4441.7	<u>4552.3</u>
4048.8	<u>4166.0</u>	4271.0	4367.7	4442.6	4554.0
4051.2	4169.0	<u>4273.1</u>	4368.2	4443.8	4555.0
4051.9	<u>4171.9</u>	<u>4275.5</u>	4369.4	4444.5	<u>4555.9</u>
4053.5	4173.5	4276.8	<u>4372.4</u>	4447.8	4558.8
<u>4056.2</u>	<u>4174.2</u>	<u>4278.2</u>	<u>4374.4</u>	4450.5	4563.8
4057.5	<u>4177.6</u>	4281.2	<u>4375.0</u>	4451.5	4564.6
4063.6	<u>4178.9</u>	4282.4	<u>4375.9</u>	4452.0	4565.7
4064.3	<u>4179.5</u>	4284.2	4383.6	<u>4455.3</u>	4568.3
4066.1	4181.7	4287.3	<u>4384.5</u>	4456.6	<u>4571.2</u>
4067.0	4183.2	<u>4287.9</u>	<u>4385.2</u>	4457.9	<u>4572.0</u>
<u>4068.6</u>	<u>4184.5</u>	4289.0	<u>4385.8</u>	<u>4459.1</u>	<u>4576.3</u>
<u>4070.0</u>	<u>4187.4</u>	4290.2	<u>4386.8</u>	4461.5	<u>4580.3</u>
4072.7	<u>4189.4</u>	4294.1	4387.9	4464.4	4582.8
4076.2	4190.3	<u>4296.6</u>	4390.4	4468.5	<u>4583.7</u>
<u>4077.6</u>	4192.0	<u>4300.1</u>	4391.0	4469.1	<u>4588.3</u>
4082.9	4195.1	4301.8	4394.1	4470.8	4590.0
4087.3	<u>4200.8</u>	<u>4303.1</u>	4395.1	4471.6	4592.1
4088.8	4202.0	<u>4305.7</u>	4395.8	<u>4472.9</u>	4593.9
4101.9	4202.4	4307.9	4398.3	4474.9	4595.6
<u>4102.9</u>	4204.1	4312.9	4399.8	<u>4481.1</u>	4600.2
4103.9	4205.0	<u>4313.9</u>	4400.4	<u>4482.1</u>	4601.4
4109.2	<u>4205.6</u>	<u>4314.3</u>	4402.8	<u>4486.3</u>	4615.1
4111.0	<u>4209.7</u>	4315.0	4404.8	4487.2	4616.6
4113.1	4210.8	4316.8	4407.6	4489.2	<u>4618.9</u>
4116.0	4211.8	<u>4319.6</u>	4408.5	<u>4491.3</u>	<u>4620.5</u>
4119.3	4215.7	<u>4320.8</u>	4409.2	<u>4493.5</u>	<u>4625.9</u>
<u>4120.8</u>	4216.2	4325.0	4409.5	4494.6	<u>4629.3</u>
<u>4123.1</u>	4221.1	4325.8	4411.1	4500.4	<u>4634.1</u>
4124.8	4224.8	<u>4326.6</u>	4411.9	4501.3	4635.3
4128.1	4225.1	4330.2	<u>4413.7</u>	<u>4508.3</u>	<u>4636.2</u>
<u>4129.1</u>	4226.7	4330.7	<u>4415.1</u>	<u>4515.1</u>	<u>4645.6</u>
<u>4130.9</u>	4227.3	4331.9	4415.6	<u>4518.3</u>	4646.9
4132.2	<u>4233.2</u>	4338.0	<u>4416.3</u>	<u>4520.2</u>	4655.8

Be-class stars, cont.

4656.9	4770.0	4883.5	5197.0	5876.0	6364.-
<u>4658.1</u>	4774.8	4884.6	5197.7	<u>5900.-</u>	6369.9
4660.5	4780.0	4889.8	5234.-	5919.3	6374.7
4663.7	4798.5	<u>4893.8</u>	5234.6	5929.3	6379.7
4664.8	4799.8	4911.2	<u>5262.1</u>	5931.8	6456.-
4666.7	4805.1	4921.9	5274.3	5953.6	6482.1
4667.4	4812.4	<u>4923.9</u>	<u>5316.2</u>	5978.5	<u>6563.2</u>
4670.2	4814.5	4993.3	5325.-	6031.1	<u>6570.4</u>
4691.5	<u>4824.0</u>	5005.1	5333.-	6172.5	6572.9
4701.5	<u>4825.7</u>	5007.3	5365.0	6220.6	6578.0
4703.0	4833.2	5015.6	5375.-	6242.5	6582.9
4708.6	4836.2	<u>5018.6</u>	5423.3	6243.4	6610.-
4713.1	4839.9	5041.1	5666.6	6280.-	6678.2
4728.0	4848.3	5047.2	5676.0	6300.1	
<u>4731.4</u>	<u>4849.2</u>	5055.6	5679.6	6317.-	
4733.5	4861.4	5127.3	5686.2	6340.7	
<u>4755.0</u>	<u>4865.5</u>	5157.2	5696.5	6344.1	
4762.7	4871.3	5168.0	5710.8	6347.1	
4763.9	4874.0	5182.0	5739.8	6355.6	
4764.6	4876.5	5194.1	5834.0	6361.4	

A-11. Ae-Fe-Ge stars

Identification wavelengths. The literature cited does not always clearly identify whether the wavelengths for these objects are as observed or as identified, hence the accuracy of the tabulated values depends on the correctness of the identification and the absence of any differential radial velocity effects. The wavelengths in the table below are "as published" values.

	<u>3770.6</u>	4002.4	4305.9	<u>4521.4</u>	5017.2
	<u>3783.4</u>	4012.4	4319.6	<u>4534.2</u>	6300.-
	<u>3797.9</u>	<u>4025.7</u>	4338.4	4541.5	6364.-
	<u>3827.4</u>	<u>4053.8</u>	<u>4340.5</u>	<u>4549.5</u>	6563.-
	<u>3835.4</u>	4068.4	<u>4353.1</u>	<u>4555.9</u>	7291.5
3682.8	3848.2	4076.6	4358.4	4572.0	7328.9
3686.8	<u>3854.9</u>	<u>4101.7</u>	4368.8	4576.3	7664.9
3691.6	<u>3863.2</u>	<u>4122.6</u>	4384.6	<u>4583.3</u>	7699.0
3697.2	<u>3878.7</u>	<u>4128.3</u>	4395.0	<u>4620.5</u>	8498.-
3703.9	3880.8	<u>4138.2</u>	4401.7	<u>4629.3</u>	8542.-
3712.0	<u>3888.8</u>	4162.5	4413.7	4635.3	8662.-
3721.9	<u>3900.5</u>	4173.4	4416.5	4639.7	
3725.3	3906.0	<u>4178.8</u>	4445.0	4657.0	
3734.4	<u>3913.5</u>	<u>4233.2</u>	4451.5	4668.6	
3741.6	<u>3933.7</u>	<u>4244.4</u>	4456.1	4733.0	
3745.4	3935.9	4258.2	4471.8	4774.7	
3746.6	3938.8	<u>4275.0</u>	<u>4481.3</u>	4814.6	
<u>3750.2</u>	3960.9	<u>4286.8</u>	<u>4490.6</u>	<u>4861.3</u>	
<u>3755.6</u>	<u>3969.5</u>	4296.6	<u>4508.3</u>	<u>4905.4</u>	
<u>3759.4</u>	<u>3974.7</u>	4303.2	<u>4515.3</u>	<u>4922.9</u>	

A-12a. T-Tauri stars

	<u>3878.3</u>	<u>4163.6</u>	<u>4351.8</u>	<u>4541.5</u>	<u>5276.0</u>
	<u>3886.3</u>	<u>4171.9</u>	<u>4359.3</u>	<u>4549.6</u>	<u>5316.6</u>
	<u>3889.0</u>	<u>4173.5</u>	<u>4375.9</u>	<u>4555.9</u>	<u>5875.6</u>
	<u>3895.7</u>	<u>4178.9</u>	<u>4383.6</u>	<u>4558.7</u>	<u>5890.0</u>
	<u>3899.7</u>	<u>4182.4</u>	<u>4385.4</u>	<u>4563.8</u>	<u>5895.9</u>
3727.6	3900.5	4202.0	<u>4395.0</u>	<u>4572.0</u>	<u>6562.8</u>
3736.8	3905.5	4215.6	<u>4399.8</u>	<u>4576.3</u>	<u>8498.0</u>
3758.2	3913.5	4226.7	<u>4404.8</u>	<u>4583.5</u>	<u>8542.1</u>
3767.2	3914.5	<u>4233.2</u>	<u>4407.7</u>	<u>4588.2</u>	<u>8662.1</u>
3770.6	3920.3	<u>4244.0</u>	<u>4415.1</u>	<u>4590.0</u>	
3775.6	<u>3922.9</u>	4246.8	<u>4417.2</u>	<u>4616.7</u>	
3783.5	<u>3927.9</u>	4250.8	<u>4427.3</u>	<u>4618.3</u>	
3787.9	<u>3933.7</u>	4254.4	<u>4435.2</u>	<u>4620.5</u>	
3795.0	<u>3938.3</u>	4258.2	<u>4443.8</u>	<u>4629.3</u>	
<u>3797.9</u>	3944.0	4271.8	<u>4450.5</u>	<u>4634.1</u>	
3807.1	3961.5	4274.8	4461.6	4666.8	
3815.8	<u>3968.5</u>	4289.9	<u>4468.5</u>	<u>4670.4</u>	
3820.4	<u>3970.1</u>	<u>4294.1</u>	<u>4474.5</u>	<u>4685.8</u>	
<u>3824.4</u>	<u>4005.2</u>	<u>4296.6</u>	<u>4482.2</u>	<u>4731.5</u>	
<u>3825.9</u>	4026.2	4300.0	<u>4489.2</u>	<u>4824.1</u>	
3827.8	4045.8	4301.9	<u>4491.4</u>	<u>4848.3</u>	
3829.4	<u>4063.6</u>	4303.2	<u>4494.6</u>	<u>4861.3</u>	
3832.3	<u>4068.6</u>	4307.9	<u>4501.3</u>	<u>4923.9</u>	
3835.4	<u>4071.7</u>	4312.9	<u>4508.3</u>	<u>5018.4</u>	
<u>3838.3</u>	4077.7	4314.1	<u>4515.3</u>	<u>5167.3</u>	
3840.4	<u>4101.7</u>	4320.8	<u>4520.2</u>	<u>5169.0</u>	
3850.0	<u>4122.6</u>	4325.4	<u>4522.6</u>	<u>5172.7</u>	
3856.4	<u>4132.1</u>	4330.7	<u>4528.6</u>	<u>5183.6</u>	
3859.1	<u>4139.9</u>	4337.9	<u>4529.5</u>	<u>5197.6</u>	
<u>3872.5</u>	<u>4143.8</u>	<u>4340.5</u>	<u>4534.0</u>	<u>5234.6</u>	

A-12b. Herbig-Haro objects

Low dispersion.

<u>3727</u>	<u>3970</u>	4248	4452	5006	<u>6548</u>
<u>3735</u>	<u>4069</u>	<u>4287</u>	<u>4572</u>	<u>5200</u>	<u>6562</u>
3868	<u>4076</u>	<u>4340</u>	<u>4657</u>	<u>5577</u>	<u>6584</u>
<u>3888</u>	<u>4101</u>	<u>4359</u>	<u>4861</u>	<u>6300</u>	<u>6717</u>
<u>3933</u>	<u>4177</u>	<u>4415</u>	<u>4958</u>	<u>6363</u>	<u>6731</u>

A-13. Ke-Me-N stars

	<u>3521.2</u>	<u>3767.2</u>	3878.0	4012.4	<u>4166.6</u>
	<u>3530.9</u>	<u>3770.5</u>	3878.4	<u>4020.9</u>	<u>4167.5</u>
	<u>3535.2</u>	<u>3774.9</u>	3882.-	<u>4025.1</u>	<u>4170.8</u>
	<u>3545.3</u>	3775.6	3884.7	<u>4030.6</u>	<u>4172.0</u>
	3550.0	3776.5	3886.1	<u>4033.0</u>	<u>4172.8</u>
3162.1	3551.4	3779.6	<u>3887.1</u>	4034.4	<u>4173.6</u>
3163.0	3554.1	3783.5	<u>3888.5</u>	4042.9	<u>4177.3</u>
3164.4	3565.4	3786.8	3889.0	4043.3	<u>4177.6</u>
3166.7	<u>3581.2</u>	<u>3787.8</u>	<u>3894.1</u>	4044.6	<u>4178.8</u>
3170.3	3587.0	<u>3789.4</u>	3898.0	<u>4045.7</u>	<u>4200.0</u>
3183.1	3608.8	3790.1	3903.0	<u>4063.6</u>	<u>4202.1</u>
3185.5	<u>3618.6</u>	3791.0	3905.6	<u>4066.5</u>	<u>4204.2</u>
3186.8	3619.3	3794.9	<u>3907.6</u>	<u>4068.7</u>	<u>4206.6</u>
3192.8	3624.7	3798.0	<u>3913.0</u>	<u>4071.3</u>	<u>4215.5</u>
3193.8	3679.6	<u>3798.5</u>	3914.5	<u>4071.8</u>	<u>4216.0</u>
3196.1	3685.2	3799.6	3917.2	4073.8	<u>4225.3</u>
3204.6	3705.4	3807.1	3922.8	<u>4077.6</u>	<u>4229.5</u>
3208.0	3709.1	3807.5	3927.8	<u>4078.9</u>	<u>4233.1</u>
3210.4	3711.3	3813.0	3930.2	4080.5	<u>4240.4</u>
3213.4	3712.0	<u>3814.0</u>	<u>3932.9</u>	<u>4081.5</u>	<u>4242.5</u>
3223.7	3719.5	3814.6	3936.0	<u>4084.5</u>	<u>4243.8</u>
3227.7	3721.3	3815.9	3938.3	<u>4086.3</u>	<u>4244.0</u>
3245.1	3721.9	<u>3816.4</u>	<u>3940.9</u>	<u>4088.0</u>	<u>4245.3</u>
3246.8	3724.4	3820.4	3941.7	<u>4090.7</u>	<u>4246.8</u>
3253.6	3726.9	<u>3820.8</u>	<u>3943.8</u>	<u>4092.1</u>	<u>4247.4</u>
3255.9	3727.4	3822.9	3945.3	4093.6	<u>4249.4</u>
3266.4	3731.6	3825.8	3950.0	4095.5	<u>4250.3</u>
3269.0	3734.4	<u>3826.2</u>	3951.2	<u>4096.8</u>	<u>4250.7</u>
3277.3	3735.3	<u>3827.8</u>	3952.2	4098.8	<u>4255.6</u>
<u>3281.3</u>	<u>3736.8</u>	<u>3829.4</u>	3952.9	<u>4101.7</u>	<u>4258.3</u>
3284.0	3739.2	3832.3	3956.1	<u>4103.1</u>	<u>4263.2</u>
3285.7	3741.6	<u>3834.1</u>	3957.9	<u>4104.5</u>	<u>4271.7</u>
3290.1	3742.3	<u>3835.4</u>	3961.4	4108.8	<u>4274.6</u>
3295.8	3743.3	<u>3838.3</u>	3967.9	<u>4114.7</u>	<u>4276.8</u>
<u>3302.9</u>	3745.3	<u>3840.4</u>	<u>3970.1</u>	<u>4118.8</u>	<u>4280.5</u>
3303.5	3748.1	3841.1	3972.2	4119.5	<u>4282.6</u>
3314.0	3748.9	<u>3844.9</u>	3977.8	<u>4121.4</u>	<u>4283.8</u>
3371.9	3749.4	3845.5	3978.7	4122.9	<u>4287.4</u>
3421.2	<u>3750.3</u>	<u>3850.0</u>	3983.0	4129.0	<u>4290.3</u>
3439.0	3751.8	3850.8	<u>3993.0</u>	4132.0	<u>4291.6</u>
3444.3	3758.0	3852.7	3997.9	4134.2	<u>4292.6</u>
3460.1	<u>3759.3</u>	<u>3858.3</u>	<u>4001.9</u>	<u>4138.6</u>	<u>4296.7</u>
3466.5	<u>3761.3</u>	<u>3861.2</u>	<u>4003.3</u>	<u>4143.9</u>	<u>4300.1</u>
3477.2	<u>3763.4</u>	3865.5	4006.9	4156.8	<u>4302.1</u>
3518.4	<u>3763.9</u>	3872.5	<u>4010.3</u>	4164.7	<u>4303.0</u>

Ke-Me-N stars, cont.

4305.8	4367.1	4413.8	4494.6	4629.3	5018.4
4307.9	4368.0	4415.2	4501.3	4633.7	5890.-
4315.0	4369.5	4416.3	4509.6	4634.9	5896.-
4319.6	4372.6	4427.3	4511.4	4639.7	6036.1
4320.5	4375.9	4434.1	4515.0	4648.0	6079.4
4325.1	4379.8	4435.0	4518.7	4672.0	6573.-
4325.8	4383.4	4442.4	4520.2	4694.6	7125.8
4331.8	4384.4	4443.9	4521.5	4715.-	7159.3
4338.2	4385.4	4452.1	4522.6	4728.0	7665.-
4340.5	4388.1	4454.4	4528.5	4756.7	7699.-
4345.3	4389.7	4455.3	4531.2	4801.3	7866.1
4346.8	4391.0	4457.0	4533.9	4803.2	
4351.8	4393.4	4457.9	4537.5	4814.5	
4352.8	4394.1	4458.8	4549.5	4838.7	
4353.3	4395.0	4461.6	4556.0	4842.3	
4355.0	4398.0	4466.6	4559.9	4861.2	
4356.6	4399.0	4474.9	4562.1	4866.4	
4358.4	4400.6	4482.1	4571.1	4888.-	
4359.4	4404.8	4488.8	4578.8	4889.6	
4361.8	4405.3	4489.7	4583.9	4923.9	

A-14. Airglow

	3302.6	3479	4155	7586	9680
	3306.0	3484.8	4161	7712	9720
	3309.8	3487.6	4170	7748.3	9788
	3314.1	3490.9	4281	7794.0	<u>9872</u>
	3318.9	3494.8	4293	7821	9915
3060	3320	3499.2	4308	7852	9939
3142	3323.5	3504.2	4401	7861	<u>9948</u>
3147.9	3324.4	3509.8	4404	7871	<u>9956</u>
3150.6	3326.6	3539.2	4438	7888	9962
3153.8	3329.9	3541.0	4578	7911.0	<u>10010</u>
3157.4	3330.2	3552.5	4643	7918.5	10083
3161.4	3366.5	3557.9	4673	7965	<u>10123</u>
3165.9	3367.5	3560.7	4675.0	8027	<u>10170</u>
3170.8	3374.6	3564.0	4819	8062	<u>10190</u>
3176.2	3377.5	3567.8	4835	8090	<u>10211</u>
3209.3	3380.9	3572.2	5200.7	8109	10223
3213.8	3384.7	3577.2	5232.8	<u>8280</u>	<u>10273</u>
3216.0	3389.1	3582.7	5270.3	8299	<u>10373</u>
3218.7	3389.7	3726.2	<u>5577.4</u>	<u>8341.7</u>	<u>10418</u>
3221.9	3392.9	3728.9	<u>5886.3</u>	<u>8399</u>	<u>10469</u>
3225.5	3393.9	3734	5890.0	8430	10525
3229.6	3396.2	3739	5893	<u>8465</u>	<u>10586</u>
3232.1	3399.1	3771	5895.9	<u>8505</u>	<u>10711</u>
3234.0	3404.9	3791.0	6099	8549	<u>10729</u>
3238.4	3414.7	3799.2	6168.6	<u>8598</u>	<u>10828</u>
3239.0	4320.4	3812	6256.0	<u>8620</u>	10924
3242.1	3423.2	3829	6300.2	<u>8629.2</u>	<u>10973</u>
3244.4	3426.6	3840	6363.9	<u>8761</u>	
3257.1	3430.6	3844	6496.5	<u>8768</u>	
3262.3	3435.1	3850	6861.7	<u>8778</u>	
3265.0	3450.1	3860	7240	<u>8824.1</u>	
3268.2	3454.9	3873.8	7274.5	<u>8886</u>	
3271.9	3456.5	3887	7284.2	<u>8920</u>	
3276.1	3457.3	3898.9	7316	<u>8958</u>	
3280.9	3460.2	3914.4	7341	<u>9001</u>	
3284.3	3463.6	3938	7369	<u>9310</u>	
3286.2	3464.2	3940	7401	<u>9324</u>	
3292	3467.2	3974	7521.5	<u>9373</u>	
3297.3	3467.5	4005	7529.1	<u>9440</u>	
3299.7	3471.9	4064	7572	<u>9477</u>	

A-15. City lights

	5881.9	6718	8681.9
	5890.0	6929.5	8704.2
	5895.9	7024.1	8780.6
	5944.8	7032.4	8783.8
	5965.5	7173.9	8853.9
3650.1	5974.6	7245.2	8865.8
<u>3662.9</u>	5975.5	7488.9	8919.5
<u>3663.3</u>	6014	7943.2	8988.6
<u>4046.6</u>	6030.0	8082.5	9148.7
4077.8	6074.3	8136.4	9201.8
4181.9	6143.1	8266.1	9665.4
4190.7	6163.6	8300.3	
4198.3	6217.3	8376.4	
4200.7	6266.5	8377.6	
4259.4	6334.4	8418.4	
4331.3	6383.0	8495.4	
<u>4358.3</u>	6402.2	8591.3	
<u>5769.6</u>	6506.5	8634.6	
5790.7	6599.0	8654.4	
5852.5	6678.3	8679.5	

A-16. Aurora

	3857.2	4176.2	4661.6	5436.8	6039.8
	3857.9	4185.5	4673.7	5512.7	6046.3
	3875.8	4189.8	4673.8	5512	6046.5
	3882.5	4192.5	4676.2	5540.7	6047
	3883.2	4199.1	<u>4709.2</u>	<u>5577.4</u>	6055
<u>2972.3</u>	3884.3	4199.8	4771	5596.9	6063
<u>3116.7</u>	3887.6	4200.0	4774.2	5601.8	6070
3136.0	3891.0	4200.5	4779.7	5609.8	6097.5
3159.3	<u>3898</u>	<u>4236.5</u>	4781.2	5616.5	6103
3168.7	<u>3914.4</u>	4241.8	4788.1	5621.3	6111
3202.5	3940.3	<u>4278.1</u>	4793.7	5625.8	6120
3268.1	3943.0	4294.8	4803.3	5630.3	6127
3285.3	3945.0	4317.1	4810.3	5631.9	6156.0
3309	3947.3	4319.6	4837.1	5666.6	6156.8
3339.3	3947.5	4328.6	4861.3	5676.0	6158.2
3340.7	3947.6	4331.1	4890.9	5679.6	6160
3367.4	3953.0	4331.5	4895.2	5686.2	6164
<u>3371.3</u>	3953.1	4336.9	4914.9	5710.8	6185.2
<u>3424.6</u>	3954.4	4340.5	4924.6	5730.7	<u>6300.2</u>
3428.7	3954.6	4345.6	4957.9	5747.3	<u>6363.9</u>
3466.4	3954.7	4347.4	4961.1	5754.8	6482.1
3466.5	3955.9	4349.4	4968.8	5804.3	6512
3467.5	3973.3	4351.3	4976.4	5812	6527
3470.4	3979.1	4357.3	4987.4	5890.0	6536
3484	3982.7	4368.3	4994.4	5892	6545
3502.0	3985.5	4414.9	5001.1	5895.9	6562.8
3515.9	3995.0	4417.0	5001.5	5899	6583.6
3536.7	3998.4	4488.1	5002.7	5906.0	6597
3563.9	4026.1	4488.2	5005.1	5936	6624
3576.9	4035.1	4507.6	5006.8	5951	6677
3582.1	4041.3	4535.5	5016.4	5959.0	6704.8
3603.0	4041.8	4564.8	5025.7	5967	6753
3671.9	4043.5	4591.0	5045.1	5973.4	6756
3686	4044.8	4596.2	5148.8	5976.6	6768
3692.4	4045.0	4599.7	5175.9	5984.2	6788.6
3710.5	4046.2	4601.5	5176.0	5987.1	6853.0
3712.8	4059.4	4602.1	5198.5	5990.5	6861
3726.2	4069.6	4607.2	5200.7	5991	6875
3728.9	4069.9	4613.9	5206.7	5993.5	7036.8
3755.4	4072.2	4621.4	5228.3	5998.0	7059
3771.6	4072.5	4630.5	5230.4	5999	7165
<u>3804.9</u>	4075.9	4641.8	5259	6002.0	7239.9
<u>3822.1</u>	4078.9	4643.1	5295.7	6007	7274.5
3835.4	4171.2	4650.8	5330.7	6016.8	7318.6
3855.1	4171.6	4651.8	5356.8	6026.4	7319.4

Aurora, cont.

7329.9	<u>7772.0</u>	7987.3	8343	<u>8703.2</u>	<u>10404.1</u>
7330.7	<u>7774.2</u>	7995.1	8369.2	<u>8711.7</u>	<u>10510.0</u>
7368	<u>7775.4</u>	8053.6	8446.4	<u>8718.8</u>	
<u>7386.6</u>	<u>7825.7</u>	<u>8184.8</u>	<u>8541.8</u>	<u>8722.3</u>	
<u>7479</u>	<u>7860</u>	8188.0	<u>8598</u>	<u>8728.9</u>	
<u>7503.9</u>	<u>7880</u>	8200.3	8629.2	<u>8747.4</u>	
<u>7580</u>	<u>7915</u>	8216.3	8660	<u>8911.6</u>	
<u>7626.2</u>	7982.0	8223.1	<u>8680.2</u>	<u>9145</u>	
<u>7684.3</u>	7982.4	8242.3	<u>8683.4</u>	<u>9431.2</u>	
<u>7753.2</u>	7987.0	8293.4	<u>8686.1</u>	<u>10395.4</u>	

A-17. Lightning

	3973.3	4414.9	5005.1	7156.8	8718.8
	3982.7	4417.0	5045.1	<u>7380</u>	<u>8728.9</u>
	3995.0	4428.0	5171	7423.6	<u>8820.5</u>
	<u>3998.4</u>	4431.8	5181	<u>7442.3</u>	9060.6
	4021.0	4432.7	5187	<u>7468.3</u>	9260.9
3159.3	4026.1	4442.0	5202	7476.5	9392.8
3290.2	4035.1	4530.4	5281.2	7593.7	
3330	4041.3	4530.8	5328.7	<u>7774.2</u>	
3360	<u>4043.5</u>	4552.5	5356.8	<u>7947.2</u>	
3371.3	<u>4044.8</u>	4553.1	5436.8	<u>7947.6</u>	
3437.2	4059.4	4599.7	5460.7	8021.1	
3532.6	<u>4073.1</u>	4607.2	5616.5	<u>8115.3</u>	
3536.7	4099.9	4621.4	5666.6	<u>8184.8</u>	
3540	<u>4101.7</u>	4651.8	5676.0	<u>8188.0</u>	
3548.9	<u>4102.2</u>	<u>4669.8</u>	5999.5	<u>8200.3</u>	
3563.9	4137.6	<u>4675.0</u>	6008.5	8216.3	
3582.1	4167.8	<u>4709.2</u>	6012.0	<u>8221.8</u>	
3587.2	4214.7	4774.2	6015.4	<u>8223.1</u>	
3592	<u>4215.9</u>	4779.7	6017.7	<u>8231</u>	
3595	<u>4244.7</u>	4781.2	6095	<u>8446.4</u>	
3755.4	4236.5	4788.1	6156.0	8446.8	
<u>3804.9</u>	<u>4236.9</u>	4861.3	<u>6158.2</u>	<u>8567.7</u>	
<u>3914.4</u>	4237.0	4914.9	<u>6437.0</u>	<u>8594.0</u>	
<u>3919.0</u>	4241.8	4935	6462	<u>8629.2</u>	
3943.0	<u>4254.7</u>	<u>4963</u>	6481.7	<u>8655.9</u>	
3947.3	4278.1	4973	6482.7	8680.2	
<u>3947.5</u>	<u>4317.1</u>	4987.4	6483.8	<u>8683.4</u>	
<u>3947.6</u>	4319.6	4994.4	6562.8	<u>8686.1</u>	
<u>3954.4</u>	4368.3	5001.1	<u>6610.6</u>	<u>8703.2</u>	
<u>3955.9</u>	<u>4402</u>	5001.5	6723.1	<u>8711.7</u>	

Table B. IDENTIFICATIONS AND INTENSITIES

This table gives the identifications of the lines listed in Table A, as proposed by the various observers. The several parts of the table are discussed below.

Wavelength (Angstroms).--The table is arranged by wavelength. The wavelength identification is the laboratory wavelength of the assumed source ion or molecule, measured in air. (When an author did not identify an emission, or when the emission is not listed in the Revised Multiplet Tables,* the observed wavelength is quoted.)

V or R following the wavelength indicates a molecular band head, with shading to violet or red, respectively.

Identification, transition, and excitation potential.--Both atomic and molecular transitions are shown in these columns.

Atomic.--The atomic transitions are denoted by the ion and multiplet number used in the Revised Multiplet Tables. Because of the limitations of tabulating by computer, the form of the multiplet notations was modified from the Revised Multiplet Tables, as follows:

$$a^2P - z^6P^{\circ} = \begin{cases} A & 2P \\ Z & 6P \ 0 \end{cases}$$

In the interest of space-saving, the low term is printed on the first line and the high term is printed immediately below it. The J-values are likewise printed on two lines.

The excitation potential (of the upper level of the transition above the ground level of the ion) is in electron volts.

A hyphen (-) alone in the identification column indicates that the source ion is not identified. A hyphen following the ion indicates that this line is not listed in the Revised Multiplet Tables.

A dash (--) in place of the multiplet number indicates that the line is listed in the Revised Multiplet Tables but is unclassified.

An F following the multiplet number indicates a forbidden line.

*Moore, Charlotte E., 1945, *A Multiplet Table of Astrophysical Interest*, revised edition, *Part I. Table of Multiplets*, Contributions from the Princeton University Observatory, No. 20, 110 pp.

Molecular.--The molecular transitions are denoted by the molecular species and band "name" (under ION AND MULTIPLY NUMBER), and vibrational transition, v',v'' , (under LEVELS).

The following notation is used to designate the molecular transitions:

AlH	The Herbig bands of AlH ($A^1\Pi - X^1\Sigma^+$)
AlO	($A^2\Sigma^+ - X^2\Sigma^+$)
N2 1P	The first positive system of N ₂ ($B^3\Pi_g - A^3\Sigma_u^+$)
N2 2P	The second positive system of N ₂ ($C^3\Pi_u - B^3\Pi_g$)
N2 VK	The Vegard-Kaplan system of N ₂ ($A^3\Sigma_u^+ - X^1\Sigma_g^+$)
N2+ 1N	The first negative system of N ₂ ⁺ ($B^2\Sigma_u^+ - X^2\Sigma_g^+$)
N2+ M	The Meinel system of N ₂ ⁺ ($A^2\Pi - X^2\Sigma_g^+$)
NO β	The β system of NO ($B^2\Pi - X^2\Pi$)
O2 H	The Herzberg system of O ₂ ($A^3\Sigma_u^+ - X^3\Sigma_g^-$)
O2 SR	The Schumann-Runge system of O ₂ ($B^3\Sigma_u^- - X^3\Sigma_g^-$)
O2 A	The atmospheric system of O ₂ ($b^1\Sigma_g^+ - X^3\Sigma_g^-$)
O2+ 1N	The first negative system of O ₂ ⁺ ($b^4\Sigma_g^- - a^4\Pi_u$)
O2+ 2N	The second negative system of O ₂ ⁺ ($A^2\Pi_u - X^2\Pi_g$)
OH M	The Meinel system of OH (vibration-rotation bands, $X^2\Pi-X^2\Pi$)

Molecular excitation potentials are given only approximately, since any one band (v', v'') is made up of rotational transitions from an appreciable range of potentials.

Intensity scale.--The tabulation of intensities of emission features presents several problems.

One problem of considerable complexity is correlating the "intensity" scales used by different observers. Older publications used a relative, or logarithmic, scale that is sufficient to rank the lines by intensity but gives little information concerning relative line fluxes in energy units. Recent publications tend to use a linear scale so that intensity numbers can be used in such matters as comparing line intensity ratios. Evaluation of several references concerning observations of the same object gives the following approximate relationship between "eye estimates" (a logarithmic intensity scale) and "intensity estimates" (a linear intensity scale, as with a photocell detector):

<u>Visual estimates</u> <u>(log scale)</u>	<u>Intensity estimates</u> <u>(linear scale)</u>
0	W (weak)
1	W (weak)
2	1
3-4	5
5-6	10
7-8	30
9	50
10	100
12	300

The intensity scale used in this table is approximately linear, and it is based on an intensity of 100 for $H\beta$, since $H\beta$ is common to nearly all object spectra. In the case of some important objects, such as χ Cygni, the hydrogen lines are not visible in emission at the time of maximum visibility of the metal and other low excitation lines. In the few cases where $H\beta$ cannot be used, we have arbitrarily selected one strong emission feature to be 100. Such is the case for observational contaminants to astrophysical spectra (airglow, city lights, aurora, and lightning).

Many intensities quoted in the literature, particularly in the older publications, are eye estimates from the actual photographic plate, uncorrected for the wavelength dependence of instrumental and photographic plate sensitivity. We have generally avoided having to estimate corrections by using the more recent publications wherein the intensities have been calibrated by use of photoelectric devices. In a few cases in which intensities are unreliable because of correction uncertainties or are unreported, the presence of that line in a given source column is indicated by an X.

Another problem in tabulating intensities is that many of the object classes and even individual objects have variable spectra. In such cases we have indicated the range to be expected. To save space, the limits of the range are printed on separate lines; that is, the range 10-50 would be printed as $\frac{10}{50}$.

In some objects the emission lines dominate the spectrum, whereas in others the lines are scarcely observable above the object continuum. We have not attempted to evaluate the visibility of the emission lines in the latter case, since this depends very strongly (in this case) on the intrinsic sharpness of the emission lines and the dispersion (resolution) of the spectrograph.

It should be emphasized that, while we have attempted to give intensities of the lines ascribed to the listed identifications, the values are primarily to be used to indicate the approximate importance of such lines in the spectrum of the object. The table would be quite unreliable if used for astrophysical analysis, such as in the study of multiplet intensities.

Two symbols indicate properties of a line blended with a neighboring line: B means that the line is most frequently observed as a blend with a neighboring line or, if the identification is uncertain, it refers to a maximum in the blended feature. M means that the line is probably a minor contributor to the blend.

Summary of notations.--The notations discussed above are summarized below for easier reference.

V or R	Indicates direction of shading of molecular band, to violet or red, respectively.
-	Source ion is not identified, or line is not listed in Moore's Revised Multiplet Tables.
-- (in lieu of multiplet)	Line is listed in the Revised Multiplet Tables but is unclassified.
F (after the multiplet)	A forbidden line.
X	Literature gives insufficient information to assign an intensity to this line.
W	Visual intensity is 0 or 1 on the log visual scale, or about .1 or less on the linear intensity scale. (A few intensities as low as .01 are listed, but these are only for gaseous and planetary nebulae that have been measured recently by photoelectric methods.)
B	This line is most frequently observed as a blend with a neighboring line; or, where the identification is uncertain, the wavelength refers to a maximum in the blended feature.
M	Probably a minor contributor to the blend.

Some further notations are given as footnotes to the tables.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J		NUC EM	NOVA GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE	AE	T-	KE	AIR	CTY	AUR	LGT
													BE	FE	TAU	ME				
1084.6	N II 1	2P2 3D 2P3 3D 0 2	1	11.39	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1215.7	H 1	1S 2S 2P 2P 0	1/2 3/2	10.15	2 5B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1215.7	H 1	1S 2S 2P 2P 0	1/2 1/2	10.15	2 5B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1260.7	SI II 4	3P 2P 0 3D 2D	1/2 3/2	9.79	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1265.0	SI II 4	3P 2P 0 3D 2D	3/2 5/2	9.80	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1304.4	SI II 3	3P 2P 0 3P2 2S	1/2 1/2	9.46	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1309.3	SI II 3	3P 2P 0 3P2 2S	3/2 1/2	9.46	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1367	-	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1393.7	SI IV 1	3S 2S 3P 2P 0	1/2 3/2	8.86	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1402.7	SI IV 1	3S 2S 3P 2P 0	1/2 1/2	8.80	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1407	O IV	.	.	.	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1443.1	-	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1449.7	-	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1488	N IV	1S 3P 0	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1508	-	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1548.2	C IV 1	2S 2S 2P 2P 0	1/2 3/2	7.97	2 10B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1548.2	C IV 1	2S 2S 2P 2P 0	1/2 1/2	7.97	2 10B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1551	C IV	.	.	.	2 10B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1564	-	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1640.4	HE II 12	2P 2P 0 3D 2D	3/2 5/2	48.16	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1640.4	HE II 12	2P 2P 0 3D 2D	1/2 3/2	48.16	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1640.4	HE II 12	2P 2P 0 3D 2D	3/2 3/2	48.16	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1663	O III -F	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1909	C III	1S 3P 0	.	.	1 6	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1934	-	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2424	NE IV	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2470	O II -F	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2506	MG VII -F	3P 1D	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2626	MG VII -F	3P 1D	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2795.5	MG II 1	3S 2S 3P 2P 0	1/2 3/2	4.41	1 10B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2802.7	MG II 1	3S 2S 3P 2P 0	1/2 1/2	4.40	1 10B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2854	A IV -F	4S 2P 0	.	.	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2869	A IV -F	4S 2P 0	.	.	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2928	MG V F	.	.	.	1 2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
2972.3	O I 2F	2P4 3P 2P4 1S	1 0	4.17	.	.	.	.	.	.	.	.	.	.	.	.	.	.	300	.
2974	NE V 2F	2P2 1D 2P2 1S	.	.	1 2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3013.1	FE III 9	A 5F Z 5F 0	5 5	14.36	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.
3023.9	FE III 10	A 5F Z 5D 0	2 2	14.37	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
3028.1	ZR II 76	A 2G Y 2G 0	7/2 7/2	5.04	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
3029.0	MN II 10	Z 5P 0 E 5S	3 2	9.43	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLYET NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J																	
					NUC EM QUAS	GAL	NOVA	LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	Symb	Oe BE PCYG	Ae FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA
3108.9	TI II 77	B 2D X 2D U	5/2 3/2	5.54	.	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	
3109.0	A III 2F	3P4 3P 3P4 1S	1 0	4.11	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	
3110.9	ZR II 5	A 4F Z 4D U	7/2 7/2	4.06	.	.	.	.WB	.	.	.	.	.	.	.	.	.	.	.	
3114.3	FE II 82	B 4D X 4D O	3/2 3/2	7.83	.	.	.	.WB	.	.	.	.	.WB	.	.	.	.	.	.	
3114.7	FE II 82	B 4D X 4D U	1/2 3/2	7.83	.	.	.	.WB	.	.	.	.	.	.	.	.	.	.	.	
3115.7	CR II 46	B 4F Y 4D U	7/2 7/2	8.01	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3116.6	FE II 82	B 4D X 4D O	5/2 3/2	7.83	.	.	.	.WB	.	.	.	.	.WB	.	.	.	.	.	.	
3116.7V	N2 2P	3 2	.	11.1	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	
3116.8	CR II 126	B 2G Y 2F O	7/2 5/2	8.71	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3118.7	CR II 5	A 4D Z 4F U	1/2 3/2	6.37	.	.	.	.WB	.	.	.	.	.	.	.	.	.	.	.	
3120.0	FE II 96	B 2F Y 4G U	7/2 7/2	7.91	.	.	.	.WB	.	.	.	.	.	.	.	.	.	.	.	
3120.4	CR II 5	A 4D Z 4F O	3/2 5/2	6.38	.	.	.	.WB	.	.	.	.	.WB	.	.	.	.	.	.	
3121.7	O III 12	3P 3S 3D 3P U	1 1	40.69	.	.	.	.	.	XB	.	.42 .84	1	.	.	.	.	.	.	
3122.6	O II 14	3P 4D O 4S 4P	5/2 5/2	29.49	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	
3123.7	FE II --	.	.	.	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3124.2	FE II 12F	A 6D B 2F	9/2 7/2	3.95	.	.	.	.WB	.	.	.	.	.WB	.	.	.	.	.	.	
3125.0	CR II 5	A 4D Z 4F O	5/2 7/2	6.39	.	.	.	.WB	.	.	.	.	.WB	.	.	.	.	.	.	
3125.3	V II 1	A 5F Z 5G O	1 2	4.27	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3125.5	CR II 55	B 4G Z 2G O	11/2 9/2	8.11	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3125.7	FE I 194	A 3H U 3G U	6 5	6.34	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3127.6	-	.	.	.	.	.	.	.	.	.	.21	.	.	.	.	.	.	.	.	
3128.6	TI II 121	Z 2F O E 2F	7/2 7/2	7.83	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3128.7	CR II 5	A 4D Z 4F O	3/2 3/2	6.37	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3129.0	FE II 96	B 2F Y 4G O	7/2 9/2	7.89	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3131.7	FE II 107	A 2I Y 2G U	11/2 9/2	8.00	.	.	.	.WB	.	.	.	.	.WB	.	.	.	.	.	.	
3132.1	CR II 125	B 2G X 4G O	7/2 7/2	8.69	.	.	.	.WB	.	.	.	.	.WB	.	.	.	.	.	.	
3132.9	O III 12	3P 3S 3D 3P O	1 2	40.67	1 2	.	.	.	.	XB	.	.6 .45	4	.	.	.	.	.	.	
3133.0	FE II 82	B 4D X 4D O	3/2 5/2	7.81	.	.	.	.WB	.	.	.	.	.WB	.	.	.	.	.	.	
3134.8	O II 14	3P 4D O 4S 4P	7/2 5/2	29.49	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	
3135.4	CR II 124	B 2G Y 2H U	7/2 9/2	8.69	.	.	.	.	.	.	.	.	.WB	.	.	.	.	.	.	
3135.4	FE II 82	B 4D X																		

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY													AIR GLW	CTY LTS	AUR ORA	LGH NING						
		LEVELS	J		NUC EM	NOVA	W-R LI	W-R N	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME													
															QUAS	GAL	NOVA				LI	CE	N	C	NEB	NEB	SYMB
3216.6	CR II 82	B 2F Z 2D U	5/2 3/2	8.23	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3217.1	TI II 2	A 4F Z 4F U	7/2 9/2	3.86	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3217.1	V II 38	B 3G Z 3H U	5 6	5.88	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3217.4	CR II 9	A 4G Z 4F U	5/2 3/2	6.37	.	.	.	W	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3217.5	FE II -F	.	.	.	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3218.7	O2 H	3 4	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
3219.1	CR II 140	C 2G X 4G U	9/2 11/2	8.75	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3221.9	O2 H	3 4	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
3222.6	FE II -F	.	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3222.8	TI II 2	A 4F Z 4F U	5/2 7/2	3.84	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3223.3	NI II -F	.	.	.	.	.	.	WB	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3223.4	FE II --	.	.	.	.	.	.	WB	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3223.7	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.
3224.2	TI II 84	A 2H Y 2G U	11/2 9/2	5.40	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3224.5	FE II 11F	A 6D B 4D	7/2 5/2	3.87	.	.	.	W	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3225.5	O2 H	3 4	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.
3226.4	FE II 17B	C 2F W 2F U	5/2 5/2	9.37	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3227.0	FE II 11F	A 6D B 4D	7/2 3/2	3.87	.	.	.	.	.	.	.	.	100B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3227.7	FE II 6	A 4P Z 4U U	5/2 7/2	5.49	.	.	.	W 2B	.	.	.	.	100B	.	.	.	W 5	.	.	.	.	.	.	.	.	.	.
3229.6	O2 H	3 4	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.
3230.2	FE II 12F	A 6D B 2F	3/2 5/2	3.93	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3231.7	ZR II 3	A 4F Z 4F U	5/2 7/2	3.86	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3231.7	FE II 80	B 4D Z 2G U	5/2 7/2	7.69	.	.	.	WB	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3232.1	O2 H	5 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
3232.3	TI II 36	A 2G Y 2F U	7/2 5/2	4.03	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3232.8	FE II 119	C 2G Z 2F U	7/2 7/2	7.95	.	.	.	W	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3233	C IV -	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3234.0	O2 H	3 4	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.
3234.0	SI III 6	4 3P U 5 3S	1 1	25.44	.	.	.	.	.	XB	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3234.1	CR II 63	A 2H Z 2G U	3/2 7/2	8.09	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3234.5	TI II 2	A 4F Z 4F U	9/2 9/2	3.86	.	.	.	.	.	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.
3234.9	FE II 1	A 4D Z 6D U	7/2 5/2	4.80	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3236.6	TI II 2	A 4F Z 4F U	7/2 7/2	3.84	.	.	.	.	.	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.
3237.4	FE II 81	B 4D Y 4F U	3/2 3/2	7.68	.	.	XB	WB	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3237.8	FE II 81	B 4D Y 4F U	1/2 3/2	7.68	.	.	XB	WB	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3238.4	O2 H	5 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.
3238.5	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3238.6	O III 9	3P 3D 3D 1D U	1 2	40.09	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3238.8	CR II 63	A 2H Z 2G U	11/2 9/2	8.11	.	.	.	WB	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3239.0	O2 H	3 4	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPL NUMBER	TRANSITION		UPPER EXC. PCT., EV	EMISSION LINE INTENSITY															
		LEVELS	J																	
					NUC EM	NOVA LIKE	H-R N	H-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME	AIR GLW	CITY LTS	AUR ORA	LIGHT NING		
3239.0	TI II 2	A 4F Z 4F D	5/2	3.82	.	.	.	WB	.	.	.	.	WB	.	.	.	.1B	.	.	.
3239.1	CR II 5F	A 6S B 4P	5/2	3.81	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	
3239.4	FE I 157	Z 7D O F 5D	4	6.22	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	
3239.7	-	.	.	.	.	.	.	.	.	.09	.	.	.	.	.	.	.	.	.	
3239.7	FE III 6F	A 5D A 3D	4 3	3.81	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	
3241.0	ZR II 4	A 4F Z 2D O	5/2	3.85	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	
3241.7	SI III 6	4 3P O 5 3S	2 1	25.44	.	.	.	.	.	.08	.	.	.	.	.	.	.	.	.	
3241.7	FE II 80	B 4D Z 2G O	7/2	7.69	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	
3241.7	NA IV -F	2P4 3P 2P4 1D	2	.	.	.	.	.	.	1.0 X	.	.	.	.	.	.	.	.	.	
3242.0	TI II 2	A 4F Z 4F O	3/2	3.81	.	.	.	.	.	.	.	.	.	.	.	.	.1B	.	.	
3242.1	O2 H	3 4	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	1.	.	
3243.3	NE II 15	3P 4D O 3D 2D	5/2	34.60	.	.	.	.	.	.12B	.	.	.	.	.	.	.	.	.	
3243.7	FE II 119	C 2G Z 2F O	9/2	7.94	.	.	.	W	.	.	.	WB	.	.	.	.	.	.	.	
3244.2	NE II 14	3P 4D O 3D 2F	5/2	34.60	.	.	.	.	.	.12B	.	.	.	.	.	.	.	.	.	
3244.2	FE II 11F	A 6D B 4D	5/2	3.89	.	.	.	W	.	.	.	WB	.	.	.	.	.	.	.	
3244.4R	O2 H	5 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	1.	.	
3245.0	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	
3246.4	-	.	.	.	.	.	.	.	.	.12 W	.	.	.	.	.	.	.	.	.	
3246.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	
3247.2	FE II 81	B 4D Y 4F O	3/2	7.67	.	.	.	1	.	.	.	WB	.	.	.	.	.	.	.	
3247.3	CR II 81	B 2F Y 4P O	5/2	8.19	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	
3247.4	FE II 119	C 2G Z 2F O	7/2	7.94	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	
3248.2	FE I 157	Z 7D O F 5D	3	6.24	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	
3250.8	CR II 61	A 2H Z 4G O	9/2	8.07	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
3251.9	TI II 2	A 4F Z 4F O	5/2	3.81	.	.	.	.	.	.	.	.	.	.	.	.	.1B	.	.	
3252.9	TI II 2	A 4F Z 4F O	1/2	3.82	.	.	.	WB	.	.	.	.	.	.	.	.	.1B	.	.	
3253.7	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	W	.	.	
3254.2	FE II 11F	A 6D B 4D	5/2	3.87	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	
3254.3	TI II 2	A 4F Z 4F O	9/2	3.84	.	.	.	1B	.	.	.	.	.	.	.	.	.1B	.	.	
3255.2	-	.	.	.	.	.	.	.	.	.01	.	.	.	.	.	.	.	.	.	
3255.9	FE II 1	A 4D Z 6D O	7/2	4.77	.	.	.	2	.	.	.	50	.	.	.	.	1 30	.	.	
3256.3	FE II 11F	A 6D H 4D	5/2	3.87	.	.	.	.	.	.	.	5B	.	.	.	.	.	.	.	
3256.7	FE II 11F	A 6D B 4D	5/2	3.87	.	.	.</													

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LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY													AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM	NOVA	W-R LIKE	W-R N	PLAN NEB	GAS NEB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
															QUAS	GAL	NOVA				
3326.2	O III 28	3P 5P U 3D 5P	1 1	49.15	.	.	.	.	.	.	.12	.	.	.	.	.	.	.	.		
3326.6	O2 H	6 6	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.		
3327.6	FE II 64	B 2G Z 4G U	7/2 9/2	7.51	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.		
3329.5	TI II 7	B 4F Z 4F U	7/2 7/2	3.84	.	.	.	.	.	.	.	W	.	.	.	1	.	.	.		
3329.9	FE III 18	B 3H Z 3H U	4 4	14.65	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.		
3329.9	O2 H	4 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.		
3330	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W		
3330.2	O2 H	6 6	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.		
3333.2	SI II 6	4 2P U 6 2S	1/2 1/2	13.73	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.		
3334.9	NE II 2	3S 4P 3P 4D U	5/2 7/2	30.75	.	.	.	.	.	.	.03 .28	.28	.	.	.	.	.	.	.		
3334.9	FE III 6F	A 5D A 3D	2 2	3.79	.	.	.	.	.	.	.28	.	.	.	.	.	.	.	.		
3335.2	TI II 7	B 4F Z 4F U	5/2 5/2	3.82	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.		
3338.5	FE II 76	B 4D Y 4P U	5/2 3/2	7.57	.	.	.	103	.	.	.	.	.	.	.	.	.	.	.		
3338.8	FE III --	.	.	.	.	.	.	.	.	.08	.	.	.	.	.	.	.	.	.		
3339.1	FE II 27F	A 4F B 2F	9/2 5/2	3.93	.	.	.	103	.	.	.	.	.	.	.	.	.	.	.		
3339.3V	N2 2P	1 1	.	11.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	20		
3339.4	FE III 7	A 5F Z 5G U	1 2	14.02	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.		
3339.8	CR II 4	A 4D Z 6D U	3/2 3/2	6.12	.	.	.	103	.	.	.	WB	.	.	.	.	.	.	.		
3340.3	TI II 7	B 4F Z 4F U	3/2 3/2	3.81	.	.	.	.	.	.	.	WB	.	.	.	2B	.	.	.		
3340.4	V II -F	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.		
3340.7	O III 3	3S 3P U 3P 3S	2 1	36.73	.	.	2B	.	XB	XB	.13 3.30	4 20	.	.	.	.	.	.	1.28		
3341.9	TI II 16	A 2F Z 2G U	5/2 7/2	4.26	.	.	.	.	.	.	.	.	.	.	.	2B	.	.	.		
3342.5	CR II 4	A 4D Z 6D U	5/2 5/2	6.14	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.		
3342.7	CL III 2F	3P3 4S U 3P3 2P U	3/2 3/2	3.69	.	.	.	.	.	.	1.06 .28	.	.	.	.	.	.	.	.		
3342.8	N III 7	3S 2P U 3P 2S	1/2 1/2	40.38	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.		
3342.9	NE III 2F	2P4 1D 2P4 1S	2 0	6.88	.	.	4B	3	.	.	.20 .28	.	.	.	.	.	.	.	.		
3345.9	NE V 1F	2P2 3P 2P2 1D	1 2	3.74	1	20	2 4	.	.	.	.43 12B .13B	1 8	.	.	.	.	.	.	.		
3345.9	NE II 10	3S1 2D 3P1 2P U	3/2 3/2	34.11	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.		
3345.9	NE II 12	3P 4D U 3D 4D	5/2 3/2	34.48	.	.	.	.	.	.	.13B	.	.	.	.	.	.	.	.		
3347.8	CR II 4	A 4D Z 6D U	3/2 1/2	6.11	.	.	.	2	.	.	.	W 5	.	.	.	.	.	.	.		
3348.1	O IV 4	3S 2P U 3P 2D	1/2 3/2	59.59	.	.	.	.	XB	XB	.	.	.	.	.	.	.	.	.		
3348.1	O III 28	3P 5P U 3D 5P	3 2	49.15	.	.	.	.	.	XB	.08	.	.	.	.	.	.	.	.		
3349.0	TI II 16	A 2F Z 2G U	7/2 9/2	4.29	.	.	.	.	.	.	.	.	.	.	.	2B	.	.	.		
3349.1	O IV 4	3S 2P U 3P 2D	3/2 5/2	59.62	.	.	.	.	.	XB	.15	.	.	.	.	.	.	.	.		
3349.4	TI II 1	A 4F Z 4G U	9/2 11/2	3.73	.	.	.	2B	.	.	.	.	.	.	.	2B	.	.	.		
3350.4	NI II 1	B 2D Z 4D U	3/2 3/2	6.62	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.		
3351.0	O III 22	3S 5P 3P 5P U	3 3	45.46	.	.	.	.	XB	.	.19	.	.	.	.	.	.	.	.		
3352.8	CO II 2	4S 5P 4P 5D U	2 1	5.91	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.		
3353.1	CR II 4	A 4D Z 6D U	7/2 9/2	6.15	.	.	.	1B	.	.	.	1	.	.	.	.	.	.	.		
3353.4	CL III 2F	3P3 4S U 3P3 2P U	3/2 1/2	3.68	.	.	.	.	.	.	.26B .08	.	.	.	.	.	.	.	.		

G-10				EMISSION LINE INTENSITY																
LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV																
		LEVELS	J		NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME	AIR GLW	CTY LTS	AUR ORA	LGHT NING		
3369.4	O III 11	3P 3S	1	40.40	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.
3370.2	O II 52	3D 3D U	1	32.22	.	.	.	.	.	.	.05	.	.	.	.	.	.	.	.	.
3370.4	S III 2	5P 4D U	3/2	21.33	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
3371.3	-	3D 3P U	2	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
3371.3V	N2 2P	4P 3P	1	11.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	50	108
3372.0	NI I 7	0	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.
3372.2	SC II 4	A 3F	3	3.82	.	.	.	.	.	.	.	.	.	.	.	.	.	.	38	.
3372.8	TI II 1	Z 3G U	4	3.68	.	.	.	.	.	.	.	.	.	.	.	.	.	.	38	.
3374.0	NI II 1	A 3D	3	3.67	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3374.1	N III 5	Z 3P U	2	39.18	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3374.6	O2 H	A 4F	9/2	3.89	.	.	.	.	.	.	.	.	.	.	.	.	.	.	9	.
3376.2	FE II 26F	B 4D	7/2	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3377.5	O2 H	Z 4G U	7/2	3.65	.	.	.	.	.	.	.	.	.	.	.	.	.	.	8	.
3378.6	NI II 5F	B 2D	5/2	6.74	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3379.8	CR II 21	Z 4D U	5/2	3.70	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3380.3	TI II 1	A 4F	9/2	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	7	.
3380.9	O2 H	Z 4G U	9/2	9.20	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3381.0	FE II 177	C 2F	5/2	57.81	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3381.3	O IV 3	X 2D U	3/2	49.10	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3382.7	O III 27	3S 4P U	5/2	3.65	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3383.8	TI II 1	3P 4D	3	49.10	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3383.9	O III 27	3D 5D	2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3384.3	O V -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3384.7	O2 H	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	6	.
3385.0	O III 27	3P 5P U	3	49.11	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3385.6	O IV 3	3D 5D	4	57.84	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3386.0	-	3S 4P U	5/2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3386.5	FE II 88	3P 4D	7/2	7.57	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3386.7	FE II --	B 2F	5/2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3387.1	FE II 26F	Y 4P U	3/2	3.87	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3387.1	S III 2	A 4F	9/2	21.31	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3387.8	-	B 4D	5/2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3389.1	O2 H	3D 3P U	1	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.
3389.7	O2 H	4P 3P	0	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
3390.3	O II 9	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3391.4	CR II 3	3P 2S U	1/2	28.82	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3392.0	ZR II 1	3D 2P	3/2	6.05	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3392.9	O2 H	A 4D	1/2	6.05	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
3393.0	CR II 21	Z 4P U	3/2	3.80	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3393.9	O2 H	A 4F	9/2	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3393.9	O2 H	B 4D	1/2	6.73	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3393.9	O2 H	Z 4D U	3/2	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	4	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J		NUC EM		NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGHT NING
					QUAS	GAL														
3393.9	CR II 21	R 4D Z 4D U	3/2 3/2	6.73	.	.	.	20B	.	.	.	.	.	.	.	.	.	.	.	.
3394.3	O III 27	3P 5P U 3D 5D	3 3	49.10	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.
3394.3	CR II 21	H 4D Z 4D U	5/2 3/2	6.73	.	.	.	20B	.	.	.	.	.	.	.	.	.	.	.	.
3394.6	TI II 1	A 4F Z 4G U	5/2 5/2	3.65	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.
3396.2	O2 H	5 6	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.
3396.7	FE III 18	B 3H Z 3H U	6 6	14.61	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.
3396.8	O IV 3	3S 4P U 3P 4D	3/2 3/2	57.79	.	.	.	.	.	XB	.11	.	.	.	.	.	.	.	.	.
3397.8	NI II 8	A 2P Z 2F U	3/2 5/2	7.22	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.
3397.9	NE II 36	3P 4S U 4S 4P	3/2 3/2	34.86	.	.	.	.	.	.	.04	.	.	.	.	.	.	.	.	.
3398.2	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
3398.4	FE II 105	A 2I Z 2I U	13/2 13/2	7.69	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.
3399.1	O2 H	3 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.
3399.5	CR II 100	A 2G Z 2G U	9/2 9/2	8.11	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
3400.3	FE V 4F	3D4 5D 3D4 3G	2 3	3.68	.	.	.	.	.	.	.05	.	W	.	.	.	.	.	.	.
3401.8	NI II 4	A 4P Z 4D U	3/2 1/2	6.69	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.
3403.3	CR II 3	A 4D Z 4P U	3/2 3/2	6.05	.	.	.	1	.	.	.	.	W	.	.	.	.	.	.	.
3403.6	O IV 2	3 2P U 3 2D	1/2 3/2	51.79	.	.	.	.	XB	280B	.10 .82	.	.	.	.	.	.	.	.	.
3404.9	O2 H	3 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.
3405.7	O III 15	3P 3P 3D 3P U	0 1	40.69	.	.	.	.	XB	.	.14 .60	.	WB	2	.	.	.	.	.	.
3406.6	FE V 4F	3D4 5D 3D4 3G	3 4	3.72	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.
3407.3	NI II 4	A 4P Z 4D U	1/2 1/2	6.69	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.	.
3407.4	O II 44	3P 2D U 4S 2D	5/2 5/2	32.01	.	.	.	20B	.	.	.11	.	.	.	.	.	.	.	.	.
3408.1	O III 15	3P 3P 3D 3P U	1 0	40.69	.	.	.	.	XB	.	.08 .24	.	.	.	.	.	.	.	.	.
3408.8	CR II 3	A 4D Z 4P U	7/2 5/2	6.09	.	.	.	10	.	.	.	.	W	.	.	.	WB	.	.	.
3409.3	-	.	.	.	.	.	.	.	.	.	.01	.	.	.	.	.	.	.	.	.
3409.8	O IV 3	3S 4P U 3P 4D	5/2 5/2	57.81	.	.	.	.	.	XB	.45	.	.	.	.	.	.	.	.	.
3409.8	TI II 1	A 4F Z 4G U	7/2 5/2	3.65	.	.	.	.	.	.	.	.	.	.	.	.	WB	.	.	.
3410.3	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
3411.4	NE II 45	3P 2P U 3D 2P	3/2 3/2	37.72	.	.	.	.	.	.	.05 .15B	.	.	.	.	.	.	.	.	.
3411.8	O IV 2	3 2P U 3 2D	3/2 5/2	51.79	.	.	.	20B	.	XB	280B	1.3B	.	W	.	.	.	.	.	.
3413.7	O IV 2	3 2P U 3 2D	3/2 3/2	51.79	.	.	.	.	XB	280B	.22B	.	.	.	.	.	.	.	.	.
3414.7R	O2 H	7 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.
3414.8	NE II 20	3P 2D U 3D 2F	3/2 5/2	34.66	.	.	.	.	.	.	.18	.	.	.	.	.	.	.	.	.
3415.3	O III 15	3P 3P 3D 3P U	1 1	40.69	.	.	.	.	.	.	.76	.	W	.	.	.	.	.	.	.
3415.8	CO II 2	4S 5P 4P 5D U	3 3	5.81	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.
3416.0	FE II 16	A 2P Z 4P U	3/2 1/2	5.88	.	.	.	2B	.	.	.	.	W	.	.	.	.	.	.	.
3416.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.
3417.7	NE II 20	3P 2D U 3D 2F	5/2 7/2	34.60	.	.	.	.	.	.	.07	.	.	.	.	.	.	.	.	.
3420.4	O2 H	7 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.
3421.2	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
3466.2	O III 25	3P 5D O 3D 5F	4 4	48.67	.	.	.	.	.058	.	.	.	.	.	.	.	.	.	.	.	.	.
3466.3	MN II 12	E 7D Y 7F O	5	13.38	.	.	108	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3466.4	N I 2F	2P3 4S O 2P3 2P O	3/2	3.56	.	.	.	.	.47	X	.	.	.	.	.	.	.	.	.	X	.	.
3466.5	FE I 24	A 5F Z 3G O	5 4	4.42	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.
3466.5	NI -F	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	20	.	.
3466.9	O III 25	3P 5D O 3D 5F	3 2	48.66	.	.	.	.	.05	.	.	.	.	.	.	.	.	.	.	.	.	.
3467.2	O2 H	4 6	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	28	.	.	.
3467.5	O2 H	2 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	28	.	.	.
3467.5V	N2+ 1N	3 4?	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	XB	.	.
3468.3	K III 1	4S 4P 4P 4P O	3/2 5/2	29.32	.	.	.	.	.04	.	.	.	.	.	.	.	.	.	.	.	.	.
3468.7	FE II 114	C 2G Z 2G O	7/2 7/2	7.69	.	.	X	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3469.9	-	.	.	.	.	.	.	.	.08	.	.	.	.	.	.	.	.	.	.	.	.	.
3470.2	FE II 89	B 2F Z 4G O	7/2 9/2	7.51	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
3470.4	O II 27	3P 2D O 4S 2P	3/2 1/2	29.67	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.
3471.1	-	.	.	.	.	.	8	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3471.4	NI II 4	A 4P Z 4D O	1/2 3/2	6.62	.	.	.	203	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3471.8	HE I 44	Z 3P O 16 3D	.	24.43	.	.	.	208	.	.03 .12	.13	.	.	.	.	.	.	.	.	.	.	.
3471.9	O2 H	2 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.
3474.0	FE II --	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.
3474.1	MN II 3	A 5D Z 5P O	2 1	5.38	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
3475.1	CR II 2	A 4D Z 6P O	3/2 3/2	5.97	.	.	.	2 50	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3475.7	FE II 4	A 4P Z 6F O	5/2 7/2	5.21	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
3476.3	-	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	.	.	.
3477.2	TI II 6	B 4F Z 4G O	5/2 7/2	3.67	.	.	.	28	.	.	.	.	.	.	W	.	.	.	.	.	.	.
3477.9	-	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
3478.7	N IV 1	3 3S 3 3P O	1 2	50.11	.	.	1	1	70 7008	.27	.	.	.	.	.	.	.	.	.	.	.	.
3479 R	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	6	.	.	.
3479.0	HE I 43	2 3P O 15 3D	.	24																		

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGH NING
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
																QUAS	GAL	NOVA				
3484.8	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.		
3484.9	N IV 1	3 3S 3 3P O	1 0	50.11	.	.	.	1B	70 700B	.	.	.	WB	.	.	.	.	.	.			
3486.0	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.			
3486.9	SI III --	.	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.			
3487.6	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	4	.	.	.		
3487.7	HE I 42	2 3P O 14 3D	.	24.41	.	.	.	.	.	.02 .14	.15	.	XB	.	.	.	.	.	.			
3488.0	FE II 4	A 4P 7 6F U	3/2 5/2	5.23	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.			
3488.2	O II 7	3P 2S O 3D 4P	1/2 1/2	28.71	.	.	.	.	.	.02	.	.	.	.	.	.	.	.	.			
3488.7	MV II 3	A 5D Z 5P U	1 1	5.38	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.			
3489.8	O IV 14	3S* 2P O 3P* 2D	3/2 5/2	67.57	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.			
3490.0	FE II 26F	A 4F B 4D	5/2 7/2	3.89	.	.	.	10 20B	.	.	.	.	.	.	.	.	.	.	.			
3490.9	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.		
3491.1	TI II 6	B 4F Z 4G O	3/2 5/2	3.65	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.			
3492.2	O IV 14	3S* 2P O 3P* 2D	1/2 3/2	67.57	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.			
3493.5	FE II 114	C 2G Z 2G O	9/2 9/2	7.66	.	.	X	10 20B	.	.	.	1 5B	.	.	.	.	.	.	.			
3494.7	FE II 16	A 2P Z 4P O	3/2 5/2	5.80	.	.	.	50B	.	.	.	1 5B	.	.	.	.	.	.	.			
3494.7	FE VI 4F	3D3 4F 3D3 2D	3/2 3/2	3.53	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
3494.7	O II 70	3D 4D 5P 4P O	5/2 5/2	32.26	.	.	.	.	.	.04	.	.	.	.	.	.	.	.	.			
3494.8	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.		
3495.4	CR II 2	A 4D Z 6P O	5/2 3/2	5.97	.	.	.	203	.	.	.	1 5B	.	.	.	.	.	.	.			
3495.6	CR II --	.	.	.	.	.	.	.	.	.	.	1 5B	.	.	.	.	.	.	.			
3495.6	FE II 115	C 2G Y 4F U	9/2 7/2	7.66	.	.	.	20B	.	.	.	1 5B	.	.	.	.	.	.	.			
3495.8	MN II 3	A 5D Z 5P O	0 1	5.38	.	.	.	20B	.	.	.	1 5B	.	.	.	.	.	.	.			
3496.1	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.			
3496.2	ZR II 1	A 4F Z 4G O	5/2 7/2	3.57	.	.	.	203	.	.	.	.	.	.	.	.	.	.	.			
3497.3	S III --	.	.	.	.	.	.	.	.	XB	.03	.06	.	.	.	.	.	.	.			
3497.5	MN II 3	A 5D Z 5P U	1 2	5.37	.	.	.	2B	.	.	.	W	.	.	.	.	.	.	.			
3498.6	HE I 40	2 3P O 13 3D	.	24.40	.	.	.	.	.	.07 .22	.15	.	W	.	.	.	.	.	.			
3499.2	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.		
3500.0	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.			
3500.3	FE III 48	E 3F Z 3G O	4 5	15.05	.	.	.	.	.	.	.	.	3B	.	.	.	.	.	.			
3501.6	FE II 26F	A 4F B 4D	5/2 5/2	3.87	.	.	.	2B	.	.	.	W 5B	.	.	.	.	.	.	.			
3501.7	CO II 2	4S 5P 4P 5D O	3 4	5.72	.	.	.	20B	.	.	.	WB	.	.	.	.	.	.	.			
3501.8	FE III 48	E 3F Z 3G O	3 4	15.05	.	.	.	.	.	.	.	.	3B	.	.	.	.	.	.			
3502.0	N2 VK	2 11	.	6.1	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.			
3502.4	HE I 39	2 3P O 13 3S	.	24.40	.	.	.	.	.	.	.15	.	.	.	.	.	.	.	.			
3503.5	FE II 4	A 4P Z 6F O	1/2 1/2	5.24	.	.	.	20B	.	.	.	W	.	.	.	.	.	.	.			
3504.0	FE II 26F	A 4F B 4D	5/2 1/2	3.87	.	.	.	203	.	.	.	1B	.	.	.	.	.	.	.			
3504.2	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.		
3504.5	FE II 26F	A 4F B 4D	5/2 3/2	3.87	.	.	.	203	.	.	.	1B	.	.	.	.	.	.	.			

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY													
		LEVELS	J															
					NUC EM	NOVA	W-R	W-R	PLAN	GAS	OE	AE	T-	KE	AIR	CTY	AUR	LIGHT
QUAS	GAL	NOVA	LIKE	N	C	NEB	NEB	SYMB	PCYG	GE	TAU	N	GLW	LTS	ORA	NING		
3504.9	TI II 88	B 2G Y 2G U	9/2 9/2	5.40	.	.	.	.	.	.	.	.	.	.	.	.	.	
3505.4	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
3505.8	FE II 25F	A 4F B 2G	9/2 9/2	3.75	.	.	.	2B	.	XB	.	.	.	.	.	.	.	
3506	N V -	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	
3506	O V -	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	
3506.9	FE III 48	E 3F Z 3G U	2 3	15.05	.	.	.	.	.	.	.	.	.	.	.	.	.	
3507.4	FE II 16	A 2P Z 4P U	1/2 3/2	5.85	.	.	.	2B	.	.	.	.	.	.	.	.	.	
3508.2	FE II 4	A 4P Z 6F U	1/2 3/2	5.23	.	.	.	2B	.	.	.	.	.	.	.	.	.	
3509.0	O VI 5	6 2U 7 2P U	.	127.5	.	.	.	.	XB	.	.	.	.	.	.	.	.	
3509.3	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
3509.8	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	
3510.8	TI II 88	B 2G Y 2G U	7/2 7/2	5.40	.	.	.	.	.	.	.	.	.	.	.	.	.	
3511.8	CR II 2	A 4D Z 6P U	7/2 5/2	5.99	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3512.5	HE I 38	2 3P U 12 3U	.	24.39	.	.	.	.	.	.	.	.	.	.	.	.	.	
3513.8	FE I 24	A 5F Z 3G U	5 5	4.37	.	.	.	.	.	.	.	.	.	.	.	.	.	
3513.9	NI II 1	B 2D Z 4U U	5/2 7/2	6.36	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3515.8	FE II 20R	D 2F V 2G U	5/2 7/2	10.28	.	.	.	2B	.	.	.	.	.	.	.	.	.	
3515.9	O2 H	6 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	
3517.3	HE I 37	2 3P U 12 3S	.	24.38	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3517.3	V II 6	A 3F Z 5D U	4 3	4.63	.	.	.	.	.	.	.	.	.	.	.	.	.	
3518.3	CU I 36	A 2F Y 2D U	5/2 3/2	4.55	.	.	.	.	.	.	.	.	.	.	.	.	.	
3521.3	FE I 24	A 5F Z 3G U	4 4	4.42	.	.	.	.	.	.	.	.	.	.	.	.	.	
3524.4	FE II 26F	A 3F B 4D	3/2 5/2	3.89	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3525.8	ZR II 9	B 4F Z 4F U	5/2 7/2	3.86	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3528.3	FE II 25F	A 4F B 2G	7/2 7/2	3.80	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3529.7	FE II -F	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3530.5	HE I 36	2 3P U 11 3U	.	24.37	.	.	.	1	.	.	.	.	.	.	.	.	.	
3530.8	V II 5	A 3F Z 3D U	2 1	4.56	.	.	.	.	.	.	.	.	.	.	.	.	.	
3530.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
3532.6V	N2+ 1N	5 4	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	
3532.7	FE II 75	B 4D Z 4S U	3/2 3/2	7.37	.	.	.	2	.	.	.	.	.	.	.	.	.	
3535.2	ZR I 59	A 1G X 1G U	4 4	4.49	.	.	.	.	.	.	.	.	.	.	.	.	.	
3535.4	TI II 98	B 2P X 2D U	3/2 5/2	5.54	.	.	.	.	.	.	.	.	.	.	.	.	.	
3535.6	FE II 75	B 4D Z 4S U	5/2 3/2	7.37	.	.	.	2	.	.	.	.	.	.	.	.	.	
3536.3	FE II 26F	A 4F B 4D																

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. PGT., EV	EMISSION LINE INTENSITY														
		LEVELS	J																
					NUC EM QUAS	GAL	NUVA	LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE SYMB	AE FE PCYG	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA
3539.2	FE II 26F	A 4F B 4G	3/2 3/2	3.87	.	.	.	20B	.	.	.	.	W 5	.	.	.	.	.	.
3539.2	O2 H	3 6	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	4	.	.
3539.9	NE II 50	3P* 2D 0 3D* 2P	3/2 3/2	37.72	.	.	.	.	.	.	.06	.	.	.	.	.	.	.	.
3540.0	N2+ 1N	4 3	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10B
3540.9	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
3541.0	O2 H	1 5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	4	.	.
3544.4	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.
3545.2	V II 5	A 3F Z 3D 0	3 2	4.57	.	.	X	.	.	.	.	.	.	.	W	.	.	.	.
3548.9V	N2+ 1N	3 2	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10B
3549.9	FE I 48	A 3F X 5F 0	2 1	5.08	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.
3551.5	NI I 5	A 3F Z 3F 0	3 4	3.64	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.
3551.9	ZR II 1	A 4F Z 4G U	7/2 7/2	3.57	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.
3552.5	O2 H	5 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	8	.	.
3554.1	FE I 23	A 5F Z 5G 0	3 2	4.43	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.
3554.4	HE I 34	2 3P 0 10 3D	2,1	24.35	.	.	.	.	.	20 50B	.14 .44B	.7B	.	.	.	.	.	.	.
3554.5	HE I 34	2 3P U 10 3D	0	24.35	.	.	.	.	.	20 50B	.14 .44B	.7B	1	.	.	.	.	.	.
3556.6	ZR II 9	B 4F Z 4F U	9/2 9/2	3.93	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.
3556.8	V II 5	A 3F Z 3D U	4 3	4.59	.	.	X	.	.	.	.	W	.	.	.	.	.	.	.
3556.9	O III 24	3S 3P 3P 3P 0	2 2	46.71	.	.	.	.	.	20 50B	.	.	.	.	.	.	.	.	.
3557.5	FE II 176	C 2F X 2F 0	7/2 7/2	9.01	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.
3557.9	O2 H	5 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	7	.	.
3558.1	FE VI 4F	3D3 4F 3D3 2D	5/2 3/2	3.53	.	.	.	.	.	.	.	1 5	.	.	.	.	.	.	.
3558.5	SC II 3	A 3D Z 3U 0	2 3	3.48	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.
3559.9	NI II 5F	A 2D A 2P	3/2 1/2	3.65	.	.	.	2 203	.	.	.	1 5	.	.	.	.	.	.	.
3560.4	O IV 12	3P 2D 3D 2F U	3/2 5/2	63.05	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
3560.7	O2 H	5 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	6	.	.
3563.0	HE I 33	2 3P U 10 3S	.	24.34	.	.	.	2b	.	.	.	.15	.	.	.	.	.	.	.
3563.4	O IV 12	3P 2D 3D 2F 0	5/2 7/2	63.08	.	.	.	.	.	20 50B	.	.	.	.	.	.	.	.	.
3563.9V	N2+ 1N	2 1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1.6	10
3564.0	O2 H	5 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.
3564.1	FE I 48	A 3F X 5F U	2 2	5.06	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.
3565.4	FE I 24	A 5F Z 3G U																	

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., eV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
3596.0	TI II 15	A 2F 7 4D 0	7/2 5/2	4.04	.	.	.	.	.	.	.	.	1	.	.	W	.	.	.	.		
3599.3	HE I 30	2 3P 0 9 3S 1	2,1 1	24.30	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.		
3600.7	Y II 9	A 3D 2 3D 0	3 3	3.61	.	.	.	.	.	.	.	.	W	.	.	W	.	.	.	.		
3600.9	FE III 36	D 3F 2 3F 0	3 3	14.60	.	.	.	.	.	.	.	.	8	.	.	.	.	.	.	.		
3601.6	AL III 1	3 2D 4 2P 0	5/2 3/2	17.74	.	.	.	.	.	XB	.	.	.4	.	.	.	.	.	.	.		
3603.0R	N2 VK	0 10	.	6.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.		
3603.8	CR II 13	A 4P 2 6D 0	1/2 3/2	6.12	.	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.		
3603.9	CR II 13	A 4P 2 6D 0	3/2 3/2	6.12	.	.	.	103	.	.	.	.	.	.	.	.	.	.	.	.		
3603.9	FE III 36	D 3F 2 3F 0	2 2	14.59	.	.	.	.	.	.	.	.	8	.	.	.	.	.	.	.		
3605.2	-	.	.	.	.	.	.	.	.	.	.14	W	.	.	.	.	.	.	.	.		
3608.9	FE I 23	A 5F 2 5G 0	1 2	4.43	.	.	.	.	.	.	.	.	W	.	.	24	.	.	.	.		
3609.0	C III 10	4 3P 0 5 3D	.	42.64	.	.	.	.	15 80B	.	.	.	.	.	.	.	.	.	.	.		
3609.6	C III 10	4 3P 0 5 3D	2 3	42.64	.	.	.	.	15 80B	.	.	.	W	.	.	.	.	.	.	.		
3612.4	AL III 1	3 2D 4 2P 0	3/2 1/2	17.73	.	.	.	.	.	.	.	W 2	2	.	.	.	.	.	.	.		
3613.2	CR II 13	A 4P 2 6D 0	1/2 1/2	6.11	.	.	.	.	.	.	.	WB W	W	.	.	.	.	.	.	.		
3613.3	CR II 13	A 4P 2 6D 0	3/2 1/2	6.11	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.		
3613.6	HE I 6	2 1S 5 1P 0	0 1	23.94	.	.	.	2B	XB	XB	.17 .86	1.4	WB	6	.	.	.	.	.	.		
3613.8	SC II 2	A 3D 2 3F 0	3 4	3.44	.	.	.	.	.	.	.	WB	2B	.	.	.	.	.	.	.		
3613.8	V II -F	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.		
3614.8	ZR II 9	B 4F 2 4F 0	5/2 5/2	3.77	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.		
3614.9	FE II 112	C 2G 2 4H 0	1/2 7/2	7.55	.	.	.	2B	.	.	.	WB	.	.	.	.	.	.	.	.		
3618.8	FE I 23	A 5F 2 5G 0	2 3	4.40	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.		
3619.4	NI I 35	A 1D 2 1F 0	2 3	3.83	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.		
3621.2	CO II 1	4S 5P 4P 5F 0	3 4	5.60	.	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.		
3621.3	FE II 144	A 2S 2 2P 0	1/2 1/2	8.00	.	.	.	103	.	.	.	.	.	.	.	.	.	.	.	.		
3622	O VI 3	6 2P 0 7 2S	.	127.25	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.		
3624.7	FE II --	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.		
3624.7	NI I 2	A 3F 2 5G 0	4 5	3.40	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.		
3624.8	TI II 52	A 2P 2 2S 0	1/2 1/2	4.62	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.		
3624.9	FE II 144	A 4S 2 2P 0	1/2 1/2	8.00	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.		
3626.2	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.		
3626.7	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.		
3626.9	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.		
3627.2	FE II 193	D 2D W 2F 0	5/2 7/2	9.33	.	.	50B	.	.	.	.	.	.	.	.	.	.	.	.	.		
3627.4	NI II 5F	A 2D A 2P	3/2 3/2	3.59	.	.	50B	.	.	.	.	W 5	.	.	.	.	.	.	.	.		
3628.7	FE II 25F	A 4F B 2G	5/2 9/2	3.75	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.		
3630.3	FE VI 5F	3D 3 4F 3D 3 2H	1/2 9/2	3.55	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.		
3630.7	SC II 2	A 3D 2 3F 0	2 3	3.41	.	.	.	.	.	.	.	.	.	.	.	1B	.	.	.	.		
3631.5	CR II 12	A 4P 2 4P 0	5/2 5/2	6.09	.	.	.	50B	.	.	.	WB	5	.	.	.	.	.	.	.		
3631.7	CR II 12	A 4P 2 4P 0	3/2 5/2	6.09	.	.	.	503	.	.	.	WB	.	.	.	.	.	.	.	.		

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY																	
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGT NING			
3632.0	S III 1	3D 3P 0 4P 3D 3	2	21.07	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.			
3632.3	FE II 112	C 2G Z 4H 0	9/2	7.53	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
3632.6	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.			
3634.2	HE I 28	2 3P 0 8 3D 3	2,1	24.27	.	.	.	20	.	.	.25 .65B	1.4B WB	5	.	.	.	.	.	.			
3634.4	HE I 28	2 3P 0 8 3D 3	0	24.27	.	.	WB	.	.	XB	.25 .65B	1.4B WB	6	.	.	.	.	.	.			
3638.7	O III 35	3P 3P 0 3D 3D 3	2	50.11	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.			
3639.8	CR I 47	A 5G U 5F 0	6	5.92	.	.	.	103	.	.	.	.	.	.	.	.	.	.	.			
3641.3	TI II 52	A 2P Z 2S 0	3/2	4.62	.	.	.	10 208	.	.	.	W	W	.	.	1B	.	.	.			
3642.8	SC II 2	A 3D Z 3F 0	1	3.39	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
3645.3	SC II 2	A 3D Z 3F 0	3	3.41	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.			
3647.0	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.			
3647.3	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
3647.4	FE I 497	C 3P T 3D 0	1	6.44	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.			
3647.6	FE I 574	Z 5D 0 F 3D 3	2	6.64	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.			
3647.9	-	.	.	.	.	.	.	.	.	.	.	W	.	6B	.	.	.	.	.			
3649.0	TI II 14	A 2F Z 2D 0	5/2	3.95	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
3649.3	-	.	.	.	.	.	.	.	.	.	.	.	6B	.	.	.	.	.	.			
3650.1	CL II 7	3D 3D 0 4P 3F 4	3	18.17	.	.	.	.	.	.	.04	.	.	.	.	.	.	.	.			
3650.1	HG I 3	6 3P 0 6 3D 3	2	8.82	.	.	.	.	.	.	.	.	.	.	.	.	100B	.	.			
3650.4	CR II 156	B 2I Z 2I 0	13/2	8.34	.	.	.	20	.	.	.	.	.	.	.	.	.	.	.			
3652.1	HE I 27	2 3P 0 8 3S 1	0	24.25	.	.	1B	.	.	.	.06	W	.	4	.	.	.	.	.			
3652.7	FE III --	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.			
3653.5	TI I 19	A 3F Y 3G 0	4	3.43	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.			
3654.5	S II 4	3S3P42P 4P2P 0	5/2	16.47	.	.	.	.	.	.	.07	.	.	.	.	.	.	.	.			
3654.7	-	.	.	.	.	.	.	.	.	.	.	.25	.	.	.	.	.	.	.			
3655.0	AL II 12	4 3P 0 5 3D 3	2	16.40	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.			
3655.6	ZR II 71	A 2G Z 2G 0	7/2	4.34	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.			
3656.1	H 38 7	2 2P 0 38 2D	.	13.53	.	.	.	.	.	.	.	W	W	.	.	.	.	.	.			
3656.7	H 37 7	2 2P 0 37 2D	.	13.53	.	.	.	.	.	.	.	.15	WB	.	.	.	.	.	.			
3657.3	H 36 7	2 2P 0 36 2D	.	13.53	.	.	.	.	.	.	.24B	.12	WB	.	.	.	.	.	.			
3657.9	H 35 7	2 2P 0 35 2D	.	13.53	.	.	.	.	.	.	.24B	.37	WB	.	.	.	.	.	.			
3658.6	H 34 7	2 2P 0 34 2D	.	13.53	.	.	.	.	.	.	.23 .41	.42	WB	.	.	.	.	.	.			
3659.4	H 33 6	2 2P 0 33 2D	.	13.53	.	.	.	.	.	.	.22 .38	.41	WB	.	.	.	.	.	.			
3659.8	TI II 75	B 2D Y 2F 0	5/2	4.95	.	.	.	50B	.	.	.	.44	W	2	.	.	.	.	.			
3660.0	FE II 10F	A 6D A 2F	9/2	3.37	.	.	.	50B	.	.	.	.	.	.	.	.	.	.	.			
3660.2	H 32 6	2 2P 0 32 2D	.	13.53	.	.	.	.	.	.	.21 .38	.49	W	W	.	.	.	.	.			
3660.7	-	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.			
3661.3	H 31 6	2 2P 0 31 2D	.	13.53	.	.	.	.	.	.	.21 .38	.53	W	W	.	.	.	.	.			
3661.4	V II 191	A 1F Y 1G 0	3	6.68	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.			
3662.2	TI II 75	B 2D Y 2F 0	3/2	4.93	.	.	.	50	.	.	.	.	.	.	.	.	.	.	.			

LAB WAVE LENGTH, Å	ION AND MULTIPLT NUMBER	TRANSITION LEVELS J	UPPER EXC. PCT., EV	EMISSION LINE INTENSITY																
				NUC EM	NOVA LIKE	W-R N	PLAN NEB	GAS NEB	Symb	OE BE	AE FE	T- TAU	KE ME	AIR GLW	CTY LTS	AUR ORA	LIGHT NING			
3662.3	H 30 6	2 2P U 30 2D	.	13.52	.	.	.	.	.24 .45	.63	W 2	W 1	.	.	.	.	.	.		
3662.9	HG I -	" "	.	.	.	.	.	.	.	.	.	.	.	.	.	.	508	.		
3663.3	HG I 2	6 3P U 6 1U	2 Z	8.81	.	.	.	.	.	.	.	.	.	.	.	.	1008	.		
3663.5	H 29 6	2 2P U 29 2D	.	13.52	.	.	.	.	.29 .50	.74	W 5	W 1	.	.	.	.	.	.		
3664.1	Fe VI 4F	3D3 4F 3D3 2D	7/2 5/2	3.51	.	.	1 10	.	.	.	1 20	.	.	.	.	.	.	.		
3664.7	H 28 6	2 2P U 28 2D	.	13.52	.	.	.	103	.27 .52	.78	W 5	W 1	.	.	.	.	.	.		
3664.7	Fe II -	" "	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.		
3665.0	Cr II 156	8 2I 2 2P U	11/2 11/2	8.33	.	.	.	103	.	.	WB	.	.	.	.	.	.	.		
3666.1	H 27 5	2 2P U 27 2D	.	13.52	.	.	.	.	.27 .69	.15 .84	W 8	W 1	.	.	.	.	.	.		
3666.9	-	" "	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3667.7	H 26 5	2 2P U 26 2D	.	13.52	.	.	.	.	.32 .63	.3	W 10	W 1	.	.	.	.	.	.		
3669.4	V II 116	B 3H 2 3H U	6 6	5.88	.	.	.	.	.	.	1	.	.	.	.	.	.	.		
3669.5	H 25 5	2 2P U 25 2D	.	13.52	.	.	.	103	.31 .72	.6 1.0	W 8	W 1	.	.	2	.	.	.		
3670.3	-	" "	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3670.6	Fe II 10F	A 6D A 2F	7/2 5/2	3.41	.	.	.	103	.	.	.	.	.	.	.	.	.	.		
3671.5	H 24 5	2 2P U 24 2D	.	13.52	.	.	.	2	.38 .71	.7 1.1	4 12	W 2	.	.	.	.	.	.		
3671.9V	N2 2P	3 5	.	11.1	.	.	.	.	.	.	.	.	.	.	.	.	4	.		
3673.8	H 23 5	2 2P U 23 2D	.	13.51	.	.	.	103	.42 .78	1.4	W 15	W 2	.	.	.	.	.	.		
3674.2	-	" "	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.		
3674.7	Zr II 9	B 4F Z 4F U	3/2 3/2	3.68	.	.	.	103	.	.	.	.	.	.	.	.	.	.		
3675.2	Fe VI 5F	3D3 4F 3D3 2H	9/2 11/2	3.60	.	.	.	.	.	.	2	.	.	.	.	.	.	.		
3676.4	H 22 4	2 2P U 22 2D	.	13.51	.	.	.	208	.46 .82	1.4	W 15	W 2	.	.	.	.	.	.		
3677.7	Cr II 12	A 4P Z 4P U	5/2 3/2	6.05	.	.	.	1 208	.	.	108	W 38	.	.	.	.	.	.		
3677.8	He II -	" "	.	.	.	.	.	.	.07	.	.	.	.	.	.	.	.	.		
3677.9	CR II 12	A 4P Z 4P U	1/2 3/2	6.05	.	.	.	1 208	.	.	108	W 38	.	.	.	.	.	.		
3677.9	CR II 12	A 4P Z 4P U	3/2 3/2	6.05	.	.	.	1 208	.	.	108	W 38	.	.	.	.	.	.		
3678.3	-	" "	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.		
3679.4	H 21 4	2 2P U 21 2D	.	13.51	.	.	.	10	.49 1.06	2.2	1 18	W 3	.	.	2	.	.	.		
3679.9	Fe I 5	A 5D Z 5F U	4 4	3.35	.	.	.	.	.	.	W	.	.	.	W	.	.	.		
3681.0	-	" "	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3681.3	HE II -	" "	.	.	.	.	.	.	.12	.	W	.	.	.	.	.	.	.		
3682.8	H 20 4	2 2P U 20 2D	.	13.51	.	.														

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR CTY AUR LGHT			
		LEVELS	J		NUC EM	NOVA GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME	GLW	LTS	ORA	NING		
3686.8	H 19 4	2 2P 0 19 2D	.	13.50	.	.	.	.58	.	.	.68	3.0	2	1	.	.	.	.	.	.		
3687.0	C IV -	.	.	.	.	.	.	.	X	.	1.22	.	22	4	.	.	.	.	.	.		
3688	CA VII 2F	3P2 1D 3P2 1S	2 0	6.05	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.		
3688.3	MO II 5	4D 4F U	5/2 7/2	6.45	.	.	.	103M	.	.	.	.	.	.	.	.	.	.	.	.		
3688.5	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3689.2	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3689.5	-	.	.	.	.	.	.	.	.	.	.05 .25 .25	.	.	.	.	.	.	.	.	.		
3690.0	HE II -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.		
3690.8	-	.	.	.	.	.	.	.	.	.	.	.	W 5	W	.	.	.	.	.	.		
3691.3	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3691.6	H 18 4	2 2P 0 18 2D	.	13.50	.	.	.	10	.	.	.82 1.63	3.0	2	1	W	.	X	.	.	.		
3692.4	-	.	.	.	.	.	.	.	.	.	.	.	W 2	.	.	.	.	.	.	.		
3692.4	O I 6	3 3S 0 5 3P	1	12.82	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.		
3692.6	HE II -	.	.	.	.	.	.	.	.	.	.15	.	.	.	.	.	.	.	.	.		
3693.4	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.		
3693.9	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3694.0	FE I 394	Z 7P 0 E 7S	2 3	6.37	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3694.2	NE II 1	3S 4P 3P 4P 0	5/2 5/2	30.39	.	.	.	.	.	.	.06 .37	.10	W 3	2	.	.	.	.	.	.		
3695.0	CR II 169	B 2D Y 2D 0	5/2 5/2	8.64	.	.	.	.	.	.	.	.	W 28	.	.	.	.	.	.	.		
3695.1	FE I 229	B 3F V 5F 0	3 4	5.92	.	.	.	.	.	.	.	.	W 28	.	.	.	.	.	.	.		
3695.4	O III 21	3S 5P 3P 5D 0	1 2	45.08	.	.	.	.	.	50 150B	.	.	.	.	.	.	.	.	.	.		
3695.6	HE II -	.	.	.	.	.	.	.	.	.	.30	.	.	.	.	.	.	.	.	.		
3697.2	H 17 3	2 2P 0 17 2D	.	13.49	.	.	10	20	.	.	.99 1.61	3.8	2	1	W	.	X	.	.	.		
3698.7	O III 21	3S 5P 3P 5D 0	2 3	45.10	.	.	.	.	.	50 150B	.	.	.	.	.	.	.	.	.	.		
3698.7	HE II -	.	.	.	.	.	.	.	.	.	.3	.	.	.	.	.	.	.	.	.		
3699.9	FE II 131	B 2D X 4D 0	5/2 5/2	7.81	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.		
3700.3	V II 116	B 3H Z 3H 0	5 5	5.84	.	.	.	.	.	.	.	.	1	W	.	.	.	.	.	.		
3701	O V 8	3P 3D 3D 3D 0	1 1	86.89	.	.	.	.	XB	50 150B	.	.	.	.	.	.	.	.	.	.		
3701.1	FE I 385	Z 7P 0 E 7F	3 4	6.32	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3702.3	HE II -	.	.	.	.	.	.	.	.	.	.3	.	.	.	.	.	.	.	.	.		
3702.6	MO II 5	4D 4F 0	1/2 3/2	6.34	.	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.		
3702.8	O III 14	3P 3P 3D 3D 0	0 1	40.40	.	.	1B	.	XB	50 150B	.21 .33	.	1	.	.	.	.	.	.	.		
3703	O V 8	3P 3D 3D 3D 0	2 3	86.91	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.		
3703.4	O III 21	3S 5P 3P 5D 0	3 4	45.11	.	.	1B	.	.	50 150B	.	.	1B	.	.	.	.	.	.	.		
3703.5	O III 12	3P 1P 3D 1P 0	1 1	42.80	.	.	.	.	.	50 150B	.	.	1B	.	.	.	.	.	.	.		
3703.9	H 16 3	2 2P 0 16 2D	.	13.49	.	.	20B	1	.	.	.02 1.77	4.58	3	1	W	.	X	.	.	.		
3704.7	O III 21	3S 5P 3P 5D 0	1 1	45.08	.	.	.	.	.	50 150B	.	.	.	.	.	.	.	.	.	.		
3705.0	HE I 25	2 3P 0 7 3D	2,1	24.21	.	.	1B	1B	40B	50 150B	.52 .93B	4.58	2	1	.	.	.	.	.	.		
3705.1	HE I 25	2 3P 0 7 3D	0	24.21	.	.	1B	1B	40B	50 150B	1.0	4.58	2	1	.	.	.	.	.	.		
3705.6	FE I 5	A 5D Z 5F 0	3 3	3.38	.	.	.	10B	.	.	.	.	2	W	.	.	.	.	.	.		

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM	NOVA	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N						
3706.0	CA II 3	4 2P U 5 2S	1/2 1/2	6.44	.	.	.	.	.	.	.	W	4B	W	.	.	.	.	.	.	.	
3706.0	HE II -	.	.	.	.	.	.	.	.	.23	.	.	.	.	.	.	.	.	.	.	.	
3706.2	TI II 73	B 2D Z 2P U	3/2 3/2	4.89	.	.	.	1B	.	.	.	3B	4B	.	.	.	.	.	.	.	.	
3707.2	O III 14	3P 3P 3D 3D U	1 2	40.40	.	.	W 1B	.	50 150B	.04 .33	.	W	.	.	.	.	.	.	.	.	.	
3707.3	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	2	.	.	.	.	.	
3709.1	FE II 10F	A 6D A 2F	5/2 5/2	3.41	.	.	.	103	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3709.2	FE I 21	A 5F Y 5F U	4 3	4.24	.	.	.	.	.	.	.	1B	WB	.	.	W	.	.	.	.	.	
3709.3	ZR II 45	A 2F Y 2D U	7/2 5/2	4.13	.	.	.	103	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3709.4	S III 1	3D 3P U 4P 3D	1 2	20.99	.	.	.	XB	XB	.	2.8B	.	.	.	.	.	.	.	.	.	.	
3709.5	O III 21	3S 5P 3P 5D U	1 0	45.07	.	.	.	.	.	.04 .07B	.	.	.	.	.	.	.	.	.	.	.	
3709.6	NE II 1	3S 4P 3P 4P U	3/2 1/2	30.44	.	.	.	.	.	.20	.01B	.	WB	.	.	.	.	.	.	.	.	
3709.9	A II 67	4P 2D U 4D 4F	3/2 3/2	23.00	.	.	.	.	.	.04 .07B	.	.	.	.	.	.	.	.	.	.	.	
3710.3	Y II 7	A 3D Z 3F U	3 4	3.51	.	.	.	.	.	.	.	1B	.	.	.	W	.	.	.	.	.	
3710.4	HE II -	.	.	.	.	.	.	.	.	.27	.	.	.	.	.	.	.	.	.	.	.	
3710.5V	N2 2P	2 4	.	11.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	
3711.3	FE I 75	A 5P V 5D U	2 2	5.51	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	
3712.0	H 15 3	2 2P U 15 2D	.	13.48	.	.	4 20	.5B	.	.04 2.60	5.2	4 30	1 7	W	.	W	.	.	.	.	.	
3712.3	FE II 10F	A 6D A 2F	7/2 7/2	3.37	.	.	.	.5 50B	.	.	.	.	.	.	.	.	.	.	.	.	.	
3712.5	O III 21	3S 5P 3P 5D U	2 2	45.08	.	.	.	.	50 150B	.	.	.	.	.	.	.	.	.	.	.	.	
3712.8	O II 3	3S 4P 3P 4S U	1/2 3/2	26.19	.	.	.	.	.	.	.	W 10B	5B	.	.	.	.	.	1	.	.	
3713.0	CR II 12	A 4P Z 4P U	1/2 1/2	6.02	.	.	.	1 50B	.	.	.	W 10B	1 6B	.	.	.	.	.	.	.	.	
3713.0	CR II 12	A 4P Z 4P U	3/2 1/2	6.02	.	.	.	1 50B	.	.	.	W 10B	1 6B	.	.	.	.	.	.	.	.	
3713.1	AL III 4	4 2P U 5 2S	3/2 1/2	21.07	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	
3713.1	NE II 5	3S 2P 3P 2D U	3/2 5/2	30.99	.	.	.	.	.	.24	.	.	5B	.	.	.	.	.	.	.	.	
3714	N IV 12	3P 3D 3D 3P U	3 2	63.14	.	.	.	XB	.	.16 .22	.	WB	.	.	.	.	.	.	.	.	.	
3714.0	O III 14	3S 3P 3D 3D U	1 1	40.40	.	.	.	XB	50 100B	.	.	WB	.	.	.	.	.	.	.	.	.	
3714.8	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
3715.0	HE II -	.	.	.	.	.	.	.	.	.7	.	.	.	.	.	.	.	.	.	.	.	
3715.1	O III 14	3S 3P 3D 3D U	2 3	40.41	.	.	W	XB	50 150B	.05 .30	.	1 5B	.	.	.	.	.	.	.	.	.	
3715.2	C III -	.	.	.	.	.	.	.	.	.5B	.	.	.	.	.	.	.	.	.	.	.	
3715.2	CR II 20	B 4D Z 4F U	7/2 9/2	6.41	.	.	.	1 20B	.	.	.	1 5B	4B	.	.	.	.	.	.	.	.	
3715.5	CR II 145	C 2F Z 2D U	5/2 3/2	8.23	.	.	.	20B	.	.	.	1 5B	4B	.	.	.	.	.	.	.	.	
3715.5A	V II 15	A 3H Z 3G U	6 5	4.89	.	.	.	1B	.	.	.	1 5B	4B	.	.	.	.	.	.	.	.	
3716.2	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
3717	O V 8	3P 3D 3D 3D U	2 2	86.90	.	.	.	.	50 150B	.	.	.	.	.	.	.	.	.	.	.	.	
3717.4	TI I 17	A 3F X 3F U	2 3	3.32	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	
3717.8	S III 6	4S 3P U 4P 3S	2 1	21.48	.	.	.	.	XB	.	.2	.	.	.	.	.	.	.	.	.	.	
3718.2	A II 131	4P' 2D U 4D' 2F	3/2 5/2	24.72	.	.	.	.	.	.04	.	.	.	.	.	.	.	.	.	.	.	
3718.2	-	.	.	.	.	.	.	.	.	.	.	W 2	.	.	.	.	.	.	.	.	.	
3719.9	FE I 5	A 5D Z 5F U	4 5	3.32	.	.	.	20B	.	.	.	5 2	2	.	.	15	.	.	.	.	.	

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING	
		LEVELS	J		NUC		NOVA	W-R	W-R	PLAN	GAS	OE	AE	T-	KE								
					EM	GAL										QUAS	NEB	NEB					SYMB
3720.4	HE II -	.	.	.	.	.	.	.	.	.	.36	.	.	.	.	.	.	.	.	.	.	.	.
3720.9	O III 21	3S 5P 3P 5D 0	3 3	45.10	.	.	.	.	.	50 150B	.	.	1	.	.	.	.	.	.	.	.	.	.
3721.1	S III 2F	3P2 3P 3P2 1S	1 0	3.35	.	.	.	1B	.	.	2.1 3.7B	4.3B	.	.	.	.	.	.	.	.	.	.	.
3721.3	FE I 75	A 5P V 5D 0	3 4	5.48	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.
3721.6	TI II 13	A 2F Z 2F U	5/2 7/2	3.89	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.
3721.9	NE II 37	3P 2P 0 3D 4F	3/2 5/2	34.69	.	.	.	.	.	.	2.8B	.	.	.	.	.	.	.	.	.	.	.	.
3722.0	O III 21	3S 5P 3P 5D 0	2 1	45.08	.	.	.	.	.	.	2.7 2.8B	.	.	.	.	.	.	.	.	.	.	.	.
3722.0	H 14 3	2 3P 0 14 3D	.	13.47	.	.	4	1.5 20B	.	.	2.3 2.8B	4.3 8.0B	3 32	1 7	W	W	W 3	.	.	.	.	.	.
3722.0	C IV -	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3722.6	FE I 5	A 5D Z 5F 0	2 2	3.40	.	.	.	50B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3723.0	-	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.	.
3723.4	CR II 144	C 2F Y 4P 0	7/2 5/2	8.24	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3723.9	-	.	.	.	.	.	.	.	.	.	.21	.	.	.	.	.	.	.	.	.	.	.	.
3724.4	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.
3725.3	FE II 130	B 2D Y 4D 0	3/2 3/2	7.77	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3725.3	O III 14	3P 3P 3D 3D 0	2 2	40.40	.	.	.	.	XB	XB	.	.	W	.	W	.	.	.	.	.	.	.	.
3725.8	O IV 6	3P 4D 3D 4F 0	1/2 3/2	61.10	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.
3725.9	FE II --	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3726.2	O II 1F	2P3 4S 0 2P3 2D 0	3/2 3/2	3.31	XB	25 140B	4 8B	20B	.	X*	8.5 147B	50 127B	4 50	.	.	100 W 120B*	.5 2B	.	1B	.	.	.	.
3726.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.
3727.0	FE II 192	D 2D X 2D 0	3/2 3/2	9.20	.	.	.	.	.	.	.	.	1 5B	.	.	.	.	.	.	.	.	.	.
3727.1	NE II 5	3S 2P 3P 2D 0	1/2 3/2	31.05	.	.	.	.	.	.	.	.	1 5B	2B	.	.	.	.	.	.	.	.	.
3727.3	O II 3	3S 4P 3P 4S 0	3/2 3/2	26.19	.	.	.	.	.	.	.	.	1 5B	2B	.	.	.	.	.	.	.	.	.
3727.4	V II 21	B 3F Z 3F 0	4 4	4.99	.	.	.	1B	.	.	.	.	1 5B	2	.	.	.	.	.	.	.	.	.
3727.4	CR II 117	B 4G Z 4G 0	.	8.07	.	.	.	1 20B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3727.6	FE I 21	A 5F Y 5F 0	3 2	4.26	.	.	.	.	.	.	.	.	1 5B	.	.	2	W 10	.	.	.	.	.	.
3728.9	O II 1F	2P3 4S 0 2P3 2D 0	3/2 5/2	3.31	XB	25 140B	4 8B	.	X*	5.0 5.6	100 127B	4 50	.	.	100 W 120B*	.5 2B	.	1B	.	.	.	.	.
3729.7	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
3729.8	TI I 17	A 3F X 3F 0	2 2	3.31	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
3730.8	CR I 2	A 7S Z 5P 0	3 2	3.31	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.
3730.9	HE I -	.	.	.	.	.	.	.	.	140B	.02	.	.	.	.	.	.	.	.	.	.	.	.
3731.1	-	.	.	.	.	.	.	.	.	.	.21	.	.	.	.	.	.	.	.	.	.	.	.
3732.8	V II 15	A 3H Z 3G 0	5 4	4.86	.	.	.	.	.	.	.	.	1 10B	.	.	.	.	.	.	.	.	.	.
3732.9	HE I 24	2 3P 0 7 3S	2,1 1	24.18	.	.	.	103	40 50B	140B	.08 .60	.13B 108	7B	.	.	.	.	.	.	.	.	.	.
3733.0	HE I 24	2 3P 0 7 3S	0 1	24.18	.	.	.	108	40 50B	140B	.	.13B 108	7B	.	.	.	.	.	.	.	.	.	.
3733.3	FE I 5	A 5D Z 5F 0	1 1	3.42	.	.	.	.	.	.	.	.	1 5B	.	.	.	.	.	.	.	.	.	.
3734.0	-	.	.	.	.	.	.	.	.	.	.05	.	.	.	.	.	.	.	.	.	.	.	.
3734 R	O2 H	1 6	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3734.4	H 13 3	2 2P 0 13 2D	.	13.46	.	.	6	2.5 50B	.	1.9 2.7	3.4 7.2	3 34	1 9	1 5	.	W 2	.	.	.	.	.	.	.
3734.9	FE I 21	A 5F Y 5F 0	5 5	4.16	.	.	.	50B	.	.	.	.	.	.	.	.	10	.	.	.	.	.	.

*Nebular emission background origin.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														
		LEVELS	J		NUC		NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE SYMB	AE FE PCYG	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGT NING
					EM	GAL													
3747	O V 8	3P 3D 3D 3D O	3 3	86.91	.	.	.	.	.	140B	.	.	.	.	.	.	.	.	
3747.7	N IV 8	3S 1P U 3P 1D	1 2	61.69	.	.	.	.	100B	.	.	.	.	.	.	.	.	.	
3747.9	S III 1	3D 3P O 4P 3D	0 1	20.96	.	.	.	.	XB	.	.	.01	.	.	.	.	.	.	
3748.0	TI II 107	B 2F X 2F U	5/2 5/2	5.88	.	.	.	.	.	.	.	2	.	.	.	.	.	.	
3748.3	FE I 5	A 5D Z 5F U	1 2	3.40	.	.	.	.	.	.	.	2 10B 2 10B	.	.	.	2	.	.	
3748.5	FE II 154	C 2D Z 2P O	5/2 3/2	8.00	.	.	.	2	.	.	.	2 10B	4	.	.	.	.	.	
3748.6	-	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	
3748.7	CR II 11	A 4P Z 6P U	5/2 5/2	5.99	.	.	.	.	.	.	.	2 10B	.	.	.	.	.	.	
3748.8	HE II -	.	.	.	.	.	.	.	.4	.	.	1	.	.	.	.	.	.	
3749.0	NI I 1	A 3F Z 5D O	4 3	3.29	.	.	.	.	.	.	.	W	.	.	.	2	.	.	
3749.5	FE I 21	A 5F Y 5F O	4 4	4.20	.	.	.	.	.	.	.	.	.	.	.	20	.	.	
3749.5	O II 3	3S 4P 3P 4S O	5/2 3/2	26.19	.	.	.	.	.	.	.	10B	.	.	.	.	.	.	
3750.2	H 12 2	2 2P O 12 2D	.	13.45	.	.	8	2	.	.	2.4 4.5	4.4 9.3	4 38	1 9	5	.	W 5	.	
3750.7	S III 1	3D 3P O 4P 3D	1 1	20.96	.	.	.	.	XB	XB	.	.01	.	.	.	.	.	.	
3750.9	V II 21	B 3F Z 3F O	3 3	4.96	.	.	.	2BM	.	.	.	.	.	.	.	.	.	.	
3751.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	
3752.9	TI I 17	A 3F X 3F O	4 4	3.34	.	.	.	.	.	.	.	.	.	.	.	W	.	.	
3753.6	TI I 17	A 3F X 3F O	3 2	3.34	.	.	.	.	.	.	.	.	.	.	.	1	.	.	
3754.6	CR II 20	B 4D Z 4F U	3/2 5/2	6.38	.	.	.	.3	.	.	.	W	1	.	.	2	.	.	
3754.6	N III 4	3S 4P O 3P 4S	3/2 3/2	38.79	.	.	.	.	100B	.	.05 .52	1 15B	W	.	.	.	.	.	
3754.7	O III 2	3S 3P O 3P 3D	1 2	36.29	.	.	X	.	.	50 100B	.20 1.4	1 15B	.	.	.	.	.	.	
3755.1	CR II 20	B 4D Z 4F U	5/2 5/2	6.38	.	.	.	.	.	.	.	1 15BM	.	.	.	.	.	.	
3755.4V	N2 2P	1 3	.	11.1	.	.	.	.	.	.	.	.	.	.	.	.	10	50	
3755.5	FE V 3F	3D4 5D 3D4 3F	1 2	3.30	.	.	1B	.	.	2	.	1 15B	.	.	.	.	.	.	
3755.6	-	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	
3755.6	FE II 154	C 2D Z 2P O	3/2 3/2	3.37	.	.	.	.	.	.	.	W	W	W	.	.	.	.	
3756.1	HE I 66	2 1P O 14 1D	1 2	24.41	.	.	.	.	.	.09	W	1B	.	.	.	.	.	.	
3757.2	O III 2	3S 3P O 3P 3D	0 1	36.28	.	.	.	2B	.	50 100B	.07 .8	5B	.	.	.	.	.	.	
3757.6	N III 11	3P 4D 3D 4P U	1/2 3/2	41.51	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	
3757.7	TI II 72	B 2D Y 2D O	3/2 3/2	4.84	.	.	.	20B	.	.	.	5BM	W	.	.	.	.	.	
3758.2	FE I 21	A 5F Y 5F O	3 3	4.24	.	.	.	.											

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. PUT., EV	EMISSION LINE INTENSITY															AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM GAL	NOVA NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N								
3761.3	TI II 13	A 2F Z 2F 0	5/2 5/2	3.85	.	.	1B 20B	1	.	.	.	.	2 10B	W 10	.	.	10	.	.	.	.		
3761.6	CA III 3	4S 3 0 4P 1	0 1	33.60	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3761.7	-	.	.	.	.	.	.	.	.	.	.04	.	.	.	.	.	.	.	.	.	.		
3761.7	CR II 11	A 4P Z 6P 0	5/2 3/2	5.97	.	.	.	.	.	.	.	.	2 10B	4B	.	.	.	.	.	.	.		
3761.9	TI II 107	B 2F X 2F 0	7/2 7/2	5.86	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.		
3761.9	CR II 11	A 4P Z 6P 0	3/2 3/2	5.97	.	.	.	.	.	.	.	.	2B	4B	.	.	.	.	.	.	.		
3761.9	CR II 11	A 4P Z 6P 0	1/2 3/2	5.97	.	.	.	.	.	.	.	.	2B	4B	.	.	.	.	.	.	.		
3762.0	-	.	.	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.		
3762.4	SI IV 3	4 2D 5 2P 0	5/2 3/2	34.14	.	.	.	.	.	.	.05B	.	W 5B	.	.	.	.	.	.	.	.		
3762.6	O II 31	3P 4S 0 4S 4P	3/2 3/2	29.47	.	.	.	.	.	.	.10B	.	1B	.	.	.	.	.	.	.	.		
3762.6	N III 11	3P 4D 3D 4P 0	3/2 3/2	41.51	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.		
3762.9	FE II 192	D 2U X 2D 0	5/2 5/2	9.21	.	.	.	20B	.	.	.	.	W	W	W	.	.	.	.	.	.		
3763.8	FE I 21	A 5F Y 5F 0	2 2	4.26	.	.	.	.	.	.	.	.	1 5B	.	.	.	15B	.	.	.	.		
3764.1	FE II 29	B 4P Z 4P 0	5/2 3/2	5.85	.	.	.	1.5	.	.	.	.	1 5B	3	.	.	.	.	.	.	.		
3764.2	FE I 74	A 5P W 5F 0	2 2	5.47	.	.	.	.	.	.	.	.	.	.	.	.	15B	.	.	.	.		
3764.8	FE V 3F	3D4 5D 3D4 3F	2 4	3.33	.	.	10B	.	.	.	.	.	1 5B	.	.	.	.	.	.	.	.		
3765.4	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.		
3765.7	-	.	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.		
3766.1	FE I 226	B 3F 1 D	3 2	5.85	.	.	.	.	.	.	.	.	W 5B	.	.	.	.	.	.	.	.		
3766.3	NE II 1	3S 4P 3P 4P 0	3/2 5/2	30.39	.	.	.	.	.	.	.06	.	.	.	.	.	.	.	.	.	.		
3766.7	CR II 20	B 4D Z 4F 0	3/2 3/2	6.37	.	.	.	.	.	.	.	.	W 5B	.	.	.	.	.	.	.	.		
3766.8	ZR II 7	B 4F Z 4G 0	7/2 9/2	3.68	.	.	.	10B	.	.	.	.	W 5B	.	.	.	.	.	.	.	.		
3767.2	FE I 21	A 5F Y 5F 0	1 1	4.28	.	.	.	10B	.	.	.	.	WB	.	.	W	5	.	.	.	.		
3768.8	HE I 65	2 1P 0 13 1D	1 2	24.40	.	.	.	.	.	.	.04 .60	.	WB	XB	.	.	.	.	.	.	.		
3769.5	NI II 4	A 4P Z 4D 0	5/2 7/2	6.36	.	.	.	3 10B	.	.	.	.	WB	5	.	.	.	.	.	.	.		
3770.6	H II 2	2 2P 0 11 2D	.	13.43	.	.	8	.5 10B	.	X*	3.4 5.4	5.4 10.5	4 38	1 7	5	5	W 5	.	.	.	.		
3771.0	V II 21	B 3F Z 3F 0	2 2	4.94	.	.	.	.5B	.	.	.	.	.	.	.	.	.	.	.	.	.		
3771.5	N III 11	3P 4D 3D 4P 0	5/2 3/2	41.51	.	.	.	.	100B	XB	.	.	.	.	.	.	.	.	.	.	.		
3771.6V	N2 VK	2 12	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.		
3771.7	TI I 17	A 3F X 3F 0	4 3	3.32	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.		
3773.1	SI IV 3	4 2D 5 2P 0	3/2 1/2	34.13	.	.	.	.	.	.	.07	.	WB	.	.	.	.	.	.	.	.		
3773.4	-	.	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.		
3774.0	O III 2	3S 3P 0 3P 3D	1 1	36.28	.	.	W	.	XB	XB	.08 .70	.	WB	.	.	.	.	.	.	.	.		
3774.3	Y II 7	A 3D Z 3F 0	2 3	3.40	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.		
3774.7	TI II 12	A 2F Z 4F 0	5/2 7/2	3.84	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.		
3774.8	FE I 73	A 5P W 5D 0	1 1	5.48	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.		
3774.9	FE VI 3F	3D3 4F 3D3 2P	3/2 1/2	3.27	.	.	.	.	.	.	.	.	1 3B	.	.	.	.	.	.	.	.		
3775.6	NI I 33	A 1D Z 3D 0	2 2	3.69	.	.	.	2	.	.	.	.	.	.	.	W	W	.	.	.	.		
3775.7	-	.	.	.	.	.	.	.	.	.	.	4	.	.	.	.	W	.	.	.	.		

*Nebular emission background origin.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY																			
		LEVELS	J		NUC		NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGHT NING				
					CM	GAL																		
3776.1	FE VI 4F	3D3 4F 3D3 2D	9/2 5/2	3.51	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.				
3776.1	TI II 72	H 2D Y 2D U	5/2 3/2	4.84	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.				
3776.5	FE I 74	A 5P W 5F 0	3 4	5.43	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.				
3777.2	NE II 1	3S 4P 3P 4P 0	1/2 3/2	30.42	.	.	.	.	.	.07 .09	.06	.	.	.	.	.	.	.	.	.				
3777.4	FE V 1F	3D4 5D 3D4 3P	0 2	3.27	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.				
3777.6	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.				
3778.4	V II 21	B 3F Z 3F 0	3 2	4.94	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.				
3778.9	FE II -	.	.	.	.	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.				
3778.9	S III 5	4S 3P 0 4P 3P	1 2	21.38	.	.	.	.	.	.05 .07	.	.	.	.	.	.	.	.	.	.				
3779.5	FE I 74	A 5P W 5F 0	1 1	5.48	.	.	.	.5B	.	.	.	1	2	.	.	2	.	.	.	.				
3779.6	-	.	.	.	.	.	.	.	.	.05	.	.	.	.	.	.	.	.	.	.				
3779.6	FE II 23	A 2D Z 4P 0	5/2 5/2	5.80	.	.	.	.5B	.	.	.	W	.	W	.	.	.	.	.	.				
3780.1	-	.	.	.	.	.	.	.	.	.	.	W	.	WB	.	.	.	.	.	.				
3781.5	FE II 130	B 2D Y 4D 0	5/2 5/2	7.74	.	.	.	.3	.	.	.	WB	WB	W	.	.	.	.	.	.				
3781.7	HE II 5	4 2F 0 21 2G	.	54.06	.	.	.	.	XB	.6	.	WB	.	.	.	.	.	.	.	.				
3782.6	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.				
3782.9	FE V 3F	3D4 5D 3D4 3F	2 3	3.31	.	.	10	.5	.	.05 .20	X	1 10B	.	.	.	.	.	.	.	.				
3783.3	FE II 14	A 2P Z 4D 0	3/2 5/2	5.53	.	.	.	2 20	.	.	.	1 10B	1 6	W	.	.	.	.	.	.				
3783.5	NI I 30	A 1D Z 5F 0	2 3	3.68	.	.	.	.	.	.	.	.	.	.	W	1 5	.	.	.	.				
3784.9	HE I 64	2 1P 0 12 1D	1 2	24.39	.	.	.	.	.	.04 .09	.15	1 5	.	.	.	.	.	.	.	.				
3785.0	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.				
3786.0	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.				
3786.0	FE I 177	A 3H Z 3I 0	5 6	5.68	.	.	.	.	.	.	.	1 5B	.	.	.	.	X	.	.	.				
3786.3	TI II 12	A 2F Z 4F 0	7/2 9/2	3.86	.	.	.	.	.	.	.	1 5B	2	.	.	.	.	.	.	.				
3786.4	FE II 15	A 2P Z 4F 0	1/2 3/2	5.59	.	.	.	.	.	.	.	1 5B	.	.	.	.	.	.	.	.				
3786.7	FE I 22	A 5F Z 3P 0	1 0	4.27	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.				
3787.2	V II 100	C 3P Z 3P 0	2 2	5.77	.	.	.	.	.	.	.	W	W	.	.	.	.	.	.	.				
3787.5	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.				
3787.9	FE I 21	A 5F Y 5F 0	1 2	4.26	.	.	.	10B	.	.	.	WB	.	.	1	5	.	.	.	.				
3788.7	Y II 7	A 3D Z 3F 0	1 2	3.36	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.				
3789.4	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.				
3790.1	FE I 22	A 5F Z 3P 0	2 1	4.24	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.				
3790.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.				
3791.0	O2 3Δ2	6 3	.	4.8	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.				
3791.0	NE II 30	3P 4S 0 3D 4D	3/2 3/2	34.48	.	.	.	.	.	.06B	.	.	.	.	.	.	.	.	.	.				
3791.3	O III 2	3S 3P 0 3P 3D	2 2	36.29	.	.	1	.	XB	.33 .65B	.	1 8B	.	.	.	.	.	.	.	.				
3791.4	SI III 5	4 3P 0 4 3D	0 1	24.88	.	.	.	.	.	.03B	.	1 8B	2	.	.	.	.	.	.	.				
3792.0	-	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.				
3792.9	N III 11	3P 4D 3D 4P 0	7/2 5/2	41.51	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.				
3794.5	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.				

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM GAL	NOVA	LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N						
3814.1	FE II 153	C 2D Z 2F O	3/2 5/2	7.95	.	.	.	10	.	.	.	.	1B	2	.	.	.	.	.	.	.	.
3814.5	FE I 22	A 5F Z 3P U	1 1	4.24	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
3814.6	U IV -	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.
3814.6	TI II 12	A 2F Z 4F O	5/2 3/2	3.81	.	.	.	.	.	.	.	.	1 10B	2	.	.	2	.	.	.	.	.
3815.1	FE VI 3F	3D3 4F 3D3 2P	3/2 3/2	3.23	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.
3815.8	FE I 45	A 3F Y 3D O	4 3	4.71	.	.	.	2B	.	.	.	.	1	.	.	2	1 5	.	.	.	.	.
3816.2	CR I 40	A 5G W 5F O	5 4	5.77	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3816.3	FE I 73	A 5P W 5D O	2 3	5.42	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
3816.8	O III 18	3P 1D 3D 1P O	2 1	41.08	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.
3817.4	-	.	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3817.6	ZR II 18	A 2D Z 2D O	3/2 3/2	3.76	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3819.3	-	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	W	.	.	.	.	.
3819.6	HE I 22	2 3P O 6 3D	2,1 0	24.11	.	.	1	1 10B	.	20B	1.1 1.8B	4.3B 20B	4 20B	W 9B	WB	.	.	.	.	.	.	.
3819.8	HE I 22	2 3P O 6 3D	0 0	24.11	.	.	.	1 103	.	20B	1.2	4.3B 20B	4 9B	W 9B	WB	.	.	.	.	.	.	.
3820.2	FE V 3F	3D4 5D 3D4 3F	3 4	3.33	.	.	.	5B	.	.	.	.	1B	.	.	.	.	.	.	.	.	.
3820.4	FE I 20	A 5F Y 5D O	5 4	4.09	.	.	.	10B	.	.	.	.	1B	W	.	3	1 10 2	.	.	.	.	.
3820.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3821.9	FE II 14	A 2P Z 4D O	1/2 3/2	5.56	.	.	.	.3	.	.	.	.	W	2	.	.	.	.	.	.	.	.
3822.1	N I 11	3S 2P 3P' 2P O	1/2 1/2	13.86	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.
3822.7	FE II --	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.
3822.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.
3823.4	ZR II 44	A 2F Z 4D O	7/2 5/2	4.03	.	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3824	N IV 10	3P 1P 4 1P O	1 1	62.59	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.
3824.2	-	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
3824.4	FE I 4	A 5D Z 5D O	4 3	3.23	.	.	.	.	.	.	.	.	1 5	.	.	.	.	.	.	.	.	.
3824.9	FE II 29	B 4P Z 4P O	5/2 5/2	5.80	.	.	.	4 20B	.	.	.	.	1B	1 5	W	.	.	.	.	.	.	.
3825.0	-	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
3825.9	FE I 20	A 5F Y 5D O	4 3	4.14	.	.	.	.	.	.	.	.	1 5	W	.	2	1 10 2	.	.	.	.	.
3826.3	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3826.8	A II 54	4P 4D O 4D 4D	5/2 5/2	22.69	.	.	.	.	.	.02	.	.	.	.	.	.	.	.	.	.	.	.
3827.1	FE II 153	C 2D Z 2F O	.	7.94	.	.	.	10	.	.	.	.	1B	2	WB	.	.	.	.	.	.	.
3827.7	FE II 128	B 2D Y 4F O	3/2 3/2	7.68	.	.	.	.	.	.	.	.	1B	.	WB	.	.	.	.	.	.	.
3827.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.
3827.8	FE I 45	A 3F Y 3D O	3 2	4.77	.	.	.	.	.	.	.	.	1B	W	.	2	1 10	.	.	.	.	.
3828.6	V I 9	A 4F Y 4D O	5/2 3/2	3.24	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.
3829 R	O2 H	2 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10B	.	.	.
3829 R	O2 H	4 8	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10B	.	.	.
3829.4	MG I 3	3 3P O 3 3D	0 1	5.92	.	.	.	.	.	.03	.	.	WB	2B	.	1	1 20	.	.	.	.	.
3829.5	V II 3	A 3F Z 5G O	2 3	4.29	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.
3829.7	V II 3	A 3F Z 5G O	3 4	4.31	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLT NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J																	
					NUC EM	NOVA GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME	AIR GLW	CTY LTS	AUR ORA	LGHT NING	
3829.8	N II 30	3P 3P 4S 3P U	1 2	24.28	.	.	.	.	.	.	.05	W	.	.	.	.	.	.		
3829.8	NE II 39	3P 2P U 3D2 2D	3/2 5/2	34.60	.	.	.	.	.	.	.06	W 5B	.	.	.	.	.			
3830.8	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.			
3831.7	NI I 31	A 1D Z 3P U	2 1	3.64	.	.	.	.	.	.	.	.	.	.	W	.	.			
3831.9	S III 5	4S 3P U 4P 3P	0 1	21.33	.	.	.	.	.	.	.03 .04	.08	.	.	.	.	.			
3832.1	C II 13	4 2P U 3P 2D	3/2 5/2	23.28	.	.	.	.	XB	.	.	.	.	.	.	.	.			
3832.3	MG I 3	3 3P U 3 3D	1 2	5.92	.	.	.	.	.	.	.	3 10B	W 3	.	W	1 30	.			
3833.0	FE II 23	A 2D Z 4P U	3/2 3/2	5.85	.	.	.	.	.	.	.	3 10B	.	.	.	.	.			
3833.6	HE I 62	2 1P U 10 1D	1 2	24.35	.	.	.	.	.	.07 .13	.13	1 15B	XB	.	.	.	.			
3833.8	HE II 4	4 2F U 18 2G	.	54.02	.	.	.	XB	20B	.6	.	1 15B	.	.	.	.	.			
3834.2	O VI 1	3 2S 3 2P U	1/2 1/2	82.23	.	.	.	XB	20B	.	.	1 15B	.	.	.	.	.			
3834.2	FE I 2Q	A 5F Y 5D U	3 2	4.17	.	.	.	.	.	.	.	1 15B	.	.	.	1 5	.			
3835.4	N2+ 1N	3 3	.	18.7	.	.	.	.	.	.	.	.	.	.	.	1	.			
3835.4	H γ (9) 2	2 2P U 9 2D	.	13.37	.	.	2 25	5 50	X	4.6 7.3	10.9 8	8 48	1 12	5	5	W 8	.			
3836.1	C II 13	4 2P U 3P 2D	1/2 3/2	23.28	.	.	.	.	XB	.	.	.	.	.	.	.	.			
3836.1	V I 44	A 4D V 4D U	3/2 3/2	4.26	.	.	.	.	.	.	.	.	.	.	W	.	.			
3836.9	FE II 8F	A 6D A 4G	9/2 5/2	3.22	.	.	.	.	.	.	.	W	.	.	.	.	.			
3837.8	S III 5	4S 3P U 4P 3P	1 2	21.33	.	.	.	XB	XB	.06B	.1	.	.	.	.	.	.			
3838.1	FE V 1F	3D4 5D 3D4 3P	2 2	3.27	.	.	.	1.5B	.	.	.	.	.	.	.	.	.			
3838.1	HE I 61	2 1P U 10 1S	1 0	24.34	.	.	.	.	.	.04 .09	.	1 15B	1B	.	.	.	.			
3838.3	MG I 3	3 3P U 3 3D	2 3	5.92	.	.	.	3	.	.09B	.	1 15B	1 4B	.	W	1 40	.			
3838.3	S III 5	4S 3P U 4P 3P	2 2	21.38	.	.	.	.	XB	.09B	.13B	.	.	.	.	.	.			
3838.4	N II 30	3P 3P 4S 3P U	2 2	24.28	.	.	.	.	.	.06B	.13B	.	5B	.	.	.	.			
3838.9	FE V 3F	3D4 5D 3D4 3F	3 3	3.31	.	.	X	1	.	.31	X	1 15B	.	.	.	.	.			
3839.0	V I 44	A 4D V 4D U	5/2 5/2	4.27	.	.	.	.	.	.	.	.	.	.	.	W	.			
3839.1	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.			
3840	O2 H	0 6	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	2B			
3840.4	FE I 20	A 5F Y 5D U	2 1	4.20	.	.	.	.	.	.	.	W	WB	.	W	W 2	.			
3841.0	-	.	.	.	.	.	.	.3	.	.	.	.	.	.	.	.	.			
3841.1	FE I 45	A 3F Y 3D U	2 1	4.81	.	.	.	.	.	.	.	.	WB	.	.	1 10	.			
3841.5	A II 54	4P 4D U 4D 4D	3/2 1/2	22.74	.	.	.	.	.	.05	.	.	.	.	.	.	.			
3842.0	CO I 33	A 2F Z 2D U	1/2 5/2																	

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGH NING
		LEVELS	J		NUC EM GAL	NOVA NOVA	W-R LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N						
3845.2	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	W	.	.	.	.	.	.
3845.2	FE II 127	B 2D Z 2D 0	3/2 3/2	7.67	.	.	10	.	.	.	.	W	2	.	.	.	.	.	.	.	.	.
3845.5	CO I 34	A 2F Y 2G 0	7/2 9/2	4.13	.	.	.	.	.	.	.	.	.	.	.	1 5	.	.	.	.	.	.
3847.1	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3847.3	V II 156	A 1P Z 1S 0	1 0	5.96	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3847.8	FE II 8F	A 6D A 4G	9/2 7/2	3.21	.	.	10B	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.
3847.9	O II 12	3P 4D 0 3D 4D	1/2 1/2	28.73	.	.	.	.	.	.	.	1 3	.	.	.	.	.	.	.	.	.	.
3848.2	MG II 5	3 2D 5 2P 0	5/2 3/2	12.03	.	.	.	.	.	.07	.	2 10	3	1	.	.	.	.	.	.	.	.
3849	-	.	.	.	.	.	.	.	.	.09	.	.	.	.	.	.	.	.	.	.	.	.
3849.1	FE VI 3F	3D3 4F 3D3 2P	5/2 1/2	3.27	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3849.6	NI II 11	A 2G Z 2F 0	7/2 5/2	7.22	.	.	1 10B	.	.	.	.	.	1 3	.	.	.	.	.	.	.	.	.
3849.8	V II 33	A 3G Z 3F 0	3 4	4.99	.	.	.	.	.	.	.	3 10BM	.	.	.	.	.	.	.	.	.	.
3850	O2 H	6 9	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.
3850.0	FE I 20	A 5F Y 5D 0	.	4.21	.	.	.	.	.	.	.	3 10BM	.	.	1	1 2	.	.	.	.	.	.
3850.4	MG II 5	3 2D 5 2P 0	3/2 1/2	12.03	.	.	204	.	.	.04	.	3 10B	1	.	.	.	.	.	.	.	.	.
3850.8	FE I 22	A 5F Z 3P 0	2 2	4.19	.	.	.	.	.	.	.	.	.	.	.	W 1	.	.	.	.	.	.
3850.8	FE V 3F	3D4 5D 3D4 3F	3 2	3.30	.	.	1	.	.	.09	X	W	.	.	.	.	.	.	.	.	.	.
3850.8	O II 12	3P 4D 0 3D 4D	3/2 5/2	28.73	.	.	.	.	.	.	.	1 5B	.	.	.	.	.	.	.	.	.	.
3851.0	O II 12	3P 4D 0 3D 4D	3/2 3/2	28.73	.	.	.	.	.	.05	.	1 5B	.	.	.	.	.	.	.	.	.	.
3852.6	FE I 73	A 5P W 5D 0	3 4	5.37	.	.	.	.	.	.	.	1	.	.	.	60	.	.	.	.	.	.
3853.1	S II 30	3D 2F 4P1 2D 0	5/2 3/2	17.38	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	.	.
3853.7	SI II 1	3P2 2D 4 2P 0	3/2 3/2	10.03	.	.	1 10B	.	.	.06	.02	1 4B	2 3	5B	.	.	.	.	.	.	.	.
3854.2	CR I 69	A 5P V 5D 0	2 2	5.90	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3854.7	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3855.1	N II 30	3P 3P 4S 3P 0	1 0	24.26	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2B	.	.
3855.4	ZR II 18	A 2D Z 2D 0	5/2 3/2	3.76	.	.	20B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3855.6	CR I 69	A 5P V 5D 0	1 2	5.90	.	.	20B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3855.7	CR I 138	B 5D T 5F 0	4 4	6.20	.	.	20B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3856.0	SI II 1	3P2 2D 4 2P 0	5/2 3/2	10.03	.	.	X 20B	.	.	.3	1.5	4 36B	5 10	5B	.	.	.	.	.	.	.	.
3856.1	HE I -	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
3856.1	N II 30	3P 3P 4S 3P 0	2 1	24.27	.	.	.	.	.	.14B*	.	.	.	.	.	.	.	.	.	.	.	.
3856.2	A II 55	4P 4D 0 3D1 2S	3/2 1/2	22.73	.	.	.	.	.	.09B*	.	.	.	.	.	.	.	.	.	.	.	.
3856.2	O II 12	3P 4D 0 3D 4D	3/2 1/2	28.73	.	.	.	.	.	.10B*	.	.	.	.	.	.	.	.	.	.	.	.
3856.4	FE I 4	A 5D Z 5D 0	3 2	3.25	.	.	.	.	.	.	.	4 36B	.	.	.	.	.	.	.	.	.	.
3857.2	-	.	.	.	.	.	.	.	.	.	.01	.	.	.	.	.	.	.	.	.	.	.
3857.2	O II 13	3P 4D 0 3D 2F	5/2 5/2	28.74	.	.	.	.	.	.04B	.	W	.	.	.	.	.	.	.	2B	.	.
3857.7	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.
3857.9V	N2+ 1N	2 2	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2B	.	.
3858.1	HE II 4	4 2F 0 17 2G	.	54.00	.	.	1 3	XB	XB	.14 .80	.	W	9	.	.	.	.	.	.	.	.	.
3858.2	-	.	.	.	.	.	.	.	.	.04B	.	.	.	.	.	.	.	.	.	.	.	.

*A good example of different authors' listing the same apparent feature with different identifications or choices of blends.

LAB WAVE LENGTH, A	ION AND MULTIPL NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT VING
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME								
3858.3	NI I 32	A 1D Z 3F U	2 3	3.62	.	.	.	.	.	.	.	.	.	1 5	.	.	.	.	.	.	.	.
3858.5	-	.	.	.	.	.	.	.5	.	.	.	.06	.	.	.	.	.	.	.	.	.	.
3859.2	FE I 175	A 3H Y 3G U	6 5	5.59	.	.	.	.	.	.	.	W 58M	.	.	.	.	.	.	.	.	.	.
3859.9	FE I 4	A 5D Z 5D U	4 4	3.20	.	.	.	103	.	.	.	.	1 5B	.	.	.	.	.	.	.	.	.
3860	O2 343	5 3	.	4.8	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.
3860.2	S II 41	4P 2S U 5S 2P	1/2 1/2	18.69	.	.	.	.	.	.03B	.	.	.	.	.	.	.	.	.	.	.	.
3860.6	S II 50	4P 4P U 4D 4P	5/2 3/2	19.26	.	.	.	.	.	.03B	.	.	.	.	.	.	.	.	.	.	.	.
3860.6	S III 5	4S 3P U 4P 3P	1 0	21.31	.	.	.	XB	.	.03B	.	.	.	.	.	.	.	.	.	.	.	.
3860.9	FE II --	.	.	.	.	.	.	103	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3861.0	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3861.2	CO I 33	A 2F Z 2D U	5/2 3/2	4.24	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
3862.3	NI VIII 2F	3D3 4F 3D3 2G	7/2 9/2	3.46	.	.	.	.	.	2	1	.	.	.	.	.	.	.	.	.	.	.
3862.6	SI II 1	3P2 2D 4 2P U	3/2 1/2	10.02	.	.	X	.5B	.	.06 .35	.7B 20	4 10	2 5B	.	.	.	.	.	.	.	.	.
3863.4	FE II 152	C 2D Y 4G U	5/2 7/2	7.91	.	.	.	.	.	.	.	.	5B	.	.	.	.	.	.	.	.	.
3863.5	O II 12	3P 4D U 3D 4D	5/2 1/2	28.73	.	.	.	.	.	.	.7B	.	.	.	.	.	.	.	.	.	.	.
3863.8	V II 33	A 3G Z 3F U	4 4	4.99	.	.	.	.	.	.	.	W WB	.	.	.	.	.	.	.	.	.	.
3864.0	FE II 127	B 2D Z 2D U	5/2 3/2	7.67	.	.	.	.53M	.	.	.	WB 5B	.	.	.	.	.	.	.	.	.	.
3864.0	FE II 152	C 2D Y 4G U	3/2 5/2	7.91	.	.	.	.	.	.	.	.	5B	.	.	.	.	.	.	.	.	.
3864.7	O II 12	3P 4D U 3D 4D	5/2 3/2	28.73	.	.	.	.	.	.07 .14	.	.	.	.	.	.	.	.	.	.	.	.
3864.9	V I 7	A 4F Y 4F U	5/2 5/2	3.21	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
3865	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.
3865.5	FE I 20	A 5F Y 5D U	1 1	4.20	.	.	.	.	.	.	.	3B	.	.	4 10	.	.	.	.	.	.	.
3865.6	CR II 167	B 2D Z 2F U	5/2 7/2	8.49	.	.	.	.5 10B	.	.	.	3B 1	.	.	.	.	.	.	.	.	.	.
3866.0	CR II 130	C 2G Z 2G U	7/2 7/2	8.09	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3866.5	CR II 130	C 2G Z 2G U	9/2 9/2	8.11	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3866.7	-	.	.	.	.	.	.	.	.	.04	.	.	.	.	.	.	.	.	.	.	.	.
3866.7	V II 11	A 3P Z 5D U	1 2	4.61	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3867.2	FE I 488	C 3P W 3P U	2 2	6.20	.	.	.	.	.	.	.	1 108M	.	.	.	.	.	.	.	.	.	.
3867.5	HE I 20	2 3P U 6 3S	2,1 1	24.07	.	.	5B	.	.	.	.	1 108	.2B	.	.	.	.	.	.	.	.	.
3867.6	HE I 20	2 3P U 6 3S	0 1	24.07	.	.	WB	20B	.	.10	W	1 108	W .2B	W	.	.	.	.	.	.	.	.
3868.7	NE III 1F	2P4 3P 2P4 1D	2 2	3.19	1 2	15 78	40 60	4 20B	3 100	19.7 .26	10 100	.	.	5 6	X	.	.	.	.	.	.	.
3870.6	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3870.8	-	.	.	.	.	.	.	.	.	.29	.	.	.	.	.	.	.	.	.	.	.	.
3871.8	FE I 429	B 3G Y 3H U	5 5	6.12	.	.	.	.	.	.	.	2 108M	.	.	.	.	.	.	.	.	.	.
3871.8	HE I 60	2 1P U 9 1D	1 2	24.31	.	.	.	.	.	.08 .40	.7	2 108	W .3	.	.	.	.	.	.	.	.	.
3872.0	-	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.
3872.5	FE I 20	A 5F Y 5D U	2 2	4.17	.	.	.	103	.	.	.	1 48	2B	.	2 5	.	.	.	.	.	.	.
3872.8	FE II 29	B 4P Z 4P U	3/2 1/2	5.88	.	.	.	2	.	.	.	1 48	3	.	.	.	.	.	.	.	.	.
3873.1	CO I 18	B 4F Z 4D U	9/2 7/2	3.62	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
3873.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPL NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	KE T- TAU N							
3874.0	CU I 18	B 4F Z 4D 0	7/2 5/2	3.70	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
3874.1	FE II 8F	A 6D A 4G	9/2 9/2	3.19	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3874.7	-	.	.	.	.	.	.	.	.	.	.04	.	.	.	.	.	.	.	.	.	.	.
3875.1	V I 7	A 4F Y 4F 0	7/2 7/2	3.22	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
3875.8	O II 13	3P 4D 0 3D 2F	7/2 5/2	28.74	.	.	.	.	.	.	.06	.	.	.	.	.	.	.	.	.	1.08	.
3876.0	FE I 22	A 5F+ Z 3P 0	1 2	4.19	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
3876.1	C II 33	3D 4F 0 4F 4G	3/2 5/2	27.35	.	.	.	2H	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3876.2	C II 33	3D 4F 0 4F 4G	9/2 11/2	27.36	.	.	.	2H	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3876.4	C II 33	3D 4F 0 4F 4G	7/2 9/2	27.35	.	.	.	2H	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3876.7	C II 33	3D 4F 0 4F 4G	5/2 7/2	27.35	.	.	.	2H	.	.	.04	WB	.	.	.	.	.	.	.	.	.	.
3876.8	CO I 17	B 4F Z 4G 0	9/2 9/2	3.61	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.
3878.0	FE I 20	A 5F Y 5D 0	3 3	4.14	.	.	.	.	.	.	.	1 5B	WB	.	3B 1	10B	.	.	.	.	.	.
3878.2	HE I 59	2 1P 0 9 1S	1 0	24.31	.	.	.	10B	.	.	.	1 5B	WB	.	.	.	.	.	.	.	.	.
3878.6	FE I 4	A 5D Z 5D 0	2 1	3.27	.	.	.	10B	.	.	.	1 4B	W	.	3B 2B	W	.	.	.	.	.	.
3878.7	V II 33	A 3G Z 3F 0	5 4	4.99	.	.	.	1	.	.	.	1 4B	2B W	.	.	.	.	.	.	.	.	.
3879.9	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3880.8	FE II -	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
3881.9	NI II 13	Z 4P 0 E 4F	5/2 7/2	11.40	.	.	.	10B	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
3882.0	C N	0	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.
3882.2	O II 12	3P 4D 0 3D 4D	7/2 7/2	28.73	.	.	.	.	.	.	.	1 10B	.	.	.	.	.	.	.	.	.	.
3882.3	TI II 34	A 2G Z 2G 0	7/2 9/2	4.29	.	.	.	.	.	.	.	1 10B	3	.	.	.	.	.	.	.	.	.
3882.5	O II 11	3P 4D 0 3D 4P	3/2 3/2	28.71	.	.	W	.	.	.11	W	1 10B	.	.	.	.	.	.	.	40B	.	.
3882.7	FE II 24F	A 4F A 2F	9/2 5/2	3.41	.	.	.	.	.	.	.	1 10B	.	.	.	.	.	.	.	.	.	.
3883.2	O II 12	3P 4D 0 3D 4D	7/2 5/2	28.73	.	.	.	.	.	.	.	1 10B	.	.	.	.	.	.	.	40B	.	.
3883.4V	CN	0 0	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.
3883.8	C III 15	4 3D 5 3F 0	1 2	42.84	.	.	.	.	.	20 130B	.	.	.	.	.	.	.	.	.	.	.	.
3884.3V	N2+ 1N	1	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	40B	.	.
3884.6	CO I 32	A 2F Y 4F 0	5/2 3/2	4.22	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
3884.8	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
3884.9	V II 33	A 3G Z 3F 0	3 3	4.96	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3885.9	-	.	.	.	.	.	W	3	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3886.0	C III 15	4 3D 5 3F 0	2 3	42.84	.	.	.	.	.	20 130B	.	.	.	.	.	.	.	.	.	.	.	.
3886.0	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
3886.3	FE I 4	A 5D Z 5D 0	3 3	3.23	.	.	.	.	.	.	.	1	1	.	5	W	.	.	.	.	.	.
3887	O2- 3Δ3	5 2	.	4.8	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	4	.	.
3887.1	FE I 20	A 5F Y 5D 0	4 4	4.09	.	.	.	.	.	.	.	.	.	.	.	1 10	.	.	.	.	.	.
3887.4	HE II 4	4 2F 16 2G	.	22.99	.	.	6B	80B	20 130B	.	.	.	.	.	.	.	.	.	.	.	.	.
3887.6R	N2 VK	0 11	.	6.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	4	.	.
3888	C IV -	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
3888.5	FE I 45	A 3F Y 3D 0	2 2	4.77	.	.	.	.	.	.	.	.	.	.	.	W 10B	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N								
3888.6	HE I 2	2 3S 3 3P U	1 0	22.91	.	.	50B	.	80B	20 130B	1.7 19B	18B	10 65B	25 50B	10B	.	W 10BM	.	.	.	.	.	
3888.6	HE I 2	2 3S 3 3P U	1 1	22.91	.	.	50B	.	80B	20 130B	1.7 19B	18B	10 65B	25 50B	10B	.	W 10BM	.	.	.	.	.	
3888.6	HE I 2	2 3S 3 3P U	1 2	22.91	.	.	50B	.	80B	20 130B	1.7 19B	18B	10 65B	25 50B	10B	.	W 10BM	.	.	.	.	.	
3889.1	H γ (8) 2	2 2P U 8 2D	.	13.33	X	5 7	20 50B	10 50	15B	X*	12 19B	18B	10 65B	14 50B	5 10B	5 10	W 10B	.	.	.	.	.	
3889.2	C III 15	4 3D 5 3F U	3 4	42.84	.	.	.	.	.	20 130B	.	.	10B	.	.	.	.	.	.	.	.	.	
3890.9	FE VI 3F	3D3 4F 3D3 2P	5/2 3/2	3.23	.	.	40B	.	.	.	.	.	2 10B	.	.	.	.	.	.	.	.	.	
3891.0	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	1	.	.	
3891.8	FE V 3F	3D4 5D 3D4 3F	4 4	3.33	.	.	50B	2	.	.	.1 .5	X	2 10B	.	.	.	.	.	.	.	.	.	
3893.5	O II 11	3P 4D U 3D 4P	3/2 5/2	28.70	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
3894.1	-	.	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
3894.1	CO I 34	A 2F Y 2G U	5/2 1/2	4.21	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	
3895.7	FE V 1F	3D4 5D 3D4 3P	3 2	3.27	.	.	1 10	2	.	.	.09 .35	X	1 5B	.	.	.	.	.	.	.	.	.	
3895.7	FE I 4	A 5D Z 5D U	1 0	3.28	.	.	.	.	.	.	.	.	1 5B	.	.	1	.	.	.	.	.	.	
3896.1	FE II 23	A 2D Z 4P U	3/2 5/2	5.80	.	.	.	.	.	.	.	.	W	1B	.	.	.	.	.	.	.	.	
3896.2	V II 10	A 3P Z 3D U	0 1	4.56	.	.	.	.5	.	.	.	.	W	1B	.	.	.	.	.	.	.	.	
3897.5	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
3898 V	N2+ 1N	O O	R	18.7	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	100	.	.	
3898.0	FE I 20	A 5F Y 5D U	1 2	4.17	.	.	.	.	.	.	.	.	W	.	.	.	W 2	.	.	.	.	.	
3898.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	
3899.1	V II 33	A 3G Z 3F U	4 3	4.96	.	.	.	.5	.	.	.	.	1 5	W	W	.	.	.	.	.	.	.	
3899.7	FE I 4	A 5D Z 5D U	2 2	3.25	.	.	.	2B	.	.	.	.	W	W	.	.	.	.	.	.	.	.	
3900.5	TI II 34	A 2G Z 2G U	9/2 9/2	4.29	.	.	X	2	.	.	.	.	W 10B	W 6	W	W	W	W	.	.	.	.	
3900.7	AL II 1	3 1P U 3 1D	1 2	10.55	.	.	.	.	.	.	.	.	10B	W	.	.	.	.	.	.	.	.	
3902.9	FE I 45	A 3F Y 3D U	3 3	4.71	.	.	.	.	.	.	.	.	W 5B	.	.	.	W 2	.	.	.	.	.	
3903.3	V II 11	A 3P Z 5D U	2 3	4.63	.	.	.	1	.	.	.	.	W 5B	2	W	.	.	.	.	.	.	.	
3903.8	-	.	.	.	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	
3904.8	TI I 56	A 1D Y 1F U	2 3	4.06	.	.	.	.	.	.	.	.	W 4	.	.	.	3	.	.	.	.	.	
3905.5	SI I 3	3P2 1S 4S 1P U	0 1	5.06	.	.	W	.	.	.	.	.	1 8B	4	X	5	40	.	.	.	.	.	
3905.6	CR II 167	B 2D Z 2F U	3/2 5/2	8.47	.	.	.	.	.	.	.	.	1 8B	.	.	.	.	.	.	.	.	.	
3905.6	FE II 8F	A 6D A 4G	7/2 7/2	3.21	.	.	.	20B	.	.	.	.	1 8B	.	.	.	.	.	.	.	.	.	
3905.8	-	.	.	.	.	.	.	.	.	.	.04	.	.	.	.	.	X	.	.	.	.	.	
3906.0	FE II 173	C 2F X 2G U	5/2 7/2	8.71	.	.	.	1 10B	.	.	.	.	W 2	W	W	.	.	.	.	.	.	.	
3907	-	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	
3907.5	O II 11	3P 4D U 3D 4P	5/2 5/2	28.70	.	.	.	.	.	.	.03	.	W	.	.	.	.	.	.	.	.	.	
3907.5	SC I 8	A 2D Y 2F U	3/2 5/2	3.16	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	
3908.5	FE II 29	B 4P Z 4P U	3/2 3/2	5.85	.	.	.	.	.	.	.	.	W 4	.	.	.	.	.	.	.	.	.	
3909.9	CO I 3	A 4F Z 6G U	9/2 11/2	3.16	.	.	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	
3909.9	V I 7	A 4F Y 4F U	9/2 7/2	3.22	.	.	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	
3910.5	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
3910.8	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	

*Nebular emission background origin.

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM QUAS	NOVA GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	DE BE PCYG	AE FE GE	T- TAU	KE ME N						
3911.1	FE V 3F	3D4 5D 3D4 3F	4 3	3.31	.	.	1	.	.	.	.07	X	WB	.	.	.	.	.	.	.	.	.
3911.3	CR II 129	C 2G 2 4G U	9/2 9/2	8.07	.	.	.	10B	.	.	.	.	.	10B	.	.	.	.	.	.	.	.
3911.8	SC I 8	A 2U Y 2F O	5/2 1/2	3.18	.	.	.	.	.	.	.	.	WB	.	.	1	.	.	.	.	.	
3912.0	O II 17	3S ⁺ 2D 3P ⁺ 2P O	5/2 3/2	28.71	.	.	.	.	.	.	.	.	1 5B	.	.	.	.	.	.	.	.	
3912.1	O II 17	3S ⁺ 2D 3P ⁺ 2P O	3/2 3/2	28.71	.	.	.	.	.	.	.	.	1 5B	.	.	.	.	.	.	.	.	
3913.0	-	.	.	.	.	.	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	
3913.0	NI I 15	A 3D Z 5D O	3 4	3.18	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	
3913.5	FE II -F	A 4D C 2G	.	.	.	.	.	.	.	.	.	.	1 10B	.	XB	.	.	.	.	.	.	
3913.5	TI II 34	A 2G Z 2G O	7/2 7/2	4.26	.	.	X	1 10B	.	.	.	.	1 10B	W 40B	5B	W	1	.	.	.	.	
3913.7	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	
3914.3	TI I 15	A 3F Y 5D O	4 4	3.20	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	
3914.3	V II 33	A 3G Z 3F O	3 2	4.94	.	.	.	3BM	.	.	.	.	1 15B	8B	.	.	.	.	.	.	.	
3914.4V	N2+ 1N	O O	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	W 50	.	300	100	
3914.5	FE II 3	A 4P Z 6D O	5/2 3/2	4.82	.	.	.	3 10B	.	.	.	.	1 15B	W .8B	5B	1	W	.	.	.	.	
3914.8	SC III 1F	3 2D 4 2S	3/2 1/2	3.15	.	.	.	.	.	.	.	.	1 15B	.	.	.	.	.	.	.	.	
3914.8	A II 2	3D 4D 4P 4D O	3/2 3/2	19.53	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	
3915.3	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
3916.4	V II 10	A 3P Z 3D O	1 2	4.57	.	.	.	X	.	.	.	.	W 4	W 1	.	.	.	.	.	.	.	
3917.2	FE I 20	A 5F Y 5D O	2 3	4.14	.	.	.	.	.	.	.	.	.	.	.	.	1 2	.	.	.	.	
3918.5	FE II 191	D 2D Y 2P O	3/2 1/2	9.03	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	
3918.7	FE II -F	A 4D C 2G	.	.	.	.	.	.	.	.	.	.	1 4B	.	.	.	.	.	.	.	.	
3918.7	-	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	
3919.0	C II 4	3 2P O 4 2S	1/2 1/2	19.41	.	.	XB	10B	XB	.07	.15	1 4B	.	.	.	5B	.	.	.	.	.	
3919.0	N II 17	3P 1P 3D 1P O	1 1	23.47	.	.	.	.	.	.	.02 .16B	WB	2	.	.	.	.	.	.	1	.	
3919.2	CR I 23	A 5D Z 5D O	4 4	4.17	.	.	.	10B	.	.	.	.	WB	.	.	.	.	.	.	.	.	
3919.2	CR I 136	B 5D T 5P O	3 3	6.15	.	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.	.	
3919.3	O II 17	3S ⁺ 2D 3P ⁺ 2P O	3/2 1/2	28.70	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	
3920.3	FE I 4	A 5D Z 5D O	0 1	3.27	.	.	.	10B	.	.	.	.	WB	W	.	1	.	.	.	.	.	
3920.7	C II 4	3 2P O 4 2S	3/2 1/2	19.41	.	.	XB	10B	XB	.08 .16	.5	WB	.	.	.	5B	.	.	.	.	.	
3921.4	TI I 14	A 3F Z 3P O	.	3.15	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	
3922.9	FE I 4	A 5D Z 5D O	3 4	3.20	.	.	.	.	.	.	.	.	1 3B	W	.	3	5	.	.	.	.	
3923.0	TI IX -F	.	.	.	.	.	.	.	.	.	.	.	1 3B	.	.	.	.	.	.	.	.	
3923.5	FE V 3F	3D4 5D 3D4 3F	4 2	3.30	.	.	.	.	.	.	.	.	1 10B	.	.	.	.	.	.	.	.	
3923.5	HE II 4	4 2F O 15 2G	.	53.94	.	.	WB	.	20B	XB	.18 .70	.	1 10B	.	.	.	.	.	.	.	.	
3923.5	S II 55	4P 2D U 4D 2F	3/2 5/2	19.27	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	
3923.9	ZR II 100	B 2D X 4D O	5/2 5/2	4.89	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	
3924.3	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
3924.4	SI III -	.	.	.	.	.	.	.	.	130B	.	.	.	.	.	.	.	.	.	.	.	
3924.5	TI I 13	A 3F Y 3D O	3 3	3.17	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	
3926.5	HE I 58	2 1P O 8 1D	1 2	24.27	.	.	3	10	.	.	.14 .30	.5	1 20	W 6	.	.	.	.	.	.	.	

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA GAL	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME							
3927.9	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
3927.9	FE I 4	A 5D Z 5D 0	1 2	3.25	.	.	.	.	.	.	W 4	W	.	1	2	.	.	.	.	.	.	.
3928.6	A II 10	4S 4P 4P 4S 0	1/2 3/2	19.88	.	.	.	.	.	.02B	.	.	.	.	.	.	.	.	.	.	.	.
3928.6	S III 8	3D 3D 0 4P 3P	3 2	21.38	.	.	.	XB	XB	.02B .06	.	.	.	.	.	.	.	.	.	.	.	.
3929.0	-	.	.	.	.	.	.	.	.	.4	.	.	.	.	.	.	.	.	.	.	.	.
3929.7	V II 10	A 3P Z 3D 0	1 1	4.56	.	.	.	.	.	.	1 15B	W	.	.	.	.	.	.	.	.	.	.
3930	-	.	.	.	.	.	7	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.
3930.3	FE I 4	A 5D Z 5D 0	2 3	3.23	.	.	.	.	.	.	1 15B	.	.	.	2	.	.	.	.	.	.	.
3930.3	FE II 3	A 4P Z 6D 0	3/2 1/2	4.83	.	3	.	.	.	.	1 15B	5	.	.	.	.	.	.	.	.	.	.
3931.4	FE II 8F	A 6D A 4G	9/2 11/2	3.14	.	.	3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3932.0	TI II 34	A 2G Z 2G 0	9/2 7/2	4.26	.	.	.	.	.	.	W 4	W 1	.	.	.	.	.	.	.	.	.	.
3932.1	-	.	.	.	.	.	1	.	.	.08	.	W	.	.	.	.	.	.	.	.	.	.
3932.7	FE II 8F	A 6D A 4G	7/2 9/2	3.19	.	.	W 3B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3932.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.
3933.3	S II 55	4P 2D 0 4D 2F	5/2 7/2	19.33	.	.	.	XB	.03B	.	.	.	.	.	.	.	.	.	.	.	.	.
3933.7	CA II 1	4 2S 4 2P 0	1/2 3/2	3.14	.	1 2	2B	.	.	.	1 15 1	3 100	5 120B*20	10 2	2	.	.	.	.	.	.	.
3934	FE V -F	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3934.4	N III 8	3P 2P 3D 2D 0	1/2 3/2	41.30	.	.	20B	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
3934.5	-	.	.	.	.	.	1	.	.	.06	.	.	.	.	.	.	.	.	.	.	.	.
3934.8	ZR II 43	A 2F Z 2D 0	5/2 5/2	3.85	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3935.0	-	.	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	.	.
3935.9	FE II 173	C 2F X 2G 0	7/2 9/2	8.68	.	.	1 10B	.	.	.	W 4B	2	W	.	.	.	.	.	.	.	.	.
3935.9	HE I 57	2 1P 0 8 1S 5 2S	1 0 1/2	24.26	.	.	10B	.	.4 .7	.15	W 4B	1	.	.	.	.	.	.	.	.	.	.
3936	C IV 2	6 2P 0 A 2F Y 4F 0	1/2 7/2 9/2	58.12	.	.	.	130B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3936.0	CO I 32	A 2F Y 4F 0	7/2 9/2	4.05	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.
3937.8	FE II 8F	A 6D A 4G	5/2 5/2	3.22	.	.	20B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3938 R	O2 H	1 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3938.3	FE II 3	A 4P Z 6D 0	5/2 5/2	4.80	.	W	8 20B	.	.	.	1 20B	1 9	1B	.	15	.	.	.	.	.	.	.
3938.4	-	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.
3938.5	N III 8	3P 2P 3D 2D 0	3/2 5/2	41.30	.	.	20B	.	.	.	1 20B	5	.	.	.	.	.	.	.	.	.	.
3939.0	FE II 190	D 2D X 2F 0	3/2 5/2	9.02	.	.	20B	.	.	.	1 20BM	1 2	1B	.	.	.	.	.	.	.	.	.
3940 R	O2 H	5 9	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3940.3V	N2 VK	7 16	.	6.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2B	.
3940.9	FE I 20	A 5F Y 5D 0	3 4	4.09	.	.	.	.	.	.	.	.	.	.	1 2	.	.	.	.	.	.	.
3941.7	CO I 17	B 4F Z 4G 0	9/2 11/2	3.56	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.
3942.8	N III 8	3P 2P 3D 2D 0	3/2 3/2	41.30	.	.	20B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
3943.0V	N2 2P	2 5	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2B	1
3944.0	AL I 1	3 2P 0 4 2S	1/2 1/2	3.13	.	.	.	.	.	.	W	.	.	W	2 10	.	.	.	.	.	.	.
3944.2	-	.	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	.	.
3945.0	O II 6	3S 2P 3P 2P 0	1/2 3/2	26.45	.	.	.	.	.	.03	1 10B	W	.	.	.	.	.	.	.	.	W	.

*Nebular as well as stellar origin.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC FM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
3958.2	TI I 13	A 3F Y 3D 0	4 3	3.17	.	.	.	.	.	.	.	.	.	.	3	.	.	.	.			
3958.2	ZR II 16	A 2D Z 2F 0	3/2 5/2	3.64	.	.	103	.	.	.	.	.	.	.	.	.	.	.	.			
3960.7	-	.	.	.	.	.	.	.	.	.02	.	.	.	.	.	.	.	.	.			
3960.9	FE II 212	D 2G V 2G 0	9/2 9/2	10.35	.	.	103	.	.	.	.	1 15B	W	W	.	.	.	.	.			
3961.5	AL I 1	3 2P 0 4 2S	3/2 1/2	3.13	.	.	.	.	.	.	.	1 15B	W	.	W	10	.	.	.			
3961.6	O III 17	3P 1D 3D 1F 0	2 3	40.96	.	.	.	XB	50B	.20	.	.	.	.	.	.	.	.	.			
3961.6	S III 8	3D 3D 0 4P 3P	1 1	21.33	.	.	.	.	XB	.05	.	.	.	.	.	.	.	.	.			
3962.9	TI I 12	A 3F Y 3F 0	2 3	3.11	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.			
3964.3	TI I 12	A 3F Y 3F 0	3 4	3.13	.	.	.	.	.	.	.	X	.	.	1	.	.	.	.			
3964.4	CR II 10	A 4P Z 6F 0	5/2 5/2	5.81	.	.	.	.	.	.	.	1 20B	.	.	.	.	.	.	.			
3964.6	FE II 29	B 4P Z 4P 0	1/2 1/2	5.88	.	.	.	.	.	.	.	1 20B	.	.	.	.	.	.	.			
3964.7	HE I 5	2 1S 4 1P 0	0 1	23.64	.	.	50B	2B	40B	50B	.6 1.2	4.3 1 30B	W	W	.	.	.	.	.			
3965.9	-	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.	.	.			
3966.1	FE I 45	A 3F Y 3D 0	2 3	4.71	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.			
3966.4	FE II 3	A 4P Z 6D 0	1/2 1/2	4.83	.	.	2.5BM	.	.	.	.	W	1	.	.	.	.	.	.			
3967.4	O II 22	3P 4P 0 3D 2P	1/2 1/2	28.83	.	.	.	.	.	.	9.3 24B	.	.	.	.	.	.	.	.			
3967.5	NE III 1F	2P4 3P 2P4 1D	1 2	3.19	1 3B	5 25B	65B 10B	2.5	.	1.0 47B	9.3 24B	5 25	.	.	18 30B	.	.	.	.			
3967.7	-	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.			
3968.0	O III -F	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.			
3968.4	HE II 3	4 2F 0 14 2G	.	53.91	.	.	.	30B	XB	X	.	.	.	.	.	.	.	.	.			
3968.5	CA II 1	4 2S 4 2P 0	1/2 1/2	3.11	.	.	.2 12B	2B	40B	.	.	1 15B	2	10	5 100	5 100B40	.	.	.			
3969.4	FE II 3	A 4P Z 6D 0	3/2 5/2	4.80	.	.	7BM	.	.	.	.	1 15BM	.	.	10BM	.	.	.	.			
3970.1	FE V 1F	3D4 5D 3D4 3P	4 2	3.27	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.			
3970.1	HE I	2 2P 0 7 2D	.	13.26	1 3B	5 25	30 65B	7B	30B	X*	13.6 47B	15 18B	15 65	1 40	18 30	5 100B10	W	.	.			
3970.2	C II 38	3D 4D 0 4F 2D	5/2 3/2	27.38	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.			
3972.2	NI I 29	A 1D Z 5G 0	2 3	3.53	.	.	.	.	.	.	.	.	.	.	W 5	.	.	.	.			
3972.4	C II 37	3D 4D 0 4F 4D	3/2 5/2	27.37	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.			
3972.5	-	.	.	.	.	.	1.5	.	.	.	.	.	.	.	.	.	.	.	.			
3973.3	O II 6	3S 2P 3P 2P 0	3/2 3/2	26.45	.	.	.	.	.	.06	XB	18	.	.	.	.	.	2	1			
3973.6	NI I 31	A 1D Z 3P 0	2 2	3.53	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.			
3973.6	V II 9	A 3P Z 5F 0	1 2	4.53	.	.	.	.	.	.	.	1 4B	W	.	.	.	.	.	.			
3974 R	O2 H	7 10	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.			
3974.2	FE II 29	B 4P Z 4P 0	3/2 5/2	5.80	.	.	4 10B	.	.	.	.	1 4B	1 5	18	.	.	.	.	.			
3975.0	FE II 191	D 2D Y 2P 0	5/2 3/2	9.04	.	.	10B	.	.	.	.	.	.	18	.	.	.	.	.			
3976.2	FE III 4F	A 5D A 3G	4 3	3.10	.	.	.	.	.	.	.4	.	.	.	.	.	.	.	.			
3976.4	-	.	.	.	.	.	.	.	.	.02	.	W	.	.	.	.	.	.	.			
3976.9	FE III 69	D 3G W 3G 0	5 5	17.22	.	.	.	.	.	.	.	2B	.	.	.	.	.	.	.			
3977.1	O IV 10	3P 4P 3D 4P 0	5/2 3/2	62.21	.	.	.	.	.	.02	.	.	.	.	.	.	.	.	.			
3977.7	FE I 72	A 5P X 5P 0	2 2	5.29	.	.	.	.	.	.	.	W 4B	.	.	15	.	.	.	.			
3977.7	V II 10	A 3P Z 3D 0	2 2	4.57	.	.	.5	.	.	.	.	W 4B	1	.	.	.	.	.	.			

*Nebular emission background origin.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	DE BE SYMB	AE FE PCYG	T- TAU	KE ME N								
															QUAS	GAL	NOVA	W-R				
4030.8	MN I 2	A 6S Z 6P 0	5/2	3.06	.	.	.	.	.	.	.	.	1 5	1	.	.	50	.	.	.	.	
4031.5	FE II 151	C 2D Y 4D 0	5/2	7.77	.	.	.	.	.	.	.	.	.	W	.	.	X	.	.	.	.	
4032.0	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4032.3	NI VIII 2F	3D3 4F 3D3 2G	7/2	3.33	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4032.9	FE II 126	B 2D Z 4G 0	5/2	7.54	.	.	.	2	.	.	.	.	W	W 3	WB	.	.	.	.	.	.	
4033.1	MN I 2	A 6S Z 6P 0	5/2	3.06	.	.	.	.	.	.	.	.	1 5B	.	.	.	24	.	.	.	.	
4033.6	NI II 4F	A 2D A 4P	5/2	3.06	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	
4034.5	MN I 2	A 6S Z 6P 0	5/2	3.06	.	.	.	.	.	.	.	.	1 5B	.	.	.	24	.	.	.	.	
4035.1	N II 39	3D 3F 0 4F 3G	2	26.08	.	.	.	.	.	.	.	.	W	.	.	.	.	.	4	50	.	
4035.1	O II 51	3D 4F 4F 2F 0	5/2	31.62	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	
4035.5	FE II 22	A 2D Z 4F 0	5/2	5.59	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	
4035.5	FE III 119	Y 7P 0 F 7S	2	23.57	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	
4035.6	V II 32	A 3G Z 3G 0	3	4.84	.	.	.	.5	.	.	.	.	WB	3	W 3	.	.	.	.	.	.	
4035.8	FE III 45	E 3F Z 3F 0	3	14.59	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	
4036.8	V II 9	A 3P Z 5F 0	2	4.53	.	.	.	.	.	.	.	.	1 4	1	.	.	.	.	.	.	.	
4037.0	-	.	.	.	.	.	.	.	.	.	.	.	W 1B	.	.	.	.	.	.	.	.	
4038.0	CR II 194	D 2G W 2H 0	9/2	9.52	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	
4038.2	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4038.7	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
4039.1	FE III 45	E 3F Z 3F 0	2	14.59	.	.	.	.	.	.	.	.	1 2B	.	.	.	.	.	.	.	.	
4039.9	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
4041.0	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4041.3	O II 50	3D 4F 4F 4F 0	5/2	31.61	.	.	.	.	.	.	.	.	.02 .18	W	.	.	.	.	6B	.	.	
4041.3	N II 39	3D 3F 0 4F 3G	4	26.10	.	.	7B	.	.	.	.	.	.02 .18	W	.	.	.	.	6B	50	.	
4041.8	FE II 13	A 2P Z 6P 0	1/2	5.39	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
4041.8	NO#	0 12	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	6B	.	.	
4043.0	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4043.4	FE III -	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	
4043.5	N II 39	3D 3F 0 4F 3G	3	26.08	.	.	.	.	.	.	.	.	.04	.	.	.	.	.	2B	50	.	
4044.0	FE II 172	C 2F Y 2F 0	5/2	8.60	.																	

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION LEVELS J	UPPER EXC. POT., EV	EMISSION LINE INTENSITY																AIR GLW	CTY LTS	AUR ORA	LGT NING	
				NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME										
															QUAS	GAL	NOVA	LIKE	N					C
4049.1	-	.	.	.	.	1	.	.	.	.	.3	.	.	.	.	.	.	.	.	.	.	.	.	.
4049.1	CR II 193	D 2G W 2F 0	7/2 5/2	9.50	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
4050.2	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
4051.2	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
4051.2	FE II 172	C 2F Y 2F 0	5/2 7/2	8.59	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.
4052.0	CR II 19	B 4D Z 6D 0	7/2 5/2	6.14	.	.	.	.	.	.	.	4	2	.	.	.	.	.	.	.	.	.	.	.
4052.1	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4053.3	FE III 119	Y 7P 0 F 7S 3	3 3	23.57	.	.	.	.	.	.	.	2B	XB	.	.	.	.	.	.	.	.	.	.	.
4053.8	TI II 87	B 2G Y 2F 0	7/2 5/2	4.93	.	.	.	2BM	.	.	.	.	1 15B 3	W	W	.	.	.	.	.	.	.	.	.
4054.1	CR II 19	B 4D Z 6D 0	5/2 5/2	6.14	.	.	.	2B	.	.	.	.	1 158	2	.	.	.	.	.	.	.	.	.	.
4056.1	C III 24	4 1D 5 1F 0	2 3	43.07	.	.	.	.	60 150B	.02	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
4056.2	TI II 11	A 2F Z 4G 0	7/2 5/2	3.65	.	.	.	.	.	.	.	WB	1	.	.	.	.	.	.	.	.	.	.	.
4057.1	-	.	.	.	.	.	.	.	.	.03	.	.	WB	WB	.	.	.	.	.	.	.	.	.	.
4057.5	FE II 212	D 2G V 2G 0	7/2 7/2	10.28	.	.	.	2B	.	.	.	.	WB	WB	WB	.	.	.	.	.	.	.	.	.
4057.8	-	.	.	.	.	.	.	.	.	.05	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4057.8	N IV 3	3 1P 0 3 1D 2	1 2	52.98	.	.	.	50 350	XB	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
4058.1	-	.	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
4058.7	-	.	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4059.3	F IV 1F	2P2 3P 2P2 1D 2	2 2	3.12	.	.	.	.	.	.03 .30	.7	W 2	.	.	.	.	.	.	.	.	.	.	.	.
4059.4V	N2 2P	0 3	.	11.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	50	.
4060.6	O II 97	3D [*] 2F 4F [*] 2G 0	7/2 9/2	34.05	.	.	.	.	.	.03B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4061.5	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4062.6	-	.	.	.	.	.	.	.	.	.4	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4062.9	N II -	.	.	.	.	.	.	.	.	.05B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4062.9	O II 50	3D 4F 4F 4F 0	9/2 9/2	31.62	.	.	.	.	.	.05B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4063.6	FE I 43	A 3F Y 3F 0	3 3	4.59	.	.	.	.	.	.	.	1 10B	2	W	15	5	.	.	.	.	.	.	.	.
4063.9	CR II 19	B 4D Z 6D 0	5/2 7/2	6.13	.	.	.	.	.	.	.	1 10B	WB	.	.	.	.	.	.	.	.	.	.	.
4064	O2 H	0 7	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.
4064.4	TI II 106	B 2F Y 2P 0	5/2 3/2	5.62	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
4064.5	-	.	.	.	.	.	2.5	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4064.8	FE II 39	B 4F Z 4P 0	3/2 1/2	5.88	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
4065.1	V II 215	D 3F X 3G 0	4 5	6.82	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
4066.3	FE II 214	Z 2D 0 E 4F	3/2 3/2	10.70	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.
4066.4	CO I 30	A 2F Y 4D 0	7/2 7/2	3.95	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10B	.	.	.
4066.5	AL H	1 0	R(5) R(6)	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10B	.	.	.
4067.0	V II 9	A 3P Z 5F 0	2 1	4.50	.	.	.	.	.	.	.	W 3B	.	.	.	.	.	.	.	.	.	.	.	.
4067.1	NI II 11	A 2G Z 2F 0	9/2 7/2	7.05	.	.	.	5	.	.	.	W 3B	3 6	WB	.	.	.	.	.	.	.	.	.	.
4067.9	C III 16	4 3F 0 5 3G 3	2 3	42.77	.	.	.	.	60 150B	.	.	5 25B	.	.	.	.	.	.	.	.	.	.	.	.
4068.6	S II 1F	3P3 4S 0 3P3 2P 0	3/2 3/2	3.03	.	5 50B	8 16	12	XB 4.5	.50 7.3	2.7 25B	5 2*	.	5 70*	20	.	.	.	.	.	.	.	.	.
4069.0	C III 16	4 3F 0 5 3G 4	3 4	42.77	.	.	8B	.	60 150B	.	.	5 25B	.	.	.	.	.	.	.	.	.	.	.	.

*Nebular as well as stellar origin.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME								
															QUAS	SAL	NOVA	W-R				
4069.6	O II 10	3P 4D 0 3D 4F 3/2	1/2	28.55	.	.	XH	.	.	.	.17 .21	.12B 5 25B	.	.	.	.	.	48	.			
4069.9	O II 10	3P 4D 0 3D 4F 5/2	3/2	28.56	.	.	.	.	.	.	.53	.12B 5 25B	.	.	.	.	.	48	.			
4070.0	FE II 22	A 2D Z 4F 0 5/2	5/2	5.57	.	.	.	.	.	.	.	.	1	W	.	.	.	.	.			
4070.3	C III 16	4 3F 0 5 3G 5	4	42.77	.	.	30B	.	.	60	.5	.3B 5 150B 25B	.	.	.	.	.	.	.			
4070.7	FE III 4F	A 5D 4 A 3G 5	4	3.03	.	.	.	.	.	.	.	.01	.	.	.	.	.	.	.			
4070.9	CR II 193	D 2G W 2F 0 7/2	9/2	9.49	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.			
4071.3	AL H	1 0	R(9)	.	.	.	.	.	.	.	.	.	.	.	.	.	5 10B	.	.			
4071.5	FE V 1F	3D4 5D 3D4 3P 1	2	3.08	.	.	8B 4	.	.	.	.22	X 1 6B	.	.	.	.	.	.	.			
4071.7	FE I 43	A 3F 2 Y 3F 0 2	2	4.63	.	.	.	.	.	.	.	1 6B	W	.	1	10B	.	.	.			
4072.0	A II 33	4S 2D 4P 2D 0 5/2	5/2	21.41	.	.	.	.	.	.	.15B	.	.	.	.	.	.	.	.			
4072.2	O II 10	3P 4D 0 3D 4F 7/2	5/2	28.57	.	.	XB	.	.	.	.10 .44	.3 1 6B	.	.	.	.	.	5B	.			
4072.5	N2 VK	2 13	.	6.1	.	.	.	.	.	.	.	.	.	.	.	.	.	5B	.			
4072.6	CR II 26	B 4P Z 4D 0 3/2	1/2	6.73	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.			
4073.1	N II 38	3D 3F 0 4F 3F 3	2	26.05	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10			
4073.8	AL H	1 0	R(10)	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.			
4073.9	-	.	.	.	.	.	.	W	.	.	.20	.	.	.	.	.	.	.	.			
4073.9	O III 23	3S 3P 3P 3D 0 2	1	46.25	.	.	.	.	60	W	.	W	.	.	.	.	.	.	.			
4075.4	SI II --	.	.	.	.	.	.	.	150B	.06	.	W	.	.	.	.	.	.	.			
4075.6	CR II 19	B 4D Z 6D 0 3/2	1/2	6.12	.	.	.	.	.	.	.	1 15B	.	.	.	.	.	.	.			
4075.9	O II 10	3P 4D 0 3D 4F 9/2	7/2	28.58	.	.	XB	.	XB	.	.58B 3.0B 1 15B	W	.	.	.	.	5B	.				
4076.0	C II 36	3D 4D 0 4F 4F 9/2	7/2	27.30	.	.	.	.	XB	W	.	1 15B	.	.	.	.	.	.	.			
4076.0	FE II 21	A 2D Z 4D 0 3/2	5/2	5.56	.	.	.	.	.	.	.	1 15B	.	WB	.	.	.	.	.			
4076.2	S II 1F	3P3 4S 0 3P3 2P 0 1/2	3/2	3.03	.	5	2	.	XB	.75	3.0B 1 2.3 15B	W	.	.	40 70	.	.	.	.			
4077.2	FE II -	.	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.			
4077.5	CR II 19	B 4D Z 6D 0 3/2	5/2	6.12	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.			
4077.7	SR II 1	5 2S 5 2P 0 3/2	1/2	3.03	.	.	.	.	.	.	.	.	.	.	5	10	.	.	.			
4077.8	HG I -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	30	.	.			
4077.8	-	.	.	.	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.			
4078.9	AL H	1 0	Q(6)	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.			
4078.9	O II 10	3P 4D 0 3D 4F 3/2																				

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME								
																QUAS	GAL	NOVA	LIKE				
4083.9	O II 49	3D 4F 4F 2G O	5/2 7/2	31.58	.	.	.	.	.	.	.05 .11	XB	.	.	.	.	.	.	.	.	.	.	.
4084.5	AL H	1 0	Q(8)	.	.	.	.	.	.	.	.	.	.	.	.	15	.	.	.	.	.	.	.
4084.7	O II 21	3P 4P O 3D 2F	5/2 7/2	28.76	.	.	.	.	.	.	.03B	.	W	.	.	.	.	.	.	.	.	.	.
4085.1	O II 10	3P 4D O 3D 4F	5/2 5/2	28.56	.	.	.	.	.	.	.03 .14B	XB	W	W	.	.	.	.	.	.	.	.	.
4085.9	-	.	.	.	.	.	.	.	.	.	.	.01	.	.	.	.	.	.	.	.	.	.	.
4086.3	CO I 58	B 4P X 4D O	5/2 7/2	4.89	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.
4086.5	CA XIII 1F	2P4 2P 2P4 2P	2 1	3.02	.	.	1	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4086.7	NE II 54	3D 4D 4F 4G O	7/2 5/2	37.48	.	.	.	.	.	.	.02B	.	.	.	.	.	.	.	.	.	.	.	.
4087.2	O II 48	3D 4F 4F 4G O	3/2 5/2	31.57	.	.	.	.	.	.	.14	.01	.	.	.	.	.	.	.	.	.	.	.
4087.3	FE II 28	B 4P Z 4F O	5/2 3/2	5.59	.	.	.	.	.	.	.	.	W	1	.	.	.	.	.	.	.	.	.
4088.0	AL H	1 0	Q(8)	.	.	.	.	.	.	.	.	.	.	.	.	15	.	.	.	.	.	.	.
4088.8	FE II 39	B 4F Z 4P O	5/2 3/2	5.85	.	.	.	.	.	.	.	.	WB	1	.	.	.	.	.	.	.	.	.
4088.9	CR II 19	B 4D Z 6D O	3/2 1/2	6.11	.	.	.	.	.	.	.	.	5B	.	.	.	.	.	.	.	.	.	.
4088.9	SI IV 1	4 2S 4 2P O	1/2 3/2	26.97	.	.	2 3	.	20	X	.08 .48*	.	5B	2	.	.	.	.	.	.	.	.	.
4089.3	O II 48	3D 4F 4F 4G O	9/2 11/2	31.60	.	.	.	.	.	.	.15	1B	5B	W	.	.	.	.	.	.	.	.	.
4089.5	CR II 164	B 2D Y 4G O	5/2 5/2	8.31	.	.	.	.	.	.	.	.	5BM	.	.	.	.	.	.	.	.	.	.
4090.7	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	2	.	.	.	.	.	.
4091.0	-	.	.	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4092.0	AL H	1 0	Q(10)	.	.	.	.	.	.	.	.	.	.	.	.	15	.	.	.	.	.	.	.
4092.4	CO I 29	A 2F Z 2F O	7/2 7/2	3.93	.	.	.	.	.	.	.	.	.	.	.	15BM	.	.	.	.	.	.	.
4092.9	O II 10	3P 4D O 3D 4F	7/2 7/2	28.57	.	.	.	.	.	.	.03 .09	.	1 5	.	.	.	.	.	.	.	.	.	.
4093.6	AL H	1 0	P(17)	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.
4094.2	O II 10	3P 4D O 3D 4F	5/2 3/2	28.55	.	.	.	.	.	.	.	.	5 15	.	.	.	.	.	.	.	.	.	.
4095.5	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.	.	.	.	.
4095.7	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4096.3	-	.	.	.	.	.	.	.	.	.	1 3	.	.	.	.	.	.	.	.	.	.	.	.
4096.8	AL H	1 0	Q(11) Q(12)	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.	.	.	.	.
4097	K VI 2F	3P2 1D 3P2 1S	2 0	5.35	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4097.3	C II -	.	.	.	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.
4097.3	N III 1	3 2S 3 2P O	1/2 3/2	30.33	.	.	W 80B	3	140 500B	XB 2.0	.5 WB	5 30B	W 5	.	.	.	.	.	.	.	.	.	.
4097.3	O II 20	3P 4P O 3D 4D	1/2 3/2	28.73	.	.	.	.	.	.	.19 1.9B	WB	5 30B	.	.	.	.	.	.	.	.	.	.
4097.3	O II 48	3D 4F 4F 4G O	7/2 9/2	31.58	.	.	.	.	.	.	.19 1.9B	WB	5 30B	.	.	.	.	.	.	.	.	.	.
4098.2	AL H	1 0	P(8)	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.	.	.	.	.
4098.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.
4099.5	-	.	.	.	.	.	.	.	.	.	.05	.	.	.	.	.	.	.	.	.	.	.	.
4099.9	N I 10	3S 2P 3P' 2D O	1/2 3/2	13.64	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	50
4100.0	HE II 3	4 2F O 12 2G	.	53.81	.	.	20B	.	140 500B	70	1.4	.	1 8	.	.	.	.	.	.	.	.	.	.
4101.7	Hg 1	2 2P O 6 2D	.	13.16	X	7 35	40 80	10 30B	175B X**	14 26.5	25 75	20 50	1 50	20 50	15 30	W 20	.	.	.	.	.	.	.
4101.7	N I -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	50
4102.2	N I -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	50

*Stellar origin reported in one case.

**Nebular emission background origin.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. PCT., EV	EMISSION LINE INTENSITY															AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N								
4102.9	SI I 2	3P2 1S 4S 3P 0	0 1	4.91	.	.	.	.	.	.	.	2	4	.	.	10B	.	.	.	.	.	.	.
4103.0	O II 20	3P 4P 0 3D 4D 1/2	1/2 1/2	28.73	.	.	.	.	.	.97B	.	.	.	.	.	.	.	.	.	.	.	.	.
4103.1	AL H	1 0	P(9)	.	.	.	.	.	.	.	.	.	.	.	.	10B	.	.	.	.	.	.	.
4103.4	N III 1	3 2S 3 2P U	1/2 1/2	30.33	.	.	80B 303	140 500B	XB	.20 1.9	20B 30	.	.	.	.	.	.	.	.	.	.	.	.
4103.9	FE III -	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.
4104.2	FE II 39	H 4F Z 4P 0	.	5.85	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4104.5	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	X	.	.	.	.	.	.	.
4104.7	O II 20	3P 4P 0 3D 4D 5/2	3/2 5/2	28.73	.	.	.	.	.	.05 .20	WB WB	.	.	.	.	.	.	.	.	.	.	.	.
4105.0	O II 20	3P 4P 0 3D 4D 3/2	3/2 3/2	28.73	.	.	.	.	.	.10	WB WB	.	.	.	.	.	.	.	.	.	.	.	.
4105.5	-	.	.	.	.	.	.	.	.	.3 .6	.	W W	.	.	.	.	.	.	.	.	.	.	.
4106.0	O II 10	3P 4D 0 3D 4F 5/2	7/2 5/2	28.56	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4107.1	O II 47	3D 4F 5/2 4F 4D 0 7/2	5/2 7/2	31.57	.	.	.	.	.	.06	.	W	.	.	.	.	.	.	.	.	.	.	.
4107.8	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4108.8	AL H	1 0	P(10)	.	.	.	.	.	.	.	.	.	.	.	.	24	.	.	.	.	.	.	.
4108.9	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4109	-	.	.	.	.	.	.	.	.	.3 .7	X	.	.	.	.	.	.	.	.	.	.	.	.
4109.6	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4109.8	CA II 17	5 2P U 7 2D 5/2	3/2 5/2	10.49	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4110.0	-	.	.	.	.	.	15	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4110.0	FE III --	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.
4110.0	N II 44	3D 1D 0 4F 3D 2	2 2	26.10	.	.	.	.	X	.03	.	.	.	.	.	.	.	.	.	.	.	.	.
4110.2	O II 37	3P ¹ 2F 0 3D ¹ 2D 3/2	5/2 3/2	31.24	.	.	.	.	.	.04B	.	WB	.	.	.	.	.	.	.	.	.	.	.
4110.4	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.
4110.8	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4110.8	O II 20	3P 4P 0 3D 4D 1/2	3/2 1/2	28.73	.	.	.	.	.	.05	.06	WB	.	.	.	.	.	.	.	.	.	.	.
4111.0	CR II 18	B 4D Z 4P U	7/2 5/2	6.09	.	.	.	1B	.	.	.	.	2B	.	.	.	.	.	.	.	.	.	.
4111.0	CR II 26	B 4P Z 4D 0	3/2 5/2	6.74	.	.	.	.	.	.	.	.	2B	.	.	.	.	.	.	.	.	.	.
4111.9	FE II 18B	D 2D W 4F U	5/2 7/2	8.93	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.
4112.0	O II 21	3P 4P 0 3D 2F 5/2	5/2 5/2	28.74	.	.	.	.	.	.04	X	.	W	.	.	.	.	.	.	.	.	.	.
4113.2	CR II 18	B 4D Z 4P U	5/2 5/2	6.09	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
4113.2	FE III --	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.
4113.8	O II 37	3P ¹ 2F 0 3D ¹ 2D 5/2	7/2 5/2	31.24	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.
4114.5	FE II 23F	A 4F B 2H	9/2 11/2	3.23	.	.	.	7 10	.	.	.15	.	.	.	.	10B	.	.	.	.	.	.	.
4114.7	AL H	1 0	P(11)	.	.	.	.	.	.	.	.	.	.	.	.	10B	.	.	.	.	.	.	.
4115	-	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4116.1	SI IV 1	4 2S 4 2P U	1/2 1/2	26.95	.	.	X	70	XB	.03 .45*	.	W	1.8	.	.	.	.	.	.	.	.	.	.
4117.6	-	.	.	.	.	.	.	.	.	X	W	.	.	.	.	.	.	.	.	.	.	.	.
4118.5	-	.	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.
4118.7	FE III -	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.
4118.8	CO I 28	A 3F Z 2G U	5/2 7/2	4.04	.	.	.	.	.	.	.	.	.	.	.	15**	.	.	.	.	.	.	.

*Stellar origin reported in one case.

**Molecular band to 4200. Also at least 25 unidentified lines between 4100 and 4200.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
4155 R	02 H	3	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	4	.	.	.
4155.0	-	9	.	.	.	.	.	.	.	.5	2	.	.	.	.	.	.	.	.	.	.	.
4155.0	FE III --	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.
4155.5	-	.	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	.	.
4156.2	ZR II 29	A 2P Z 4F U	1/2 3/2	3.68	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4156.5	C III 21	3P 3D 5 3F U	2 3	42.84	.	.	.	.	X	.3	.	WB	.	.	.	.	.	.	.	.	.	.
4156.5	O II 19	3P 4P U 3D 4P	5/2 3/2	28.71	.	.	.	.	60B	.3	.4	WB	.	.	.	.	.	.	.	.	.	.
4156.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.
4156.8	N II 50	3D 3D U 4F 3D	1 1	26.11	.	.	.	.	.	.	.15B	.	.	.	.	.	.	.	.	.	.	.
4156.8	N II 51	3D 3D U 4F 1D	1 2	26.11	.	.	.	.	.	.	.15B	.	.	.	.	.	.	.	.	.	.	.
4157.9	FE II 37F	A 4D B 2F	7/2 7/2	3.95	.	.	.	.	.	.	.	1 4	.	.	.	.	.	.	.	.	.	.
4158.5	FE II 12	A 2P Z 6F U	3/2 3/2	5.23	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4158.8	O V 11	3P 3S 3D 3P U	1 2	87.36	.	.	.	.	60B	.	.	.	.	.	.	.	.	.	.	.	.	.
4159.2	-	.	.	.	.	.	.	.	.	.02	.	.	.	.	.	.	.	.	.	.	.	.
4160.6	FE II 39	B 4F Z 4P U	5/2 5/2	5.80	.	.	.	.	.	.	.	W	1	.	.	.	.	.	.	.	.	.
4160.8	N II 50	3D 3D U 4F 3D	2 1	26.11	.	.	.	.	.	.	.15B	.	.	.	.	.	.	.	.	.	.	.
4160.8	N II 51	3D 3D U 4F 1D	2 2	26.11	.	.	.	.	.	.	.15B	.	.	.	.	.	.	.	.	.	.	.
4161	02 H	5	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	4	.	.	.
4161.1	-	10	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	.	.
4161.2	ZR II 42	A 2F Z 4F U	5/2 3/2	3.68	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4161.5	FE III -	.	.	.	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
4161.5	TI II 21	A 2D Z 4D U	5/2 7/2	4.05	.	.	.	.	.	.	.	W	4	WB	.	.	.	.	.	.	.	.
4162.7	S II 44	4P 4D U 4D 4F	7/2 9/2	18.84	.	.	.	10B	.	XB	.	.	.	.	.	.	.	.	.	.	.	.
4162.7	S II 65	4P' 2F U 4D' 2G	7/2 9/2	20.29	.	.	.	103	.	XB	.	.	.	.	.	.	.	.	.	.	.	.
4162.8	C III 21	3P 3D 5 3F U	3 4	42.84	.	.	.	.	60B	.	.	W	.	.	.	.	.	.	.	.	.	.
4163.6	TI II 105	B 2F X 2D U	7/2 5/2	5.54	.	.	X	2.5	.	.	.	1 15	7	1	5	.	.	.	.	.	.	.
4164.1	-	.	.	.	.	.	.	.	.	.	.	.	.3	.	.	.	.	.	.	.	.	.
4164.6	NI I 28	A 1D Z 5D U	2 2	3.38	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
4164.8	FE III --	.	.	.	.	.	.	.	.	.	.	W 2B	.	.	.	.	.	.	.	.	.	.
4164.9	FE III -	.	.	.	.	.	.	.	.	.	.	W 2B	.	.	.	.	.	.	.	.	.	.
4165.6	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4166.0	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.
4166	K V 1F	3P3 4S U 3P3 2D U	3/2 3/2	2.96	.	.	.	.	.	.02 .50	.2	W 3	.	.	.	.	.	.	.	.	.	.
4166.4	ZR I 45	A 5F Y 5F U	4 5	3.65	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.
4166.6	-	.	.	.	.	.	.	.	.	.	.07	.	.	.	.	.	.	.	.	.	.	.
4166.9	FE III 118	Y 7P U F 7D	4 4	23.51	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.
4167.5	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.
4167.8	C N	3	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	108*
4168.4	AL II 61	4 1F U 10 1G	3 4	18.20	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
4168.4	FE III 118	Y 7P U F 7D	4 3	23.51	.	.	.	.	.	.	.	1B	X	.	.	.	.	.	.	.	.	.

*Molecular band to 4200. Also at least 25 unidentified lines between 4100 and 4200.

[illegible]

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. PUT., EV	EMISSION LINE INTENSITY																	
		LEVELS	J		NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGHT NING			
4204.2	V II 25	A 5P Z 5D 0	3 3	4.63	.	.	.	.	.	.	.	2 8B	1	.	.	.	.	.	.			
4205.1	V II 37	B 3G Z 3F 0	4 3	4.96	.	.	.	.	.	.	.	1 3	2	.	.	.	.	.	.			
4205.5	FE II 22	A 2D Z 4F 0	3/2 5/2	5.57	.	.	.	.	.	.	.	W	2	.	.	.	.	.	.			
4205.8	-	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.			
4206.6	-	.	.	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.	.			
4206.7	FE I 3	A 5D Z 7P 0	3 3	2.99	.	.	.	.	.	.	.	.	.	.	.	15	.	.	.			
4207	-	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.	.			
4207.5	-	.	.	.	.	.	.	.	.	.3 .5	.	.	.	.	.	.	.	.	.			
4208.2	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.			
4209.0	-	.	.	.	.	.	7	.	.	.	.	.	.	.	.	.	.	.	.			
4209.7	V II 25	A 5P Z 5D 0	1 1	4.60	.	.	.	.	.	.	.	1 3	W	.	.	.	.	.	.			
4210.6	ZR II 97	B 2D Y 4D 0	3/2 5/2	4.59	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.			
4210.8	-	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.			
4210.9	FE III --	.	.	.	.	.	3	.	.	.	.	2B	.	.	.	.	.	.	.			
4210.9	FE II -F	A 4F B 2H	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
4211.5	FE III 104	X 5P E 7D	1 2	18.18	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.			
4211.8	FE II 21	A 2D Z 4D 0	3/2 3/2	5.56	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
4212.4	SI IV 5	5 2D 6 2F 0	.	38.92	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.			
4213.6	-	.	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.			
4214.1	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.			
4214.7	N I 5	3S 4P 4P 4P 0	3/2 5/2	13.21	.	.	.	.	.	.	.	.	.	.	.	.	.	.	50			
4215.5	SR II 1	5 2S 5 2P 0	1/2 1/2	2.93	.	.	.	2	.	.	.	1 4B	2	W	8	30	.	.	.			
4215.7	N III 6	3S 2P 0 3P 2D	3/2 3/2	39.62	.	.	.	.	40 150B	XB	.03	1 4B	.	.	.	.	.	.	.			
4215.9	N I 5	3S 4P 4P 4P 0	1/2 3/2	13.21	.	.	.	.	.	.	.	.	.	.	.	.	.	.	50			
4216.2	FE I 3	A 5D Z 7P 0	4 4	2.93	.	W	.	.	.	.	.	W	2	.	.	10	.	.	.			
4216.9	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
4217	C IV 11	6 2S 8 2P 0	1/2	60.80	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.			
4217.1	O I 33	4 3P 3D 3P 0	.	15.23	.	.	.	.	.	.	.03B	.	.	.	.	.	.	.	.			
4217.2	NE II 52	3D 4D 4F 4D 0	7/2 5/2	37.39	.	.	.	.	.	.	.03B	.	.	.	.	.	.	.	.			
4217.6	LA II 78	Z 3F 0 E 3D	4 3	4.86	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.			
4219.2	-	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.			
4219.8	NE II 52	3D 4D 4F 4D 0	7/2 7/2	37.38	.	.	.	.	.	.08 .6	.	.	.	.	.	.	.	.	.			
4220.1	CA II 16	5 2P 0 8 2																				

*Probable blend with C IV 4217.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. PGT., EV	EMISSION LINE INTENSITY																
		LEVELS	J																		
					NUC EM	NOVA	W-R	W-R	PLAN	GAS	OE	AE	T-	KE	AIR	CTY	AUR	LIGHT			
					QUAS	GAL	NOVA	LIKE	N	C	NEB	NEB	SYMB	PCYG	GE	TAU	N	GLW	LTS	ORA	NING
4240.4	ZR I 45	A 5F Y 5F U	1 1	3.51	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.	.
4241.0	-	.	.	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.
4241.7	ZR I 45	A 5F Y 5F U	3 3	3.56	.	.	.	.	.	.	.	.	.	.	.	.	20B	.	.	.	.
4241.8	N II 47	3D 3D O 4F 1F	2 3	26.05	.	.	10F.	.	Xb	.	.07 .10	X	.	.	.	.	.	.	.	15B	.
4241.8	N II 48	3D 3D O 4F 3F	3 4	26.06	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	15B 50	.
4242.4	CR II 31	A 4F Z 4D U	3/2 1/2	6.76	.	.	.	453	.	.	.	.	1 8	W 7	1	.	.	.	.	.	.
4242.5	AL H	O	R(16)	.	.	.	.	.	.	.	.	.	.	.	.	.	20B	.	.	.	.
4243.1	AL H	O	R(17)	.	.	.	.	.	.	.	.	.	.	.	.	.	70B	.	.	.	.
4244.0	FE II 21F	A 4F A 4G	3/2 11/2	3.14	.	.	3	45H	.	.	W	.3	3 5B	1 3B	5 10B	2 6B	70B	.	.	.	.
4244.2	NE II 62	3D 2D 4F 4F U	5/2 5/2	37.51	.	.	.	.	.	.	.06	.	.	.	.	.	.	.	.	.	.
4244.5	Fe II 12	A 2P Z 6F U	1/2 1/2	5.24	.	.	.	.	.	.	.	.	3 5B	.	.	.	W	.	.	.	.
4244.8	FE II 21F	A 4F A 4G	7/2 7/2	3.21	.	.	10 50	45H	.	.	.	.	3 5B	1 2B	5 10B	6B	40B	.	.	.	.
4244.8	NI II 9	A 2G Z 4F U	1/2 5/2	6.92	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.
4245.2	AL H	O	R(18)	.	.	.	.	.	.	.	.02	.	.	.	.	.	40B	.	.	.	.
4245.7	-	.	.	.	.	.	.	.	.	.	.07	.	.	.	.	.	.	.	.	.	.
4245.9	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
4246.4	CR II 31	A 4F Z 4D U	3/2 5/2	6.74	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.
4246.8	SC II 7	A 1D Z 1G U	2 2	3.22	.	.	1B	6	.	.	.	.	WB	3	W	1	W	.	.	.	.
4147.4	AL H	O	R(19)	.	.	.	.	.	.	.	.	.	.	.	.	.	35	.	.	.	.
4247.6	C III 11	3P 1P 5 1P U	1 1	42.37	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.
4248.4	-	.	.	.	.	.	.	.	.	.	.	.2	.	.	.	.	.	.	.	.	.
4248.8	-	.	.	.	.	.	.	.	.	.	.02	.	.	.	.	.	.	.	.	.	.
4249	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	15B X	.	.	.	.
4249.1	FE II 36F	A 4D B 4D	7/2 7/2	3.89	.	.	.	6 10B	.	.	.	.	1 4B	.	.	.	15B	.	.	.	.
4249.5	NI II 4F	A 2D A 4P	3/2 5/2	3.09	.	.	.	103	.	.	.	.	.	.	.	.	.	.	.	.	.
4250	-	.	.	.	.	.	.	.	.	.	.6	.	.	.	.	.	.	.	.	.	.
4250.0	FE III --	.	.	.	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.
4250.1	FE I 152	Z 7D U E 7D	2 3	5.36	.	.	.	.	.	.	.	.	1 4B	.	.	.	.	.	.	.	.
4250.3	AL H	O	R(20)	.	.	.	.	.	.	.	.	.	.	.	.	.	25	.	.	.	.
4250.7	NE II 52	3D 4D 4F 4D U	3/2 5/2	37.39	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.
4250.8	FE I 42	A 3F Z 3G U	3 3	4.45	.	.	.	.	.	.	.	.	W	W	.	2	5	.	.	.	.
4251.2	-	.	.	.	.	.	.	.	.	.	.03	W	.	.	.	.	.	.	.	.	.
4251.5	FE II 12	A 2P Z 6F U	1/2 3/2	5.23	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.
4252.6	CR II 31	A 4F Z 4D O	5/2 5/2	6.74	.	.	.	6HM	.	.	.	.	WB	W 2	.	.	.	.	.	.	.
4253.0	MN II 7	A 5F Z 5D U	3 2	8.26	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.
4253.6	S III 4	4S 3P U 4P 3D	2 3	21.07	.	.	.	.	XB	X	.04	.06B	.	.	.	.	.	.	.	.	.
4253.7	O II 101	3D' 2G 4F' 2H O	3/2 11/2	34.09	.	.	X	.	.	.	.08	.06B	.	XB	.	.	.	.	.	.	.
4254.0	O II 101	3D' 2G 4F' 2H O	7/2 9/2	34.09	.	.	.	.	.	.	.09	.06B	.	XB	.	.	.	.	.	.	.
4254.3	CR I 1	A 7S Z 7P O	3 4	2.90	.	.	.	.	.	.	.	.	WB	W	.	1	W	.	.	.	.
4254.4	V II 18	B 3F Z 3D U	3 2	4.57	.	.	.	.	.	.	.	.	W	1	.	.	.	.	.	.	.

[illegible]

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., eV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
4309.1	A II 36	3D 2F 4P 2D 0	5/2 3/2	21.40	.	.	.	.	.	.05	.	.	.	.	.	.	.	.	.	.	.	.
4309.6	Y II 5	A 3D Z 3P 0	3 2	3.04	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
4310.4	FE III 121	E 7F 0 E 7G	6 7	25.63	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.
4311	-	.	.	.	.	.	.	.	.	.	.3	.	.	.	.	.	.	.	.	.	.	.
4311.5	-	.	.	.	.	.	.	.	.	.02	.	W	.	.	.	.	.	.	.	.	.	.
4312.9	TI II 41	A 4P Z 4D 0	5/2 5/2	4.04	.	.	1	.	.	.	.	8	5B*	2	1B	.	.	.	.	.	.	.
4313.0	FE II 220	X 4D 0 E 4F	7/2 9/2	10.63	.	.	.	.	.	.	.	.	WB*	.	.	.	.	.	.	.	.	.
4313.4	O II 78	3D 2F 4F 4F 0	7/2 9/2	31.62	.	.	.	.	.	.04B	.	.	.	.	.	.	.	.	.	.	.	.
4313.5	C II 28	3P 4P 4S 4P 0	3/2 5/2	25.88	.	.	.	.	XB	.04B	.	.	.	.	.	.	.	.	.	.	.	.
4314.1	SC II 15	A 3F Z 5D 0	4 3	3.48	.	.	.	.	.	.	W	W 10	WB	5	1B	.	.	.	.	.	.	.
4314.2	SI IV 4	5 2P 0 6 2S	1/2 1/2	37.00	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4314.3	FE II 32	A 3H Z 4F 0	9/2 7/2	5.52	.	.	2.5 10B	.	.	.	1 2 8	W 4	WB	.	.	.	.	.	.	.	.	.
4315.0	TI II 41	A 4P Z 4D 0	1/2 1/2	4.02	.	.	2.5 10B	.	.	.	1 8	4	5B	.	1 2B	.	.	.	.	.	.	.
4315.1	FE I 71	A 5P Z 5S 0	.	5.05	.	.	.	.	.	.	1 8B	.	.	.	1 2B	.	.	.	.	.	.	.
4315.4	O II 79	3D 2F 4F 2F	7/2 5/2	31.62	.	.	.	.	.	.04B	.	.	.	.	.	.	.	.	.	.	.	.
4316.8	TI II 94	B 2P Z 2P 0	1/2 1/2	4.90	.	.	.	.	.	.	1 4B	3	.	.	.	.	.	.	.	.	.	.
4317.1	O II 2	3S 4P 3P 4P 0	1/2 3/2	25.73	.	10B	.	XB	.05 .13	4	1 4B W	W	.	.	.	.	.	.	W	1	.	.
4317.5	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4318.7	S II 49	4P 4P 0 4D 4D	5/2 5/2	18.92	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4319.6	FE II 21F	A 4F A 4G	5/2 7/2	3.21	.	25B	15 503	.	.	.03B	1 4B	W 2	W	.	5 30	.	.	.	.	.	.	.
4319.6	O II 2	3S 4P 3P 4P 0	3/2 5/2	25.74	.	10 25B	.	.	.06	.03B	1 4B	W	.	.	.	.	.	W	1	.	.	.
4319.7	FE II 220	X 4D 0 E 4F	5/2 7/2	10.67	.	50B	.	.	.	.	1 4B	.	.	.	W	.	.	.	.	.	.	.
4320.5	AL H	0 0	P(18)	.	.	.	.	.	.	.	.	.	.	.	30	.	.	.	.	.	.	.
4320.7	SC II 15	A 3F Z 3D 0	3 2	3.46	.	.	.	.	.	.	WB	5B	W	W	W	.	.	.	.	.	.	.
4321.0	TI II 41	A 4P Z 4D 0	3/2 1/2	4.02	.	.	.	.	.	.	WB	5B	.	2	.	.	.	.	.	.	.	.
4322.0	C II 28	3S 4P 4S 4P 0	3/2 3/2	25.87	.	.	.	XB	.05	.	.	.	.	.	.	.	.	.	.	.	.	.
4323.9	N III 10	3P 4D 3D 4D 0	3/2 3/2	41.09	.	.	25 175B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4323.9	N III 10	3P 4D 3D 4D 0	3/2 5/2	41.09	.	.	25 175B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4325.0	SC II 15	A 3F Z 3D 0	2 1	3.45	.	.	.	.	.	.	W	1 3	W	W	W	.	.	.	.	.	.	.
4325.7	C III 7	3S 1P 0 3P 1D	1 2	41.12	.	.	.	50 140B	.12 14**	.	WB	.	.	.	.	.	.	.	.	.	.	.
4325.7	FE I 2	A 5D Z 7F 0	4 3	2.85	.	.	.	.	.	.	WB	.	.	.	15B	.	.	.	.	.	.	.
4325.8	FE I 42	A 3F Z 3G 0	2 3	4.45	.	.	10B	.	.	.	.	4	.	5	15B	.	.	.	.	.	.	.
4325.8	O II 2	3S 4P 3P 4P 0	1/2 1/2	25.72	.	.	.	.	.02 10	.	WB	.	.	.	.	.	.	.	.	.	.	.
4325.9	C II 28	3P 4P 4S 4P 0	5/2 3/2	25.87	.	.	.	.	.	.3B	.	.	.	.	.	.	.	.	.	.	.	.
4325.9	C II 28	3P 4P 4S 4P 0	3/2 1/2	25.87	.	.	.	.	.	.3B	.	.	.	.	.	.	.	.	.	.	.	.
4326.1	AL H	0 0	P(19)	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
4326.2	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4326.6	-	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.
4326.9	NI II 3F	A 2D B 2D	5/2 5/2	2.85	.	.	10B	.	.	3B	.	.	.	.	.	.	.	.	.	.	.	.
4327.0	FE II 20	A 2D Z 6P 0	5/2 3/2	5.39	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.

*A good example of different authors' listing the same apparent feature with different identifications or choices of blends.

**Stellar origin reported in one case.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J		NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGT NING	
4345.3	AL H	0	P(22)	.	.	.	.	.	.	.	.	.	.	30	.	.	.	.		
4345.6	O II 2	3S 4P 3P 4P 0	3/2 1/2	25.72	.	.	.	.	.	.	.15 .16	.5 4	1 W .3	.	.	.	10	.		
4346.1	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.		
4346.9	FE II 21F	A 4F A 4G A 5D Z 7F 0	1/2 11/2 4 4	3.14	.	.	X	8B	.	.	.08 .03	1 4	W	.	.	W 108 W 108	.	.	.	.
4347.2	FE I 2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4347.4	FE II 36F	A 4D B 4D	5/2 5/2	3.87	.	.	.	8B	.	.	.	.	.	.	.	.	.	.	.	
4347.4	O II 16	3S ⁺ 2D 3P ⁺ 2D 0	3/2 3/2	28.39	.	.	.	.	.	.	.07 .3B	W	.	.	.	.	5	.		
4348.4	N III 10	3P 4D 3D 4D 0	1/2 7/2	41.09	.	.	.	.	W	XB	.09 .3B	W	.	.	.	.	.	.	.	
4349.4	O II 2	3S 4P 3P 4P 0	5/2 5/2	25.74	.	.	XB	.	.	XB	.08 .15	.5 1 3	W	.	.	.	10	.		
4350.0	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	
4350.8	TI II 94	B 2P Z 2P 0	3/2 3/2	4.89	.	.	.	.	.	.	.	1 3	2	.	.	.	.	.	.	
4351.3	O II 16	3S ⁺ 2D 3P ⁺ 2D 0	5/2 5/2	28.39	.	.	XB	.	.	XB	.04 .02	2 4B	W	.	.	.	5	.		
4351.8	FE II 27	B 4P Z 4D 0	3/2 5/2	5.53	.	.	X	20B	.	.	.	5 20B	W 15	5 10B	15 10	.	.	.	.	
4351.8	FE II 36F	A 4D B 4D	5/2 3/2	3.87	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	
4351.9	MG I 14	3 1P 0 6 1D	1 2	7.16	.	.	.	.	.	.	.	5 20B	.	.	.	.	.	.	.	
4352.1	-	.	.	.	.	.	.	.	.	.	.	.	.8	.	.	X	.	.	.	.
4352.7	FE I 71	A 5P Z 5S 0	1 2	5.05	.	.	.	.	.	.	.	.	.	.	.	10B	.	.	.	.
4352.8	FE II 21F	A 4F A 4G	5/2 9/2	3.19	.	.	X	20B	.	.	.30	W	W	5 10B	.	10B	.	.	.	.
4353.1	AL H	1 1	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.
4353.3	-	.	.	.	.	.	.	.	.	.	2	5	.	.	.	.	.	.	.	.
4353.3	AL H	1 1	R(5) R(6)	.	.	.	.	.	.	.	.	.	.	.	.	45	.	.	.	.
4353.7	N III 10	3P 4D 3D 4D 0	7/2 5/2	41.09	.	.	.	.	.	XB	.	3	.	.	.	.	.	.	.	.
4354.3	-	.	.	.	.	.	20	.	.	.	.	.	.	.	.	.	.	.	.	.
4354.4	FE II 213	Y 4D 0 F 4D	7/2 7/2	10.45	.	.	.	.	.	.	.	W	.	10B	.	.	.	.	.	.
4354.6	SC II 14	A 3F Z 3F 0	3 4	3.44	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.
4354.6	S III 7	3D 3D 0 4P 3D	2 3	21.07	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
4355.0	AL H	1 1	R(7)	.	.	.	.	.	.	.	.	1	.	.	.	25	.	.	.	.
4355.0	FE II 202	C 4F W 2G 0	7/2 7/2	9.03	.	.	.	10B	.	.	.	.	.	.	.	.	.	.	.	.
4355.5	-	.	.	.	.	.	.	.	.	.	.02	.	.	.	.	.	.	.	.	.
4356.1	FE II 22F	A 4F B 2P	5/2 3/2	3.18	.	.	.	10B	.	.	.									

LAB WAVE LENGTH, A	ION AND MULTIPL NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGHT NING
4361.2	FE II --	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
4361.5	S III 4	4S 3P O 4P 3D	2 2	20.99	.	.	.	XB	XB	.	.	.	W	.	.	.	.	.	.
4361.7	AL H	1 1	R(11)	.	.	.	.	.	.	.	.	W	.	.	20	.	.	.	.
4361.9	C III --	.	.	.	.	.	.	40B	.	.	.	.	.	.	.	.	.	.	.
4362.1	NI II 9	A 2G Z 4F O	9/2 7/2	6.84	.	.	.	6	.	.	.	W	W 2	.	.	.	.	.	.
4363.2	O III 2F	2P2 1D 2P2 1S	2 0	5.33	1 3	10 40	20 240	20B	.	.	.94 30.7	1.55 4.3	10 60	X	.	X	.	.	.
4363.6	MO II 3	4D 4P O	5/2 3/2	5.93	.	.	.	20B	.	.	.	.	.	.	.	.	.	.	.
4364.7	LA II 53	A 3P Y 3P O	0 1	3.48	.	.	.	20B	.	.	.	.	.	.	.	.	.	.	.
4364.8	-	.	.	.	.	.	.	3	.	.	W	.	.	.	.	.	.	.	.
4365.6	-	.	.	.	.	.	.	.	.	X	.	W	.	.	.	.	.	.	.
4365.6	FE III 4	A 5P Z 5P O	1 1	11.05	.	.	.	.	.	.07	.	.	1B	.	.	.	.	.	.
4365.7	NE II 57	3D 4F 4F 4G O	7/2 5/2	37.48	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.
4366.9	O II 2	3S 4P 3P 4P O	5/2 3/2	25.73	.	.	.	.	.	.10 .16	.5	W	W	.	.	.	.	.	.
4367.1	AL H	1 1	Q(6)	.	.	.	.	.	.	.	.	.	W	.	50B	.	.	.	.
4367.7	TI II 104	B 2F Y 2G O	7/2 9/2	5.60	.	.	.	.	.	.	.	1 4	W 4	1	.	.	.	.	.
4367.9	FE I 41	A 3F Z 5G O	2 2	4.43	.	.	.	.	.	.	.	.	.	.	50B	.	.	.	.
4368.1	C II 45	3D 4P O 4F 4D	.	27.37	.	.	.	XB	.12 .16	.5	.	.	.	.	.	.	.	.	.
4368.1	C III --	.	.	.	.	.	.	40B	.	.	.	1B	.	.	.	.	.	.	.
4368.3	FE II --	.	.	.	.	.	.	.	.	.	.	WB	W	WB	.	.	.	.	.
4368.3	O I 5	3 3S O 4 3P	1	12.31	.	.	.	.	.	.17	.	WB	.	.	.	.	30	50	.
4368.4	-	.	.	.	.	.	.	.	.	.	.09	.	.	.	.	.	.	.	.
4369.3	O II 26	3P 2D O 3D 2D	3/2 3/2	28.94	.	.	.	.	.	.05	.5B	WB	.	.	.	.	.	.	.
4369.4	FE II 28	B 4P Z 4F O	1/2 3/2	5.59	.	.	.	3 10B	.	.	.	1 8B	1 7	WB	.	.	.	.	.
4369.6	AL H	1 1	.	.	.	.	.	.	.	.	.	.	.	.	50	.	.	.	.
4370.3	V II 31	A 3G Z 5D O	5 4	4.63	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
4370.8	A II 39	3D 2D 4P 2D O	3/2 3/2	21.40	.	.	.	.	.	.	.	.	.6	.	.	.	.	.	.
4371.1	FE III 4	A 5P Z 5P O	3 2	11.03	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.
4371.6	C II 45	3D 4P O 4F 4D	1/2 3/2	27.37	.	.	.	.	XB	.	.	W	.	.	.	.	.	.	.
4371.7	O II 76	3D 2F 4F 4G O	7/2 9/2	31.58	.	.	.	.	.	.05	.	.	.	.	.	.	.	.	.
4372.1	A II 86	4P 4S O 5S 2P	3/2 1/2	22.70	.	.	.	.	.	.04B	.	.	.	.	.	.	.	.	.
4372.2	FE II 33	A 4H Z 4D O	9/2 7/2	5.49	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.
4372.4	FE II 21F	A 4F A 4G	3/2 7/2	3.21	.	.	.	9	.	.	.3	.	.	.	.	.	.	.	.
4372.4	TI I 277	A 1P V 1P O	1 1	5.30	.	.	.	.	.	.	.	.	.	.	75B	.	.	.	.
4372.4	.	.	.	.	.	.	.	.	.	.	.	W 5B	W 1	.	.	.	.	.	.
4372.5	C II 45	3D 4P O 4F 4D	3/2 5/2	27.37	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.
4372.5	AL H	1 1	Q(8)	.	.	.	.	.	.	.	.	.	.	.	75B	.	.	.	.
4372.7	-	.	.	.	.	.	.	.	.	.	.	1 6	.	.	X	.	.	.	.
4374.3	C II 45	3D 4P O 4F 4D	5/2 7/2	27.37	.	.	.	.	.	.	.	5B	.	.	.	.	.	.	.
4374.5	SC II 14	A 3F Z 3F O	4 4	3.44	.	.	.	2B	.	.	.	5B	2 3	1	W	.	.	.	.
4374.8	TI II 93	B 2P Y 2D O	3/2 5/2	4.87	.	.	.	2B	.	.	.	5B	3	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY																AIR GLW	CTY LTS	AUR ORA	LGHT NING																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
																QUAS	GAL	NOVA	LIKE	N					C	NEB	NEB	SYMB	PCYG	GE	TAU	N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
4375	-	.	.	.	.	.	2B	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.

*A good example of different authors' listing the same apparent feature with different identifications or choices of blends.

[illegible]

LAB WAVE LENGTH, A	ION AND MULTIPLT NUMBER	TRANSITION		UPPER EXC., POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J																	
					NUC EM	NOVA GAL	NOVA LIKE	W-R N	W-R C	PLAN NER	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME	AIR GLW	CTY LTS	AUR ORA	LGT NING
4443.0	ZR II 88	A 2H Z 2G U 7/2	9/2	4.26	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	
4443.1	O II 35	3P' 2F O 3D' 2F 5/2	5/2	31.01	.	.	.	.	.	.	.	WB	XB	.	.	.	.	.	.	
4443.8	TI II 19	A 2D Z 2F O 5/2	3/2	3.85	.	.	X	1.5 103	.	.	.	1 12	W 5	W 5	5	2	.	.	.	
4444.6	FE II 201	C 4F W 4F O 9/2	5/2	8.97	.	.	.	103	.	.	.	WB	.	.	.	.	.	.	.	
4444.6	TI II 31	A 2G Z 2F O 7/2	7/2	3.89	.	.	.	.	.	.	.	WB	3	5 20	.	.	.	.	.	
4446.0	-	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	
4446.2	FE II 187	D 2D X 2G O 7/2	5/2	8.71	.	.	X	.	.	.	.	1 4	.	W	.	.	.	.	.	
4446.5	NE II 56	3D 4F 4F 4F O 3/2	5/2	37.47	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	
4447.0	N II 15	3P 1P 3D 1D O 2	1	23.10	.	.	10B	.	.	.	W	1 5	.7	.	.	.	.	.	.	
4447.2	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	
4447.7	FE I 68	A 5P X 5D O 1	1	4.99	.	.	.	.	.	.	.	W	1	.	.	.	.	.	.	
4447.8	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
4447.8	O III 33	5P 5S O 3D 5P 2	2	49.15	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	
4448.2	O II 35	3P' 2F O 3D' 2F 7/2	7/2	31.01	.	.	10B	.	.	.06	.	WB	.	.	.	.	.	.	.	
4448.5	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
4449.7	FE II 222	Y 4G O E 4F 7/2	9/2	10.67	.	.	.	.	.	.	.	.	.	5B	.	.	.	.	.	
4450.5	TI II 19	A 2D Z 2F O 5/2	5/2	3.85	.	.	.	.	.	.	.	2 10	W 5	1	1	W	.	.	.	
4451.5	FE II --	.	.	.	.	.	10B	.	.	.	.	W	2	5B	.	.	.	.	.	
4451.6	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	
4452.1	FE II 7F	A 6D A 6S 5/2	3/2	2.88	.	.	8 40B	20	.	.3 .5B	.06 .15B	WB	W 3	5B	5	40	.	.	.	
4452.4	O II 5	3S 2P 3P 2D O 3/2	3/2	26.11	.	.	10B	.	XB	.05	.01 .15B	WB	.	.	.	.	.	.	.	
4452.6	-	.	.	.	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	
4452.8	-	.	.	.	.	.	.	.	.	.14	.	.	.	.	.	.	.	.	.	
4453	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
4453.3	-	.	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	
4454.4	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	
4454.7	-	.	.	.	.	.	3	.	.	.	.	W	.	.	.	.	.	.	.	
4455.3	FE II --	.	.	.	.	.	10B	.	.	.	.	W	W	1	.	.	.	.	.	
4455.3	TI I 113	B 3F V 3F O 3	3	4.21	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	
4455.3	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	40	.	.	.	
4456.4	S II 43	4P 4D O 5S 4P 1/2	1/2	18.55	.	.	.	.	XB	.03	.	.	.	.	.	.	.	.	.	
4456.7	TI II 115	C 2D X 2F O 5/2	5/2	5.88	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	
4457.0	-	.	.	.	.	.	.	.	.	.	4	.	.	.	.	X	.	.	.	
4457.0	NE II 61	3D 2D 4F 4D O 3/2	3/2	37.39	.	.	.	.	.	.04	.	.	.	.	.	.	.	.	.	
4458.0	FE II 6F	A 6D B 4F 7/2	7/2	2.82	.	.	8 40B	18	.	.	.15	W 5	W 1	W	.	40	.	.	.	
4458.7	-	.	.	.	.	.	.	.	.	.12	.	.	.	.	.	X	.	.	.	
4459.0	-	.	.	.	.	.	.	.	.	.04	.	.	.	.	.	.	.	.	.	
4459.1	FE I 68	A 5P X 5D O 3	3	4.93	.	.	.	.	.	.	.	X	W	.	.	X	.	.	.	
4460.0	N II 21	3P 3D 3D 3P O 0	1	23.32	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	
4461.0	-	.	.	.	.	.	.	.	.	.3	.	.	.	.	.	.	.	.	.	

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LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY																	
		LEVELS	J																			
					NUC EM	NOVA	W-R	W-R	PLAN	GAS		OE	AE	T-	KE	AIR	CTY	AUR	LGT			
				QUAS	GAL	NOVA	LIKE	N	C	NEB	NEB	SYMB	PCYG	GE	TAU	N	GLW	LTS	ORA	NING		
4461.4	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	
4461.4	FE II 26	B 4P Z 6P 0	5/2 5/2	5.34	.	.	.	25	.	.	.	1 12B	8B	.	.	.	.	.	.	.	.	
4461.5	NI II 10F	A 4F A 2G	5/2 7/2	4.01	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	
4461.6	O III 33	3P 5S 0 3D 5P	2 3	49.14	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	
4461.7	FE I 2	A 5D Z 7F 0	2 3	2.85	.	.	.	.	.	.	.	1 12B	8B	.	2	30	.	.	.	.	.	
4461.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	
4463.6	S II 43	4P 4D 0 5S 4P	7/2 5/2	18.64	.	.	.	2B	.	.	.	W	.	.	.	.	.	.	.	.	.	
4464.1	-	.	.	.	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	
4464.3	-	.	.	.	.	.	.	.	.	.03	.	.	.	.	.	.	.	.	.	.	.	
4464.4	S II --	.	.	.	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	
4464.5	-	.	.	.	.	.	.	.	1	.3	W	.	.	.	.	.	.	.	.	.	.	
4464.5	TI II 40	A 4P Z 2D 0	1/2 3/2	3.92	.	.	.	3B	.	.	.	1 8	4	.	.	.	.	.	.	.	.	
4465.4	O II 94	SP33S6S° SP33P6P	5/2 7/2	33.06	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4465.5	N II 21	3P 3D 3D 3P 0	1 1	23.32	.	.	.	.	.	.03 .06	W	.	.	.	.	.	.	.	.	.	.	
4466.2	-	.	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	
4466.6	FE I 2	A 5D Z 7F 0	1 0	2.87	.	.	.	.	.	.	.	.	.	.	1	5	.	.	.	.	.	
4467.8	S III 7	3D 3D 0 4P 3D	2 2	20.99	.	.	.	.	XB	.	.05B	.	.	.	.	.	.	.	.	.	.	
4467.9	O II 94	SP33S6S° SP33P6P	5/2 5/2	33.06	.	.	.	.	.	.05B	W	.	.	.	.	.	.	.	.	.	.	
4468.5	TI II 31	A 2G Z 2F 0	9/2 7/2	3.89	.	.	.	2	.	.	.	2 15B	6	10B	5	2	.	.	.	.	.	
4468.5	FE II -F	A 4D B 2G	.	.	.	.	.	.	.	.	.	2 15B	.	.	.	.	.	.	.	.	.	
4469.2	TI II 18	A 2D Z 4F 0	5/2 7/2	3.84	.	.	.	.	.	.	.	1 3B	2	.	.	.	.	.	.	.	.	
4469.3	O II 59	3P 2P 0 4D 4D	3/2 5/2	31.47	.	.	.	.	.	.	.	1 3B	.	.	.	.	.	.	.	.	.	
4469.3	O II 94	SP33S6S° SP33P6P	5/2 3/2	33.06	.	.	.	.	.	.	.	1 3B	.	.	.	.	.	.	.	.	.	
4470.3	FE II 6F	A 6D B 4F	5/2 3/2	2.84	.	.	.	15BM	.	.	.	5 20BM	.	.	.	.	.	.	.	.	.	
4470.9	TI II 40	A 4P Z 2D 0	3/2 3/2	3.92	.	.	.	.	.	.	.	5 20BM	2	.	.	.	.	.	.	.	.	
4471.5	HE I 14	2 3P 0 4 3D	2,1 0	23.63	.	.	4 16B	10 20B	60B	15 210B	3.2 6	.04 15B	5 20B	8 40B	10B	5	.	.	.	.	.	
4471.7	HE I 14	2 3P 0 4 3D	0 0	23.63	.	.	4 16B	10 20B	60B	15 210B	5 15B	4.6 15B	5 20B	14 40B	.	.	.	.	.	.	.	
4472.9	FE II 37	B 4F Z 4F 0	5/2 3/2	5.59	.	.	.	15BM	.	.	.	1 10	10	10B	.	.	.	.	.	.	.	
4474.2	FE II 171	C 2F Y 2D 0	5/2 3/2	8.30	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4474.9	FE II 7F	A 6D A 6S	1/2 5/2	2.88	.	.	10 30	12	.	.	.	W	W	10B	4	25	.	.	.	.	.	
4475.0	O III 37	3P 1D 0 3D 1D	2 2?	55.39	.	.	.	.	XB	.	.	W	.	.	.	.	.	.	.	.	.	
4476.2	-	.	.	.	.	.	.	.	.	.	.	1	5	.	.	.	.	.	.	.	.	
4476.6	N V -	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	
4477.7	N II 21	3P 3D 3D 3P 0	2 1	23.32	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4477.9	O II 88	3D 2P 4F 4G 0	3/2 5/2	31.57	.	.	.	.	.	.	.05	.	.	.	.	.	.	.	.	.	.	
4479	N IV 6	3S 3P 0 3P 3S	0 1	60.19	.	.	.	.	7B	.	.	.	.	.	.	.	.	.	.	.	.	
4479.3	CA II 6	5 2S 6 2P 0	1/2 1/2	9.20	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4480.0	AL III 8	4 2F 0 5 2G	7/2 5/2	23.45	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	
4480.7	FE II --	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.	
4481.1	MG II 4	3 2D 4 2F 0	3/2 5/2	11.58	.	.	X	1B	7B	XB	.10B	1 20B	3 20B	1	.	.	.	.	.	.	.	

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. PGT., EV	EMISSION LINE INTENSITY													
		LEVELS	J		NUC EM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE 8E SYMB	AE FE PCYG	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGT NING
4569	C III -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4569.5	O III 36	3P 10 D 3D 1F	2 3?	55.33	.	.	.	.	XB	.	.	W	.	.	.	.	.	.
4569.7	-	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.
4571.0	CR I 173	A 3F Y 3G U	4 3	5.81	.	.	.	10B	.	.	.	.	.	.	.	.	.	.
4571.1	MG I 1	3 1S 3 3P O	0 1	2.70	.	.	.	6 10B	.	.	.13 1.9	.3B 4	1 2	.	.5	200	.	.
4571.1	CR I 125	B 5D Y 3D O	3 3	5.70	.	.	.	10B	.	.	.	.	.	.	.	.	.	.
4572.0	TI II 82	A 2H Z 2G U	9/2 7/2	4.26	.	.	.	6 10B	.	.	.	1 8	W 8	1 5	2	.	.	.
4573.9	FE III 3F	A 5D A 3F	4 2	2.70	.	.	X	.	.	.3B .3B	.	W	.	.	.	.	.	.
4574.8	SI III 2	4 3S 4 3P U	1 0	21.62	.	.	WB	.	.	XB	.	WB .8	.	.	.	.	.	.
4576.3	FE II 38	B 4F Z 4D O	5/2 5/2	5.53	.	.	WB	10	.	.	.	1 12B	W 14	1 5	1	.	.	.
4577.8	FE II 54	B 2P Z 4P O	3/2 1/2	5.88	.	.	.	.	.	.	.	1 12B	.	.	.	.	.	.
4578 R	O2 H	0 9	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.
4579	-	.	.	.	.	.	.	.	.	.3	.	.	.	.	X	.	.	.
4579.5	FE II --	.	.	.	.	.	.	.	.	.	.	3B	.	.	.	.	.	.
4580.1	FE II 26	B 4P Z 6P O	5/2 7/2	5.27	.	.	.	7B	.	.	.	3B 8	1 8	1	.	.	.	.
4580.8	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.
4580.8	CR II 3F	A 6S A 4P	5/2 3/2	2.69	.	.	.	8B	.	.	.	.	.	.	.	.	.	.
4580.9	CR II 3F	A 6S A 4P	5/2 1/2	2.69	.	.	.	8B	.	.	.	.	.	.	.	.	.	.
4581.2	CR II 3F	A 6S A 4P	5/2 5/2	2.69	.	.	.	8B	7B	.	.	W	.	.	.	.	.	.
4582.1	FE II 19	A 2D Z 6F O	5/2 5/2	5.23	.	.	.	.	.	.	.	1 8	.	.	.	.	.	.
4582.8	FE II 37	B 4F Z 4F O	5/2 7/2	5.52	.	.	.	25B	.	.	.	5 30B	6 20B	30B	.	W	.	.
4583.8	FE II 38	B 4F Z 4D O	9/2 7/2	5.49	.	.	.	8 16B	25B	.	.	5 30B	5 20B	10 30B	10 10	.	.	.
4584	-	.	.	.	.	.	.	.	.	.4 .6	.	.	.	.	.	.	.	.
4584.0	FE II 26	B 4P Z 6P O	3/2 3/2	5.49	.	.	X	.	.	.	.	5 30B	XB	.	.	X	.	.
4585.1	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.
4588.2	CR II 44	B 4F Z 4D O	7/2 5/2	6.74	.	.	.	9	.	.	.	1 8	1 6	1	2	.	.	.
4589.9	CR II 44	B 4F Z 4D O	3/2 5/2	6.74	.	.	.	.	.	.	.	1 5B	W	.	.	.	.	.
4590.0	TI II 50	A 2P Z 2D O	3/2 3/2	3.92	.	.	.	.	.	.	.	1 5B	4	1	5	.	.	.
4590.7	-	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.
4591.0	O II 15	3S* 2D 3P* 2F O	5/2 7/2	28.24	.	.	10	.	.	.07 .10	.04	W	.	.	.	.	W	.
4591.3	FE II 17	A 2H Z 6F O	11/2 9/2	5.20	.	.	.	.	.	.	.	1 4B	.	.	.	.	.	.
4592.																		

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., eV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGH NING
		LEVELS	J		NUC EM GAL	NOVA	W-R LIKE	W-R N	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
4596.1	-	.	.	.	.	.	1.5B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4596.2	O II 15	3S' 2D 3P' 2F 0	3/2 5/2	28.24	.	X	.	.	.08	.	1 4B	.	.	.	.	.	.	.	.	.	W	.
4597.1	-	.	.	.	.	.	.	.	.08	.	.	.	.	.	.	.	.	.	.	.	.	.
4598.4	-	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
4598.5	FE II 219	X 4D 0 F 4D	7/2 7/2	10.45	.	.	2B	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.
4599.7V	N2+ 1N	2 4	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	18
4600.2	V II 56	A 3D Z 3F 0	1 2	4.94	.	.	.	.	.	.	W	1	.	.	.	.	.	.	.	.	.	.
4601.3	FE II 43	A 6S Z 4D 0	5/2 3/2	5.56	.	.	.	.	.	.	WB	W 1	.	.	.	.	.	.	.	.	.	.
4601.5	N II 5	3S 3P 0 3P 3P	1 2	21.07	.	.	W	.	.4	.3	WB	2.6	.	.	.	.	.	.	.	.	W	.
4602.1	O II 93	3D 2D 4F 2F U	3/2 5/2	31.62	.	.	.	.	.08	.	.	.	.	.	.	.	.	.	.	.	W	.
4602.8	FE II 19	A 2D Z 6F 0	5/2 7/2	5.21	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4602.9	FE I 39	A 3F Y 5F 0	4 5	4.16	.	.	.	.	.	.	W	1	.	.	.	.	.	.	.	.	.	.
4603.2	N V 1	3 2S 3 2P 0	1/2 3/2	58.99	.	.	XB	70 280B	.05 1	.5B	W	.	.	.	.	.	.	.	.	.	.	.
4604.3	-	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4604.5	FE II 5F	A 6D A 4H	9/2 7/2	2.68	.	.	.	.	.04	.	.	.	.	.	.	.	.	.	.	.	.	.
4605.8	LA II 52	A 3P Z 1P 0	1 1	3.39	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4607.0	FE III 3F	A 5D A 3F	4 3	2.68	.	.	XB	.	.35 .5	.3	.	W 1.4	.	.	.	.	.	.	.	.	.	.
4607.2	N II 5	3S 3P 0 3P 3P	0 1	21.06	.	.	20	.	XB	XB	.05 .06	W 2	.	.	.	.	.	.	.	.	.3 .8B	10
4609.4	O II 93	3D 2D 4F 2F 0	5/2 7/2	31.62	.	.	.	.	XB	.05 .17B	.	.	.	.	.	.	.	.	.	.	.	.
4610	A V 2F	3P2 1D 3P2 1S	2 0	4.69	.	.	X	.	.	.	.4	.	.	.	.	.	.	.	.	.	.	.
4610.1	O II 92	3D 2D 4F 4F 0	3/2 5/2	31.61	.	.	.	.	XB	.17B	.	.	.	.	.	.	.	.	.	.	.	.
4610.9	-	.	.	.	.	.	.	.	.11	.	W	.	.	.	.	.	.	.	.	.	.	.
4613.9	N II 5	3S 3P 0 3P 3P	1 1	21.06	.	.	3	.	XB	.	.4	W	1.3	.	.	.	.	.	.	.	W	.
4616.3	-	.	.	.	.	.	.	.	.	.	.	2 3	.	.	.	.	.	.	.	.	.	.
4616.6	CR II 44	B 2F Z 4D 0	3/2 3/2	6.73	.	.	XB	2	.	.	.	5 1 5	5	1	1	.	.	.	.	.	.	.
4618.8	CR II 44	B 2F Z 4D 0	5/2 3/2	6.73	.	.	XB	2B	.	.	.	1 8B 8	W 3	5B	1	.	.	.	.	.	.	.
4618.9	C II 50	3D 2F 0 4F 2G	.	27.36	.	.	.	.	XB	.05	.	1 8B 1	.	.	.	.	.	.	.	.	.	.
4619.4	N V 1	3 2S 3 2P 0	1/2 1/2	58.98	.	.	XB	70 280B	.12	.	.	158 1	.	.	.	.	.	.	.	.	.	.
4620.5	FE II 38	B 4F Z 4D 0	7/2 7/2	5.49	.	.	3	4 10	.	.	.	1 158	W 14	1 5	2	.	.	.	.	.	.	.
4620.9	-	.	.	.	.	.	.	.	.07	.	.	.	.	.	.	.	.	.	.	.	.	.
4621.4	N II 5	3S 3P 0 3P 3P	1 0	21.06	.	.	3	.	XB	.	.4 .6	4B	W	1.5	.	.	.	.	.	.	W	10
4621.5	C I 2F	2P2 3P 2P2 1S	1 0	2.67	.	.	.	.	.	.	4B	.	.	.	.	.	.	.	.	.	.	.
4623.1	-	.	.	.	.	.	1	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4625.9	FE II 186	D 2D Y 2F 0	5/2 5/2	8.60	.	.	.	.	.	.	W	W	.	.	.	.	.	.	.	.	.	.
4626.9	-	.	.	.	.	.	.	.	.07	.	.	.	.	.	.	.	.	.	.	.	.	.
4627.9	FE II 54	B 2P Z 4P 0	3/2 3/2	5.85	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
4628.8	-	.	.	.	.	.	.	.	W	.3	.	.	.	.	.	.	.	.	.	.	.	.
4628.8	NI II 3F	A 2D B 2D	3/2 5/2	2.85	.	.	.	25B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4629.3	FE II 37	B 4F Z 4F 0	9/2 9/2	5.46	.	.	15 80	50	.	.	.	5 30	1 20	5	10	15	.	.	.	.	.	.
4629.3	TI II 38	A 4P Z 4F 0	5/2 7/2	3.84	.	.	.	.	.	.	.	1	10	.	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY																		
		LEVELS	J		NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME	AIR GLW	CTY LTS	AUR ORA	LGHT NING				
																				QUAS	GAL	NOVA	LIKE
4629.4	-	.	.	.	.	.	.	25B	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.
4630.5	C II 49	3D 2F 0 4F 4G 9/2	7/2	27.36	.	.	.	.	.	XB	.05B	.	.	.	.	.	.	.	.	.	.	.	.
4630.5	N II 5	3S 3P 0 3P 3P 2	2	21.07	.	.	W	.	XB	.	.05B	.3	W	3.3	.	.	.	.	.	.	W	.	.
4631.4	SI IV 6	5 2F 0 6 2G	.	38.92	.	.	6 9	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4632.1	-	.	.	.	.	.	.	.	.	.	.43	.	.	.	.	.	.	.	.	.	.	.	.
4632.4	-	.	.	.	.	.	.	.	.	.	.7	.	.	.	.	.	.	.	.	.	.	.	.
4632.8	-	.	.	.	.	.	.	.	.	.	.27	.	.	.	.	.	.	.	.	.	.	.	.
4633.7	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.
4634.1	CR II 44	B 2F Z 4D 0	3/2	6.72	.	.	.	4 108*	.	.	.	.	2 25B	W 4	1	2	.	.	.	.	.	.	.
4634.2	N III 2	3 2P 0 3 2D	1/2	32.99	.	.	16 60H	48 280B	70 280B	X	.37 7*	.04 .10	2 25B	W 5	.	.	.	.	.	.	.	.	.
4634.6	FE II 25	B 4P Z 6F 0	5/2	5.23	.	.	.	.	.	.	.	.	WBM	.	.	.	.	.	.	.	.	.	.
4634.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.
4635.3	FE II 186	D 2D Y 2F 0	5/2	8.59	.	.	.	4 10B	.	.	.	.	W	1 2	1	.	.	.	.	.	.	.	.
4635.9	-	.	.	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.
4636.3	TI II 38	A 4P Z 4F 0	3/2	3.82	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4637.6	C II -	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	W 10B	.	.
4637.9	-	.	.	.	.	.	.	.	.	.	.	W	W	.	.	.	.	.	.	.	.	.	.
4638	FE III -F	.	.	.	.	.	.	.	.	.	.	.	W 5	.	.	.	.	.	.	.	.	.	.
4638.9	C II -	.	.	.	.	.	.	.	.	XB	.	.08	.	.	.	.	.	.	.	.	W 10B	.	.
4638.9	O II 1	3S 4P 3P 4D 0	1/2	25.53	.	.	.	20B	.	.	.05 .12	1	W	.	.	.	.	.	.	.	.	.	.
4639.7	FE II 4F	A 6D B 4P	3/2	2.77	.	.	30B	15 20B	.	.	.	.	5	.	1	.	25	.	.	.	.	.	.
4640	-	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4640.6	N III 2	3 2P 0 3 2D	3/2	32.99	.	.	16 100B	5B 280B	70 280B	XB	.5 3.6*	10 40	5 10B	.3 10B	.	.	.	.	.	.	.	.	.
4641.8	O II 1	3S 4P 3P 4D 0	3/2	25.54	.	.	.	.	.	.	.9 .5	.5	1 10B	.	.	.	.	.	.	.	.	10B	.
4641.9	N III 2	3 2P 0 3 2D	3/2	32.99	.	.	60B	18B 280B	70 280B	XB	.6*	.	1 10B	W 10B	.	.	.	.	.	.	.	.	.
4643	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.
4643.1	N II 5	3S 3P 0 3P 3P 1	2	21.06	.	.	X	.	XB	.	.26*	.	1 4	2.5	.	.	.	.	.	.	.	XB	.
4643.9	-	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.
4644.1	FE II 31	A 4H Z 6P 0	7/2	5.34	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4645	N V -	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4646.6	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4647	C IV 6	5 2D 6 2F 0	.	58.19	.	.	.	.	.	300 1500B	.	.	W	.	.	.	.	.	.	.	.	.	.
4647.3	NE II 72	4S 4P 4F 4F 0	5/2	37.47	.	.	.	.	.	.	.18	.	.	.	.	.	.	.	.	.	.	.	.
4647.4	C III 1	3 3S 3 3P 0	1	32.05	.	.	.	.	.	300 1500B	.4 1.2*	10 15	1 15	.	.	.	.	.	.	.	.	.	.
4647.5	LA II 77	Z 3F 0 E 1F	4	4.59	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4648.2	AL 0	1 0	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.
4648.2	FE II 38	B 4F Z 4D 0	5/2	5.49	.	.	.	.	.	.	.	.	1 4B	.	.	.	.	.	.	.	.	.	.
4648.9	FE II 25	B 4P Z 6F 0	5/2	5.83	.	.	.	.	.	.	.	.	1 4B	2	.	.	.	.	.	.	.	.	.
4649.1	O II 1	3S 4P 3P 4D	5/2	25.55	.	.	20B	.	.	.	.29 .87	.3	1 15	X	.	.	.	.	.	.	.	.	.
4650	-	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.

*Stellar origin reported in one case.

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM GAL	NOVA	W-R LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N						
4650.2	C III 1	3 3S 3 3P 0	1	32.05	.	20B	3	.	300 1500B	.54*	10B	1 10B	.	.	.	.	.	.	.	.	.	
4650.8	C II -	.	.	.	.	.	.	.	.	.	10B	.	.	.	.	.	.	.	.	.	.	
4650.8	O II 1	3S 4P 3P 4D	1/2	25.52	.	XB	.	.	XB	.17	.08	1 10B	.	.	.	.	.	.	.	XB	.	
4651.4	C III 1	3 3S 3 3P 0	1	32.05	.	.	.	.	300 1500B	.35*	10B	1 10	.	.	.	.	.	.	.	.	.	
4651.8V	N2+ 1N	1 3	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	20B	80	
4653.2	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4654.0	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4655.8	TI II 38	A 4P Z 4F 0	1/2 3/2	3.81	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
4656.3	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4656.7	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	
4656.8	SI II --	.	.	.	.	.	.	.	.	.	.	1 8B	.	.	.	.	.	.	.	.	.	
4657.0	FE II 43	A 6S Z 4D 0	5/2 5/2	5.53	.	.	10B	.	.	.	.	1 8B	3 8B	1	.	.	.	.	.	.	.	
4657.2	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4657.2	TI II 59	B 4P Z 2F 0	5/2 7/2	3.89	.	.	.	.	.	.	.	.	8B	.	.	.	.	.	.	.	.	
4657.7	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4657.9	A II 15	4S 2P 4P 2P 0	3/2 1/2	19.72	.	.	.	.	.	.18	.	.	.	.	.	.	.	.	.	.	.	
4658.0	FE II 170	C 2F X 4F 0	7/2 7/2	8.19	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4658.1	FE III 3F	A 5D A 3F	4 4	2.65	.	1 5B	W 80	10B	.	1.6 17B	1.2 3	1 10B	1 10**	.	4	.	.	.	.	.	.	
4658.6	C IV 8	5 2F 0 6 2G	.	52.20	.	1 5B	10	.	300 1500B	3* 1.5	1 10B	.	.	.	.	.	.	.	.	.	.	
4660	C IV 9	5 2G 6 2F 0	.	58.19	.	.	.	.	300 1500B	1	.	5	.	.	.	.	.	.	.	.	.	
4660.5	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
4661.6	O II 1	3S 4P 3P 4D 0	3/2 3/2	25.53	.	.	XB	1B	.	.15 .30	.08	W	X	.	.	.	.	.	.	5	.	
4662.5	LA II 8	A 3F Z 3D 0	2 1	2.65	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4662.7	TI II 38	A 4P Z 4F 0	3/2 3/2	3.81	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	
4662.7	TI II 38	A 4P Z 4F 0	5/2 5/2	3.81	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	
4663.1	AL II 2	3 1D 4 1P 0	2 1	13.20	.	.	.	.	.	.	.	1 1B	.	.	.	.	.	.	.	.	.	
4663.3	CR I 186	Z 7F 0 F 7D	1 1	5.73	.	.	1B	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4663.5	C III 5	3S 3P 0 3P 3P 0	1 0	40.69	.	.	.	.	300 1500B	.06	.	.	.	.	.	.	.	.	.	.	.	
4663.7	FE II 44	A 6S Z 4F 0	5/2 7/2	5.52	.	.	6B	.	.	.	.	1 5	1 3B	.	.	.	.	.	.	.	.	
4664.5	FE II 4F	A 6D B 4P	1/2 1/2	2.77	.	.	6B	.	.	.	.	2	.	.	.	.	.	.	.	.	.	
4664.8	NA I 12	3 2P 0 6 2D	1/2 3/2	4.74	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4665	C IV 7	5 2F 0 6 2D	.	58.19	.	.	.	.	300 1500B	.	.	.	.	.	.	.	.	.	.	.	.	
4665.0	FE II 5F	A 6D A 4H	9/2 11/2	2.65	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4665.6	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4665.8	FE II 26	B 4P Z 6P 0	3/2 5/2	5.34	.	.	.	.	.	.	.	WB	XB	.	.	.	.	.	.	.	.	
4665.9	C III 5	3S 3P 0 3P 3P	2 2	40.69	.	.	.	.	300 1500B	.13	.	WB	.	.	.	.	.	.	.	.	.	
4666.8	FE II 37	B 4F Z 4F 0	7/2 9/2	5.46	.	.	9B	.	.	.	.	1 8B	W 10B	1	2	.	.	.	.	.	.	
4667.0	FE III 3F	A 5D A 3F	3 2	2.70	.	.	XB	9B	.	.	.4	1 8B	W 10B	.	.	.	.	.	.	.	.	
4668.8	-	.	.	.	.	.	.	.	.	.4 .8	.	3	.	.	.	.	.	.	.	.	.	
4668.9	LA II 76	Z 3F 0 E 3F	3 2	4.42	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	

*Stellar origin reported in one case.

**Nebular origin reported in one case.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC., POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGHT NING																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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4669.2	-	.	.	.	.	.	.	.	.	.	.22	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.</

*Stellar origin reported in one case.

**Nebular origin reported in one case.

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N								
4701.8	O II 58	3P ¹ 2P 0 3/2 3D ¹ 2P 1/2		31.33	.	.	.	.	.	2.3	.	.	.	.	.	.	.	.	.	.	.	.	
4703.0	MG I 11	3 1P 0 1 5 1D 2		6.95	.	.	.	2B	.	.	.	WB	1	.	.	.	.	.	.	.	.	.	
4703.2	O II 40	3P ¹ 2D 0 3/2 3D ¹ 2F 5/2		31.01	.	.	XB	.	XB	.	XB	WB	.	.	.	.	.	.	.	.	.	.	
4705.4	O II 25	3P 2D 0 5/2 3D 2F 7/2		28.76	.	.	XB	.	XB	.	.4	1 4 W	W	.	.	.	.	.	.	.	7	.	
4706.9	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4708.6	-	.	.	.	.	.	.	.	.	.	W	.	1	.	.	.	.	.	.	.	.	.	
4708.7	TI II 49	A 2P 3/2 Z 2F 0 5/2		3.85	.	.	.	.	.	.	.	1 4B	1	.	.	.	.	.	.	.	.	.	
4708.9	FE II -	.	.	.	.	.	.	.	.	.	.	1 4B	.	.	.	.	.	.	.	.	.	.	
4709.2V	N2+ 1N	0 2	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	60	1	
4710.0	O II 24	3P 2D 0 3/2 3D 4D 5/2		28.73	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4711.4	A IV 1F	3P3 4S 0 3/2 3P3 2D 0 5/2		2.62	.	2 10	20B 6	.	.	.1 5.3	W 1.1B	W 5	.	.	.	.	.	.	.	.	.	.	
4711.6	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4712.2	-	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.	.	.	.	
4713.1	HE I 12	2 3P 0 2,1 4 3S 1		23.49	.	.	5 8B	4B	.	XB	.58 1.0	1.1 2.3B	5 30B	W 7B	WB	.	.	.	.	.	.	.	
4713.4	HE I 12	2 3P 0 0 4 3S 1		23.49	.	.	5 8B	4B	.	XB	1 15B	1.1 2.3B	5 30B	W 7B	WB	.	.	.	.	.	.	.	
4714	NE IV 1F	2P3 2D 0 5/2 2P3 2P 0 1/2		7.38	.	.	20B	.	.	.8	.	5 30BM	.	.	.	.	.	.	.	.	.	.	
4715.5	AL 0	4 3	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	
4716	NE IV 1F	2P3 2D 0 5/2 2P3 2P 0 3/2		7.38	.	.	.	.	.	.5	.	W	.	.	.	.	.	.	.	.	.	.	
4716.2	S II 9	4S 4P 3/2 4P 4S 0 3/2		16.18	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4716.4	FE II 5F	A 6D 7/2 A 4H 9/2		2.66	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4716.7	SI III --	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4718.2	CA II 7	4 2D 3/2 5 2F 0		9.63	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4718.4	N II 68	3P 5D 0 4 3D 5D 4		30.23	.	.	5B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4720	NE IV 1F	2P3 2D 0 3/2 2P3 2P 0 1/2		7.38	.	.	20	.	.	1.3	.	W 10 1 3	.	.	.	.	.	.	.	.	.	.	
4723.6	-	.	.	.	.	.	.	.	.	1	.	1 3	.	.	.	.	.	.	.	.	.	.	
4724.1	FE II 17	A 2H 9/2 Z 6F 0 11/2		5.18	.	.	X	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4724.3	-	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4724.8	-	.	.	.	.	.	.	.	.	.07 2	.4	W	.	.	.	.	.	.	.	.	.	.	
4725.1	-	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.	.	
4725.4	NE IV -F	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4727.2	C II 48	3D 2F 0 5/2 4F 2F 5/2		27.29	.	.	.	.	XB	.	.	.	XB	.	.	.	.	.	.	.	.	.	
4728.1	-	.	.	.	.	.	20	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4728.1	FE II 4F	A 6D 5/2 B 4P 3/2		2.69	.	.	10 25B	20	.	.	.	W 5	W 2	.	.	.	15	.	.	.	.	.	
4731.4	FE II 43	A 6S 5/2 Z 4D 0 7/2		5.49	.	.	W 7	.	.	.	.	1 12	2 10	1 5	2	.	.	.	.	.	.	.	
4733	N IV 11	3P 3D 2 3D 3D 0 2		62.41	.	.	.	XB	.	W	.	.	.	.	.	.	.	.	.	.	.	.	
4733.9	FE III 3F	A 5D 2 A 3F 2		2.70	.	.	XB 1.5	.	.	W .3	5	W*	.	.	.	.	.	.	.	.	.	.	
4734.1	FE II -	.	.	.	.	.	.	.	.	.	.	.	.	5B?	.	.	.	.	.	.	.	.	
4734.8	C II 48	3D 2F 0 7/2 4F 2F 7/2		27.29	.	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	
4735.0	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
4737	C IV 12	6 2D 8 2P 0		60.80	.	.	1	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	

*Nebular origin reported in one case.

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. PGT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM GAL	NOVA	NUVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N						
4738.1	C II 1	2P 2P 3 2P 0	1/2 1/2	16.26	.	.	.	.	XB	.	.	.	.	.	.	W 108	.	.	.	.		
4739.6	MG II 18	4 2D 8 2F 0	.	14.12	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.		
4740	N IV 11	3P 3D 3D 3D 0	2 1	62.40	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.		
4740.3	A IV 1F	3P3 4S 0 3P3 2D 0	3/2 3/2	2.60	.	2 12	20 6	.	.	1.6 16	.6	.	.	.	.	.	.	.	.	.		
4740.8	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.		
4745.0	C II 1	2P 2P 3 2P 0	3/2 3/2	16.26	.	.	.	.	XB	.	.	.	.	.	.	W 1008	.	.	.	.		
4745.5	FE II 20F	A 4F B 4F	9/2 5/2	2.83	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.		
4747.9	NA I 11	3 2P 0 7 2S	1/2 1/2	4.69	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.		
4750.6	FE II 5F	A 6D A 4H	5/2 1/2	2.68	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.		
4751	N V 5	6 2P 0 7 2D	.	90.55	.	.	.	XB	.	.5 1	.	W	.	.	.	.	.	.	.	.		
4751	O VI 10	7 2S 8 2P 0	1/2	129.85	.	.	.	XB	XB	.	.	.	.	.	.	.	.	.	.	.		
4752	N IV 11	3P 3D 3D 3D 0	3 3	62.41	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.		
4752.1	CR I --	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.		
4754.0	-	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.		
4754.7	FE III 3F	A 5D A 3F	3 4	2.65	.	.	3	.	.	.4	.5	W	W*	.	.	.	.	.	.	.		
4759	-	.	.	.	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.		
4762	N IV 11	3P 3D 3D 3D 0	3 2	62.41	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.		
4762.8	TI II 17	A 2D Z 4G 0	5/2 7/2	3.67	.	.	.	.	.	.	.	W	2	.	.	.	.	.	.	.		
4763.8	TI II 48	A 2P Z 4F 0	1/2 3/2	3.81	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.		
4764.3	CR I 231	A 3D X 3F 0	3 4	6.13	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.		
4764.5	TI II 48	A 2P Z 4F 0	3/2 5/2	3.82	.	.	1	.	.	.	.	.	2	.	.	.	.	.	.	.		
4768.2	-	.	.	.	.	.	.	.	.	.11	.	.	.	.	.	.	.	.	.	.		
4768.7	CL II 40	4S ¹ 3P 0 4P ¹ 3D	1 2	19.60	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.		
4769.4	FE III 3F	A 5D A 3F	2 3	2.68	.	.	2 8B	.	.	W	.3	XB	W*	.	.	.	.	.	.	.		
4771	N2 VK	5 17	.	6.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.		
4772.1	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.		
4772.4	NI VIII 1F	3D3 4F 3D3 4P	9/2 5/2	3.05	.	.	.	.	.	.3	.	.	.	.	.	.	.	.	.	.		
4772.6	O IV 9	3P 4P 3D 4D 0	1/2 3/2	61.67	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.		
4774.2	N II 20	3P 3D 3D 3D 0	1 2	23.14	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	WB	WB		
4774.7	FE II 20F	A 4F B 4F	9/2 7/2	2.82	.	.	2 8B	10	.	.	W	WB	W	W	.	.	.	.	.	.		
4776.9	-	.	.	.	.	.	.	.	.	.1	W	.	.	.	.	.	.	.	.	.		
4777.7	FE III 3F	A 5D A 3F	1 2	2.70	.	.	X	.	.	.	.7	1 2	W	.	.	.	.	.	.	.		
4779.1	S II 8	4S 4P 4P 2D 0	5/2 5/2	16.20	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.	.		
4779.1	O IV 9	3P 4P 3D 4D 0	1/2 1/2	61.66	.	.	.	.	.	.	.	1 8	.	.	.	.	.	.	.	.		
4779.7	N II 20	3P 3D 3D 3D 0	1 1	23.14	.	.	XB	.	.	.11	.	.	.	.	.	.	.	.	WB	WB		
4780.0	TI II 92	B 2P Z 2S 0	1/2 1/2	4.62	.	.	.	.	.	.	.	1 8	4	.	.	.	.	.	.	.		
4781.2	N II 20	3P 3D 3D 3D 0	2 3	23.15	.	.	XB	.	.	.1 .46	.	.	.	.	.	.	.	.	WB	WB		
4782.0	NE II 71	4S 4P 4F 4D 0	5/2 3/2	37.39	.	.	.	.	.	.56	.	.	.	.	.	.	.	.	.	.		
4783.1	CR I 283	Z 5D 0 E 5F	3 4	6.74	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.		
4783.4	O IV 9	3P 4P 3D 4D 0	3/2 5/2	61.67	.	.	XB	.	XB	50 140B	.	1	.	.	.	.	.	.	.	.		

*Nebular origin reported in one case.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM		NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME						
					QUAS	GAL																
4784.3	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4786.5	NI I 98	Z 5G U	5	5.98	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4788.1	N II 20	E 5F 3P 3D	5 2	23.14	.	.	XB	.	.	.	.	W	.	.	.	.	.	.	.	W	W	.
4789	C IV 5	3D 3D O	2	58.12	.	.	12B	.	XB	50 140B	.	.	.	.	.	.	.	.	.	.	.	.
4789.5	F II 1F	5 2D 6 2P O 2P4 3P 2P4 1D	. 2 2	2.58	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.
4790.3	CR I 31	A 5G Y 5F U	4 5	5.11	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4790.4	-	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.
4792.0	S II 46	4P 4P O	5/2	18.64	.	.	12B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4792.6	AU I 3	5S 4P 6 2P U	5/2 3/2	7.66	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4793.7	N II 20	6 2D 3P 3D 3D 3D U	5/2 2 1	23.14	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.
4794.2	O IV 9	3P 4P 3D 4D U	3/2 3/2	61.67	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.
4798.3	FE II 4F	A 6D B 4P	1/2 3/2	2.69	.	.	.	5B	.	.	.	1B	.	.	.	.	.	.	.	.	.	.
4798.3	O IV 9	3P 4P 3D 4D O	5/2 7/2	61.68	.	.	XB	.	XB	50 140B	.	1B	.	.	.	.	.	.	.	.	.	.
4798.5	TI II 17	A 2D Z 4G O	3/2 5/2	3.65	.	.	.	.	.	.	.	W	3	.	.	.	.	.	.	.	.	.
4799.3	FE II 4F	A 6D B 4P	9/2 5/2	2.57	.	.	.	5B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4801.3	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4801.6	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4802.4	-	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4802.8	S III --	.	.	.	.	.	.	.	XB	XB	.	.	.	.	.	.	.	.	.	.	.	.
4803.2	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4803.3	N II 20	3P 3D 3D 3D O	3 3	23.15	.	.	10	.	.	.	.	W	.	.	.	.	.	.	.	W	.	.
4804.0	LA II 37	A 3D Z 3P O	1 1	2.80	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4804.1	S II 8	4S 4P 4P 2D O	3/2 3/2	16.13	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.
4805	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4805.1	-	.	.	.	.	.	.	.	.	.97	.	X	.	.	.	.	.	.	.	.	.	.
4805.1	TI II 92	B 2P Z 2S O	3/2 1/2	4.62	.	.	.	.	.	.	.	1 4	5	.	.	.	.	.	.	.	.	.
4807.5	FE VI 2F	3D3 4F 3D3 2G	5/2 9/2	2.63	.	.	.	.	.	.6 2	.	.	.	.	.	.	.	.	.	.	.	.
4809.0	LA II 37	A 3D Z 3P O	1 0	2.80	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4810.3	N II 20	3P 3D 3D 3D O	3 2	23.14	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.
4812.4	CR II 30	A 4F Z 4F O	7/2 9/2	6.41	.	.	.	.	.	.	.	W	2	.	.	.	.	.	.	.	.	.
4813.1	O IV 9	3P 4P 3D 4D O	5/2 5/2	61.67	.	.	.	.	.	.	.	1 5B	.	.	.	.	.	.	.	.	.	.
4813.3	SI III 9	4 3F O 5 3G	2 3	28.42	.	.	.	.	.	XB	.	1 5B	.	.	.	.	.	.	.	.	.	.
4813.9	FE III 3F	A 5D A 3F	1 3	2.68	.	.	XB	.	.	.	.8B	1 5B	.	.	.	.	.	.	.	.	.	.
4814.6	FE II 20F	A 4F B 4F	9/2 9/2	2.79	.	.	10 30	25	.	.	.8B	1	W 3	W 1	.	25	.	.	.	.	.	.
4819 R	O2 H	4 .12	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.
4819.4	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
4819.7	SI III 9	4 3F O 5 3G	3 3	28.42	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.
4823.9	O IV 9	3P 4P 3D 4D O	5/2 3/2	61.67	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4824	SC VII 1F	3P3 2D O 3P3 2P O	3/2 3/2	2.56	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4824.1	CR II 30	A 4F Z 4F U	9/2 9/2	6.41	.	.	X	5	.	.	.	1 12	W 6	5	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPL NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	DE BE PCYG	AE FE GE	T- TAU	KE ME N						
4824.1	FE III 3F	A 5D A 3F	2 4	2.65	.	.	XB	.	.	.6 1	.	.	.	.	1	.	.	.	.	.	.	
4824.1	LA II 50	A 3P Y 3D 0	0 1	3.21	.	.	.	1 28	.	.	.	.	.	.	.	.	.	.	.	.	.	
4825.1	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4825.7	FE II 30	A 4H Z 6F 0	13/2 11/2	5.18	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	
4828.9	SI III 9	4 3F 0 5 3G	4	28.42	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	
4829.2	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4830.5	-	.	.	.	.	.	.	.	.	.7 1	.	W	.	.	.	.	.	.	.	.	.	
4833.2	FE II 30	A 2H Z 6F 0	11/2 9/2	5.20	.	.	18	.	.	.	.	W	2	.	.	.	.	.	.	.	.	
4834.3	-	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	
4835 R	O2 H	2 11	.	5.1	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	
4836.2	CR II 30	A 4F Z 4F 0	5/2 7/2	6.39	.	.	1	.	.	.	.	2	.	.	.	.	.	.	.	.	.	
4837.1	N2 VK	2 15	.	6.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.	
4838.7	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	X	.	.	.	.	.	
4839.3	TI II 110	A 2S Z 4P 0	1/2 1/2	5.18	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4840.0	FE II 30	A 2H Z 6F 0	9/2 7/2	5.21	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	
4842.1	AL 0	0 0	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	
4848	-	.	.	.	.	.	.	.	.	4	.5	.	.	.	.	.	.	.	.	.	.	
4848.2	CR II 30	A 4F Z 4F 0	7/2 7/2	6.39	.	.	4	.	.	.	.	1 8	1 4	1	1	.	.	.	.	.	.	
4849.2	TI II 29	A 4G Z 4G 0	9/2 7/2	3.67	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	
4850.9	FE VI 2F	3D3 4F 3D3 2G	3/2 5/2	2.54	.	.	.	.	.	2 3	.	.	.	.	.	.	.	.	.	.	.	
4852.7	FE II 20F	A 4F B 4F	7/2 3/2	2.84	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4854.5	-	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.	
4855.9	-	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	
4858.7	N III 9	3P 4D 3D 4F 0	1/2 3/2	40.77	.	.	.	140 280B	.	.	.	.	.	.	.	.	.	.	.	.	.	
4858.9	N III 9	3P 4D 3D 4F 0	3/2 5/2	40.77	.	.	.	140 280B	.	.	.	.	.	.	.	.	.	.	.	.	.	
4859.3	HE II 2	4 2F 0 8 2G	.	53.34	.	.	100B	.	100B	100B	100B	.	.	.	.	.	.	.	.	.	.	
4861.3	H ϵ 1	2 2P 0 4 2D	.	12.69	100	100	100	100	100*	100*	100	100	100	100	100	100	100**	.	1 20	1	.	
4861.3	N III 9	3P 4D 3D 4F 0	5/2 7/2	40.78	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	
4862.6	C II -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	
4864.1	-	.	.	.	.	.	.	.	.	.	.08	.	.	.	.	.	.	.	.	.	.	
4864.3	CR II 30	A 4F Z 4F 0	5/2 5/2	6.38	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4865.6	TI II 29	A 3G Z 4G 0	7/2 5/2	3.65	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
4866.1	AL 0	1 1	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.	.	
4867.1	C II -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	
4867.2	N III 9	3P 4D 3D 4F 0	7/2 9/2	40.79	.	.	.	140 280B	.	.	.	WB	.	.	.	.	.	.	.	.	.	
4867.7	FE II 30	A 4H Z 6F 0	11/2 11/2	5.18	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	
4868	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	
4868.2	FE IV -	.	.	.	.	.	8	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4869.3	F II 1F	2P4 3P 2P4 1D	1 2	2.58	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
4871.3	FE II 25	B 4P Z 6F 0	3/2 5/2	5.23	.	.	.	.	.	.	.	W	3	.	.	.	.	.	.	.	.	

*Nebular emission background origin.

**H β is sometimes not present at all; hence the value should be given as 0-100.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY													
		LEVELS	J		NUC EM GAL	NOVA	W-R LIKE N	W-R C	PLAN NEB	GAS NEB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGT NING
4871.6	O II 57	3P ¹ 2P 0 3D ¹ 2D	3/2 5/2	31.24	.	.	.	.	.	.	W	.	.	.	.	.	.	.
4874.0	TI II 114	C 2D Y 2P 0	3/2 1/2	5.61	.	.	.	.	.	.	.	W	3	.	.	.	.	.
4874.5	FE II 20F	A 4F B 4F	7/2 5/2	2.83	.	.	.	10	.	.	.	.	W	.	.	.	.	.
4876.5	CR II 30	A 4F Z 4F 0	7/2 5/2	6.38	.	.	.	108	.	.	.	1 4	4	.	.	.	.	.
4877.3	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
4880.0	-	.	.	.	.	.	.	.	.	.	6	W	.	.	.	.	.	.
4881.0	FE III 2F	A 5D A 3H	4 4	2.53	.	.	XB	.	.	.	2B	.5	.	.	.	.	.	.
4883.4	V II 209	D 3F Y 3D 0	4 3	6.31	.	.	.	.	.	.	.	.	W	.	.	.	.	.
4883.9	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
4883.9	FE III 1F	A 5D A 3P	2 0	2.62	.	.	XB	.	.	.	2B	.	.	.	.	.	.	.
4884.1	N III 9	3P 4D 3D 4F 0	7/2 1/2	40.78	.	.	.	.	XB	XB	.	.	.	.	.	.	.	.
4884.6	CR II 30	A 4F Z 4F 0	5/2 3/2	6.37	.	.	.	1	.	.	.	.	1	.	.	.	.	.
4885.6	S II 15	4S 2P 4P 2P 0	1/2 3/2	16.47	.	.	.	108	.	.	.	.	.	.	.	.	.	.
4886.4	-	.	.	.	.	.	.	.	.	.	.12	.	.	.	.	.	.	.
4886.9	FE II 54	B 2P Z 4P 0	1/2 3/2	5.85	.	.	.	.	.	.	.	W	.	.	.	.	.	.
4888.4	AL 0	2 2	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.
4889.6	FE II 4F	A 6D B 4P	7/2 5/2	2.57	.	.	.	20	.	.	.	.	W	2B	.	.	10	.
4889.7	FE II 3F	A 6D A 2D	3/2 3/2	2.63	.	.	3 6	108	.	.	.	.	1 2B	W	.	.	.	.
4890.9	O II 28	3P 4S 0 3D 4P	3/2 1/2	28.71	.	.	.	.	.	.	.9 2	.	.	.	.	.	3	.
4893.8	FE II 36	B 4F Z 6P 0	7/2 5/2	5.34	.	.	.	2B	.	.	.	.	2	.	.	.	.	.
4893.9	FE VII 2F	3D2 3F 3D2 3P	2 1	2.52	.	.	.	.	.	.	.6 3	.	3 10	.	.	.	.	.
4895.2	N II 1	2P3 1D 0 3P 1P	2 1	20.32	.	.	.	.	.	.	.	.	W	.	.	.	3	.
4895.4	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.
4896.7	N III 9	3P 4D 3D 4F 0	7/2 5/2	40.77	.	.	.	.	XB	XB	.	.	.	.	.	.	.	.
4898.0	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.
4898.7	-	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.
4899.6	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.
4900.2	FE IV -	.	.	.	.	.	.	8	.	.	.	.	.	.	.	.	.	.
4901.4	-	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.
4903	-	.	.	.	.	.	.	.	.	.	.7 3	.	.	.	.	.	.	.
4905.4	FE II 20F	A 4F B 4F	7/2 7/2	2.82	.	.	.	15	.	.	.	.	1	1	W	.	.	.
4906.2	FE IV -	.	.	.	.	.	.	4	.	.	.	.	5	.	.	.	.	.
4906.9	O II 28	3P 4S 0 3D 4P	3/2 3/2	28.71	.	.	.	.	.	.	.16	.	.	.	.	.	.	.
4910	-	.	.	.	.	.	.	.	.	.	2 4	.	.	.	.	.	.	.
4911.2	TI II 114	C 2D Y 2P 0	5/2 3/2	5.62	.	.	.	.	.	.	.	.	W	3	.	.	.	.
4912.5	CR II 190	D 2G X 2G 0	7/2 7/2	8.97	.	.	.	1	.	.	.	.	.	.	.	.	.	.
4913.4	FE II 218	X 4D 0 32	3/2 5/2	10.35	.	.	.	1	.	.	.	.	.	.	.	.	.	.
4914.9	N I 9	3S 2P 4P 2S 0	1/2 1/2	13.14	.	.	.	.	.	.	.	.	.	.	.	.	5	1
4915	-	.	.	.	.	.	.	.	.	.	1.5	.	.	.	.	.	.	.
4920.4	HE I 49F	2 1P 0 4 1F 0	1 3	23.64	.	.	.	.	.	.	.	.	W	.	.	.	.	.

[illegible]

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY																AIR GLW	CTY LTS	AUR ORA	LGH NING
		LEVELS	J		NUC. EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME										
5035.5	FE II 4F	A 6D B 4P	1/2 5/2	2.57	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5035.8	FE II --	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5039.9	-	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	
5040.1	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
5041.1	SI II 5	4 2P 0 4 2D	1/2 3/2	12.47	.	.	8B	WB	.	.	.7 5	.6	1 4	1 3	.	.	.	.	.	.	.	.	.	
5042.0	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
5043.5	FE II 20F	A 4F B 4F	3/2 5/2	2.83	.	.	.	8	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
5045.1	N II 4	3S 3P 0 3P 3S	2 1	20.85	.	.	XB	WB	.	.	.	.	5	.	.	.	.	.	.	.	3	W	.	
5047.2	C II 35	3D 4D 0 4P 4P	5/2 3/2	26.71	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	
5047.7	HE I 47	2 1P 0 4 1S	1 0	23.57	.	.	8 16B	WB	.	XB	2 5	.7 2	5 15	1	.	.	.	.	.	.	.	.	.	
5047.9	FE II -	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5048.2	FE II -F	.	.	.	.	.	.	6B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5049.3	FE II 20F	A 4F B 4F	5/2 9/2	2.79	.	.	.	6B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5053.7	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5056.0	SI II 5	4 2P 0 4 2D	3/2 5/2	12.47	.	.	12	4B	.	.	.	.	1B	1 5B	.	.	.	.	.	.	.	.	.	
5056.4	SI II 5	4 2P 0 4 2D	3/2 3/2	12.47	.	.	8B	4B	.	.	1.5	2	1B	.	.	.	.	.	.	.	.	.	.	
5058.6	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5060.3	FE III 1F	A 5D A 3P	1 1	2.55	.	.	XB	4B	.	.	.	.8B	1	W	.	.	.	.	.	.	.	.	.	
5063.7	FE III 2F	A 5D A 3H	2 4	2.53	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5065.4	TI II 19F	B 4F B 2F	9/2 5/2	2.59	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5067	N V 6	6 2D 7 2P 0	.	90.49	.	.	.	.	.	.	.3	.	.	.	.	.	.	.	.	.	.	.	.	
5071.0	FE II --	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5071.6	CR IV 4F	3D3 4F 3D3 2D	9/2 5/2	2.55	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5072.4	FE II 19F	A 4F A 4H	9/2 9/2	2.66	.	.	.	5	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
5073.6	N II 10	3S 1P 0 3P 3S	1 1	20.85	.	.	.	.	.	.	.3	.	.	1 8	.	.	.	.	.	.	.	.	.	
5074.1	FE II 205	D 2F X 2D 0	7/2 5/2	9.21	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5075.8	FE II --	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5076.6	FE II 20F	A 4F B 4F	3/2 7/2	2.82	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5078.7	CR I --	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5083.7	FE II 35F	A 4D A 2F	7/2 5/2	3.41	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5084.8	FE III 1F	A 5D A 3P	0 1	2.55	.	.	.	1	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	
5086.7	FE III 5	B 2D Z 5P 0	2 1	11.05	.	.	.	.	.	.	.	.	W	1.6	.	.	.	.	.	.	.	.	.	
5087.3	FE II --	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5090	-	.	.	.	.	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.	
5090.8	C IV -	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5091.9	-	.	.	.	.	.	.	.	.	.	4	.	W	.	.	.	.	.	.	.	.	.	.	
5094	-	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.	.	.	
5097.3	CR II 24	B 4P Z 6D 0	1/2 3/2	6.12	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5097.8	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
5100.7	FE II 35	B 4F Z 6F 0	9/2 7/2	5.21	.	.	.	.	.	.	.	.	1 4B	2B	.	.	.	.	.	.	.	.	.	

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY													
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE	AE	KE	AIR	CTY	AUR	LGT	
											BE	FE	T- TAU					ME N
5151.1	C II 16	3S 4P 0 3P 4P	5/2 3/2	23.02	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
5151.9	FE III 1F	A 5D A 3P	4 2	2.40	.	.	.	.	.	.	5B	.	.	.	.	.	.	.
5153	-	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.
5153.5	CR II 24	B 4P Z 6D 0	3/2 5/2	6.14	.	.	.	5B	.	.	.	.	.	.	.	.	.	.
5154.1	TI II 70	B 2D Z 2D 0	3/2 5/2	3.95	.	.	.	.	.	.	.	2B	.	.	.	.	.	.
5154.4	FE II 35	B 4F Z 6F 0	5/2 5/2	5.23	.	.	.	5B	.	.	.	2B	.	.	.	.	.	.
5156.0	FE III 5	B 5D Z 5P 0	4 3	11.00	.	.	.	.	.	.	.	5	.	.	.	.	.	.
5156.1	FE II --	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
5157.0	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.
5158.0	FE II 18F	A 4F B 4P	7/2 3/2	2.69	.	.	.	45B	.	.	.	1 108M	X	.	.	.	.	.
5158.3	FE VII 2F	3D2 3F 3D2 3P	3 1	2.52	.	2 5B	10	.	.	5	.7B	1 10B	.	.	.	.	.	.
5158.8	FE II 19F	A 4F A 4H	9/2 13/2	2.62	.	.	.	5B	.	.	2B	.7B 1 10B	1 3	.	.	.	.	.
5161.7	-	.	.	.	.	.	.	.	.	.6B	.	.	.	.	.	.	.	.
5163.9	FE II 35F	A 4D A 2F	7/2 7/2	3.37	.	.	.	10	.	.	.	.	.	.	.	.	.	.
5167.3	MG I 2	3 3P 0 4 3S	0 1	5.09	.	.	.	.	.	.	.	.	3B	10B	W	.	.	.
5167.5	FE I 37	A 3F Z 3D 0	4 3	3.87	.	.	.	.	.	.	.	.	3B	.	.	.	.	.
5168	-	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.
5169.0	FE II 42	A 6S Z 6P 0	5/2 7/2	5.27	.	.	10 18B	50	.	.	.	1 5	10 50	1 15	.	.	.	.
5171	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W
5171.6	FE II 35	B 4F Z 6F 0	9/2 11/2	5.18	.	.	.	.	.	.	.	1	.	.	.	.	.	.
5172.6	AL III 18	5 2G 7 2F 0	.	25.83	.	.	.	.	.	XB	.	.	.	.	.	.	.	.
5172.7	MG I 2	3 3P 0 4 3S	1 1	5.09	.	.	.	.	.	.	.	.	3	10B	W	.	.	.
5175.9	N II 66	3P 5D 0 3D 5F	3 4	29.99	.	.	10B	.	.	.	.	.	.	.	.	.	.	W
5176.0	O II 32	3P 2P 0 3D 2P	3/2 1/2	28.83	.	.	.	.	.	.	.	.	.	.	.	.	.	W
5177.0	FE VI 2F	3D3 4F 3D3 2G	9/2 9/2	2.63	.	.	50	.	.	.	.9 5	W 5	.	.	.	.	.	.
5179.5	N II 66	3P 5D 0 3D 5F	4 5	29.99	.	.	10B	.	.	.	1.5	.	.	.	.	.	.	.
5181	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W
5182.0	FE II 18F	A 4F B 4P	3/2 1/2	2.77	.	.	.	W 10	.	.	.	.	.	.	.	.	.	.
5183.4	LA II 36	A 3D Z 3D 0	3 3	2.78	.	.	.	1B	.	.	.	.	.	.	.	.	.	.
5183.6	MG I 2	3 3P 0 4 3S	2 1	5.09	.	.	.	1B	.	.	.	W 4	1B	1	.	.	.	.
5184.6	CR I 201	Z 7D 0 F 7D	3 4	5.78	.	.	.	1B	.	.	.	.	.	.	.	.	.	.
5184.8	FE II 19F	A 4F A 4H	7/2 7/2	2.68	.	.	.	4	.	.	.	.	.	.	.	.	.	.
5185.6	-	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
5185.9	TI II 86	B 2G Z 2G 0	7/2 7/2	4.26	.	.	.	.	.	.	.	1	.	.	.	.	.	.
5186.2	-	.	.	.	X	.	.	.	.	.	1.5 5	.	.	.	.	.	.	.
5186.5	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
5187	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W
5188.2	LA II 95	Z 3H 0 E 3G	6 5	4.82	.	.	.	1	.	.	.	.	.	.	.	.	.	.
5188.7	TI II 70	B 2D Z 2D 0	5/2 5/2	3.95	.	.	.	.	.	.	.	2	1B	.	.	.	.	.
5191.4	A III 3F	3P4 1D 3P4 1S	2 0	4.11	.	.	.	.	.	.	.9	.	.	.	.	.	.	.

LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM QUAS	GAL	NOVA	LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N					
5192.0	-	.	.	.	.	.	.	.	.	.	3 5	.7	.	.	.	.	.	.	.	.	.	.
5193.9	FE III 5	B 5D Z 5P U	3 3	11.00	.	.	.	.	.	.	.	.	1.1	.	.	.	.	.	.	.	.	.
5194.4	FE III 5	B 5D Z 5P O	2 3	11.00	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.	.
5197.6	FF II 49	A 4G Z 4F O	5/2 3/2	5.59	.	.	3 6	18B	.	.	.	.	1 16	8	3	.	.	.	.	.	.	.
5198.5	N I 1F	2P3 4S O 2P3 2D O	3/2 3/2	2.37	.	W 25	20B	.	XB	.	1.2 6	4	1B	.	.	9B	.	.	.	5 20	.	.
5199.2	FE II 35F	A 4D A 2F	5/2 5/2	3.41	.	.	.	18B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5200.7	N I 1F	2P3 4S O 2P3 2D O	3/2 5/2	2.37	.	.	20B	.	XB	.	.3	.	1B	.	.	9B	.	W 5	.	5 20	.	.
5202	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W
5202.5	SI II --	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
5204.1	LA II 96	Z 3H O E 3F	5 4	4.67	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5206.7	O II 32	3P 2P O 3D 2P	3/2 3/2	28.82	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.
5207.2	-	.	.	.	.	.	.	.	W	.	W 3	.	.	.	.	.	.	.	.	.	.	.
5210.9	CR II 38	A 2F Z 4F O	5/2 7/2	6.39	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5211.5	TI II 103	B 2F Y 2F O	7/2 7/2	4.95	.	.	.	WB	.	.	.	.	W	.	.	.	.	.	.	.	.	.
5212	-	.	.	.	.	.	.	.	.	.	3	.6	.	.	.	.	.	.	.	.	.	.
5216	-	.	.	.	.	.	.	1	.	.	.5 3	.	.	.	.	.	.	.	.	.	.	.
5218.3	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.
5220.1	FE II 19F	A 4F A 4H	7/2 9/2	2.66	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5221	-	.	.	.	.	.	.	.	.	.	.	.6	.	.	.	.	.	.	.	.	.	.
5226.5	TI II 70	B 2D Z 2D O	3/2 3/2	3.92	.	.	.	.	.	.	.	WB	2	1	.	.	.	.	.	.	.	.
5227.2	FE I 37	A 3F Z 3D O	3 2	3.91	.	.	.	.	.	.	.	WB	2	.	.	.	.	.	.	.	.	.
5228.3V	N2+ 1N	0 3	.	18.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.
5230.4	N2 VK	5 18	.	6.1	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.
5232.8	OH M	9 2	.	3.2	.	.	.	.	.	.	.4	.	.	.	.	.	.	W	.	.	.	.
5234.6	FE II 49	A 4G Z 4F O	7/2 5/2	5.57	.	.	9 12	16B	.	.	.	5	W 16	10B	5	.	.	.	.	.	.	.
5235.3	FE III 113	E 7D Y 7P O	4 4	20.54	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.
5236	-	.	.	.	.	.	.	.	.	.	.	.6	.	.	.	.	.	.	.	.	.	.
5236.6	FE VI 1F	3D3 4F 3D3 4P	5/2 5/2	2.42	.	.	.	.	.	.	3	.	1	.	.	.	.	.	.	.	.	.
5237.3	CR II 43	B 4F Z 4F O	9/2 9/2	6.41	.	.	.	16B	.	.	.	.	2	10B	.	.	.	.	.	.	.	.
5239.0	CR I 59	A 5P X 5P O	2 1	5.05	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5243.3	FE III 113	E 7D Y 7P O	5 4	20.54	.	.	.	.	.	.	.	.	1.4	.	.	.	.	.	.	.	.	.
5243.6	-	.	.	.	.	.	.	.	.	.	.	.6	.	.	.	.	.	.	.	.	.	.
5244.5	C III 4	3S 3P O 3P 3S	0 1	40.39	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.
5249.4	CR II 23	B 4P Z 4P O	3/2 5/2	6.09	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5249.6	C III 23	4 1D 5 1P O	2 1	42.37	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.
5250.0	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5252	-	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.	.
5253.6	C III 4	3S 3P O 3P 3S	1 1	40.39	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.
5254.9	CR I 201	Z 7D O F 7D	3 2	5.74	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5254.9	FE II 49	A 4G Z 4F O	5/2 5/2	5.57	.	.	.	8B	.	.	.	.	6	.	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY													
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE SYMB	AE FE PCYG	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGT NING
5363.6	C II -	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
5363.6	C III -	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
5366	-	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
5375	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
5376.5	FE II 19F	A 4F A 4H	3/2 7/2	2.68	.	.	.	25	.	.	.	.	3	.	.	.	.	.
5381.0	TI II 69	B 2D Z 2F 0	3/2 5/2	3.85	.	.	X	.	.	.	.	.	1	.	.	.	.	.
5387	-	.	.	.	.	.	.	.	.	1 4	.	.	.	.	.	.	.	.
5387.1	FE II --	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
5390.1	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.
5395	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.
5396.3	TI II 80	A 2H Z 4F 0	11/2 9/2	3.86	.	.	.	WB	.	.	.	.	1	.	.	.	.	.
5396.6	TI II 102	B 2F Y 2D 0	5/2 5/2	4.87	.	.	.	WB	.	.	.	.	1	.	.	.	.	.
5397	-	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
5397.1	FE I 15	A 5F Z 5D 0	4 4	3.20	.	.	.	.	.	.	.	8	.	.	.	.	.	.
5398.8	TI IV	5 2S 5 2P 0	1/2 3/2	28.51	.	.	.	.	.	XB	.	.	.	.	.	.	.	.
5402	-	.	.	.	.	.	.	.	.	1 10	.	.	.	.	.	.	.	.
5405.8	FE I 15	A 5F Z 5D 0	2 1	3.27	.	.	.	.	.	.	.	5	.	.	.	.	.	.
5407.6	CR II 23	B 4P Z 4P 0	5/2 5/2	6.09	.	.	.	5	.	.	.	.	1	.	.	.	.	.
5408.8	FE II 184	D 2D X 4F 0	5/2 5/2	8.21	.	.	.	W	.	.	.	.	3	.	.	.	.	.
5410	O VI 13	7 2D 8 2P 0	.	129.85	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
5411.5	HE II 2	4 2F 0 7 2G	.	53.08	.	.	1 8	5 280	70 280	XB 25	.	1 10	.	.	.	.	.	.
5412.6	FE II 17F	A 4F A 2D	5/2 3/2	2.63	.	.	.	W 158	.	.	.	.	1	.	.	.	.	.
5413.2	C IV -	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
5413.2	TI IV -	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
5413.3	FE II 16F	A 4F A 2H	9/2 11/2	2.51	.	.	.	158	.	.	.	.	.	.	.	.	.	.
5414.1	FE II 48	A 4G Z 4D 0	7/2 7/2	5.49	.	.	.	.	.	.	.	108	6	.	.	.	.	.
5417	O V 13	3P 3P 3D 3P 0	1 2	87.36	.	.	.	.	.	XB	.	.	.	XB	.	.	.	.
5418	-	.	.	.	.	.	.	.	.	1 3	.	.	.	.	.	.	.	.
5418.8	TI II 69	B 2D Z 2F 0	5/2 5/2	3.85	.	.	.	.	.	.	.	.	1	.	.	.	.	.
5420.4	MN I 4	A 6D Y 6P 0	7/2 5/2	4.41	.	.	.	.	.	.	.	.	1	.	.	.	.	.
5420.9	CR II 23	B 4P Z 4P 0	3/2 1/2	6.02	.	.	.	.	.	.	.	.	3	.	.	.	.	.
5423.3	FE III -	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.
5425.3	FE VI 1F	3D3 4F 3D3 4P	5/2 3/2	2.34	.	.	1 208	.	.	.	2	.	XB	.	.	.	.	.
5425.3	FE II 49	A 4G Z 4F 0	9/2 9/2	5.46	.	.	3	8	.	.	.	.	5	12	W	.	.	.
5427.8	FE II --	.	.	.	.	.	.	W	.	.	.	.	2	.	.	.	.	.
5428.6	FE VI 1F	3D3 4F 3D3 4P	7/2 5/2	2.42	.	.	1 208	.	.	.	XB	.	XB	.	.	.	.	.
5429.7	FE I 15	A 5F Z 5D 0	3 3	3.23	.	.	.	.	.	.	.	.	1	.	.	.	.	.
5432	O V 13	3P 3P 3D 3P 0	2 1	87.38	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
5432.5	MN I 1	A 6S Z 8P 0	5/2 5/2	2.27	.	.	.	.	.	.	.	.	W	.	.	.	.	.
5433.0	FE II 55	B 2H Z 4F 0	9/2 7/2	5.52	.	.	.	.	.	.	.	.	8	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J		NUC		NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGH NING
					EM	GAL														
5433.2	FE II 18F	A 4F B 4P	7/2 5/2	2.57	.	.	.	18	.	.	.	.	.	.	.	.	.	.	.	.
5434.5	FE I 15	A 4F Z 5D 0	1 0	3.28	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.
5435.4	-	.	.	.	.	.	.	.	.	1.5 3	.	.	.	.	.	.	.	.	.	.
5435.8	FE II 48	A 4G Z 4D 0	5/2 7/2	5.49	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.
5436.8	O I 11	3 5P 6 5S 0	3 2	12.96	.	.	.	.	.	.	.	.	.	.	.	.	.	2	1	.
5440.5	FE II 16F	A 4F A 2H	7/2 9/2	2.57	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
5446.9	FE I 15	A 5F Z 5D 0	2 2	3.25	.	.	.	W	.	.	.	5	.	.	.	.	.	.	.	.
5453.8	S II 6	4S 4P 4P 4D 0	5/2 7/2	15.88	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
5454	-	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.
5455.0	-	.	.	.	.	.	.	6	.	.	.	.	.	.	.	.	.	.	.	.
5457.6	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.
5460.7	HG I 1	6. 3P 0 7 3S	2 1	7.70	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.
5462.6	N II 29	3P 3P 3D 3P 0	1 1	23.32	.	.	.	4 6H	XB	.	.	.	.	.	.	.	.	.	.	.
5466.9	FE II --	.	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.
5468	-	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.
5469.6	C IV -	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.
5470.9	-	.	.	.	.	.	.	.	.	.4	.	.	.	.	.	.	.	.	.	.
5473	O V 13	3P 3P 3D 3P 0	2 2	87.36	.	.	.	XB	XB	.	.	.	.	.	.	.	.	.	.	.
5473.6	S II 6	4S 4P 4P 4D 0	1/2 1/2	15.78	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
5477.3	FE II 34F	A 4D B 2P	3/2 1/2	3.32	.	.	.	103	.	.	.	.	.	.	.	.	.	.	.	.
5477.7	FE II 49	A 4G Z 4F 0	7/2 9/2	5.46	.	.	.	108	.	.	.	.	.	.	.	.	.	.	.	.
5478.1	N II 29	3P 3P 3D 3P 0	1 2	23.31	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.
5478.4	CR II 50	B 4G Z 4F 0	11/2 9/2	6.41	.	.	.	1	.	.	.	.	2	.	.	.	.	.	.	.
5480.1	N II 29	3P 3P 3D 3P 0	2 1	23.32	.	.	.	48	.	.	.	W	.	.	.	.	.	.	.	.
5485.7	FE VI 1F	3D3 4F 3D3 4P	5/2 1/2	2.31	.	.	.	.	.	.5 2	.	X	.	.	.	.	.	.	.	.
5486.9	-	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
5488	-	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.
5491.7	-	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.	.
5492.4	TI IV --	.	.	28.47	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.
5494	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
5495.7	N II 29	3P 3P 3D 3P 0	2 2	23.31	.	.	.	48	XB	.	.	WB	.	.	.	.	.	.	.	.
5495.8	FE II 17F	A 4F A 2D	3/2 3/2	2.63	.	.	.	1 3	8	.	.	.	WB	.	.	.	.	.	.	.
5497.5	FE I 15	A 5F Z 5D 0	1 2	3.25	.	.	.	W	.	.	.	5	.	.	.	.	.	.	.	.
5497.7	FE II 204	D 2F X 2F 0	5/2 5/2	9.02	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
5498.2	FE II 24	B 4P Z 6D 0	5/2 3/2	4.82	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
5502.1	CR II 50	B 4G Z 4F 0	9/2 7/2	6.39	.	.	.	W	.	.	.	.	2	.	.	.	.	.	.	.
5503.2	CR II 50	B 4G Z 4F 0	5/2 3/2	6.37	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.
5505	-	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.
5506.8	FE I 15	A 5F Z 5D 0	2 3	3.23	.	.	.	.	.	.	.	1	1	.	.	.	.	.	.	.
5508.1	O III 16	3P 1D 3D 1D 0	2 2	40.09	.	.	.	.	X	XB	.	W	.	.	.	.	.	.	.	.

					EMISSION LINE INTENSITY																
LAB WAVE LENGTH, A	ION AND MULTIPL NUMBER	TRANSITION		UPPER EXC. POT., EV																	
		LEVELS	J		QUAS	NUC EM GAL	NOVA NOVA	W-R LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGT NING
5647.0	S II 14	4S 2P. 4P 2D 0	1/2 3/2	16.13	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.
5648	CA XV 1F	2P2 3P 2P2 3P	0 1	2.19	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5648.1	C II 15	3S 4P 0 3P 4S	3/2 3/2	22.80	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.
5650.4	FE II 2F	A 6D A 2P	5/2 3/2	2.27	.	.	.	6B	.	.	.	.	.	.	.	.	.	.	.	.	.
5650.9	FE II 39F	A 4P B 4D	3/2 3/2	3.87	.	.	.	6B	.	.	.	.	.	.	.	.	.	.	.	.	.
5654.9	FE II 17F	A 4F A 2D	5/2 5/2	2.53	.	.	.	30B	.	.	.	.	.	.	.	.	.	.	.	.	.
5657.9	FE II 57	A 2F Z 4F 0	5/2 3/2	5.59	.	.	.	6B	.	.	.	.	5B	.	.	.	.	.	.	.	.
5657.9	SC II 29	A 3P Z 3P 0	2 2	3.68	.	.	.	6B	.	.	.	.	5B	.	.	.	.	.	.	.	.
5658	-	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.	.
5659.8	FE II 33F	A 4D A 4G	5/2 5/2	3.22	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.
5661.0	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.
5662.1	O VI -	.	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.
5665.9	FE II -F	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.
5666.6	N II 3	3S 3P 0 3P 3D	1 2	20.56	.	.	30B	W	XB	.	.	1	1.4	.	.	.	.	2	W	.	.
5667	C III -	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.
5667.2	SC II 29	A 3P Z 3P 0	1 1	3.67	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.
5668.8	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.
5669.0	SC II 29	A 3P Z 3P 0	1 0	3.67	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.
5669.3	-	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
5670.5	-	.	.	.	.	.	X.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5671.7	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
5672.7	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.
5673.2	FE II -F	.	.	.	.	.	.	8	.	.	.	.	.	.	.	.	.	.	.	.	.
5676.0	N II 3	3S 3P 0 3P 3D	0 1	20.56	.	.	XB	WB	XB	.	.	.	1	.	.	.	.	4B	1	.	.
5678.0	FE VI 1F	3D3 4F 3D3 4P	9/2 5/2	2.42	.	.	12 30B	2B	.	.	1.1 3	W	.	.	.	.	.	.	.	.	.
5678.4	CR II 189	D 2G Y 2G 0	7/2 7/2	8.63	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.
5679.6	N II 3	3S 3P 0 3P 3D	2 3	20.58	.	.	30	WB	XB	.	1.5 3	.7 .01	5 2.8	.	.	.	.	4B	.	.	.
5682.9	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5683.6	FE II 33F	A 4D A 4G	5/2 7/2	3.21	.	.	12B	.	.	.	.	.	.	.	.	.	.	.	.	.	.
5684.2	SC II 29	A 3P Z 3P 0	2 1	3.67	.	.	X	.	.	.	.	.	1	.	.	.	.	.	.	.	.
5686.2	N II 3	3S 3P 0 3P 3D	1 1	20.56	.	.	XB	WB	XB	.	.	.	1	.	.	.	.	1	.	.	.
5688.2	NA I 6	3 2P 0 4 2D	3/2 3/2	4.27	.	.	.	WB	.	.	.	.	1	.	.	.	.	.	.	.	.
5688.2	NA I 6	3 2P 0 4 2D	3/2 5/2	4.27	.	.	.	WB	.	.	.	.	1	.	.	.	.	.	.	.	.
5692	-	.	.	.	.	.	.	.	.	.	.	.7	.	.	.	.	.	.	.	.	.
5694.5	HE II 8	LIMIT	.	54.18	.	.	.	.	.	.	.	.	1B	.	.	.	.	.	.	.	.
5694.7	CR I 239	Z 5F 0 F 5D	4 3	6.01	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.
5695.0	NI I 220	Z 1P 0 E 1P	1 1	6.24	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.	.
5696.0	C III 2	3 1P 0 3 1D	1 2	34.13	.	.	X	.	.	X	W	.	1B	W 5	.	.	.	.	.	.	.
5696.4	FE I 2F	A 5D A 5P	4 3	2.17	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.	.	.
5696.5	AL III 2	4 2S 4 2P 0	1/2 3/2	17.74	.	.	.	.	.	XB	.	.	.	W	.	.	.	.	.	.	.

					EMISSION LINE INTENSITY																
LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV																	
		LEVELS	J		NUC EM	NOVA GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGT NING	
5702	-	.	.	.	.	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	
5708.4	SI I 10	4S 3P 0 5P 3P 2	2 2	7.09	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	
5709.4	FE I 686	Z 5F 0 E 5D 4	4 4	5.52	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	
5710.8	N II 3	3S 3P 0 3P 3D 2	2 2	20.56	.	.	XB	.	XB	.	.	.	.9	.	.	.	.	.	W	.	
5711.1	MG I 8	3 1P 0 5 1S 0	1 0	6.49	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	
5712.2	FE I 686	Z 5F 0 E 5D 3	2 3	5.56	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	
5712.7	CR III 1F	3D4 5D 3 3D4 3P 2	3 2	2.20	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.	
5712.8	CR I 119	B 5D 4 Y 5D 0 4	4 4	5.16	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	
5713.4	FE II 2F	A 6D 3/2 A 2P 3/2	3/2 3/2	2.27	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	
5715.1	NI I 231	Y 3F 0 F 3F 4	4 4	6.23	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	
5719.8	CR I 119	B 5D 3 Y 5D 0 4	3 4	5.16	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	
5720	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
5720.9	FE VII 1F	3D2 3F 2 3D2 1D 2	2 2	2.16	.	W 20B	16	.	.	.	1.2 4	10	.	.	.	.	.	.	.	.	
5721.4	FE II 33F	A 4D 7/2 A 4G 11/2	7/2 11/2	3.14	.	.	.	4B	.	.	.	.	.	.	.	.	.	.	.	.	
5722.6	FE II 58	A 2F 7/2 Z 4D 0 5/2	7/2 5/2	5.53	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	
5722.7	AL III 2	4 2S 1/2 4 2P 0 1/2	1/2 1/2	17.73	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	
5725.9	FE II 39F	A 4P 1/2 B 4D 3/2	1/2 3/2	3.87	.	.	.	4B	.	.	.	.	.	.	.	.	.	.	.	.	
5726.0	FE II 57	A 2F 5/2 Z 4F 0 5/2	5/2 5/2	5.57	.	.	.	4B	.	.	.	.	.	.	.	.	.	.	.	.	
5730.7	N II 3	3S 3P 0 3P 3D 1	2 1	20.56	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	
5732	-	.	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	
5732.7	FE II 57	A 2F 7/2 Z 4F 0 7/2	7/2 7/2	5.52	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	
5734.8	-	.	.	.	.	.	.	.	.	.	1.5 4	.	.	.	.	.	.	.	.	.	
5737.7	FE II 58	A 2F 5/2 Z 4D 0 3/2	5/2 3/2	5.56	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	
5739.8	SI III 4	4 1S 0 4 1P 0 1	0 1	21.79	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	
5745.7	FE II 17F	A 4F 3/2 A 2D 5/2	3/2 5/2	2.53	.	.	.	20B	.	.	.	.	5B	.	.	.	.	.	.	.	
5747.0	FE II 34F	A 4D 5/2 B 2P 3/2	5/2 3/2	3.18	.	.	.	20B	.	.	.	.	1	.	.	.	.	.	.	.	
5747.3	N II 9	3S 1P 0 3P 3D 2	1 2	20.56	.	.	.	2B	.	.	.	.	5B	.2	.	.	.	.	W	.	
5747.9	FE II 164	C 2F 5/2 Z 2G 0 7/2	5/2 7/2	7.69	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	
5754.8	N II 3F	2P2 1D 2 2P2 1S 0	2 0	4.04	.	W 20	1 200	30B	.	.	.2 30 1.5 3	8 5 .9	5 10 5	2 5	.	.	.	.	2	.	
5765.5	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
5766	-	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	
5769.6	HG I 5	6 1P 0 6 3D 2	1 2	8.81	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	
5771.7	C III -	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	
5772.6	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	
5775	-	.	.	.	.	.	.	W	.	.	X	.	.	.	.	.	.	.	.	.	
5777	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	
5783.4	MN VI 2F	3D2 3F 3 3D2 3P 1	3 1	2.23	.	.	.	1.5B	.	.	.8	.	.	.	.	.	.	.	.	.	
5785.0	FE II 215	Y 4F 0 E 4D 7/2	9/2 7/2	9.81	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	
5785.4	CR III 1F	3D4 5D 4 3D4 3P 2	4 2	2.20	.	.	.	1.5B	.	.	.	.	.	.	.	.	.	.	.	.	
5787.5	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														
		LEVELS	J		NUC		NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME	AIR GLW	CTY LTS	AUR ORA	LGT NING
					EM	GAL													
5790.7	HG I 4	6 1P 0 6 1D 2	1	8.81	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.
5794	N IV 15	3P 3P 3D 3P 0	1	63.14	.	.	.	.	15B	.	.	1	.	.	.	.	.	.	.
5801.5	C IV 1	3 2S 3 2P 0	1/2	39.51	.	.	4	7	40	.	2	.	5B	.	.	.	.	.	.
5804.3	-	.	.	.	.	.	.	.	350B	.	4	.	.	.	.	.	.	2	.
5809.4	FE II 33F	A 4D A 4G	1/2 5/2	3.22	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
5812	N IV 15	3P 3P 3D 3P 0	1 2	63.14	.	.	.	.	15B	.	.	.	5B	.	.	.	.	.7	.
5812.1	C IV 1	3 2S 3 2P 0	1/2	39.51	.	.	4B	2	40	XB	1.1	.	5	.	.	.	.	.	.
5813.7	FE II 163	C 2F Z 2D 0	5/2	7.67	.	.	.	.	350B	.	.	.	W	.	.	.	.	.	.
5814.6	TI II 79	A 2H Z 4G 0	11/2	3.70	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
5820.7	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.
5823.2	FE II 164	C 2F Z 2G 0	7/2	7.66	.	.	3B	.	.	.	.	.	.	.	.	.	.	.	.
5824.4	FE II 58	A 2F Z 4D 0	5/2	5.53	.	.	3B	.	.	.	.	.	.	.	.	.	.	.	.
5824.6	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
5827.1	C III 22	4 1D 3D 1F 0	2 3	42.14	.	.	.	.	.	X	.	.	W	.	.	.	.	.	.
5828	N IV 15	3P 3P 3D 3P 0	2 1	63.14	.	.	.	.	15B	.	.	1	.	.	.	.	.	.	.
5829	-	.	.	.	.	.	.	.	.	.	8	.	5	.	.	.	.	.	.
5833.7	FE III 114	E 7S Y 7P 0	3 4	20.54	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
5834.4	-	.	.	.	.	.	1	.	.	.	.	.	W	.	.	.	.	.	.
5834.9	FE II 57	A 2F Z 4F 0	5/2	5.52	.	.	.	8B	.	.	.	.	.	.	.	.	.	.	.
5835.4	FE II 58	A 2F Z 4D 0	7/2	5.49	.	.	3B	.	.	.	.	.	.	.	.	.	.	.	.
5835.6	FE II -F	.	.	.	.	.	.	8B	.	.	.	.	.	.	.	.	.	.	.
5836.5	HE II -	.	.	.	.	.	.	.	.	.	.3	.	.	.	.	.	.	.	.
5839	-	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
5843.9	FE II 34F	A 4D B 2P	3/2	3.18	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
5844.6	CR I 119	B 5D Y 5D 0	3 2	5.11	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
5846	N IV 15	3P 3P 3D 3P 0	2 2	63.14	.	.	.	.	15B	.	.	.	5	.	.	.	.	.	.
5847	-	.	.	.	.	.	.	.	.	.	.	.9	.	.	.	.	.	.	.
5852.5	NE I 6	3S 4 0 3P 10	1 0	18.88	.	.	.	.	.	.	.	.	.	.	.	.	20	.	.
5853	-	.	.	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.
5853.9	-	.	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.
5860.6	-	.	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.
5863.1	MN V 2F	3D3 4F 3D3 2G	7/2	2.21	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.
5867.2	FE I 2F	A 5D A 5P	1 1	2.21	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.
5871.6	C III 20	3P 3D 3D 3P 0	2 1	41.98	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.
5875.6	HE I 11	2 3P 0 3 3D	2	22.97	.	1	40	45	50	XB	24	31	30	10	.	W	.	.	.
						35B	120		100B		77	140	100B	80					
5875.6	HE I 11	2 3P 0 3 3D	1	22.97	.	1	40	45	50	XB	24	31	30	10	.	W	.	.	.
						35B	120		100B		77	140	100B	80					
5876.0	HE I 11	2 3P 0 3 3D	0	22.97	.	1	.	.	50	XB	25	.	30	55B	.	W	.	.	.
						35B			100B		40B		100B						
5881.9	NE I 1	3S 1 0 3P 9	2 1	18.65	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.
5885	-	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.
5886	-	.	.	.	.	.	.	4	.	.	2	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGHT NING	
5941.7	N II 28	3P 3P 3D 3D O	2 3	23.15	.	.	8 20B	W	.	.	.	.	W	1	.	.	.	.	.	
5944.8	NE I 1	3S 1 C 3P 8	2 2	18.62	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	
5945.0	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
5951	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	
5951	N2 1P	8 4	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	
5951.3	S II 21	3D 4D 4P 4S O	5/2 3/2	16.18	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
5952.4	N II 28	3P 3P 3D 3D O	2 2	23.14	.	.	XB	W	.	.	.	.	3	.	.	.	.	.	.	
5952.6	FE II 182	D 2D Z 2P O	5/2 3/2	8.00	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	
5953.0	HE II -	.	.	.	.	.	.	.	.	.9	.	.	.	.	.	.	.	.	.	
5953.7	FE III 115	E 5D W 5P O	4 3	20.78	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	
5954.4	-	.	.	.	.	.	.	.	.	2	1	.	.	.	.	.	.	.	.	
5956.5	FE II --	.	.	.	.	.	W	WB	.	.	.	.	.	.	.	.	.	.	.	
5957.6	SI II 4	4 2P O 5 2S	1/2 1/2	12.09	.	.	.	W	.	.	.01	1	1 1.4	.	.	.	.	.	.	
5959.0V	N2 1P	8 4	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	
5960.9	N II 28	3P 3P 3D 3D O	2 1	23.14	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	
5960.9	-	.	.	.	.	.	.	.	.	.	.	.	.5	.	.	.	.	.	.	
5964.2	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
5965.5	NE I --	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	
5967.0	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
5967	N2 1P	8 4	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	
5970.7	-	.	.	.	.	.	.	.	.	1.5	.	.	.	.	.	.	.	.	.	
5971.1	-	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	
5973.4	O2+ 1N	1 1	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	
5974.6	NE I 28	3P 5 4D 8 O	2 3	20.62	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	
5975.5	NE I --	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	6	.	.	
5976.6	O2+ 1N	0 0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	
5977.1	HE II -	.	.	.	.	.	.	.	.	.7	.	.	.	.	.	.	.	.	.	
5978.9	FE III 117	E 5S W 5P O	2 1	20.80	.	.	.	.	.	.	.	.	18	.	.	.	.	.	.	
5979	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	
5979.0	SI II 4	4 2P O 5 2S	3/2 1/2	12.09	.	.	X	8	.	.	5	W	5 6	.	.	.	.	.	.	
5981.3	BA II 13	7 2P O 8 2D	3/2 5/2	8.24	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
5983.8	-	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
5984.2	O2+ 1N	0 0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	18	.	
5987.1	O2+ 1N	0 0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	18	.	
5987.6	-	.	.	.	.	.	.	3 4	.	.	.	.	.	.	.	.	.	.	.	
5990.5	O2+ 1N	0 0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	18	.	
5991	N2 1P	7 5	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	18	.	
5991.4	FE II 46	A 4G Z 6F O	11/2 9/2	5.20	.	.	20B	W 10	.	.	.	W	20 5	.	.	.	.	.	.	
5993.5	O2+ 1N	0 0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	
5998.0	O2+ 1N	0 0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.	

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J																	
					NUC EM	NOVA	W-R	W-R	PLAN	GAS	OE	AE	KE	AIR	CTY	AUR	LGT			
				QUAS	GAL	NOVA	LIKE	N	C	NEB	NEB	SYMB	PCYG	GE	T- TAU	N	GLW	LTS	ORA	NING
5998.6	-	.	.	.	.	.	.	.	.	.	.	.	2.4	.	.	.	.	.	.	.
5999	N2 1P	7 3	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
5999.3	FE III 117	E 5S W 5P U	2 2	20.79	.	.	W	.	.	.	.	.	1	.	.	.	.	.	.	.
5999.5	N I 16	3P 2S O	1/2	13.61	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	108
5999.9	BA II 13	4D 2P 7 2P O 8 2D	1/2 3/2 3/2	8.23	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
6000.0	FE I 2F	A 5D A 5P	1 3	2.17	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
6001	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
6002	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
6002.0	O2+ 1N	0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
6003.0	-	0	.	.	.	.	.	.	.	2	.9	1	.	.	.	.	.	.	.	.
6005	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6007	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6007	N2 1P	7 3	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.
6007.3	NI II 8F	A 4F A 4P	9/2 5/2	3.09	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
6008.5	N I 16	3P 2S O 4D 2P	1/2 3/2	13.61	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	108
6009	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
6012.0	N I -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	108
6014	N2 1P	7 3	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.
6015.3	ZR II 12F	B 4F B 2F	5/2 5/2	2.41	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
6015.4	N I -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	108
6016.8	O2+ 1N	0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
6017.7	N I -	0	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	108
6018.0	-	.	.	.	.	.	.	.	.	.2	.	.	.	.	.	.	.	.	.	.
6021.2	FE II 24	B 4P Z 6D O	1/2 3/2	4.82	.	.	20B	.	.	.	.	.	.	.	.	.	.	.	.	.
6021.8	MN I 27	Z 6P O E 6S	7/2 5/2	5.11	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.
6026.4V	O2+ 1N	0	.	18	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
6028.3	-	0	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
6030.0	NE I 3	3S 2 O 3P 9	1 1	18.65	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.
6031.4	-	.	.	.	.	.	.	.	.	.	.	.	5.1	.	.	.	.	.	.	.
6032.3	FE III 117	E 5S W 5P O	2 3	20.78	.	.	W	.	.	.	.	.	1	.	.	.	.	.	.	.
6034.3	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6035	-	.	.	.	.	.	.	.	.	2	.2	.	.	.	.	.	.	.	.	.
6036.1	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.	.
6036.7	HE II 8	5 2G 21 2H O	.	54.06	.	.	3	.	X	1.1	.	.	.	.	.	.	.	.	.	.
6039.8	O2+ 1N	0 0 0	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
6040.4	-	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.
6042	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6044.5	FE II 46	A 4G Z 6F U	11/2 11/2	5.18	.	.	6 9	10	.	.	.	.	.	.	.	.	.	.	.	.
6045.5	FE II 200	C 4F X 4F O	3/2 3/2	8.22	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
6045.8	-	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY													AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME						
																QUAS	GAL				
6046.2	-	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.			
6046.3	O I 22	3 3P 6 3S U	1 1	12.98	.	.	.	.	.	.	.	.	.	.	.	.	18	.			
6046.5	O I 22	3 3P 6 3S U	0 1	12.98	.	.	.	.	.	.	.	.	.	.	.	.	18	.			
6046.5	O I 22	3 3P 6 3S U	2 1	12.98	.	.	.	.	.	.	.	.	.	.	.	.	18	.			
6047	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.			
6047	N2 1P	6 2	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	1	.			
6049	-	.	.	.	.	.	.	.	2 3	1	.	.	.	.	.	.	.	.			
6053.5	CR II 105	C 4D Z 4D U	7/2 7/2	6.76	.	.	.	W	.	.	.	W	.	.	.	.	.	.			
6055	N2 1P	6 2	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	1	.			
6062.4	-	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.			
6063	N2 1P	6 2	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	1	.			
6064	-	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.			
6064.6	TI I 69	A 3P Z 3S U	0 1	3.08	.	.	.	.	.	.	1	.	.	.	.	.	.	.			
6065.5	FE I 207	B 3F Y 3F U	2 2	4.63	.	.	.	.	.	.	1	.	.	.	.	.	.	.			
6066.8	-	.	.	.	.	.	.	.	.9 3	1	.	.	.	.	.	.	.	.			
6068.0	CR II 197	C 4F Y 2H U	9/2 11/2	8.69	.	.	.	WB	.	.	.	.	.	.	.	.	.	.			
6069.7	CR II 197	C 4F Y 2H U	9/2 9/2	8.69	.	.	.	WB	.	.	.	.	.	.	.	.	.	.			
6070	N2 1P	6 2	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	1	.			
6072.4	-	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.			
6073.0	-	.	.	.	.	.	.	.	1 3	.	.	.	.	.	.	.	.	.			
6073.9	-	.	.	.	.	.	W	.	2	.	.	.	.	.	.	.	.	.			
6074.1	HE II 8	5 2G 20 2H U	.	54.05	.	.	.	X	X	1.1	.	.	.	.	.	.	.	.			
6074.3	NE I 3	3S 2 U 3P 6	1 0	18.63	.	.	.	.	.	.	.	.	.	.	.	.	10	.			
6078	-	.	.	.	.	.	.	.	.9 4	.	.	.	.	.	.	.	.	.			
6079.4	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	X	.	.			
6081.4	V I 34	A 4D Z 4P U	3/2 3/2	3.08	.	.	.	.	.	.	1	.	.	.	.	.	.	.			
6081.5	CR II 188	D 2G Z 2F U	9/2 7/2	8.49	.	.	.	WB	.	.	.	.	.	.	.	.	.	.			
6084	-	.	.	.	.	.	.	.	.8 1	.	.	.	.	.	.	.	.	.			
6084.1	FE II 46	A 4G Z 6F U	9/2 7/2	5.21	.	.	30B 6	.	.	.	8	.	.	.	.	.	.	.			
6085	S III -F	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.			
6085.2	TI I 69	A 3P Z 3S U	1 1	3.08	.	.	.	.	.	.	1	.	.	.	.	.	.	.			
6085.5	FE VII 1F	3D2 3F 3D2 1D	3 2	2.16	.	5 35B	30 60B	.	.	2.5 4B	5 30	.	.	.	.	.	.	.			
6085.9	CA V 1F	3P4 3P 3P4 1D	1 2	2.32	.	5 35B	30B	.	.	2.5 4B	30B	.	.	.	.	.	.	.			
6089.7	CR II 187	D 2G Z 2H U	9/2 11/2	8.49	.	.	.	W	.	.	.	.	.	.	.	.	.	.			
6090.2	V I 34	A 4D Z 4P U	7/2 5/2	3.10	.	.	.	.	.	.	W	.	.	.	.	.	.	.			
6092.1	S II 20	3D 4D 4P 2D U	5/2 3/2	16.13	.	.	.	W	.	.	.	.	.	.	.	.	.	.			
6095	NE -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	W			
6096.0	TI II 9F	A 4F B 2P	7/2 3/2	2.05	.	.	.	W	.	.	.	.	.	.	.	.	.	.			
6097	-	.	.	.	.	.	.	.	1	.8	.	.	.	.	.	.	.	.			
6097.5	N2 1P	5 1	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	1	.			

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J																	
					NUC EM	NOVA	W-R	W-R	PLAN	GAS	OE	AE	T-	KE	AIR	CTY	AUR	LGHT		
				QUAS	GAL	NOVA	LIKE	N	C	NEB	NEB	SYMB	PCYG	GE	TAU	N	GLW	LTS	ORA	NING
6099	-	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	X	.	.	.
6101.1	K IV 1F	3P4 3P 3P4 1D	2 2	2.02	.	.	18 24	5	.	3 4	.	.	.	.	.	.	.	.	.	.
6102.7	CA I 3	4 3P 0 5 3S	0 1	3.89	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.	.
6103	N2 1P	5 1	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
6103.5	FE II 200	C 4F X 4F U	5/2 5/2	8.21	.	.	.	W	.	.	.	.	2	.	.	.	.	.	.	.
6105.5	-	.	.	.	.	.	6 10	.	.	.	.	.	.	.	.	.	.	.	.	.
6111	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6111	N2 1P	5 1	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
6111.6	V I 34	A 4D Z 4P 0	1/2 1/2	3.06	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.
6113.3	FE II 46	A 4G Z 6F U	7/2 5/2	5.23	.	.	.	W	.	.	.	.	4	.	.	.	.	.	.	.
6116.0	FE II 46	A 4G Z 6F 0	5/2 3/2	5.23	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.
6118.2	-	.	.	.	.	.	.	.	.	.8 2	.8	.	.	.	.	.	.	.	.	.
6118.2	HE II 8	5 2G 19 2H 0	.	54.03	.	.	.	XB	X	1.1	.	.	.	.	.	.	.	.	.	.
6119.5	V I 34	A 4D Z 4P 0	5/2 3/2	3.08	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.
6120	N2 1P	5 1	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
6122.2	CA I 3	4 3P 0 5 3S	1 1	3.89	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.
6122.4	MN II 13	E 5D X 5F 0	4 5	12.16	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.
6123.4	S II 13	4S 2P 4P 4P 0	3/2 1/2	16.02	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
6124.6	TI II 22F	A 2F B 2F	5/2 5/2	2.59	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
6124.9	SI I 30	3D 3D 0 5F 3D	1 1	7.60	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.
6125.0	SI I 30	3D 3D 0 5F 3D	1 2	7.60	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.
6125.9	MN II 13	E 5D X 5F 0	3 4	12.16	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.
6127	N2 1P	5 1	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.
6128.7	MN II 13	E 5D X 5F 0	2 3	12.16	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.
6129.0	MN II 13	E 5D X 5F 0	2 2	12.16	.	.	.	.	.	.	.	.	WB	.	.	.	.	.	.	.
6129.2	CR II 105	C 4D Z 4D 0	5/2 5/2	6.74	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.
6129.7	FE II 46	A 4G Z 6F 0	9/2 9/2	5.20	.	.	.	4	.	.	.	.	W	.	.	.	.	.	.	.
6130.4	-	.	.	.	.	.	.	.	.	2 3	.	.	.	.	.	.	.	.	.	.
6130.8	MN II 13	E 5D X 5F 0	1 2	12.16	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
6131.9	MN II 13	E 5D X 5F U	0 1	12.16	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.
6133.8	-	.	.	.	.	.														

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGH NING
		LEVELS	J		NUC EM	NOVA	W-R LIKE	W-R N	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
6188.6	FE II 44F	A 2G B 2F	9/2 7/2	3.95	.	.	7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6190	-	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.
6191.6	FE I 169	A 3H Z 3G O	.	4.42	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6194	-	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.
6195.2	CR II 105	C 4D Z 4D O	3/2 3/2	6.73	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6199.2	FE II 162	C 2F Z 4G O	7/2 5/2	7.54	.	.	W	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
6200	-	.	.	.	.	X	.	.	.	.8	.	.	.	.	.	.	.	.	.	.	.	.
6205	-	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.
6205.6	C III -	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6211	-	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.
6212.3	TI II 108	A 2S Z 2S O	1/2 1/2	4.62	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6217	-	.	.	.	.	.	.	30	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6217.3	NE I 1	3S 1 O 3P 4	2 1	18.53	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.
6219.5	FE II 34	B 4F Z 6D O	5/2 3/2	4.82	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6220.6	FE III -	.	.	.	.	.	WB	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
6220.7	MN V 1F	3D3 4F 3D3 4P	5/2 1/2	2.03	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
6229.2	K VI 1F	3P2 3P 3P2 1D	2 2	2.34	.	.	.	.	1 3	.	.	.	.	.	.	.	.	.	.	.	.	.
6229.3	FE II 34	B 4F Z 6D O	7/2 5/2	4.80	.	.	4	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6230.7	FE I 207	B 3F Y 3F O	4 4	4.53	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
6233.5	FE II --	.	.	.	.	.	WB	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6233.8	HE II 7	5 2G 17 2H O	.	54.00	.	.	.	X	X	.9	.	.	.	.	.	.	.	.	.	.	.	.
6235.5	C II -	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6238.4	FE II 74	B 4D Z 4P O	3/2 3/2	5.85	.	.	X	128	.	.	.	W	10	1	X	.	.	.	.	.	.	.
6239.4	FE II 34	B 4F Z 6D O	9/2 7/2	4.77	.	.	.	128	.	.	.	2	.	.	.	.	.	.	.	.	.	.
6240.0	FE II 74	B 4D Z 4P O	1/2 3/2	5.85	.	.	.	.	.	.	.	2	.	X	.	.	.	.	.	.	.	.
6242.5	-	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.
6242.5	N II 57	3D 1F O 4P 1D	3 2	25.35	.	.	4B	.	.	1.5	.	W	.	.	.	.	.	.	.	.	.	.
6243.4	AL II 10	4 3P O 4 3D	2 3	15.00	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
6244	-	.	.	.	.	.	X	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.
6246	-	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.
6247.6	FE II 74	B 4D Z 4P O	5/2 3/2	5.85	.	.	3 4B	18	.	.	.	1	14	10	X	.	.	.	.	.	.	.
6248.9	FE II --	.	.	.	.	.	.	18	.	.	.	W	6	.	.	.	.	.	.	.	.	.
6252.6	FE I 169	A 3H Z 3G O	6 5	4.37	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
6253	-	.	.	.	.	.	.	.	W 2	.	.	.	.	.	.	.	.	.	.	.	.	.
6254.4	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6256.0	OH M	9 3	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	<1	.	.	.
6256.1	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6257.2	C II -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	58	.	.	.	.
6259.4	-	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.	.	.	.
6259.6	C II -	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	58	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															AIR GLW	CTY LTS	AUR ORA	LGHT NING					
		LEVELS	J		NUC EM GAL	NOVA	W-R LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N												
6262.3	LA II 33	A 3D Y 3F 0	3 4	2.37	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6265.1	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6266.5	NE I 5	3S 3D 3P 7	0 1	18.61	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.					
6267	-	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.					
6269.3	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6269.9	-	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.					
6271.8	CR II 196	C 4F X 4F 0	7/2 7/2	8.58	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6274.3	S II 19	3D 4D 4P 4P 0	3/2 5/2	16.07	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6274.9	CR II 196	C 4F X 4F 0	7/2 9/2	8.58	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6280.7	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6285	-	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.					
6286.4	S II 19	3D 4D 4P 4P 0	5/2 5/2	16.07	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6291	-	.	.	.	.	.	6	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6291.5	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6291.8	-	.	.	.	.	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.					
6292.9	V I 19	A 6D Z 6D 0	7/2 5/2	2.25	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.					
6296.1	LA II 47	A 3P Z 3D 0	2 2	2.73	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6296.1	LA II 68	B 1D Y 3D 0	2 1	3.21	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6296.5	V I 19	A 6D Z 6D 0	9/2 7/2	2.26	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.					
6300.2	O I 1F	2P4 3P 2P4 1D	2 2	1.96	.	15 150B	10 30B	20	.	.	.1 50	6	5	20*	X	400	.	30	.	100	.	.	.					
6302.5	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6305.3	FE II 200	C 4F X 4F 0	9/2 9/2	8.15	.	.	6	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6305.5	S II 19	3D 4D 4P 4P 0	7/2 5/2	16.07	.	.	10B	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6305.7	SC I 2	A 2D Z 2D 0	5/2 5/2	1.98	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.	.	.	.	.					
6310.2	S III 3F	3P2 1D 3P2 1S	2 0	3.35	.	15 150B	16B	10	.	.	1 25	5 8	1 5B	.	.	.	.	.	.	.	.	.	.					
6310.8	HE II 7	5 2G 16 2H 0	.	53.97	.	.	.	15	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6312.7	S II 26	3D 2F 4P 2D 0	5/2 3/2	16.13	.	.	10B	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6316.6	K V 2F	3P3 2D 0 3P3 2P 0	5/2 3/2	4.95	.	.	8	.	.	.	.	10	.	.	.	.	.	.	.	.	.	.	.					
6317.3	-	.	.	.	.	.	2 6	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.					
6318.0	FE II -	.	.	.	.	.	12	.	.	.	.	5	30	.	.	.	.	.	.	.	.	.	.					
6318.1	-	.	.	.	.	.	.	.	.	.	.	5	.	.	.	.	.	.	.	.	.	.	.					
6318.2	MG I 23	4 3S 6 3P 0	1 2	7.04	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6318.8	MG I 23	4 3S 6 3P 0	1 2	7.04	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6320	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6329	O V 14	3P 1D 3D 1F 0	2 3	88.02	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6331.2	-	.	.	.	.	.	.	.	.	1.5 3	.	.	.	.	.	.	.	.	.	.	.	.	.					
6332.0	FE II 199	C 4F X 4G 0	5/2 7/2	8.14	.	.	W	.	.	.	.	.	1	5	.	.	.	.	.	.	.	.	.					
6334.4	NE I 1	3S 1D 3P 3	2 2	18.50	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	.					
6335	-	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.					
6336.1	-	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.					

*Nebular origin reported in one case.

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LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION LEVELS	UPPER EXC. POT., EV	EMISSION LINE INTENSITY													AIR GLW	CTY LTS	AUR ORA	LGT VING
				NUC EM	NOVA	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME							
6340.7	N II 46	3D 3D 0 4P 3P 2	3 2	25.09	.	.	3B	.	.	.	.	W	.	.	.	.	.	.	.	
6342	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	
6344.1	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
6344.5	-	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	
6346.5	Zr II 128	8 2F 7 2G 0	7/2 9/2	4.34	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	
6346.7	Mg II 16	4 2D 6 2F 0	.	13.46	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	
6347.1	Si II 2	4 2S 4 2P 0	1/2 3/2	10.03	.	.	3 15B	10	.	.	.8 2	2	5	8.7 10	W	.	.	.	.	
6349.5	K V 2F	3P3 2D 0 3P3 2P 0	3/2 1/2	4.91	.	.	X	.	.	.	10	.	.	.	.	.	.	.	.	
6350.8	C III -	.	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	
6352	-	.	.	.	.	.	.	8	.	.	1	.	.	.	.	.	.	.	.	
6355.6	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
6357.0	N II 46	3D 3D 0 4P 3P 0	1 0	25.08	.	.	.	.	.	.	.8	.	.	.	.	.	.	.	.	
6357.0	N II 46	3D 3D 0 4P 3P 1	2 1	25.08	.	.	.	.	.	.	.8	.	.	.	.	.	.	.	.	
6361.4	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
6363.9	O I 1F	2P4 3P 2P4 1D 2	1 2	1.96	.	5 50	40B	.	.	.2 25	3	W	2*	X	150 400B	10	.	30	.	
6364.0	-	.	.	.	.	.	.	8	.	.	.	.	.	.	.	.	.	.	.	
6365.5	Ni II 8F	A 4F A 4P	7/2 5/2	3.09	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
6369.5	Fe II 40	A 6S Z 6D 0	5/2 3/2	4.82	.	.	8 18B	123	.	.	.	W	7	18	.	.	.	.	.	
6371.4	Si II 2	4 2S 4 2P 0	1/2 1/2	10.02	.	.	8 18B	123	.	.	.3	1	1	4.3	18	.	.	.	.	
6372.9	Fe X 1F	3P5 2P 0 3P5 2P 0	3/2 1/2	1.94	.	.	.	.	.	.	10B	.	.	.	.	.	.	.	.	
6374	-	.	.	.	.	.	X	X	.	.	.	.	.	.	.	.	.	.	.	
6374.5	Fe X 1F	3P5 2P 0 3P5 2P 0	3/2 1/2	1.94	.	15	40 240B	.	.	.	10B	.	.	.	.	.	.	.	.	
6374.7	-	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	
6376.0	Fe II --	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
6379.6	N II 2	3S 3P 0 3P 1P 1	1 1	20.32	.	.	X	.	.	.	.	W	.	.	.	.	.	.	.	
6380	-	.	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	
6383	N IV 2	3 1S 3 1P 0	0 1	49.94	.	.	.	.	7	.	.	.	.	.	.	.	.	.	.	
6383.0	Ne I 3	3S 2 0 3P 4	1 1	18.53	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.	
6383.7	Fe II --	.	.	.	.	.	.	10	.	.	.	5	11	.	.	.	.	.	.	
6385.5	Fe II --	.	.	.	.	.	.	.	.	.	1	5	.	.	.	.	.	.	.	
6386	-	.	.	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	
6393.6	Fe I 168	A 3H Z 5G 0	5 4	4.35	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	
6396.2	Mn V 1F	3D3 4F 3D3 4P	9/2 5/2	2.10	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	
6396.3	Fe II 44F	A 2G B 2F	7/2 7/2	3.95	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
6398	-	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	
6400.0	Fe I 816	Z 5P 0 E 5D	3 4	5.52	.	.	.	W	.	.	5	.	.	.	.	.	.	.	.	
6401.5	Ni III -F	.	.	.	.	.	.	.	.01	.	.	.	.	.	.	.	.	.	.	
6401.7	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	
6402.2	Ne I 1	3S 1 0 3P 2	2 3	18.48	.	.	.	.	.	.	.	.	.	.	.	.	20	.	.	
6403.5	-	.	.	.	.	.	.	.	2 6	.	.	.	.	.	.	.	.	.	.	

*Nebular origin reported in one case.

LAB WAVE LENGTH, A	ION AND MULTIPLT NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J																	
					NUC EM	SAL	NOVA	LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE	AE FE	T- TAU	KE ME	AIR GLW	CTY LTS	AUR ORA
6853.0	N2+ M	3 0	.	16.8	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.
6855.4	CA X -	.	.	.	.	.	3 5	.	.	.	.	.	.	.	.	.	.	.	.	.
6857	-	.	.	.	.	.	.	8	.	.	.	.	.	.	.	.	.	.	.	.
6857.3	C III 19	3P 3D 3D 3D O	1 1	41.67	.	.	.	.	.	18	.	.	.	.	.	.	.	.	.	.
6861	N2 1P	3 0	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.
6861.7	OH M	7 3	.	3.1	.	.	.	.	.	.	.	.	.	.	.	.	.	W	.	.
6862.9	C III 19	3P 3D 3D 3D O	2 2	41.67	.	.	.	.	.	18	.	.	.	.	.	.	.	.	.	.
6871.7	C III 19	3P 3D 3D 3D O	3 3	41.67	.	.	.	.	.	18	.	.	.	.	.	.	.	.	.	.
6872.2	FE II 31F	A 4D B 4F	5/2 5/2	2.83	.	.	.	6H	.	.	.	.	.	.	.	.	.	.	.	.
6873.9	FE II 43F	A 2G B 2G	9/2 9/2	3.75	.	.	.	6B	.	.	.	.	.	.	.	.	.	.	.	.
6875	N2 1P	3 0	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.
6880	-	.	.	.	.	.	.	6	.	.	.	.	.	.	.	.	.	.	.	.
6881.9	C III -	.	.	.	.	.	.	.	.	1 20B	.	.	.	.	.	.	.	.	.	.
6890	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
6890.9	HE II 7	5 2G 12 2H O	.	53.81	.	.	.	.	.	X	2.1	.	.	.	.	.	.	.	.	.
6893	-	.	.	.	.	.	.	5	.	.	.	.	.	.	.	.	.	.	.	.
6895.3	O II 45	3D 4F 4P 4D O	9/2 7/2	30.37	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.
6899.4	C III -	.	.	.	.	.	.	.	.	1 20B	.	.	.	.	.	.	.	.	.	.
6919	A XI 1F	2P4 3P 2P4 3P	2 1	1.78	.	.	5	.	.	.	.	.	.	.	.	.	.	.	.	.
6929.5	NE I --	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.
6942.9	N II 53	3D 3P U 4P 3P	2 2	25.09	.	.	.	2B	.	.	.	.	.	.	.	.	.	.	.	.
6944.9	FE II 43F	A 2G B 2G	7/2 7/2	3.80	.	.	.	8	.	.	.	.	.	.	.	.	.	.	.	.
6945.2	N I 29	3P 4P O 5S 4P	5/2 5/2	13.57	.	.	.	1B	.	.	.	.	.	.	.	.	.	.	.	.
6950	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
6952.8	O III -	.	.	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.
6959.3	C II -	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.
6966.3	FE II 31F	A 4D B 4F	3/2 3/2	2.84	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
6966.9	FE II 198	C 4F Z 2F O	3/2 5/2	7.95	.	.	.	WB	.	.	.	.	.	.	.	.	.	.	.	.
6972	-	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.
6991.8	TI III 3F	A 3F A 1G	2 4	1.78	.	.	.	4B	.	.	.	.	.	.	.	.	.	.	.	.
7001.9	O I 21	3 3P 4 3D O	1	12.70	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.	.
7006.3	A V 1F	3P2 3P 3P2 1D	2 2	2.01	.	.	X	4B	.	.	14	.	.	.	.	.	.	.	.	.
7024.1	NE I 6	3S 4 O 3P 4	1 1	18.53	.	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.
7032.4	NE I 1	3S 1 O 3P 1	2 1	18.30	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.
7036.8	N2+ M	4 1	.	16.8	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.
7036.9	C III -	.	.	.	.	.	.	.	.	100B	.	W	.	.	.	.	.	.	.	.
7059	N2 1P	8 6	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	3	.
7060	C IV -	.	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.
7065.2	HE I 10	2 3P O 3 3S	2,1 1	22.62	.	.	6B	6 40	.	100B 20	25 60	308 60	10 60	.	.	.	.	.	.	.
7067.4	FE II --	.	.	.	.	.	6B	.	.	.	.	.	.	.	.	.	.	.	.	.

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW	CTY LTS	AUR ORA	LGT NING
		LEVELS	J		NUC EM	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE	AE FE	T- TAU	KE ME								
															QUAS	GAL	NOVA	W-R				
7535	-	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.			
7550.4	O III -	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.			
7572	OH M	4 0	.	3.0	.	.	.	.	.	.	.	.	.	.	.	.	1	.	.			
7576.6	C III -	.	.	.	.	.	60	.	.	.	.	.	.	.	.	.	.	.	.			
7578.2	C III -	.	.	.	.	.	200B XB	.	.	.	.	.	.	.	.	.	.	.	.			
7580	N2 1P	3 1	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	60B	.			
7585.8	C III -	.	.	.	.	.	60	.	.	.	.	.	.	.	.	.	.	.	.			
7586	OH M	4 0	.	3.0	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.			
7586.4	C III -	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.			
7592.3	C III -	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.			
7592.7	HE II 6	5 2G 10 2H D	.	53.64	.	.	XB	.	.	XB	1.2	.	.	.	.	.	.	.	.			
7593.7V	O2 A	0 0	.	1.7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1			
7595.3	C III -	.	.	.	.	.	60	.	.	.	.	.	.	.	.	.	.	.	.			
7612.6	C III -	.	.	.	.	.	200B	.	.	.	.	.	.	.	.	.	.	.	.			
7612.6	C III -	.	.	.	.	.	60	.	.	.	.	.	.	.	.	.	.	.	.			
7626.2V	N2 1P	3 1	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	60B	.			
7625.9	C III -	.	.	.	.	.	60	.	.	.	.	.	.	.	.	.	.	.	.			
7656.7	SI IV -	.	.	.	.	.	200B 1	.	.	.	.	.	.	.	.	.	.	.	.			
7664.9	K I 1	4 2S 4 2P D	1/2 3/2	1.61	.	.	.	.	.	.	.	X	.	.	X	.	.	.	.			
7678.3	SI IV -	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.			
7680	-	.	.	.	.	.	9	.	.	.	.	.	.	.	.	.	.	.	.			
7684.3	O2 A	1 1	.	1.7	.	.	.	.	.	.	.	.	.	.	.	.	.	100	.			
7686.2	FE II 14F	A 4F A 2G	5/2 9/2	1.96	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.			
7694.8	NI II 7F	A 4F B 2D	5/2 5/2	2.85	.	.	18	.	.	.	.	.	.	.	.	.	.	.	.			
7699.0	K I 1F	4 2S 4 2P D	1/2 1/2	1.60	.	.	.	.	.	.	.	X	.	.	X	.	.	.	.			
7704.5	C IV -	.	.	.	.	.	60 100	.	.	.	.	.	.	.	.	.	.	.	.			
7707.4	C III -	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.	.			
7708	C IV -	.	.	.	.	.	XB	.	.	.	.	.	.	.	.	.	.	.	.			
7711.7	FE II 73	B 4D Z 4D U	7/2 7/2	5.49	.	.	4 10	.	.	.	.	.	.	.	.	.	.	.	.			
7712	OH M	9 4	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	25B	.	.			
7724.7	S I 3F	3P4 1D 3P4 1S	2 0	2.74	.	.	.	.	X	.	X	.	.	.	.	.	.	.	.			
7733.5	-	.	.	.	.	.	.	.	.	.	10 60	.	.	.	.	.	.	.	.			
7726	C IV -	.	.	.	.	.	XB	.	X	.	.	.	.	.	.	.	.	.	.			
7748.3	OH M	9 4	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	20	.	.			
7751.0	A III 1F	3P4 3P 3P4 1D	1 2	1.73	.	.	XB	8	.	20	X	.	.	.	.	.	.	.	.			
7752.9	CR II 11F	A 6D B 4D	5/2 3/2	3.09	.	.	.7B	.	.	.	.	.	.	.	.	.	.	.	.			
7753.2V	N2 1P	2 0	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	200B	.			
7754.1	-	.	.	.	.	.	.7B	.	.	.	.	.	.	.	.	.	.	.	.			
7764.7	FE II 30F	A 4D B 4P	7/2 5/2	2.57	.	.	1.5	.	.	.	.	.	.	.	.	.	.	.	.			
7771.1	C III -	.	.	.	.	.	.	.	40 60B	.	.	.	.	.	.	.	.	.	.			
7772.0	O I 1	3 5S D 3 5P	2 3	10.69	.	.	40B	.	.	.	.	5 40B	.	.	.	.	.	170B	.			

				EMISSION LINE INTENSITY																
LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	NUC EM		NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	OE BE SYMB	AE FE PCYG	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR DRA	LGHT NING	
		LEVELS	J		QUAS	GAL														
8454.1	-	.	.	.	.	.	.7	.	.	.	.	.	.	.	.	.	.	.	.	
8461.9	O III -	.	.	.	.	.	.	W	.	.	.	.	.	.	.	.	.	.	.	
8465	OH M	6 2	.	3.1	.	.	.	.	.	.	.	.	.	.	.	50	.	.	.	
8467.3	H(P) 10	3 2D 17 2F U	.	13.49	.	.	2 5	.	.	.	X	.	X	.	.	.	.	.	.	
8480.3	-	.	.	.	.	.	.7B	.	.	.	.	.	.	.	.	.	.	.	.	
8482.4	-	.	.	.	.	.	.7B	.	.	.	.	.	.	.	.	.	.	.	.	
8489.9	-	.	.	.	.	.	.7B	.	.	.	.	.	.	.	.	.	.	.	.	
8494	-	.	.	.	.	.	25B	.	.	.	.	.	.	.	.	.	.	.	.	
8495.4	NE I 18	3P 3 3D 4 U	2 3	19.95	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.	
8497.1	-	.	.	.	.	.	.7B	.	.	.	.	.	.	.	.	.	.	.	.	
8497.7	C III -	.	.	.	.	.	.	60	.	.	.	.	.	.	.	.	.	.	.	
8498.0	CA II 2	3 2D 4 2P U	3/2 3/2	3.14	.	.	12B	.	.	.	.	1 10B	.	X	25	.	.	.	.	
8499.4	-	.	.	.	.	.	12B	.	.	.	.	.	.	.	.	.	.	.	.	
8500.3	C III -	.	.	.	.	.	.	X	.	.	.	1 10B	.	.	.	.	.	.	.	
8502.5	H(P) 10	3 2D 16 2F U	.	13.49	.	.	6 25B	.	.	.6	X	1 10B	X	.	.	.	.	.	.	
8505	OH M	6 2	.	3.1	.	.	.	.	.	.	.	.	.	.	.	20	.	.	.	
8505.8	-	.	.	.	.	.	1.5	.	.	.	.	.	.	.	.	.	.	.	.	
8508.6	-	.	.	.	.	.	3B	.	.	.	.	.	.	.	.	.	.	.	.	
8508.9	KR I -	.	.	.	.	.	3B	.	.	.	.	.	.	.	.	.	.	.	.	
8523.4	-	.	.	.	.	.	1.5	.	.	.	.	.	.	.	.	.	.	.	.	
8528.6	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	
8536.4	-	.	.	.	.	.	.7	.	.	.	.	.	.	.	.	.	.	.	.	
8541.8V	N2 1P	3 2	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	50	.	.	
8542.1	CA II 2	3 2D 4 2P U	5/2 3/2	1.69	.	.	XB 2	.	.	.	.	1 5B	.	X	50	.	.	.	.	
8545.4	H(P) 10	3 2D 15 2F U	.	13.48	.	.	4 7	.	.	.8	X	1 5B	X	.	.	.	.	.	.	
8549	OH M	6 2	.	3.1	.	.	.	.	.	.	.	.	.	.	.	15	.	.	.	
8567.6	SC II 2F	A 3D A 1S	2 0	1.45	.	.	1.5B	.	.	.	.	.	.	.	.	.	.	.	.	
8567.7	N I 8	3S 2P 3P 2P U	1/2 3/2	12.07	.	.	1.5B	.	.	.	.	.	.	.	.	.	.	100	.	
8574.4	-	.	.	.	.	.	.7	.	.	.	.	.	.	.	.	.	.	.	.	
8591.3	NE I 30	3P 7 3D 9 U	1 2	20.05	.	.	.	.	.	.	.	.	.	.	.	.	4	.	.	
8594.0	N I 8	3S 2P 3P 2P U	1/2 1/2	12.07	.	.	4	.	.	.	.	W	.	.	.	.	.	100	.	
8598	O2 A	0 1	.	1.7	.	.	.	.	.	.	.	.	.	.	.	100	200	.	.	
8598.4	H(P) 9	3 2D 14 2F U	.	13.47	.	.	5 7	.	.	2.6	X	.	X	.	.	.	.	.	.	
8599.4	-	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.	.	.	
8608.8	-	.	.	.	.	.	.7	.	.	.	.	.	.	.	.	.	.	.	.	
8611.9	C II -	.	.	.	.	.	.	40	.	.	.	.	.	.	.	.	.	.	.	
8617.0	FE II 13F	A 4F A 4P	9/2 5/2	1.66	.	.	10 15	.	.	.	.	.	.	.	.	.	.	.	.	
8620	O2 A	0 1	.	1.7	.	.	.	.	.	.	.	.	.	.	.	100	.	.	.	
8629.2	N I 8	3S 2P 3P 2P U	3/2 3/2	12.07	.	.	8 6	.	.	.	.	.	.	.	.	XBM	X	100	.	
8634.6	NE I 23	3P 4 3D 7 U	1 2	19.96	.	.	.	.	.	.	.	.	.	.	.	.	6	.	.	

[illegible]

					EMISSION LINE INTENSITY																	
LAB WAVE LENGTH, Å	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	NUC EM		NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGHT NING		
		LEVELS	J		QUAS	GAL																
8768	OH M	7 3	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	130	.	.	.		
8778	OH M	7 3	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	120	.	.	.		
8780.6	NE I 27	3P 5 3D 8 U	2 3	19.96	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.		
8783.8	NE I --	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.		
8784.3	-	.	.	.	.	.	4	.	.	.	.	.	.	.	.	.	.	.	.	.		
8805.2	-	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.		
8819.6	-	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.	.		
8820.5	O I 37	3S' 1D U 3P' 1F	2 3	14.07	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10		
8824.1	OH M	7 3	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	200	.	.	.		
8826.0	CR II 19F	A 4D A 2D	7/2 5/2	3.87	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.	.		
8830.7	FE III 8F	A 3P A 3D	2 1	3.79	.	.	3B	.	.	.	.	.	.	.	.	.	.	.	.	.		
8831.9	CR II 18F	A 4D A 4F	5/2 5/2	3.84	.	.	1 3B	.	.	.	.	.	.	.	.	.	.	.	.	.		
8838.2	FE III 8F	A 3P A 3D	2 2	3.79	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.		
8853.9	NE I 27	3P 5 3D 5 D	2 2	19.95	.	.	.	.	.	.	.	.	.	.	.	.	.	7	.	.		
8862.8	H(P) 9	3 2D 11 2F U	.	13.43	.	.	2 5	.	.	3.5 X	.	.	.	.	.	.	.	.	.	.		
8865.8	NE I 8	3P 1 4S 4 D	1 1	19.69	.	.	.	.	.	.	.	.	.	.	.	.	.	5	.	.		
8875.9	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.		
8886	OH M	7 3	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	150	.	.	.		
8891.9	FE II 13F	A 4F A 4P	7/2 3/2	1.69	.	.	3	.	.	.	.	.	.	.	.	.	.	.	.	.		
8906.4	-	.	.	.	.	.	2	.	.	.	.	.	.	.	.	.	.	.	.	.		
8911.6V	N2 1P	1 0	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	300	.		
8912.4	-	.	.	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.	.	.		
8916.3	-	.	.	.	.	.	3	.	.	.	.	.	.	.	.	.	.	.	.	.		
8919.5	NE I 27	3P 5 3D 2 U	2 1	19.94	.	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.		
8920	OH M	7 3	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	170	.	.	.		
8929.9	CR II 18F	A 4D A 4F	7/2 9/2	3.85	.	.	4	.	.	.	.	.	.	.	.	.	.	.	.	.		
8958	OH M	7 3	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	120	.	.	.		
8988.6	NE I 8	3P 1 4S 3 D	1 0	19.68	.	.	.	.	.	.	.	.	.	.	.	.	.	2	.	.		
9001	OH M	7 3	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	80	.	.	.		
9014.9	H(P) 9	3 2D 10 2F U	.	13.40	.	.	1	.	.	5.5 X	.	.	.	.	.	.	.	.	.	.		
9060.6	N I 15	3P 2S D 3D 2P	1/2 3/2	12.91	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	.		
9069.4	S III 1F	3P2 3P 3P2 1D	1 2	1.40	.	.	.	.	.	40 72	.	.	.	.	.	.	.	.	.	.		
9122.6	-	.	.	.	.	.	.5	.	.	.	.	.	.	.	.	.	.	.	.	.		
9145	N2+ M	1 0	.	16.8	.	.	.	.	.	.	.	.	.	.	.	.	.	.	400	.		
9148.7	NE I 30	3P 7 3D 7 U	1 2	19.96	.	.	.	.	.	.	.	.	.	.	.	.	.	6	.	.		
9201.8	NE I 30	3P 7 3D 6 U	1 1	19.95	.	.	.	.	.	.	.	.	.	.	.	.	.	6	.	.		
9229.0	H(P) 8	.	.	13.37	.	.	.	.	.	5.8	.	.	.	.	.	.	.	.	.	.		
9260.9	O I 8	3 5P 3 5D U	1 .	12.03	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.		
9310	OH M	8 4	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	608	.	.	.		
9324	OH M	8 4	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	608	.	.	.		

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY														AIR GLW LTS	CTY LTS	AUR ORA	LGHT NING
		LEVELS	J		NUC EM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	OE BE PCYG	AE FE GE	T- TAU	KE ME N							
9373	OH M	8 4	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	50	.	.	.		
9392.8	N I 7	3S 2P 3P 2D 0	3/2 5/2	11.96	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.		
9431.2	N2+ M	2 1	.	16.8	.	.	.	.	.	.	.	.	.	.	.	.	.	200	.	.		
9440	OH M	8 4	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	100	.	.	.		
9477	OH M	8 4	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	150	.	.	.		
9532.1	S III IF	3P2 3P 3P2 1D	2 2	1.40	.	.	.	.	.	.	180	.	.	.	.	.	.	.	.	.		
9546.0	H(P) 8	3 2D 8 2F 0	.	13.33	.	.	.	.	.	.	8	.	.	.	.	.	.	.	.	.		
9665.4	NE I 8	3P 1 4S 1 0	1 2	19.58	.	.	.	.	.	.	.	.	.	.	.	.	.	10	.	.		
9680	OH M	3 0	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	40	.	.	.		
9697	C III -	.	.	.	.	.	.	.	X*	.	.	.	.	.	.	.	.	.	.	.		
9702	C III -	.	.	.	.	.	.	.	X**	.	.	.	.	.	.	.	.	.	.	.		
9720	OH M	3 0	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	30	.	.	.		
9788	OH M	3 0	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	100	.	.	.		
9872	OH M	3 0	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	40	.	.	.		
9915	OH M	3 0	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	60	.	.	.		
9939	OH M	9 5	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	70	.	.	.		
9948	OH M	9 5	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	50	.	.	.		
9956	OH M	9 5	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	30	.	.	.		
9962	OH M	3 0	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	30	.	.	.		
10010	OH M	9 5	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	200	.	.	.		
10049.4	H(P) 8	3 2D 7 2F 0	.	13.26	.	.	.	.	.	.	10	.	.	.	.	.	.	.	.	.		
10083	OH M	9 5	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	100	.	.	.		
10123	OH M	9 5	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	150	.	.	.		
10123.6	HE II 2	4 2F 0 5 2G	.	52.02	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.		
10170	OH M	9 5	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	180	.	.	.		
10190	OH M	4 1	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	200	.	.	.		
10211	OH M	4 1	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	100	.	.	.		
10223	OH M	9 5	.	3.2	.	.	.	.	.	.	.	.	.	.	.	.	140	.	.	.		
10273	OH M	4 1	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	400	.	.	.		
10291	C III -	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.		
10373	OH M	4 1	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	150	.	.	.		
10395.4	N I 3F	2P3 2D 0 2P3 2P 0	5/2	3.56	.	.	.	.	.	.	.	.	.	.	.	.	.	.	>200B	.		
10404.1	N I 3F	2P3 2D 0 2P3 2P 0	3/2	3.56	.	.	.	.	.	.	.	.	.	.	.	.	.	.	>200B	.		
10418	OH M	4 1	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	300	.	.	.		
10469	OH M	4 1	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	200	.	.	.		
10506	C III -	.	.	.	.	.	.	.	X	.	.	.	.	.	.	.	.	.	.	.		
10510.0	N2 1P	0 0	.	7.2	.	.	.	.	.	.	.	.	.	.	.	.	.	.	>200	.		
10525	OH M	4 1	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	120	.	.	.		
10586	OH M	4 1	.	2.9	.	.	.	.	.	.	.	.	.	.	.	.	80	.	.	.		
10711	OH M	5 2	.	3.0	.	.	.	.	.	.	.	.	.	.	.	.	300	.	.	.		

*Six lines between 9697 and 9758 (not measured).

**Six lines between 9706 and 9720 (not measured).

LAB WAVE LENGTH, A	ION AND MULTIPLY NUMBER	TRANSITION		UPPER EXC. POT., EV	EMISSION LINE INTENSITY															
		LEVELS	J		NUC EM GAL	NOVA LIKE	W-R N	W-R C	PLAN NEB	GAS NEB	SYMB	DE BE PCYG	AE FE GE	T- TAU	KE ME N	AIR GLW	CTY LTS	AUR ORA	LGHT NING	
10729	OH M	5 2	.	3.0	.	.	.	.	.	.	.	.	.	.	.	.	200	.	.	.
10828	OH M	5 2	.	3.0	.	.	.	.	.	.	.	.	.	.	.	.	600	.	.	.
10830.3	HE I 1	2 3S 2 3P U	1 1	20.87	.	.	.	.	.	.	70 250	.	.	.	.	.	.	.	.	
10830.3	HE I 1	2 3S 2 3P U	1 2	20.87	.	.	.	.	.	.	70 250	.	.	.	.	.	.	.	.	
10924	OH M	5 2	.	3.0	.	.	.	.	.	.	.	.	.	.	.	.	400	.	.	.
10938.1	HIP) 8	3 2D 6 2F U	.	13.16	.	.	.	.	.	.	20	.	.	.	.	.	.	.	.	
10973	OH M	5 2	.	3.0	.	.	.	.	.	.	.	.	.	.	.	.	500	.	.	.

Lab wave- length A	Ion and multiplet number	Transition		Upper exc. pot., eV	Emission line intensity	
		Levels	J		Quasars	Ke-Me-N*
1238.8	N V 1	2s 2S 2p 2Po	1/2 3/2	9.97	X	
1242.8	N V 1	2s 2S 2p 2Po	1/2 1/2	9.93	X	
1406	O IV -F	--	--	--	X	
1808.0	Si II 1	3p 2Po 3p2 2D	1/2 3/2	6.83	X	
1817.4	Si II 1	3p 2Po 3p2 2D	3/2 3/2	6.83	X	
2326	C II -F	--	--	--	X	
5341.1	Mn I 4	a6D y6Po	9/2 7/2	4.42		4.8
5407.4	Mn I 4	a6D y6Po	7/2 7/2	4.42		2.21
5420.4	Mn I 4	a6D y6Po	7/2 5/2	4.41		W
5457.5	Mn I 4	a6D y6Po	5/2 7/2	4.42		.34
5470.6	Mn I 4	a6D y6Po	5/2 5/2	4.41		W
5481.3	Mn I 4	a6D y6Po	5/2 3/2	4.41		.24
5505.9	Mn I 4	a6D y6Po	3/2 5/2	4.41		W
5516.8	Mn I 4	a6D y6Po	3/2 3/2	4.41		.33
5537.8	Mn I 4	a6D y6Po	1/2 3/2	4.41		.23
5668.3	Mn I -F	a6S a6D	1/2 1/2	--		.56
5690.4	Mn I -F	a6S a6D	1/2 3/2	--		.26
5728.6	Mn I -F	a6S a6D	1/2 5/2	--		.69
5784.8	Mn I -F	a6S a6D	1/2 7/2	--		.34
5862.7	Mn I -F	a6S a6D	1/2 9/2	--		2.16

*Intensities are based upon H β = 23.0 in equivalent width.

Table C. LIST OF OBJECTS TABULATED IN TABLE B

<u>Source class</u>	<u>Object</u>	<u>References*</u>	<u>Comments</u>
1. Quasars	3C 9	56	
	3C 47	56, 164	
	3C 48	56, 87	
	3C 138	56, 114	
	3C 147	56, 164	
	3C 181	56	
	3C 186	56, 114	
	3C 191	54, 56, 167	
	3C 196	56, 114	
	3C 204	56	
	3C 208	56	
	3C 215	56, 114	
	3C 245	56, 116	
	3C 249.1	56	
	3C 254	56	
	3C 261	56, 114	
	3C 270.1	56	
	3C 273	56, 87	
	3C 275.1	56, 114	
	3C 277.1	56	
	3C 279	56, 116	
	3C 280.1	56, 114	
	3C 286	56, 116	
	3C 287	56	
	3C 298	56, 114	
	3C 334	56, 116	
	3C 336	56, 114	
	3C 345	56, 116	
	3C 351	56, 116	
	3C 380	56, 116	
	3C 432	56	
	3C 446	56	

*References are listed in Table E.

<u>Source class</u>	<u>Object</u>	<u>References</u>	<u>Comments</u>
<i>Quasars, cont.</i>	3C 454	56	
	4C 20.33	56, 114	
	4C 39.25	56, 114	
	0106+0154a	56	
	0922+14	56, 114	
	0952+18	56	
	0957+00	56, 114	
	1116+12	56, 115	
	1217+02	56, 114	
	1252+11	56, 116	
	BSO 1	56	
	CTA 102	56	
	CTD 141	56	
	MSH 03-19	56	
	MSH 14-121	56	
	PHL 938	56	
2. Nuclear emission galaxies	NGC 1068	150, 165, 166	
	NGC 1275	166	
	NGC 3516	165, 166	
	NGC 4051	166	
	NGC 4151	165, 166	
	NGC 7469	166	
3. Novae (Supernovae)	NGC 4382	32	
	NGC 4564	32	
	Anon. Gal.	32	
(Novae)	DO Aql (1925)	211	
	V356 Aql (1936)	119	
	T CrB	36, 38, 39, 40, 42, 100, 112, 113, 117, 179, 190	
	DQ Her (1934)	71, 86, 113, 131, 153, 185, 190, 191, 218, 219	

<u>Source class</u>	<u>Object</u>	<u>References</u>	<u>Comments</u>
<i>Novae, cont.</i>	V446 Her (1960)	77, 123, 155	
	Her (1963)	16, 37, 63, 64	
	BT Mon (1939)	161	
	RS Oph (1898-1933- 1958)	72, 100, 109, 110, 113, 117, 156, 190	1
	RR Pic (1925)	191	
	T Pyx (1890-1902- 1920-1944)	100, 107	
	Sct (1949)	66	
	Sco (1950)	159	
	RT Ser (1909)	1, 84, 185, 191	
 (Old novae)	V368 Aql (1936)	118	
	V603 Aql (1918)	100, 113, 117, 118	
	T Aur (1891)	100, 113	
	B Cas (1572)	100	
	Q Cyg (1876)	100	
	V476 Cyg (1920)	100, 117, 118	
	DM Gem (1903)	100	
	DN Gem (1912)	100, 117, 118	
	CP Lac (1936)	118	
	DI Lac (1910)	100, 113, 117	
	DK Lac (1950)	118	
	HR Lyr (1919)	100	
	V841 Oph (1848)	100, 113	
	GK Per (1901)	100, 113, 117	
	CP Pup (1942)	118	
	WZ Sge (1913-1946)	100, 113, 117	
	V1017 Sgr (1901-1919)	100, 113	
	V1059 Sgr (1898)	100	
	CK Vul (1670)	100	
	MacRae +43°1	85	

¹McLaughlin (Stellar Atmospheres, 1960) classifies this as a recurrent nova. Star has shown three maxima since observations were begun late in 18th century, exhibits one of the strongest emission spectra known. Spectral type in Kukarkin-Parenago catalog is Ocp.

Source class	Object	References	Comments
4. Nova-like stars	η Car	81, 204, 206	2
	RR Tel	89, 204, 205	
5. Wolf-Rayet N stars	HD 151932	157, 175	WN7
	HD 190918	4	+35°3953;WN5
	HD 191765	4	+35°4001;WN6
	HD 192163	4, 147, 175, 209	+37°3821;WN6
	HD 193077	4, 147	+36°3987;WN6
	HD 193576	4, 29, 147	+38°4010;WN5,WN6
	HD 193928	147	WN6
6. Wolf-Rayet C stars	HD 68273	190	γ Vel
	HD 164270	73, 175, 183	WC8
	HD 168206	175, 183	WC7
	HD 184738	15, 73, 183, 210	+30°3639;WC8
	HD 192103	4, 73, 169, 183, 209, 210	+35°4013;WC7
	HD 192641	4, 147, 210	+36°3956;WC7
	HD 193793	4, 147, 210	+43°3571;WC6
7. Planetary nebulae	NGC 1535	5, 14	Crab Nebula
	NGC 1952	148	
	NGC 2022	14	
	NGC 2165	5, 47	
	NGC 2392	48, 220	
	NGC 2440	5, 223	
	NGC 3242	5, 170	
	NGC 3918	76	
	NGC 6210	170	
	NGC 6302	76	
	NGC 6309	5, 14	
	NGC 6326	76	
	NGC 6543	31, 170, 181, 182	

²Called both nova and nova-like. Joy (Stellar Atmospheres, 1960) called it nova-like. Amplitude of light curve is typically that of a nova, but the star has undergone increases of brightness on a number of occasions.

<u>Source class</u>	<u>Object</u>	<u>References</u>	<u>Comments</u>
<i>Planetary nebulae, cont.</i>	NGC 6567	5	
	NGC 6572	9, 31, 46, 50, 51, 170, 182, 190, 223	
	NGC 6720	14	
	NGC 6741	5, 14, 223	
	NGC 6790	5	
	NGC 6803	5	
	NGC 6807	5	
	NGC 6826	31	
	NGC 6833	5	
	NGC 6884	5	
	NGC 6886	5	
	NGC 6891	5	
	NGC 6905	5	
	NGC 7009	8, 31, 223	
	NGC 7026	5	
	NGC 7027	6, 7, 13, 31, 46, 47, 48, 49, 50, 51, 128, 168, 170, 190, 223	
	NGC 7354	5	
	NGC 7662	11, 31, 47, 48, 51, 190, 223	
	IC 351	5, 14	
	IC 418	10, 48, 170, 223	
	IC 2003	5	
	IC 2149	5	
	IC 2165	223	
	IC 3568	5, 180	
	IC 4593	5, 31	
	IC 4634	5	
	IC 4732	5	
	IC 4776	76	
	IC 4846	5	
	IC 4997	9, 18, 168, 170	
	IC 5117	5	
	IC 5217	5, 223	
	Anon. 0 ^h 25 ^m	5	

Source class	Object	References	Comments
<i>Planetary nebulae, cont.</i>	Anon. 16 ^h 14 ^m	76	
	Anon. 18 ^h 15 ^m	5	
	Anon. 18 ^h 47 ^m	5	
	Anon. 19 ^h 36 ^m	5	
	Anon. 21 ^h 29 ^m	14	VV 267
	Anon. 21 ^h 31 ^m	5	
	Anon. 22 ^h 29 ^m	5	
	Anon. 23 ^h 24 ^m	5	
	J 320	5	
	J 900	5	
8. Gaseous nebulae	NGC 1976	12, 17, 48, 65, 67, 74, 75, 79, 111, 170, 202, 223	Orion Nebula
	NGC 2070	104	30 Dor
9. Symbiotic stars	Z And	41, 70, 135, 138, 184, 189, 190, 192, 193, 194, 199	3
	R Aqu	184	
	T CBr	42, 190	
	W Cep	2	
	VV Cep	193	M2epIa + B9
	BF Cyg	42, 68, 69, 70, 132, 138, 192, 193, 198	
	CI Cyg	130, 138, 184, 190, 192, 198	
	WY Gem	2, 184, 187	42474; M2epIab + B
	RW Hya	130, 138, 190	gMep
	AG Peg	55, 139, 190, 192, 195, 197	BD +11°4673, HD 207757; Bep+M
	AX Per	42, 80, 130, 184, 188, 189, 190, 192, 198, 200	gM3ep + 0
	RX Pup	188, 190	

³Sahade (Stellar Atmospheres, 1960) terms this the prototype of the symbiotic star group defined by a combination of low temperature absorption and high excitation emission features. Amplitude of light curve is small enough to exclude it from nova and nova-like classes.

Source class	Object	References	Comments
<i>Symbiotic stars, cont.</i>	FR Sct	34, 42	M3e
	W Ser	88	cG2e
	Boss 1985	177, 187	-14°1971; K5 + Be
	MWC 143	130	
	MWC 373	78	M H α 328-116
	MWC 603	208	M3ep + 0
10. Oe-Be-P Cyg stars	κ^1 Aps	172	
	π Aqr	172	B1ne
	ι Ara	172	B3ne
	κ CMa	103, 172	B2ne; HD 50013
	ω CMa	103, 172	B3ne; HD 56139
	Z CMa	190	Beq
	27 CMa	172	
	29 CMa	172	B3; HD 57192
	α Cam	221	09e
	II Cam	172	B3e; HD 32343
	γ Cas	23, 172, 201	B0ne
	δ Cen	172	B3ne
	μ Cen	172	B3ne
	λ Cep	172, 221	06ne
	19 Cep	221	09; HD 209975
	o CetB	172	gN6e
	υ Cyg	172	B3ne
	P Cyg	28, 30, 53, 171, 184, 186, 190	⁴
	68 Cyg	221	
	κ Dra	172	B5e
	4 Her	19	B9e
	ρ Leo	146	B1I
	17 Lep	171, 216	A0
	β Lyr	172	Bep
	β Mon	172	B3ne

⁴Prototype of a class. Spectroscopic details concerning differences among P Cyg, WR, Oe, and Be groups are given in Underhill (Stellar Atmospheres, 1960). Bleq in Kukarkin-Parenago catalog.

Source class	Object	References	Comments
<i>Oe-Be-P Cyg stars, cont.</i>	χ Oph	53, 172	B3e
	XX Oph	70, 140, 142	5
	δ Ori	221	B0
	ϵ Ori	221	B0ne
	ζ Ori	221	B03
	π^4 Ori	146	B2IIII
	25 Ori	172	B3ne; HD 35439
	AR Pav	158	MWC 600
	γ Peg	178	B2
	31 Peg	172	B3e; HD 212076
	AG Peg	139, 184	HD 207757, BD +11°4673
	ξ Per	221	
	ϕ Per	154, 172	B0ne; HD 10516
	ψ Per	172	B5ne
	β Psc	172	B5e
	r Pup	103, 172	05V; HD 68980
	RY Sct	126, 184, 196	B0ep
	9 Sge	184, 222	07; HD 188001
	15 Sgr	222	B0Ia
	ζ Tau	44, 83, 172	B3e
	HD 20336	172	B3e
	HD 42054	103	B5n
	HD 45677	127, 172	B2e
	HD 50138	172	B8e
	HD 51585	20	
	HD 57150	103	B3ne
	HD 58155	103	B5n
	HD 151804	222	08f
	HD 152408	222	07f
	HD 157857	222	07f
	HD 163181	172	B1e; V453 Sco
	HD 166734	222	08f
	HD 169582	222	05f

⁵This shell star, though closely related to the P Cyg group, exhibits properties that set it outside the group. Nevertheless, it is definitely of early spectral type. Bep in the Kukarkin-Parenago catalog.

Source class	Object	References	Comments
<i>Oe-Be-P Cyg stars, cont.</i>	HD 190073	184	A0ep
	HD 192639	222	08f
	HD 193514	222	07f
	HD 195592	222	09.5Ia
	HD 217050	44	B3ne
	HD 228779	222	09.5Ib
	BD +40°4227	222	06f
	BD +14°3887	173	Bep; MWC 314
	BD +6°1309	172	
	BD -6°1391	172	
	BD -22°1874	172	
	MWC 300	143	
11. Ae-Fe-Ge stars	SU Aur	97	
	S Dor	215	6
	U Gem	97	
	FU Ori	97	
	V380 Ori	98, 149	BD -6°1253
	λ Pup	176	cA2ep
	W Ser	25	cG2e
	S Sge	94	F6Ib-G5Ib
	υ Sgr	133	Aev
	HD 117043	19	dG6
	HD 117555	97	
	HD 217476	163	G0Ia
12. T-Tauri stars	RW Aur	90, 97, 106	G5Ve
	UY Aur	106	G5Ve
	S CrA	106	dGe
	RU Lup	106	G5
	R Mon	98, 106	A-Fpe
	T Tau	92, 97, 106	K1Ve

⁶Wesselink (*Monthly Notices of the Royal Astronomical Society* 116:3-9, 1956) classifies this as a P Cyg star, but Underhill (*Stellar Atmospheres*, 1960) places it in the F supergiant category. Spectral type in the Kukarkin-Parenago catalog is A0eq.

Source class	Object	References	Comments
<i>T-Tauri stars,</i> <i>cont.</i>	UZ Tau	106	G5e
	H-H 1	43, 93	
	H-H 2	93	
13. Ke-Me-N stars	β And	33	M0III
	R And	136	S6e
	W And	33	M7-8
	ST And	160	R3e
	UV Aur	162	Ne
	VY CMa	213	cM3e
	S Cam	160	Se
	X Cas	160	Ne
	S Cep	160	N8e
	W Cep	2	K0ep
	RW Cep	145	M0Ia
	$\circ$ Cet	26, 105, 108, 174, 203	M5e-M93 (Mira)
	R CrB	91	cFep
	V CrB	160	N2e
	χ Cyg	95, 137, 141	Se
	U Cyg	160	Npe
	RS Cyg	160	N0pe
	YY Gem	217	dMe
	α Her	99	M5II
	30 Her	33	gM6
	R Hya	134	gM7e
	RY Hya	160	Ne
	R Lep	160	N6e
	δ^2 Lyr	33	M4III
	R Lyr	33	gM5
	V Oph	160	N3e
	α Ori	33	M2Iab
	β Peg	33	
	ρ Per	33	M4II-III
	Y Per	160	c4e
	19 Psc	33	cb

<u>Source class</u>	<u>Object</u>	<u>References</u>	<u>Comments</u>
<i>Ke-Me-N stars,</i> <i>cont.</i>	α Ser	33	K2IIIp
	R Ser	102	M6e-M83
	V348 Sgr	96	
	BD +20°2465	217	
	Boss 5650	2	
	HD 42474	2	
	HD 88230	19	K7V

<u>Source class</u>	<u>Object</u>	<u>References*</u>	<u>Comments</u>
1. Quasars	4C 17.46	224	
	4C 19.31		
	4C 25.40		
	4C 37.43		
	4C 42.01		
	4C 49.22		
	5C 2.10		
	5C 2.56		
	0155-10		
	LB 8755		
	PHL 1027		
	PHL 1186		
	PHL 1194		
	PHL 1222		
	PHL 1226		
	PHL 3375		
	PHL 3632		
	RS 13		
13. Ke-Me-N stars	x Cyg	225	

*References are listed on page E-8.

Table D. INSTRUMENTATION

The type of instrument used to record the spectrum of the object is important in the resulting spectrum. It is well recognized that the emission spectrum of an object is more prominent at higher dispersion. This effect is most noticeable when the natural width of the emission line is small. In this case a low-dispersion spectrograph will have an instrumental resolution poorer than the emission line width. As a consequence, the emission line will be diluted by the instrumental profile, and if an absorption line is adjacent to the emission line, one may observe neither the emission nor the absorption line.

Although the above effect can seriously reduce the visibility of an emission spectrum superimposed on a stellar absorption or continuum spectrum, the relative intensities of the emission lines will not be affected, except by nearby absorption or emission features in the spectrum. It is nevertheless very important that emission-line stars be observed with the maximum dispersion possible considering the brightness of the object.

The range of instrumental parameters used in the observations listed in this catalog are discussed in this table.

Source class	References	Wavelengths*	Instrument; dispersion; exposure times**
1. Quasars			
	54	3200-7136 3407-5640	120-in. Lick, 36-, 84-in. Kitt Peak Natl. Obs.; 110-330 A/mm
	87	3239-6510	200-in. Hale; 190-400 A/mm
	114	3239-7106	84-in. Kitt Peak Natl. Obs.
	115	4298-6926	36-, 84-in. Kitt Peak Natl. Obs.; 110-330 A/mm
	116	4298-6926	36-, 84-in. Kitt Peak Natl. Obs.; 110-330 A/mm
	164	3986-7136	200-in. Hale; 400 A/mm
	167	3200-5400	84-in. Kitt Peak Natl. Obs.; 125 A/mm
2. Nuclear emission galaxies			
	150	3312-7330	82-in. McDonald; 200-400 A/mm prism
	165	3727-5007	60-in. Mt. Wilson
	166	3726-6732	60-, 100-in. Mt. Wilson; 36-200 A/mm prism, 400 A/mm grating
3. Novae			
		3150-8750	
	1	4233-5018	60-, 100-in. Lick; 90 A/mm prism
	16	5802-7726	120-cm Haute-Provence; 230 A/mm prism
	36	3800-5100	120-cm Haute-Provence; prism
	37	3150-6900	81-, 120-, 200-cm Haute-Provence; 9-76 A/mm prism
	38	3888-6563	120-cm Haute-Provence; 41-318 A/mm prism
	39	5169-6563	120-cm Haute-Provence; prism
	40	4000-6500	120-cm Haute-Provence; prism
	42	3663-6563	120-cm Haute-Provence; prism

63	6700-8700	122-cm Asiago; 70-630 A/mm; 1 ^m -2 ^h
64	3546-8690	122-cm Asiago; 12-135 A/mm
66	3690-8630	Burrell Tel. Warner-Swasey; 283-566 A/mm prism
71	4048-5152	Lyon
72	3900-5200	200-cm Haute-Provence; 18-300 A/mm prism
77	3500-8750	200-cm Haute-Provence; 9.6-39 A/mm; 15 ^m -40 ^m
84	3345-6563	82-in. McDonald; 100-200 A/mm
107	3700-8650	100-in. Mt. Wilson; 220 A/mm prism
109 [†]	3700-7000	60-in. Mt. Wilson; 40-120 A/mm
112	3727-5007	100-in. Mt. Wilson, 200-in. Hale; 10 A/mm grating
119	4222-6364	36-in. Lick, 77-120 A/mm prism; 60-, 100-in. Mt. Wilson, 38-70 A/mm; 37-in. Ann Arbor, 40 A/mm
131	5150-6550	100-in. Mt. Wilson; 33.6 A/mm grating
155	3758-6782	122-cm Asiago; 42-74 A/mm; 5 ^m -4 ^h ₄
159	3768-6572	154-cm Bosque Alegre; 42 A/mm grating
161	3864-5007	60-in. Mt. Wilson; 103 ^m
179	3312-5876	82-in. McDonald; 53-125 A/mm prism
185	3311-6584	82-in. McDonald; 40-100 A/mm prism
190	3340-5876	82-in. McDonald; 40 A/mm
191	3346-5574	82-in. McDonald; 40-100 A/mm
211	4102-5021	various
218	3950-5018	25-in. Newhall, 15-in. Huggins; 14-39 A/mm prism
219	3916-4267	Cambridge

*Range, in Angstroms; figures in italics indicate limits for each section.

**Not always listed in papers.

†Most complete papers.

Source class	References	Wavelengths	Instrument; dispersion; exposure times
4. Nova-like stars			
	81 ⁺	3028-9123	154-cm Bosque Alegre; 42 A/mm grating
	89	3028-6966	10-in. Lamont-Hussey; 370-450 A/mm
	204 ⁺	4860-6550	74-in. Radcliffe; 22-48 A/mm prism; 2 ^m -2 ^h
	205	3678-8863	74-in. Radcliffe; 22-48 A/mm prism; 1 ^m -4 ¹ / ₂ ^h
	206	3343-8446	74-in. Radcliffe; 22-48 A/mm prism; 1 ^m -4 ¹ / ₂ ^h
		6667-9123	74-in. Radcliffe; 31 A/mm grating
5. Wolf-Rayet N stars			
		3166-11000	
	4	3203-5876	Crossley Lick; slitless prism
	27	3342-7117	various
	29	3925-6678	Dominion Ap. Obs.; 30-238 A/mm prism
	147	9000-11000	69-in. Perkins; 512 A/mm prism
	157	3889-6680	82-in. McDonald; 55 A/mm
	175	3888-6561	Yerkes-McDonald
	209 ⁺	3166-6682	72-in. Dominion Ap. Obs.; 16-127 A/mm; 41 ^m -6.3 ^h
6. Wolf-Rayet C stars			
		3128-11000	
	4	3203-6235	Crossley Lick; slitless prism
	27	3342-7117	various
	73	3360-8663	various
	147	9000-11000	69-in. Perkins; 512 A/mm prism
	183	6500-8800	82-in. McDonald; 312-370 A/mm; 20 ^m -4 ^h
	209 ⁺	3128-6740	72-in. Dominion Ap. Obs.; 16-127 A/mm; 1 ^h -8 ^h
	210	3719-6494	Dominion Ap. Obs.; 46-90 A/mm prism

7. Planetary nebulae

		3047-9069	
5	3133-5008	36-in. Lick, and various	
6	3133-9069	100-in. Mt. Wilson, 200-in. Hale; 6.7-56 A/mm coude grating; 5 ^h -21 ^h	
7	3047-4740	100-in. Mt. Wilson; coude	
8+	3121-4959	60-, 100-in. Mt. Wilson, 120-in. Lick, 74-in. Mt. Stromlo; scans and plates; 2 ^h -15 ^h	
9+	3070-4940	100-in. Mt. Wilson, 36-in. Crossley; 0.3-8 ^h	
10+	3100-5000	120-in. Lick, 100-in. Mt. Wilson	
11+	3120-5047	100-in. Mt. Wilson, 20 A/mm; 200-in. Hale, 39 A/mm	
13	6300-9069	100-in. Mt. Wilson; 3 ^h -6 ^h	
14	3132-6563	120-in. Lick; Lallemand electronic camera 65 A/mm	
31	3346-5007	Crossley; 51 A/mm slitless prism; 5 ^m -2 ^h	
46	3203-7330	various	
47	3346-8046	200-in. Hale; 9-57 A/mm grating	
50	4227-5200	36-in. Lick; three-prism; 12 ^h -17 ^h	
51	3712-6730	36-in. Lick; 130-200 A/mm prism; 0.3-13 ^h	
71	4048-5152	Lyon	
76	3727-6730	74-in. Radcliffe; 49 A/mm	
129	4471-6300	various	
148	3727-6728	100-in. Mt. Wilson; prism	
168	4711-6400	36-in. Lick; 350-1000 A/mm prism; 4 ^h -8 ^h	
170	5412-7326	36-in. Lick; 150-350 A/mm prism; 30 ^m -10 ^h	
182	6548-8665	82-in. McDonald; 310-380 A/mm grating	
190	3340-5895	various; 40 A/mm	
220	3346-5008	100-in. Mt. Wilson; 10.4 A/mm coude grating	
223	3704-6755	36-in. Lick; 90-200 A/mm prism; 3 ^h -15 ^h	

Source class	References	Wavelengths	Instrument; dispersion; exposure times
8. Gaseous nebulae			
		3187-11000	
	12	3726-10938	Curtis Schmidt; photoel. spectrophotom; 30 A/min scan
	17	5755-9016	120-cm Haute-Provence; 750 A/mm prism; $14\frac{1}{2}^h$
	65	3450-5000	200-cm Haute-Provence; 39 A/mm; 3^h
	74	9069-9531	13-cm Crimean Ap. Obs.; image converter, 95 A/mm; 3^h
	75	9000-11000	64-cm Crimean Ap. Obs.; 250 A/mm prism; 40^m
	79	3587-7330	100-in. Mt. Wilson; 86-176 A/mm grating; 1^h-5^h
	104	3727-5007	74-in. Mt. Stromlo; 298 A/mm prism; 1^h
	111+	3187-5016	100-in. Mt. Wilson, 120-in. Lick; 16-20 A/mm coudé; 4^h-20^h
	170	5412-7326	36-in. Lick; 150-350 A/mm prism; 30^m-10^h
	202	3704-6583	120-cm Haute-Provence; 36-330 A/mm prism
	223	3704-6755	36-in. Lick; 53-200 A/mm prism; $3\frac{1}{2}^h-15^h$
9. Symbiotic stars			
		3100-8703	
	2	4179-4463	various
	42	3663-6563	120-cm Haute-Provence; prism
	55	3662-4924	82-in. McDonald; $17\frac{1}{2}$ A/mm
	68	6676-8703	200-cm Haute-Provence
	70	3300-5100	100-in. Mt. Wilson; 10-20 A/mm
	78	3670-8237	Burrell Tel. Warner-Swasey; 1100 A/mm obj. prism
	88	4000-4584	Mt. Wilson; 10 A/mm
	130	3869-5007	10-in. Mt. Wilson; 70-350 A/mm obj. prism
	132	3722-6678	100-in. Mt. Wilson; 40-78 A/mm prism
	135	3312-6678	100-in. Mt. Wilson; 10-20 A/mm

138	3132-6678	100-in. Mt. Wilson; 30-80 A/mm
139	3255-6678	100-in. Mt. Wilson; 10 A/mm grating, 35 A/mm prism
177	3115-3935	82-in. McDonald; 10 A/mm
184	3327-6678	82-in. McDonald; prism
187	3100-3450	82-in. McDonald; 57 A/mm
188	3426-6678	82-in. McDonald
189	3420-6563	82-in. McDonald
190	3426-6678	82-in. McDonald
192	3121-6563	82-in. McDonald
193	3103-3889	82-in. McDonald
194+	3700-6563	120-cm Haute-Provence; 41-318 A/mm prism; 40^m-5^h
195+	3691-6563	120-, 80-cm Haute-Provence; 24-320 A/mm prism; $10^m-11\frac{1}{2}^h$
197+	3835-6563	120-cm Haute-Provence
208+	3662-6678	60-in. Mt. Wilson; 18-38 A/mm

10. Oe-Be-P Cyg stars

19	3011-8800	193-cm Haute-Provence; 39 A/mm; 1^h-5^h
20	6500-8750	50-cm Crimean Ap. Obs.; 200 A/mm prism
23	4176-6563	University of Michigan; 40 A/mm prism
28	3660-4924	various
	4254-6563	
30	3266-6678	Dominion Ap. Obs.; 4-391 A/mm prism
44	3664-4861	Crimean Ap. Obs.; 23.4 A/mm
53	6563-8800	82-in. McDonald; 35 A/mm grating
83	3300-5100	200-cm Haute-Provence; 10 A/mm grating
103	3700-4861	60-in. Bosque Alegre; 42 A/mm
126	4340-4770	100-in. Mt. Wilson; prism

Source class	References	Wavelengths	Instrument; dispersion; exposure times
<i>Oe-Be-P Cyg stars, cont.</i>	127	4068-6563	100-in. Mt. Wilson; prism
	129	4471-6300	various
	139	3255-6678	100-in. Mt. Wilson; 10 A/mm grating, 35 A/mm prism
	140	3661-6563	100-in. Mt. Wilson; 15 A/mm
	142	3656-5019	100-in. Mt. Wilson, 200-in. Hale; 10-15 A/mm
	143	4068-4861	200-in. Hale; 9 A/mm; 6 ^h
	146	3945-4922	100-in. Mt. Wilson; 2.8 A/min scan
	154	3734-4921	120-cm Asiago; 40 A/mm
	158	3670-6563	60-in. Bosque Alegre; 42 A/mm
	171	3265-4026	82-in. McDonald
	172	3970-4861	various
	173	3370-6600	82-in. McDonald; 40-100 A/mm; 1 ^h .7-7 ^h .3
	178	4352-5156	82-in. McDonald and others
	184	3327-6678	82-in. McDonald; prism
	186	3011-3291	82-in. McDonald; 40-100 A/mm prism
	190	3340-5876	82-in. McDonald; 40 A/mm
	196	4271-6563	200-cm Haute-Provence
	201	3964-4692	Lyon; 47-137 A/mm obj. prism
	216	3685-6563	Mt. Wilson
	221	4600-5730	various
	222	4485-4503	120-in. Lick; 16 A/mm grating

11. Ae-Fe-Ge stars

	3250-8750		
19	6500-8750	193-cm Haute-Provence; 39 A/mm; 1^h-5^h	
25	3682-5018	82-in. McDonald; 40 A/mm prism; $1\frac{1}{2}^h-3^h$	
97	3250-6600	various; 10-430 A/mm	
98+	3905-6563	various	
133	8498-8662	100-in. Mt. Wilson; 20 A/mm	
163	3700-5000	200-in. Hale; 9.5 A/mm grating	
	5000-6800	200-in. Hale; 13.5 A/mm grating	
176	3302-6563	McDonald; various	
215+	3837-6560	74-in. Radcliffe; 29-49 A/mm prism	

12. T-Tauri stars

	3250-8662		
43	3700-6800	Crossley Lick; 430 A/mm	
90	4132-6800	various, 36-in. Crossley Lick; 75 A/mm	
92	3723-4340	82-in. McDonald; 330 A/mm prism	
93+	3727-6727	Crossley Lick; 430 A/mm; up to $2\frac{1}{2}^h$	
97	3250-6600	various; 10-430 A/mm	
106+	3727-8662	60-, 100-in. Mt. Wilson; 35-220 A/mm prism	

13. Ke-Me-N stars

	3162-8750		
2	4179-4463	various	
19	6500-8750	193-cm Haute-Provence; 39 A/mm; 1^h-5^h	
26	4500-4900	various	
33	3277-3303	120-in. Lick	
91	3236-5896	82-in. McDonald; 75-176 A/mm prism	
95	4066-4405	various	

Source class	References	Wavelengths	Instrument; dispersion; exposure times
<i>Ke-Me-N stars, cont.</i>			
	96	3727-6579	36-in. and Crossley Lick; 130-430 A/mm prism
	99	3162-3295	82-in. McDonald; 16 A/mm prism
	102	7000-8500	69-in. Perkins; 27 A/mm
	105	3750-5018	60-, 100-in. Mt. Wilson; 35 A/mm prism
	108	3735-4511	60-, 100-in. Mt. Wilson, 200-in. Hale; 2.3-75 A/mm
	136	3227-5018	100-in. Mt. Wilson; 10-20 A/mm
	137	3227-4924	100-in. Mt. Wilson; 10-20 A/mm
	141	3227-5018	200-in. Hale; 9-38 A/mm; 3 ^h -23 ^h
	145	3653-3999	200-in. Hale; 9-13.5 A/mm
	160	4340-6563	various
	162	3969-4308	various
	174	3277-4308	82-in. McDonald, various
	217	3200-3500	82-in. McDonald; 65 A/mm
		3000-15000	
14. Airglow	24	4277-6364	120 A/mm prism
	35	5577-6600	78 A/mm grating
	58	3143-3583	23 A/mm grating
	59	3698-4916	21 A/mm grating
	60+	3000-15000	various
	62	7200-9000	70 A/mm grating; 20 ^h
	101	3700-6900	39-80 A/mm; 20 ^h -70 ^h
	120	7200-8900	247 A/mm; 12 ^h
	121	5500-6600	385 A/mm; 55 ^h

15. City lights	122	5500-9000	247 A/mm; 12 ^h
	124	3000-15000	various
16. Aurora	21	3500-6800	240 A/mm grating; 1 ^h -6 ^h
		3000-15000	
	3	3300-8700	various
	60+	3000-15000	various
17. Lightning	124	3000-15000	various
	152	3373-8800	85 A/mm grating
	212	3150-9800	22-66 A/mm grating

The information below refers to the new material on pages B-133 and C-12.

Source class	References	Wavelengths	Instrument; dispersion; exposure times
1. Quasars	224	1238-1817	120-in. Lick, prime focus, image tube, 440 A/mm; 36-in. Crossley, prime focus, 380 A/mm, 430 A/mm
13. Ke-Me-N stars	225	5341-5862	120-in. Lick, coudé, 16 A/mm

Table E. BIBLIOGRAPHY

In the bibliography below, the classes of objects covered are indicated by the numbers in parentheses, as follows:

- (1) Quasars
- (2) Nuclear emission galaxies
- (3) Novae
- (4) Nova-like stars
- (5) Wolf-Rayet N stars
- (6) Wolf-Rayet C stars
- (7) Planetary nebulae
- (8) Gaseous nebulae
- (9) Symbiotic stars
- (10) Oe-Be-P Cygni stars
- (11) Ae-Fe-Ge stars
- (12) T-Tauri stars
- (13) Ke-Me-N stars
- (14) Airglow
- (15) City lights
- (16) Aurora
- (17) Lightning

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